CENSUS OF INDIA, 1911.

VOLUME V.

BENGAL, BIHAR AND ORISSA AND SIKKIM.

PART I.

REPORT

BY

L. S. S. O'MALLEY,

OF THE INDIAN CIVIL SERVICE,
FELLOW OF THE ROYAL STATISTICAL SOCIETY,
SUPERINTENDENT OF CENSUS OPERATIONS, BENGAL.

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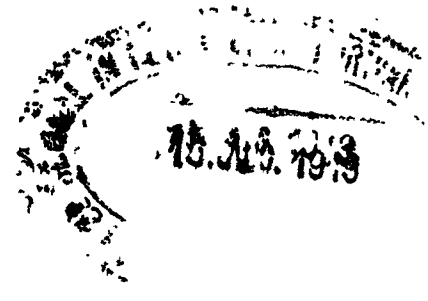
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PREFACE.

THE report to which this is a preface deals with the results of the census of 1911 in the Presidency of Bengal, the Province of Bihar and Orissa and the State of Sikkim, which have an aggregate area of nearly 200,000 square miles and a population of 85 millions, or over one-fourth of the total population of India.

The late appearance of the report is due to causes beyond my control. Since the census of 1901 Bengal has undergone two partitions, and Sikkim has been detached from it—changes which have involved the preparation of fresh statistics for the census of 1911 and also for each preceding census. It is perhaps almost superfluous to explain that if the census figures of any given area are to be of value for comparative purposes, those of previous censuses must be accessible. Accordingly, before the census of 1911 took place, figures were compiled for Bengal as constituted after the partition of 1905, for the only available figures related to the province as it stood in 1901. After the census of 1911 was concluded, tables showing its results were prepared for the two provinces of Bengal and Eastern Bengal and Assam as then existing. While this work was in progress, the repartition was announced, and the figures both of the census of 1911 and of each census since 1872 had to be worked out afresh for the Presidency of Bengal and the new Province of Bihar and Orissa. All this involved additional labour and delayed the preparation of the tables and report. On this account also I have not been able to take up all the subjects prescribed for the report, and have had to curtail the discussion of some of those with which I could deal.

ARRANGEMENTS FOR THE CENSUS.

The arrangements for the census followed the lines laid down in 1901, and a brief summary of them is all that is required. The first step was to obtain an accurate and up-to-date record of all inhabited areas, *i.e.*, to prepare a register in which every occupied village or collection of houses was entered. This is not altogether an easy task in some areas, where the villages are small clusters of houses scattered through the jungle, which are commonly deserted by their primitive inhabitants when anything untoward happens. In such cases it is believed that the village has incurred the anger of some evil spirit, and the people abandon the site and build their houses elsewhere. The village register having been compiled, each district was parcelled out into census divisions. The smallest unit was the block, which consisted of 40 to 50 houses, for which one enumerator was responsible. The blocks were grouped together by circles, each of which was under a supervisor: ordinarily about 10 to 15 circles, *i.e.*, 400 to 600 houses, were assigned to each supervisor. The circles again were grouped together by charges, which, as a rule, corresponded to police-stations. The Charge Superintendents, who were responsible for the operations throughout each charge, were themselves subordinate to the Subdivisional Officers and to the District Census Officers, who were appointed for each district.

CENSUS AGENCY.

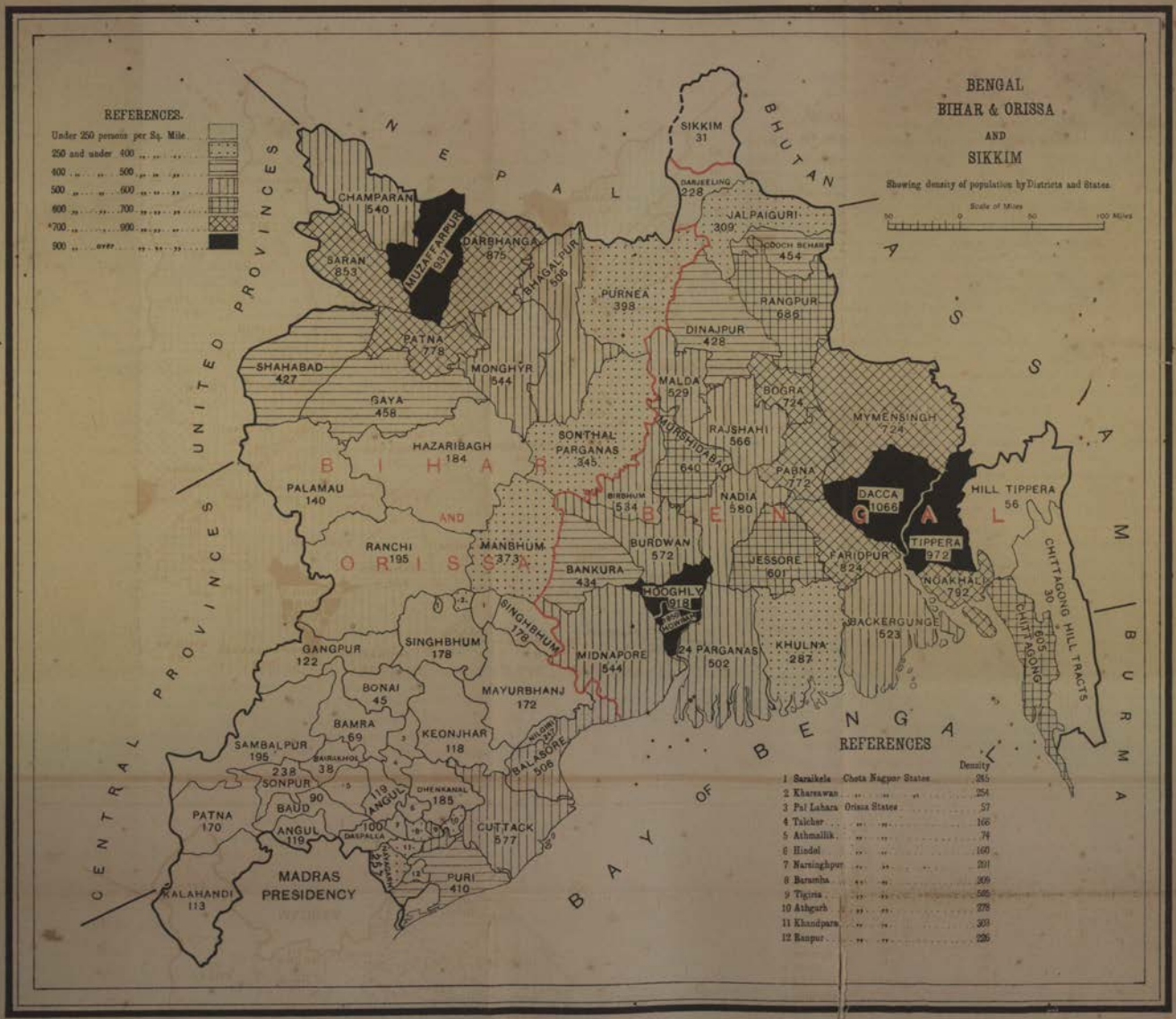
The actual enumeration was conducted by an improvised agency of supervisors and enumerators, who were drawn from the ordinary population. In Bengal as constituted at the time of the census 23,000 supervisors and 326,000 enumerators were employed. In many localities it was no easy matter to obtain a sufficient supply of men who could read and write, and a long training was necessary before they could understand the duties required of them. Even where suitable men were available, their natural reluctance to serve without pay had to be overcome, and the lesson brought home to them that the census was conducted by, through and for the people, and that, perhaps for the first time in their lives, they would be actively discharging a public duty.

HOUSE-NUMBERING.

When they had been appointed, their first duty was to number every house, a house being defined as the residence of a commensal family. The application of this definition gave rise to numerous knotty questions, which illustrate very forcibly the differences in local conditions in a large and heterogeneous province. How, for instance, were the village dormitories, in which aboriginal boys and girls sleep, to be treated? Was a separate number to be given to each house in the Paharia villages, on the summits of the Rajmahal Hills, where each married couple has a separate hut, but the whole family have their meals together in another hut. Again, how were the numbers to be affixed? The Paharia huts, for instance, are mere shanties made of branches and poles fixed in the ground, and no number could be painted on them. So, the numbers had to be put on slabs of wood, which were either stuck on the walls or hung from the eaves. The Paharias did not leave them there, but carefully wrapped them up in rags and kept them in the bamboo baskets which form their strong boxes.

THE ACTUAL CENSUS.

In February 1911, after they had been methodically trained in their duties, the enumerators conducted the preliminary enumeration, *i.e.*, they entered all the necessary particulars in the schedules for every person ordinary resident in each house. The record thus prepared was systematically checked by superior officers in order to ensure its accuracy and also its absolute completeness. The final census was held on the night of the 10th March 1911, the enumerators going round to each house and revising the preliminary record so as to make it correspond to the state of affairs as then existing. In other words, they struck out the entries for all persons who had died or gone away since the preliminary enumeration, and added fresh entries for new-comers and for infants who had been born in the interval. Too high praise cannot be given to them for the thorough manner in which they fulfilled their duties, and for their assiduity in making sure that there were no omissions. As an instance in point, I may quote the experience of a small aboriginal tribe, who feared that some mysterious evil might befall them if they were counted, and fled from forest to forest, but failed to escape the enumerators. Mistakes were of course made, some of which show how simple-minded a few of the enumerators are. One man solemnly entered an idol in his schedule, and gave particulars of age, viz., 200 years, and of language, viz., Hindi. Another recorded the language of a deaf-mute as


BENGAL
BIHAR & ORISSA
AND
SIKKIM
Showing density of population by Districts and States.
Scale of Miles
50 0 50 100 Miles
REFERENCES.
Under 250 persons per Sq. Mile
250 and under 400
400 " " 500
500 " " 600
600 " " 700
700 " " 900
900 " over
NEPAL
BHUTAN
ASSAM
BURMA
BAY OF BENGAL
CENTRAL PROVINCES
UNITED PROVINCES
MADRAS PRESIDENCY
SIKKIM 31
DARJEELING 228
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TIPPERA 972
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SAMBALPUR 195
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6 Hindol " " 160
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9 Tigiria " " 505
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atpat (an onomatopœic word describing the sounds he emitted). A globe-trotter described his occupation or means of livelihood as that of a tourist.

In many cases the enumerators had to face great difficulties, not the least of which was the danger of work in plague-infected areas, where they felt that they were fulfilling their duties at the risk of their lives. Many died of plague while engaged in the preliminary enumeration, and others on the day of the census itself. The local officers had the greatest difficulty in finding successors, often at a moment's notice; and as those who were appointed could scarcely be expected to take over the dead men's papers, the work in some cases had to be done over again. Elsewhere enumerators were exposed to attacks from wild animals in the jungles. One man who had left his home on account of the ravages of a man-eating tiger, but volunteered to show the enumerator the way to his old village, was carried off by the tiger just as he reached its outskirts.

PROVISIONAL TOTALS.

The day after the census the preparation of the provisional totals was taken in hand, *i.e.*, the entries on the schedules were totalled up to ascertain the number of males and females and the aggregate population. This may seem a simple matter, but in Bengal (as constituted at the time of the census) it involved the collection of over a third of a million men at various centres and the compilation of figures for each census division. It is by no means easy to do this quickly where distances are great and the means of communication few. Nor is it easy to ensure accuracy when, as among the Santals, the enumerators are ignorant of the elementary principles of calculation, many not being able to post figures above 100. Lastly, there is the danger of loss or destruction of papers. In one district the provisional totals were delayed by one enumerator not having compiled his totals. The Subdivisional Officer went to the spot and found that the enumerator's house had been burnt on the night of the census, and with it the census papers and two of his wives. His block had of course to be censused again.

In spite of these and other difficulties, the provisional totals for the whole of Bengal were despatched within a week after the census. Their speedy collection and despatch are due to the excellent organization effected by the District Officers, the employment of all available means of transport (one of the most useful being the bicycle), and the indomitable energy of the census staff. In the Patna State the figures had to be despatched by couriers to the nearest telegraph station, a distance of 76 miles, and in Kalahandi they had to be sent 114 miles: in the latter State the authorities had, in the words of the Political Agent, "to deal with 1,200 square miles of the most awful country inhabited by the wildest of Khonds." The totals of these two States were wired on the 13th and 15th March, respectively. In the Darjeeling district the figures for the people living on a mountain 12,000 feet high and 50 miles distant from head-quarters reached Darjeeling within 36 hours of the census. In Singhbhum again many of the census staff had to march at night through forests infested by man-eating tigers carrying torches to scare them away, and completed in 10 hours marches which, as a rule, take two days. Bad weather had to be faced

in some places. The District Census Officer of one Bengal district reports that the men had to travel through a violent hail-storm and came in late at night, "all with hurts and bruises, but with the circle summaries and enumeration books dry. I had to give up the idea of sending any special messenger to head-quarters on such a fearful night, and early next morning I acted as special messenger myself and plied my bike on a muddy road of 32 miles and carried in the provisional totals safe." In this and other cases the reports are reminiscent of "How they brought the good news from Ghent to Aix."

COMPILATION AND TABULATION.

The final results were obtained by compilation and tabulation, of which there are three stages, viz., slip-copying, sorting and compilation.

Slip-copying is the process of copying the entries in the schedules on to slips. A separate slip is used for each of the persons enumerated, so that 85 million slips had to be used for the population dealt with in this report; but much labour is saved by the use of slips of different colours to indicate religion, of symbols to indicate sex and of abbreviations for the actual entries in the schedules. As in 1901, the greater part of the slip-copying was carried out in district offices. The slips when ready were sent to the central offices, where the sorting and subsequent operations required for the preparation of the final tables were carried out. In the central offices arrangements were made for copying the slips of the districts in which they were situated, together with some or all of the slips for other districts where there were difficulties in the way of getting the work done locally. For instance, the slips of Angul and the Orissa Feudatory States were copied at the Cuttack central office besides those for Cuttack: at this office no less than 6 million slips were copied. At one time the copying staff in the Bengal offices (excluding those in Eastern Bengal) numbered over 3,000, the total outturn in the week being nearly 10 million slips, and the daily average per man 590. The average is remarkably high, considering that the work had to be done in the height of the hot weather, and in some cases, where buildings could not be secured, in tents or verandahs.

Sorting is the process of arranging the slips under the heads required for the various final tables, counting the slips as thus arranged and entering the number on forms provided for the purpose, which are called sorters' tickets. This operation was performed in the central offices, of which there were seven for the area dealt with by me. They were situated at the following places and sorted for the population noted against each, viz., Berhampore (9 millions), Bhagalpur ($7\frac{3}{4}$ millions), Cuttack (9 millions), Gaya ($6\frac{1}{2}$ millions), Hazaribagh ($5\frac{3}{4}$ millions), Hooghly (10 millions) and Patna ($9\frac{1}{4}$ millions). While sorting was in progress, inquiry was made into doubtful entries, which often raise questions which it is difficult to solve. One tribe, for instance, was entered as Jhar Manjhi, *i.e.*, men of the woods, or Makarkhia, *i.e.*, monkey-eaters. On inquiry it was ascertained that they went by no other name, and that they did not know their original habitat and could only give an account of their recent wanderings. Specimens of their language were then obtained, and it was ascertained that they were Birhors, which also means men of the woods.

Sorting was followed by compilation, or the process of combining the figures in the sorters' tickets, so as to obtain the totals for the district. The compilation registers, in which the figures were entered, were despatched as soon as they were ready to my office, where a detailed examination of the statistics was carried out, doubtful entries checked and discrepancies inquired into. This is an operation of the greatest importance, as errors in compilation affect not merely units, but hundreds and thousands. After this the tables were prepared and the report was written. The statistics are numerous enough, but some desired to utilize the census record still further. One Bengali gentleman wanted copies of all the entries—1½ million in number—that were made in the schedules for members of his caste. Another Bengali gentleman calmly asked for the names and addresses of all literate persons in the province, in order that advertising circulars might be sent to them. He naively pointed out that this would swell the postal revenue besides developing trade.

COST OF THE CENSUS.

The accounts of census expenditure were maintained separately for the provinces of Bengal and Eastern Bengal and Assam during the two years 1910-12, and jointly for Bengal and Bihar and Orissa in 1912-13. Divided proportionately according to the population of the two provinces as now constituted, the aggregate cost is Rs. 2,67,322-10-8 for Bengal and Rs. 2,05,550-10-11 for Bihar and Orissa, representing an incidence of Rs. 5-12-4 and Rs. 5-5-7 respectively per head of the population. The average cost is higher than in 1901 owing mainly to the fact that wages have risen in the interval, so that the staff engaged in compilation and tabulation had to be paid more. Additional expenditure, moreover, had to be incurred in consequence of the repartition of Bengal, which necessitated the compilation of revised statistics for the two provinces.

ACKNOWLEDGMENTS.

I cannot acknowledge too fully the services of the large body of private individuals and Government officers who conducted the census and brought it to a successful conclusion. The census laid a heavy burden on the District Officers and their subordinate stat., which they loyally bore without increase of establishment. Their self-sacrifice materially helped to keep down the cost of the operations. My grateful acknowledgments are further due to a large number of official and non-official gentlemen for interesting reports on ethnological and sociological questions, which have been freely drawn upon in this report. I also desire to mention the good work done by the Presidency Jail Press, under the supervison of Mr. J. Gray, in printing and despatching the forms which were used at various stages of the operations. Their aggregate number exceeded 70 millions, and they had to the printed in six different characters, viz., English, Bengali, Kaithi, Devanagari, Oriya and Nepali Hindi: some Tebetan forms were also printed by the Secretariat Press at Darjeeling.

My special thanks are due to the following Deputy Magistrates and Deputy Collectors for the services which they rendered as Deputy Superintendents of Census in charge of the central census offices mentioned against their names:—Babu Brajendra Nath Ray (Hooghly), Babu Nilmani Dey (Bhagalpur), Babu Manmatha Nath Sen (Cuttack), Maulvi Abdul Kadir Khan (Patna), Babu Anadi Ranjan Bose (Gaya) and

Babu Krishna Gopal Ghosh (Berhampore). Much useful work was also done by my Head Clerk, Babu Harendra Krishna Mitra, whose previous experience was a valuable asset. Lastly, I am greatly indebted to Babu Naba Gauranga Basak, Deputy Magistrate and Deputy Collector, who, as my Personal Assistant, was closely associated with me in the preparation of the tables and of the report. Two of the chapters and part of a third were written in collaboration with him, and all the diagrams, maps and subsidiary tables were prepared under his supervision.

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REPORT

ON THE CENSUS OF

BENGAL, BIHAR AND ORISSA,

AND

SIKKIM, 1911.

CHAPTER I.

DISTRIBUTION OF THE POPULATION.

ADMINISTRATIVE CHANGES.

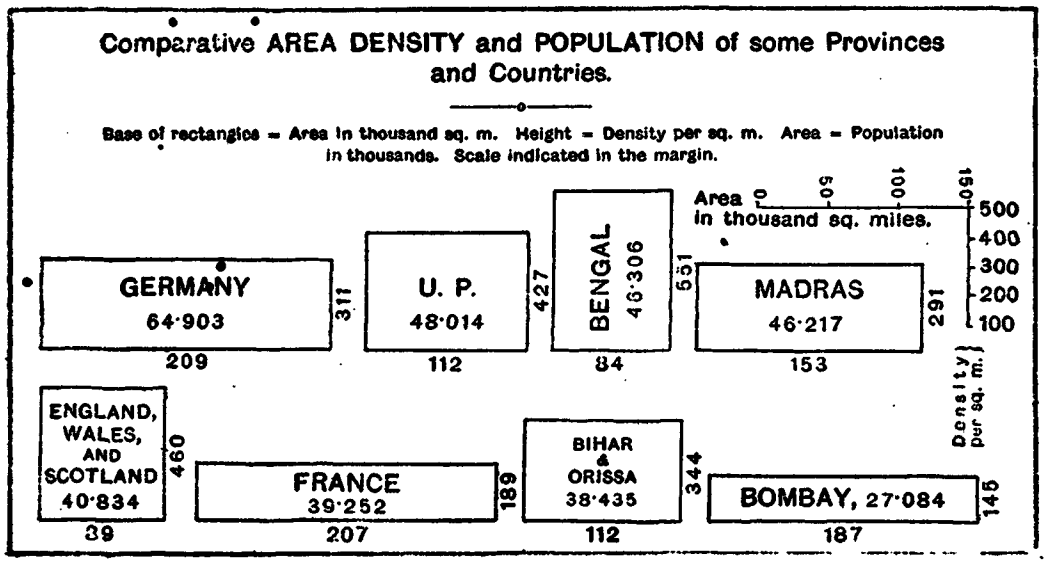
At the census of 1901 the Presidency of Bengal and the Province of Bihar and Orissa (except Sambalpur and five Feudatory States) were included in the Lieutenant-Governorship of Bengal, a Province extending over 196,408 square miles and containing a population of 78,493,410 persons. The government of such a large and populous territory had long been a task of increasing difficulty owing to the growth of population—in 30 years it had increased by over 26 millions—the commercial, industrial and educational development of the country, and the increase in the number and complexity of its administrative problems. It had been realized for some years that the Province was too large for a single administration, and eventually in 1905 a partition, accompanied by transfers of territory between it and the adjoining Provinces, was carried into effect. A new Province, called Eastern Bengal and Assam, was constituted, which included Assam and a considerable portion of the old Province of Bengal, viz., the Divisions of Dacca, Chittagong and Rajshahi (except Darjeeling), the district of Malda and the State of Hill Tippera. The five Feudatory States of Jashpur, Surguja, Udaipur, Korea and Changbhakar were at the same time transferred to the Central Provinces, while the district of Sambalpur (with the exception of two zamindaris) and the Feudatory States of Patna, Sonpur, Kalahandi, Bamra and Rairakhol were transferred from the Central Provinces to Bengal. The Province of Bengal, as constituted after these changes, extended over 148,592 square miles, and contained a population, according to the census of 1911, of 57,206,430 persons. The results anticipated from the partition not having been altogether realized, another scheme of reconstruction, accompanied by organic changes in the system of government, was determined upon. By this second partition which took effect on 1st April 1912, Bihar, Chota Nagpur and Orissa (*i.e.*, the whole Province of Bengal as constituted after 1905 with the exception of the Burdwan and Presidency Divisions, the district of Darjeeling and the State of Cooch Behar) were formed into a separate Province under a Lieutenant-Governor in Council, Assam was restored to its former position as a Chief Commissionership, and the remainder of the Province of Eastern Bengal and Assam, the Presidency and Burdwan Divisions, the district of Darjeeling and the State of Cooch Behar were created a Presidency under a Governor in Council.

Sikkim was under the Government of Bengal until 1906, when the control of its affairs was taken over by the Government of India. The figures for this State are therefore given separately and not included in those for Bengal, as in 1901.

AREA AND POPULATION.

BENGAL.

2. The Presidency of Bengal contains a population of 46,305,642 persons, and extends over 84,092 square miles, of which 5,393 square miles are in the States of Cooch Behar and Hill Tippera, and the remainder constitute the Burdwan, Presidency, Rajshahi, Dacca and Chittagong Commissionerships or administrative Divisions. Though somewhat smaller than Great Britain, it contains nearly a million more inhabitants than the whole of the British Isles. Compared with other Provinces in India, it is a little larger than the Central Provinces excluding Berar, and it has the greatest population next to the United Provinces (48,014,080), being closely followed by Madras with 46,217,245 inhabitants. Throughout almost its whole extent it is a low-lying alluvial plain, the southern portion of which is formed by the united deltas of the Ganges and Brahmaputra, while the northern portion consists of the valleys of these great rivers and their tributaries. Whether physical conditions, the character of the people and their language are considered, the Presidency is, with a few exceptions, remarkably homogeneous, but for practical purposes it may be treated as consisting of four natural Divisions, viz., Western, Central, East and North Bengal, of which a brief account will be found later in this chapter.



BIHAR AND ORISSA.

3. The Province of Bihar and Orissa has an area of 111,829 square miles and a population of 38,435,293 persons. Its area is slightly smaller than that of Austria (115,903 square miles), and a little greater than that of Italy (110,550 square miles), while its population is very little less than that of France (39,252,245). The largest Province in India next to Burma, Madras and Bombay, its population is only exceeded by that of Bengal, Madras and the United Provinces. It consists of the Divisions or Commissionerships of Patna, Tirhut, Bhagalpur, Orissa and Chota Nagpur, and of the Orissa and Chota Nagpur Feudatory States. The five Divisions, which have an area of 83,181 square miles and a population of 34,490,084, are the same as in 1901, with the following exceptions. The Orissa Division had the district of Sambalpur added to it in 1905, when the first partition of Bengal was effected. The district of Malda was at the same time transferred from the Bhagalpur Division to the Province of Eastern Bengal and Assam, whence it has been retransferred to Bengal. In 1908 the Patna Division was reconstituted, the North-Gangetic districts of Saran, Champaran, Muzaffarpur and Darbhanga being formed into a new Division known as Tirhut, while the districts of Patna, Gaya and Shahabad on the south of the Ganges were formed into another Commissionership, which retained the name of the Patna Division. Owing to the transfers which took place in 1905, the Orissa Feudatory States, which in 1901 contained 17 States, now contain 24 States, the five States of Patna, Kalahandi, Sonpur, Bamra and Rairakhol being added from the Central Provinces, and two other States, Gangpur and Bonai,

from the Chota Nagpur States. Owing to this loss and to the transfer to the Central Provinces of Jashpur, Surguja, Udaipur, Korea and Changbhakar, the Chota Nagpur States now consist only of the two small States of Kharsawan and Saraikela. The effect of these changes is shown in the marginal table.

	Area in square miles.		Population.	
	1901.	1911.	1901.	1911.
Orissa Feudatory States ...	14,387	28,046	1,947,802	3,796,563
Chota Nagpur States ...	16,014	602	1,001,429	148,646

Unlike Bengal, the Province of Bihar and Orissa is wanting in homogeneity. It consists of three sub-provinces, Bihar, Chota Nagpur and Orissa, which have different languages and land systems, while their physical configuration and the character of their peoples are entirely distinct.

BIHAR.

4. Bihar, which for administrative purposes is divided between the Patna, Tirhut and Bhagalpur Divisions, extends over 42,361 square miles, and has a population of 23,752,968 persons, or nearly 10 millions more than the Central Provinces and Berar. Physically it consists of the eastern portion of the Gangetic valley, which is bounded on the north by the lower spurs of the Himalayas and on the south by the Chota Nagpur plateau. It is an alluvial plain watered and drained by the Ganges and its tributaries, such as the Gandak, Son, Gogri and Kosi, which sometimes sweep down in disastrous floods. The climate is drier than in Bengal, and the rainfall is not only lighter, but more capricious, its vicissitudes exposing the country, especially to the north of the Ganges, to periods of scarcity, which occasionally culminate in famine. Rice is the main harvest, but heavy crops of maize, wheat and barley are also raised. The people are sturdy cultivators, clinging to their lands with grim tenacity; their industries and manufactures are of little economic importance. A little over 30 years ago they were described by the Lieutenant-Governor of Bengal as "poor helpless discontented men, bound down to a state of extreme depression and misery, tenants of the richest Province in Bengal, yet the poorest and most wretched class we find in the country." The monopoly of agriculture, the excessive dependence of the cultivators on the winter rice crop, the precariousness of that crop, rack-renting and the oppression of landlords, made it difficult for the people to accumulate reserves on which they could fall back in a year of crop failure. Since then, the security of tenure, and the protection against arbitrary enhancement of rents, afforded by the Bengal Tenancy Act and the preparation of a record-of-rights, together with the extension of railways, have worked what may almost be described as an agrarian and economic revolution. Bihar has now passed from a condition of recurring famines to one in which, though the pinch of high prices is felt, widespread famine due to the actual absence of food is only a grim memory. In times of scarcity both cultivators and labourers display staying powers which were previously unknown, while the pressure of high prices is relieved by the annual migration of landless labourers to centres of industry, and by the remittances made by them to their families.

CHOTA NAGPUR.

5. Chota Nagpur, which includes the Division of that name and the petty States of Kharsawan and Saraikela, extends over 27,679 square miles, and has 5,754,008 inhabitants, or nearly half as many as Burma. It is an upland plateau which forms the north-eastern portion of the table-land of Central India. The surface is undulating and hilly, and a large part is still covered by jungle, in which the *sal* tree (Shorea robusta) predominates. Cultivation is mainly confined to the valleys and the depressions between the ridges, which are enriched by the detritus washed down from above: laborious terracing is necessary to make rice cultivation possible on the slopes. The rainfall is about the same as in Bihar, but owing to the broken undulating surface the rain runs off rapidly, and artificial irrigation is necessary to bring the rice crop to maturity. Failures of the harvest occur periodically, but scarcity does not press severely on the people, as they are mostly hardy aboriginals, who, even in times of prosperity, have recourse to edible jungle products, such as the fruit of the *mahua* tree, and can manage to subsist under conditions which would result in famine among the cultivators of the plains. For centuries this hilly tract remained almost a *terra incognita*, outside the sphere of administration of the Mughal Government, which saw little hope of revenue from its barren

forest-clad hills. It was part of the *Jharkhand* or jungle land, a name given to the whole country stretching from Birbhum and Manbhum to Central India, and from the fort of Rohtasgarh in Shahabad to the borders of Orissa. The Mughals exercised only a nominal suzerainty over the native chiefs and, except for a few punitive expeditions, rarely penetrated its recesses, remaining content with a tribute of a few diamonds from the Chief of Kokrah (Ranchi).* It is still the home of non-Aryan tribes, who were never completely subjugated till the advent of the British, and, as stated in the last Census Report, "have preserved an individuality in respect of tribal organization, religion and language, which their congeners in the plain have long since lost."

ORISSA.

6. Orissa, with an area of 41,789 square miles and a population of 8,928,316 persons, is by far the largest but most sparsely populated sub-province. It corresponds to the Orissa Division and the Orissa Feudatory States, and consists of two distinct portions, viz., a low-land tract along the sea board and a hilly interior. The former tract, which comprises the districts of Cuttack, Balasore and Puri, is a delta formed by the Mahanadi, Baitarani, Brahmani and other rivers debouching into the Bay of Bengal. The greater part of this alluvial delta was under the direct control of the Mughals, and formed the Mughalbandi or crown lands, from which the Mughals obtained a regular revenue. On the collapse of their power, it passed under the domination of the Marathas, and did not come under British rule until 1803. It has been throughout its history a tract difficult of access, having little communication with the rest of north-eastern India; it is only within recent years that it has been connected by rail with Madras on the south and Bengal on the north. Owing largely to this isolation, the people have social characteristics and a caste system different both from that of Bengal and that of Madras, while their language (Oriya) has but few foreign elements. The interior of Orissa forms an elevated plateau with occasional higher hills, some of which reach sub-temperate altitudes. In this hinterland there are two British districts, viz. Sambalpur, the suzerainty of which was ceded by the Marathas in 1826, but which only came under direct British rule in 1849, and Angul, part of which was annexed in 1847, and the remainder, known as the Khondmals, in 1855. The rest of the country is under the rule of Feudatory Chiefs, who, protected from invasion by the nature of the country, were nominally subject to the Mughals and Marathas, but otherwise remained independent. Their subjects are mainly forest and hill tribes, or semi-Hinduized aborigines, who have been but little affected by outside influences.

SIKKIM.

7. Sikkim, with an area of 2,818 square miles is smaller than an average district in Bengal, Bihar and Orissa, while its population is less than that of any district in the two Provinces. Within its narrow limits it presents almost every conceivable condition of climate, tropical, temperate and alpine, the mountains running up beyond the snow line, the valleys sinking to 1,000 feet above sea level. At the higher altitudes arctic cold prevails; the valleys have a moist, hot and enervating climate. The rainfall in the south is very heavy: at the capital, Gangtok, it averages 133 inches in the year, but in the drier valleys to the north it falls to 20 inches or less. From about 7,000 up to 14,000 feet which is the limit of tree growth, the country is under virgin forest and uninhabited, except for occasional settlements of graziers. The population is almost exclusively confined to the valleys, slopes, and ridges below 7,000 feet, that being the highest level at which maize, the staple food of the people, comes to maturity. This and other crops, such as millets and pulses, are generally raised by means of *jhuming*. The forest is cut or burnt down, and a crop raised from the denuded surface for two years in succession, the land lying fallow for the succeeding eight years. When all the forest on his holding has been destroyed, the peasant resumes cultivation on the patch he first cleared, cutting and burning down any forest growth and scrub that may have sprung up in the meantime. At the lower levels, *i. e.*, below 4,000 feet, rice is grown by means of wet-terracing. The hill side is carved out into terraces, the outer edge of which is banked up to a height of about one foot; a channel is led from the nearest stream to the topmost terrace, from which it runs down to the terraces below, each of which is irrigated

* H. Blochmann, Notes from Muhammadan Historians, J. A. S. B., Vol. XL., 1871.

in turn. The population is a mixed one, consisting of Bhotias, Lepchas and Nepalese. The Bhotias, who are mainly graziers, live at the higher elevations. The Lepchas are a timid peaceful race of cultivators found in the lower part of the Tista valley and its affluents. Unable to face the cold, they favour the warmth of the valleys, where they lead a somewhat lazy life. The Nepalese, who are found mainly in the south and west of Sikkim, are the most thrifty and enterprising cultivators in the State, and are far more energetic and virile than the Lepchas, who give way to them whenever they come into contact.

NATURAL DIVISIONS.

8. In the Imperial Tables the districts have been grouped together by the administrative divisions known as Commissionerships, but this arrangement is not altogether convenient for discussing the main results of the census, the limits of Divisions having been fixed for administrative purposes and without regard to physical and ethnological considerations. In the Orissa Division, for instance, the Angul district is entirely different from the seaboard districts. The former is a hilly district with a non-Aryan population largely composed of Kandhs (Khonds) who till about 60 years ago practised human sacrifice. The latter are deltaic districts long under the domination of Brahmans, with an Aryan population, an old civilization and a peculiar caste system. The Bhagalpur Division again contains the Sonthal Parganas, which physically is a part of the Chota Nagpur Plateau, while its people are mainly Animistic Santals and Sauria Paharias, the latter of whom cling to the hill tops and are one of the most primitive races in the Province. In this report, therefore, and in the subsidiary tables attached to it, the districts are grouped together by natural divisions with physical and ethnical affinities. These divisions are the same as those adopted in 1901, except that Sikkim, being no longer under the Government of Bengal, is not included in North Bengal, and that the Chota Nagpur Plateau now comprises the district of Sambalpur and the States transferred from the Central Provinces, while it excludes the States which, as mentioned in paragraph 3, have been detached from Bengal.

9. The following statement shows the districts and States included in each natural division :—

Natural Division.		Districts and States.	Divisions and States.
Bengal	West Bengal	Burdwan	Burdwan.
		Birbhum	
		Bankura	
		Midnapore	
		Hooghly	
		Howrah	
	Central Bengal	24-Parganas	Presidency.
		Calcutta	
		Nadia	
		Murshidabad	
		Jessore	
	North Bengal	Rajshahi	Rajshahi.
		Dinajpur	
		Jalpaiguri	
		Darjeeling	
		Rangpur	
		Bogra	
		Pabna	
		Malda	
		Cooch Behar	Cooch Behar.
	East Bengal	Dacca	Dacca.
		Mymensingh	
		Faridpur	
		Backergunge	
		Tippera	Chittagong.
		Noakhali	
		Chittagong	
		Chittagong Hill Tracts	
		Khulna	Presidency.
		Hill Tippera	Hill Tippera.

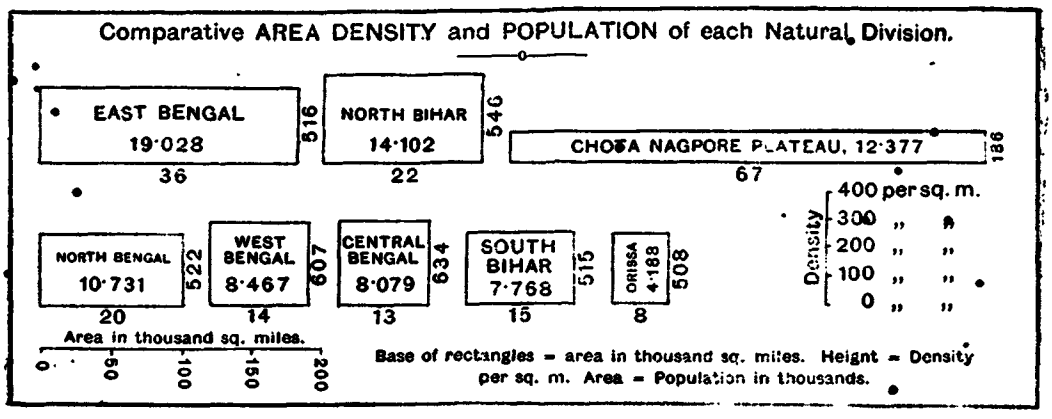
Natural Division.		Districts and States.		Divisions and States.
Bihar and Orissa.	North Bihar	...	Saran	Tirhut.
			Champaran ...	
			Muzaffarpur ...	
			Darbhanga ...	
			Bhagalpur ...	Bhagalpur.
			Purnea	
	South Bihar	...	Patna	Patna.
			Gaya	
			Shahabad ...	
			Monghyr ...	... Bhagalpur
	Orissa ...	...	Cuttack	Orissa.
			Balasore	
			Puri	
	Chota Nagpur Plateau		Hazaribagh ...	Chota Nagpur.
			Ranchi	
			Palamau	
			Manbhum ...	
			Singhbhum ...	
			Sonthal Parganas	... Bhagalpur.
			Angul	Orissa.
			Sambalpur ...	
			Athgarh ...	Orissa Feudatory States.
			Athmallik ...	
			Bamra	
			Baramba	
			Baud	
			Bonai	
			Daspalla	
			Dhenkanal ...	
			Gangpur	
			Hindol	
			Kalahandi ...	
			Keonjhar ...	
			Khondpara ...	
			Mayurbhanj ...	
			Narsinghpur ...	
			Nayagarh ...	
			Nilgiri	
			Pal Lahara ...	
			Patna	
			Rairakhol ...	
			Ranpur	
			Sonpur	
			Talcher	
			Tigiria	
			Saraikela	Chota Nagpur States.
			Kharsawan ...	

DENSITY OF POPULATION.

10. Statistics of the area, population, towns, villages and occupied houses of each district are given in Imperial Table I. Provincial Table I, which will be found at the end of the volume of Imperial Tables, gives similar figures for thanas, together with percentages of variation since 1891 and the density per square mile in 1911. Attached to this chapter are seven subsidiary tables showing—(*i*) statistics of density, water-supply and crops, (*ii*) the distribution of the population classified according to density, (*iii*) the distribution of the population between towns and villages, (*iv*) the number per mille

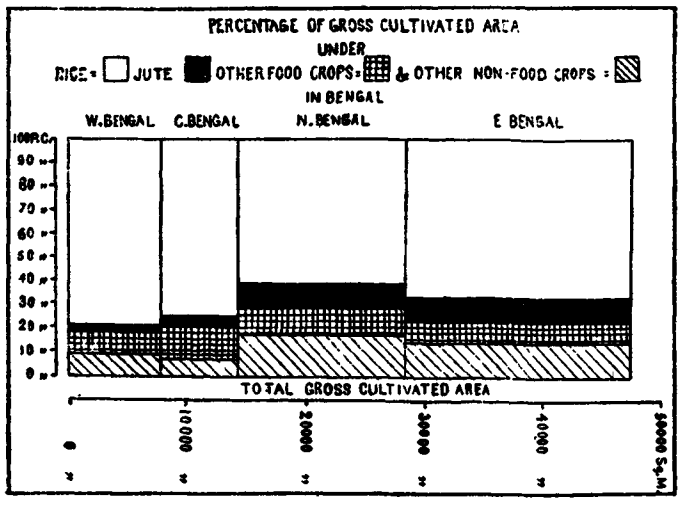
of the total population and of each main religion who live in towns,-(*v*) towns classified by population, (*vi*) density and variations in the population of cities, and (*vii*) the number of persons per house and of houses per square mile. As indicated in paragraph 6, these subsidiary tables show the districts by natural divisions, the area and population of which are given in the margin.

Locality.	Area in square miles.	Population.
Bengal	**84,092**	**46,305,642**
West Bengal	13,948	8,467,314
Central Bengal	12,734	8,078,555
North Bengal	20,542	10,731,254
East Bengal	36,868	19,028,519
Bihar and Orissa	**111,829**	**38,435,293**
North Bihar	21,822	14,102,314
South Bihar	15,076	7,767,682
Orissa	8,238	4,188,109
Chota Nagpur Plateau	66,693	12,377,188



11. Bengal, with an average density of 551 persons to the square mile is far more thickly populated than any European country, except Belgium and England. Its density would be even greater were it not for the large area occupied by hills, rivers, swamps and estuaries, which cause the most extraordinary variations within comparatively narrow limits. One district in the same natural division may be densely populated, and another support less than half the number: even in the same district one thana may contain a teeming population, and another have a few inhabitants scattered over its surface. For instance, the 24-Parganas supports 502 persons per square mile, but the adjoining district of Khulna only 287; if we exclude the uninhabited forest area in the Sundarbans, a labyrinth of tidal rivers, swampy forests and half-submerged islands, their density is 776 and 515 respectively. In Eastern Bengal the district of Dacca has 1,066 persone per square mile, but in ths Chittagong Hill Tracts each square mile supports only 30 people. In Western Bengal, again, 1,850 persons per square mile are found in Howrah, and 434 in Bankura. Owing to these variations the conditions of each natural division and district will be discussed in some detail.

Bengal.



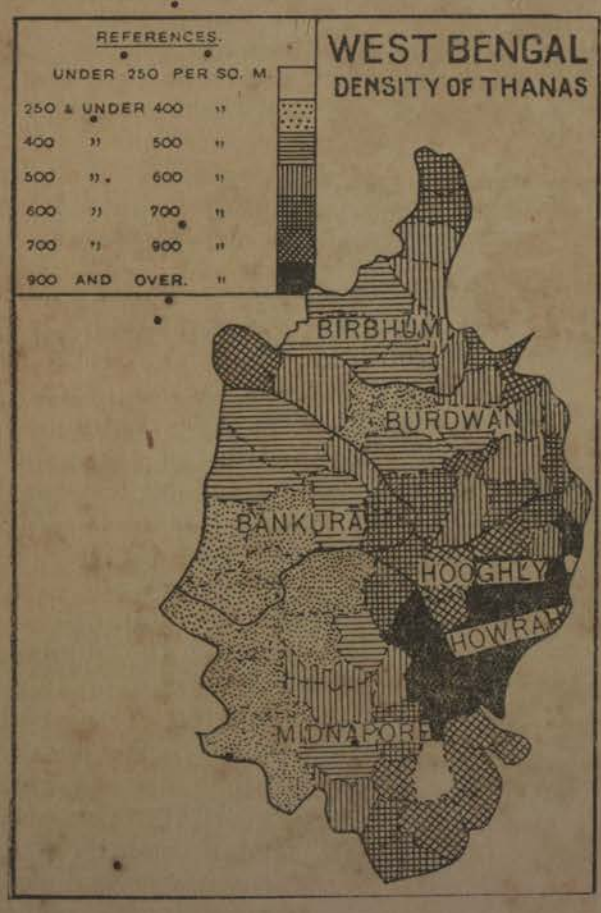
12. In the Presidency as a whole the pressure on the soil is less in North Bengal and East Bengal, where there are few big cities or large industries, than in Central Bengal and West Bengal, which contain the populous cities of Calcutta and Howrah and the metropolitan districts of Hooghly and the 24-Parganas. Central Bengal, with 634 persons per square mile, has the greatest density of population, and is closely followed by West Bengal with

607 to the square mile. In North Bengal the ratio is 522, and in East Bengal 516 per square mile. No less than one-fourth, however, of the area included in the latter division is accounted for by Hill Tippera and the Chittagong Hill Tracts, whose jungle-covered hills contribute only one-fiftieth of its population. If these two tracts are left out of account, the ratio in East Bengal rises to 674 to the square mile and is higher than in any other division.

The most congested district is Howrah, in which, as already stated, each square mile supports 1,850 persons. The only other district with a density of over 1,000 is Dacca, but two more districts (Hooghly and Tippera) have over 900 persons per square mile. Five districts, and the two States of Cooch Behar and Hill Tippera, support less than 500 persons per square mile. The scantiest population is found in the Chittagong Hill Tracts, where, density does not exceed 30 per square mile.

13. West Bengal extends along the right bank of the Bhagirathi (or Hooghly, as the river is called in its lower reaches), and approaches the Chota Nagpur Plateau on the west, while on the south it is bounded by the Bay of Bengal. It consists of two distinct zones, one a semi-aquatic rice plain, the other a rolling upland country. The portion included in the Hooghly and Howrah districts and in the east of Burdwan, Midnapore and Bankura, is an alluvial plain formed by the Bhagirathi, Damodar, Ajay and Rupnarayan rivers. The soil is fertile and bears heavy crops of rice, but the climate is damp and enervating, jungle grows thick and fever is rife. The country is but little raised above sea level, and is intersected by rivers, many of which are now silted up, while the beds of others are being gradually raised by the annual deposition of silt. Between the rivers are swampy depressions, and a large area is waterlogged. Urban or semi-urban conditions prevail along the bank of the Hooghly from Chinsura on the north to Howrah on the south. In this riparian strip of land town follows town with scarcely a break, and the development of manufactures in the last half century has converted it into a busy industrial centre. The remainder of the tract is higher in elevation and consists of rolling country with a laterite soil, which in the west includes the eastern fringe of the Chota Nagpur Plateau. A poor ferruginous soil and hard beds of laterite here take the place of the fertile deltaic detritus, and wide expanses of scrub-jungle are found instead of the closely-tilled village lands of the east. Instead of containing a well-educated population of Hindus and Muhammadans, this western tract is comparatively thinly inhabited by races or castes of a less advanced type, into whose constitution an aboriginal or semi-Hinduized element enters. Rural conditions generally prevail, there being few towns or industries of any importance except in the Asansol subdivision of the Burdwan district, where the coal-fields have attracted a large immigrant population.

WEST BENGAL.



14. The most populous districts are Howrah and Hooghly, which are entirely alluvial, and then follow, in order, Burdwan, Midnapore, Birbhum and Bankura, in all of which there are under 600 persons to the square mile. Howrah city alone contains nearly one-fifth of the total population of the Howrah district, but even if it is excluded, the average per square mile is

1,525 and exceeds that returned by any other district in the Presidency. This teeming population is due to the neighbourhood of Calcutta and the number of jute mills, cotton mills, engineering and other industrial works clustered along the bank of the Hooghly. Density is high even in rural tracts, no thana having less than 1,293 persons per square mile, while Dumjor has 2,212—an astonishingly high figure, for, though this thana adjoins Howrah city, part of the land is marshy and uncultivable. In the Sadar subdivision, which contains the city of Howrah and most of the manufacturing works, density is more than twice as great as in the Uluberia subdivision to the south, which is more purely agricultural. The density of the district has risen by nearly 50 per cent. since 1872, and even in the last decade there has been an increase of 182 persons per square mile. This growing pressure on the soil is due not only to industrial expansion, but also to the drainage of swampy areas, a large scheme affecting over half its area having been carried out since 1891. Dumjor and Jagatballabhpur thanas, which have especially benefited by it, show an increase of 268 and 198 per square mile in these twenty years, though the former was already densely populated. The population is most scanty in Amta, the drainage of which was proposed as long ago as 1873, but which still remains partially waterlogged and liable to inundation.

In Hooghly the highest density is found in the Serampore subdivision, where conditions are similar to those in Howrah, and where part of the land has been rendered cultivable by the Dankuni drainage scheme. The Serampore thana, which is a riparian strip containing five municipal towns, has as many as 5,098 persons per square mile, and, of the remaining four thanas, two have over 1,000 and two over 900 to the square mile. In the Hooghly subdivision to the north the people congregate thickly along the Hooghly as far as Tribeni, but in the low-lying, unhealthy country inland the average falls below 550. In the Arambagh subdivision the density varies according to the nature of the soil, the alluvial tract having an average density of 838 and the laterite uplands of Goghat only 677.

15. The relatively low density characteristic of lateritic soil is further exemplified in the Asansol subdivision of Burdwan, where the ratio is under 400 in Ausgram and Kaksa. In other lateritic areas the development of the coal-fields has led to a large influx of labour, and Asansol, with 887 persons to the square mile, is the most thickly populated thana in the district. In the alluvial portion the population is fairly evenly distributed, the pressure on the soil being greatest in the Katwa and then in the Kalna subdivision, both of which lie along the Bhagirathi.

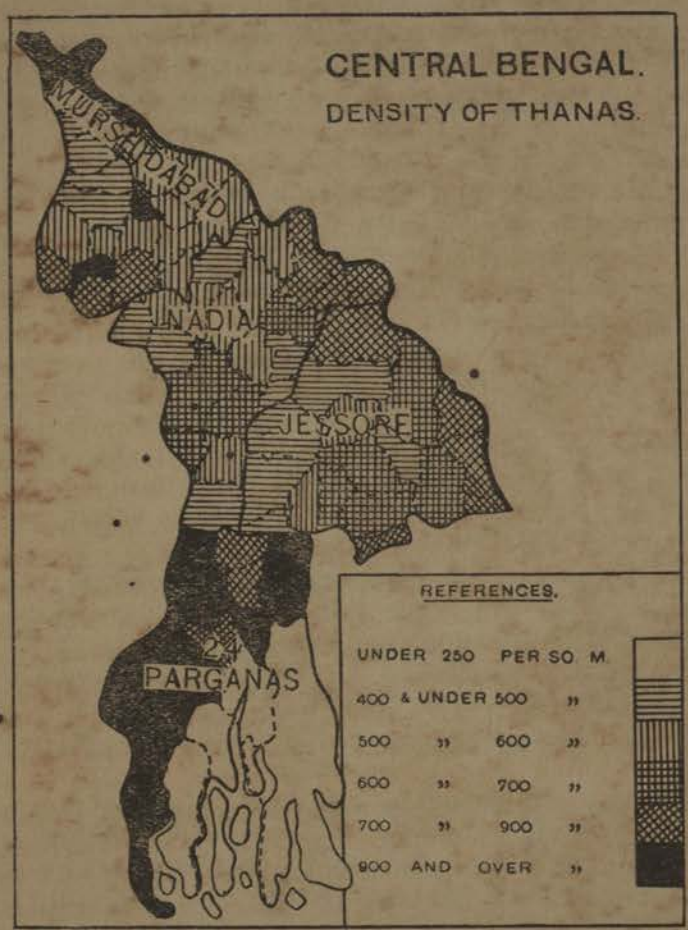
There is a very uneven distribution of the people over the 5,186 square miles that make up the Midnapore district. The eastern half of the district, which is alluvial, is thickly populated, while the west of the district, which has a lateritic soil covered here and there with jungle, is sparsely inhabited. The most populous areas lie along the bank of the Rupnarayan and the estuary of the Hooghly, the maximum density being found in the Tamluk and Ghatal subdivisions (921 and 811 respectively), which lie along the Rupnarayan and consist of fertile rice plains. Contai, with 728 persons to the square mile, lies on the sea-coast to the west of Tamluk, and there are large tracts of sandy or salt-impregnated soil. Conditions in the Sadar subdivision are very different. Two-thirds of it form part of the lateritic plateau running down from Manbhum, which cannot support a large population, for considerable areas are covered by *sal* forest and jungle. In this subdivision there are only 397 persons per square mile, while five thanas, which still retain the old name of Jungle Mahals, and cover 1,827 square miles or more than half the total area, have a mean density of under 300 to the square mile.

In Birbhum the density of population decreases towards the west on the borders of the Sonthal Parganas, where the surface is barren and undulating, and increases towards the east, which is an alluvial flat. It rises to over 600 in the Rampur Hât subdivision, which is mainly a fertile rice plain, and it is less than 500 in the Sadar subdivision, the minimum (449) being reached in the Suri thana, which is an undulating tract with a sterile soil. The difference between the density of the alluvial flats to the east and the uplands to the west is equally marked in Bankura. The Bankura subdivision, which is hilly and undulating, with large jungle tracts, has a density of only 389.

whereas in the Vishnupur subdivision, which is part of the deltaic country and almost entirely under rice cultivation, there is an average of 560 to the square mile.

CENTRAL BENGAL.

16. Central Bengal consists of a portion of the delta in which the process of land formation has ceased. It is an alluvial plain intersected by numerous rivers, which formerly received a supply of water from the Ganges, and in their turn supplied and enriched the land with annual deposits of silt. The influx of fresh water from the Ganges has ceased except in the rains, the result being that for the greater part of the year they have no current, but merely contain long stagnant stretches of water covered with vegetation. The banks of the rivers having been raised above the surrounding country by the accumulation of silt, depressions are found between them, the fall from all directions being towards the centre. Many of these depressions are of small size, but others are practically inland lakes. Some are mere accumulations of water upon low-lying ground, while others are natural drainage basins, the level of which does not admit of drainage. In some places these basins are on a fairly high level, and the central depression is under regular cultivation. Other depressions are water-logged, but can still be used for growing rice, while others again are always under water.



In the Division, as a whole, there are 634 persons to the square mile, but the density is increased by the large population of Calcutta; if the latter be excluded, there are 565 persons per square mile. This is a high figure, when it is remembered that in the 24-Parganas the uninhabited forest area in the Sundarbans extends over 1,711 square miles or more than one-third of the whole district. Excluding this area, the 24-Parganas has an average density of 777 per square mile. In the remaining districts the pressure on the soil does not vary very greatly, there being 640 persons to the square mile in Murshidabad, 601 in Jessore and 580 in Nadia.

17. In the 24-Parganas numerous towns with busy jute and cotton mills stretch along the whole length of the Hooghly from Garden Reach northwards. Away from its banks, however, the population is almost entirely rural and devoted to agriculture. Density in the different subdivisions varies accordingly, being as high as 1,540 in the Barrackpore subdivision, which is a narrow riparian strip crowded with municipal towns, factories and mills. In the Diamond Harbour subdivision the density is less than a third of this, and in the Basirhat subdivision there are only 223 persons per square mile. Both these subdivisions, however, lie to the south and merge in the Sundarbans. The difference between conditions in the north and south is even more plainly seen in the thana returns; no less than 19 thanas have more than 1,000 persons per square mile, while in two (Mathurapur and Husainabad), which extend into the Sundarbans, there are less than 100 per square mile.

In Murshidabad the people cluster more closely in the alluvial country to the east of the Bhagirathi than to the west, where the country is slightly undulating and the level is higher. The most densely populated thanas lie

along the banks of the Bhagirathi, four of them having a density of over 1,000, while four thanas to the west have under 500 persons per square mile. In Jessore the average density of population is least in the Bangaon subdivision to the south-west, where the silting up of rivers has deprived the country of the fertilising deposits it formerly received. It gradually increases as one proceeds from west to east, and reaches the maximum of 740 in the Narail subdivision in the south-east. Here the rivers still have a flowing current, and one thana contains no less than 897 persons per square mile. The Kushtia subdivision, a fertile tract lying between the Ganges and Mathabhanga, is by far the most populous part of Nadia. The scantiest population in that district is found in the Ranaghat subdivision in the extreme south-east, in spite of the fact that it contains a larger urban population than any other subdivision. Elsewhere the inhabitants are fairly evenly distributed, varying only from 521 to 554 per square mile.

North Bengal.

18. North Bengal, lying from east to west between Purnea and the Brahmaputra, and from north to south between the lower spurs of the Himalayas and the Ganges, a remarkably homogeneous area, except for the hills portion of the Darjeeling district, which forms part of the Himalayas and is inhabited by Mongolian races, mostly descendants of immigrants from Nepal. In the north there is an unhealthy submontane tract, the Tarai, from which the country gradually slopes southwards in a wide alluvial plain watered by the rivers flowing southwards from the Himalayas and broken only by the Barind. This is a comparatively high belt of laterite formation lying on the confines of Dinajpur, Malda, Rajshahi and Bogra, which grows only one crop, late rice, and which in consequence is subject to scarcity in years of deficient rainfall. The rest of the area has a rich soil of sandy loam, mainly under rice, which accounts for two-thirds of the cultivated area. Jute is also grown extensively and covers one-tenth of the cropped area; tea is raised in the submontane tract, known as the Duars, in Jalpaiguri and the Tarai in Darjeeling, as well as on the hills in the latter district. Except in the south-east, communication is mainly by land, and the demand for speedy transport has resulted in large railway extensions.



19. In most of the districts the distribution of the people is affected by the changes in the river system which have taken place since 1787. The Tista once flowed south through the centre of North Bengal to meet the Ganges, but in that year it changed its course and cut out a new channel by which it found its way to the Brahmaputra. Owing to the vagaries of this great river, North Bengal is full of silted river beds, which obstruct drainage and are largely responsible for the unhealthiness which prevails. Density is highest in the districts bordering on the Brahmaputra, viz., Pabna (772), Bogra (724) and Rangpur (686); the only other districts with over 500 per square mile are Rajshahi and Malda lying along the Ganges. In the case of both the Ganges and Brahmaputra the mean density of the districts rises according to their situation along the rivers' downward course, and it has been suggested that this is due to their fertilizing powers increasing with the fall of the level

of the land. The least populous tracts are Cooch Behar, Jalpaiguri and Darjeeling to the north, of which the two latter have large areas under forest.

20. Density in the Pabna district is highest in the Sirajganj subdivision, a fertile jute-growing tract, in which the drainage has not been obstructed by deserted river beds to the same extent as in the headquarters subdivision. Its average density (867 per square mile) is one-third greater than that of the latter subdivision, but there are extraordinary variations, the Shahzadpur thana supporting 1,209, and the Raiganj thana only 490 persons to the square mile. The latter is an unhealthy tract over which malaria has a hold, and, moreover, the large lake known as Chalan Bil occupies a considerable portion of it. The only thana in the Sadar subdivision in which the district average is exceeded is Mathura, which, lying in the angle between the Padma and the Brahmaputra, is specially benefited by the deposit of silt brought down by the rivers.

In Bogra the scantiest population is found in the west of the district, which forms part of the elevated *quasi*-laterite tract known as the Barind. The minimum (457) is reached in the Sherpur thana to the extreme south-west, where a large area is still overgrown with jungle. The mean density is double as high as this in the adjoining thana of Dhunot, which is traversed by flowing rivers and is one of the most fertile tracts in the district. Generally speaking, the population is very dense in the east of the district between the Karatoya and Dakopa rivers, where there is a rich alluvial soil in which jute is the main crop.

21. Rangpur consists of a wide alluvial plain unbroken by natural elevations of any kind. In the north there are extensive sandy plains, admirably suited to the cultivation of tobacco: the density of the Nilphamari subdivision, which is comprised in this tract, is 758 per square mile. In the east the Kurigram and Gaibandha subdivisions are enriched by the deposits of silt brought down by the Brahmaputra: the density in the former is slightly below and in the latter considerably above the district average. The Gaibandha thana supports 1,188 persons per square mile, while less than half that number are found in the Shaghatta thana which adjoins it on the south. The least populous part of the district is the Sadar subdivision, which is at once more unhealthy and less fertile than the other subdivisions.

Rajshahi is composed of four tracts with distinct agricultural conditions, viz.—(1) the tract along the bank of the Padma, which is subject to its direct fluvial action, (2) the Barind, on the north-west, with a *quasi*-laterite soil and a high undulating surface (3) a swampy water-logged depression on the east, and (4) the remainder of the district, which has neither the special advantages nor the disadvantages of the other three areas. The Sadar subdivision, which includes the whole of the first tract and portions of the second and fourth tracts, has the highest density in the district. Next comes Noagaon, in which two thanas are comprised in the Barind and two in the fourth tract. In the Nator subdivision, which includes the whole of the swampy tract and very small portions of the first and second tracts, the effects of unfavourable agricultural conditions are accentuated by the prevalence of malaria. This is consequently the least populous part of the district.

22. Density varies greatly in the three tracts making up the Malda district, viz., (1) the area locally known as the Diara, which is fertilized by the Padma, (2) the older alluvium, which is not so fertile as the Diara, and (3) the Barind, which is still less fertile. The thanas are not exactly conterminous with these natural divisions, but the marginal statement clearly indicates the influence of the character of the soil on the distribution of population. The State of Cooch Behar is thinly populated, density nowhere rising above 552. Dinhata in the south and Haldibari in the extreme west support, on the average, 545 persons per square mile, but the mean density is reduced to 421 in the remainder of the State, and tails off to 359 in Tufanganj to the north-east, which is as yet not fully developed. In Dinajpur conditions are not favourable to any great density of population. It is not watered by either the Ganges or the Brahmaputra: its climate is unhealthy, and the south is covered by the high

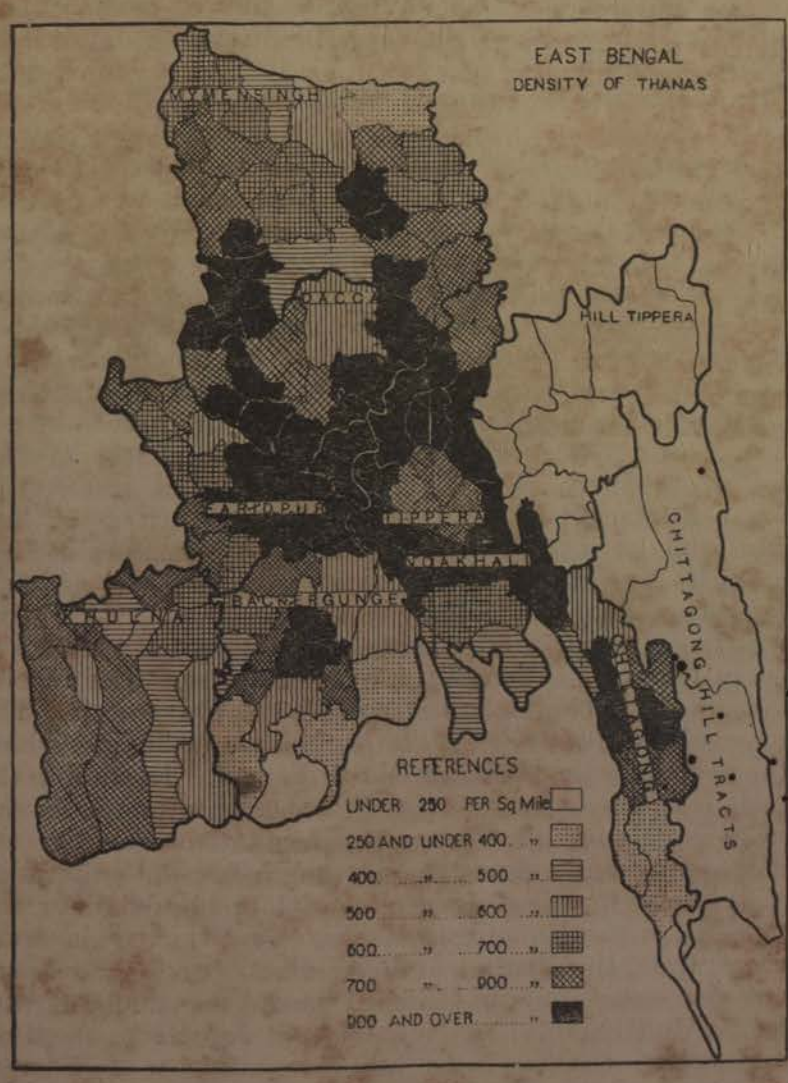
Tract.	Thana.	Density.
1	Kaliachak	845
	Sibganj	808
2	English Bazar	739
	Kharba	584
3	Gumastapur	397
	Malda	357
	Gajol	356

undulating ridges of the Barind, in which cultivation is either impossible or unprofitable. The Balurghat subdivision, which includes the greater portion of the Barind, supports only 380 persons per square mile. In the district as a whole the population is fairly evenly distributed, the lowest thana density being 315 in Parsa, which lies in the Barind, and the highest 553 in Dinajpur, which includes the district headquarters.

23. In Jalpaiguri, which lies at the foot of the Himalayas, the average is reduced by the large area under forest, the reserved forests alone accounting for more than one-sixth of its total area. If these forests and the Baikanthpur forest are left out of account, the average rises to 381 per square mile. The Sadar subdivision, where cultivation is most advanced, supports nearly twice as many persons per square mile as the Alipur subdivision, where the average falls as low as 162 in the Alipur thana to the east. The latter subdivision is, however, by far the more progressive of the two, as the waste lands available for settlement are being rapidly taken up and reclaimed, the result being that since 1901 its average density has been nearly doubled. In Darjeeling also the space available for settled habitation is very greatly reduced by the area under forest. Reserved forests extend over more than a third of the district, while a considerable area is taken up by tea gardens. Excluding the forests, there is an average density of 369 persons per square mile, which is very little less than the corresponding figure for Jaipaiguri. The most populous part of the district is the Siliguri subdivision in the Tarai. The sparsest population is found in Kalimpong, in which, however, the greatest development has taken place, the ratio rising from 65 to 120 per square mile during the last 20 years.

EAST BENGAL.

24. East Bengal is for the greater part a deltaic plain, composed of the upper and lower portions of the deltas of the Ganges and Brahmaputra. It forms a wide alluvial expanse, broken, in the north-east, by an elevated tract known as the Madhupur Jungle and, in the east, by hills which run south into and through Chittagong. The greater part is a fertile semi-aquatic plain covered by a network of rivers and waterways, which form almost the only means of communication. They are heavily charged with silt, and their banks, raised by its annual deposition, are higher than the surrounding country. The depressions between the high river banks form large marshes, which are especially numerous in the south of Faridpur and the west and north-west of Backergunge, where the whole country is a succession of basins, full of water in the rains, but partially or wholly dry in the winter months. The largest of these



depressions is the Chalan Bil, which has a water area varying from about 20 square miles in the dry season to 150 square miles in the rains. The greater part of the country is annually enriched by the silt brought down by the rivers, which in the rains overflow their banks and spread over the low-lying lands between them. The rainfall is heavy, the monsoon consisting of a series of cyclonic depressions which follow each other up the Bay of Bengal. Unlike the cultivators of Bihar, therefore, the inhabitants of this favoured region have no reason to complain of lack of moisture for their crops.

25. While this is the general character of the country, East Bengal really consists of three distinct parts. The central portion, which may be designated East Bengal proper, forms the upper portion of the delta of the Ganges and Brahmaputra, and comprises the districts of Dacca, Mymensingh, Faridpur and Tippera. It is a land of rivers and plains with a fertile soil, abundant crops and a salubrious climate. Plague, the terror of Bihar, is unknown, and malarial fever, the scourge of Central Bengal, is not wide-spread. Rice is the staple crop, but in recent years the area under jute has expanded enormously and is now equal to nearly a third of that devoted to rice. The land supports a teeming population, pressure on the soil being limited mainly by the *bils* or marshes stretching away from the river banks and the laterite formation in the north-east. The marshes are partially dry and covered with rice in the cold weather, but are under water in the rains, when they form an almost unbroken fresh-water sea bordered by the river banks, which stand only a few feet above the flooded country. In the north-east are outcrops of laterite, rising to a small height above the alluvium and usually covered with *sal* forest, which constitute the Madhupur Jungle. These forests are gradually yielding to the axes of the Santals and other pioneers of cultivation. The average density of population rises to 1,066 in Dacca, which is a remarkably high figure, considering that, though the town of Dacca contains over 100,000 inhabitants, there is only one other town of over 20,000. It is closely followed by the Tippera district, in which there are 972 persons to the square mile, in spite of the fact that there are only two towns with over 20,000 inhabitants.

26. The second portion is the lower delta, which comprises the districts of Backergunge, Noakhali, Khulna and Chittagong on the shore of the Bay of Bengal. It is a region of tidal waters, which, distributed through an endless chain of connected channels, eventually merge into the brackish estuaries of the Meghna and Haringhata. Here rice predominates to the almost entire exclusion of other crops, occupying no less than 79 per cent. of the cultivated area. The most populous district is Noakhali with 792 persons to the square mile, but elsewhere, owing to the large uninhabited area included in the mangrove forests and swampy islands of the Sundarbans, density falls off considerably, the general average being reduced to 488, while Khulna has only 287 persons per square mile. In this latter district, however, 2,089 square miles, or nearly half the total area, consist of uninhabited reserved forests in the Sundarbans.

The third portion consists of Hill Tippera and the Chittagong Hill Tracts, two hilly hinterlands, in which there is a scanty population, practising primitive methods of cultivation among their native jungle, and averaging only 42 per square mile.

27. The district of Dacca supports a teeming population, but there is a wide range of variations in density. In two of the four subdivisions the density is below the district mean of 1,066, and in two considerably above it. The greater part of the Sadar subdivision is covered by the Madhupur Jungle, which in its general character resembles the Barind of North Bengal. Excluding the Dacca thana, which includes Dacca city, and also Nawabganj, which is in the alluvial area, the average is only 661. The Narayanganj subdivision on the east has a more fertile soil, which is admirably suited to the cultivation both of rice and jute, and every thana supports over 1,100 persons to the square mile. On the south in the Munshiganj subdivision, which is hemmed in on three sides by the Padma, the Meghna and the Dhaleswari, there are as many as 1,996 persons per square mile in thana Srinagar and 1,600 persons per square mile in thana Munshiganj. Its inhabitants depend to a much smaller extent on agriculture than in any other subdivision of Eastern Bengal: only 51 per cent. of its population are supported by ordinary cultivation, while a very large number of males find employment elsewhere

in clerical and other work. Agricultural conditions in the Manikganj subdivision are as favourable as in Munshiganj, but it suffers from an unhealthy climate, and during the past decade two thanas have lost considerably by diluvion.

28. Tippera enjoys exceptional advantages in regard to both climate and soil. The Meghna, which sweeps past the western border, enriches a large tract of land, on which fine jute is produced, while a number of smaller streams, bringing down silt from the hills, spread it over the greater portion of the district. While the soil is exceedingly fertile, the rainfall is abundant and well distributed. The result of these favourable conditions is the high average of 972 persons per square mile. The range of variations is small, the density falling below the district average in only five thanas, three of which, viz., Chandina, Laksham and Hajiganj, are low-lying and suffer from occasional floods. The most densely populated part of Faridpur is the Madaripur subdivision, which consists of a flat alluvial plain, subject to the fertilizing action of large rivers and comparatively free from the ravages of malaria. In this subdivision density rises to over 1,000, while none of the others support over 800 per square mile.

29. For considering the local variations of density, the Mymensingh district may be divided into five parts. The western part, including the Tangail subdiyision and a portion of the Jamalpur subdivision, is watered and drained by the system connected with the Jamuna. This tract resembles the flat alluvial tract on the other side of the Brahmaputra in the Rangpur Bogra and Pabna districts, density in the Tangail subdivision being 989 as against 815 in Gaibandha, 724 in Bogra and 867 in Serajganj. The central portion, which includes the greater part of the Madhupur Jungle with its extensive forest reserve, corresponds to thanas Mymensingh, Fulbaria, Gaffargaon and Muktagacha of the Sadar subdivision, in all of which the density is below the district average (724). The third tract, comprising thanas Nandail, Iswarganj and part of Fulpur in the Sadar subdivision, Netrakona and part of Kendua in the Netrakona subdivision, and Kishoreganj and Kathiadi in the Kishoreganj subdivision, consists of rich alluvial lands, on which heavy crops of rice and jute are raised. The density in this tract is considerably above the district average, that of Nandail (1,171) being the highest in the district. The fourth and easternmost tract is much lower in level than the third tract and is intersected by large marshes or *haors*, where people gather during the fair weather to catch fish, graze cattle and grow *boro* rice. The density in this tract is generally below the district average. The fifth tract lying along the foot of the Garo Hills includes portions of thanas Dewanganj, Nalitabari, Sherpur, Fulpur, and Durgapur, in which the population is very sparse, the density of Durgapur (333) being the lowest in the district.

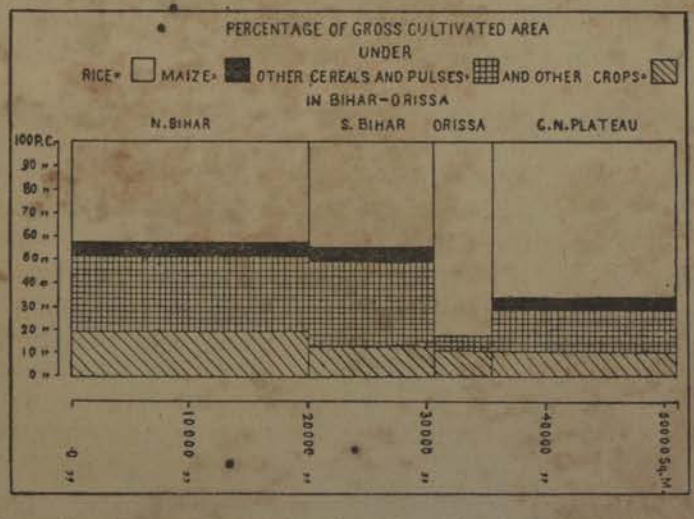
30. Noakhali consists of a mainland tract and a number of islands, the largest of which are Sandwip and Hatia. The estuary of the Meghna, which is here about 7 miles broad, sweeps past the western and southern sides of the mainland, where alluvial changes occur with surprising rapidity. The thanas which are most subject to fluvial action are Sudharam, Sandwip and Hatia, where the density falls considerably below the district average. Old lands are being constantly cut away in this area and new lands formed, which take time to develop into cultivation. In the remainder of the district, the variation is very small, the lowest average being 954 in Feni and the highest 1,223 in Ramganj, where the growth of betelnuts and cocoanuts brings the cultivators large profits. Only 40 per cent. of the total area of Chittagong is under cultivation, the uncultivable area comprising jungle-covered hills, which traverse the district, and deltaic mangrove swamps along the coast. The proportion of land under cultivation is much higher in the Sadar subdivision, where the average density is 789 persons per square mile, than in the Cox's Bazar subdivision, where it is as low as 277 persons per square mile. The land, where under cultivation, is fairly fertile, and if the uncultivated area be excluded, the density comes to 1,511 persons per square mile, a ratio which is exceeded in Eastern Bengal only by the Faridpur and Dacca districts.

31. In Backergunge the distribution of the population is limited by the uncultivable area included in the Sundarbans. The tract bordering the Sundarbans, although fertile, is also backward in cultivation and sparsely populated, the people being exposed to the dangers of cyclones and storm-waves. As a result of these conditions, the density in thanas Matbaria, Amtali, Golachipa and Baranadi falls below 400, reaching the minimum of 232 in Amtali. The density gradually rises northward with variations due to local causes, till the highest figure is reached in the Sadar subdivision. Backergunge is one of the most important rice-producing tracts in Eastern Bengal and is capable of supporting a much larger population than it does at present. In Khulna the pressure on the soil gradually increases as one proceeds from east to west, Satkhira on the east being the most and Bagerhat on the west the least populous subdivision. A number of thanas merge in the Sundarbans, and four contain 500 persons or less per square mile.

32. Both the Chittagong Hill Tracts and the Hill Tippera consist largely of hills, which are unsuitable for plough cultivation. In the former only 5 per cent. of the total area is actually under cultivation, the aboriginal inhabitants living chiefly by *jhuming*. Its population has increased during the past decade by 23 per cent., but in the three circles comprising the district the density varies only from 24 to 42 persons per square mile. In Hill Tippera cultivation is more advanced, and agricultural conditions in the narrow strip of low land along the north-western and southern boundaries are similar to those in the adjoining British territory. The density varies from 25 in Kaila Sahar to 126 in the Sadar subdivision, the average in the whole State being 56 per square mile.

33. The Province of Bihar and Orissa, with 344 persons per square mile, though not so thickly populated as the British Isles (where there are on the average 30 more persons per square mile), has a denser population than Germany. In British territory the mean density rises to 415 to the square mile, but the average for the Province is considerably reduced by the Orissa Feudatory States, which are as big as Ireland and as thinly peopled, the average density (135) being exactly the same. The most populous district is Muzaffarpur (937), which is closely followed by Darbhanga (875), while the scantiest population is found in Angul (119). Density throughout the Province is determined by the physical nature of the country and, in particular, by hills, forest and water. Bihar is an alluvial tract, the northern portion of which is traversed by rivers debouching from the Himalayas: considerable areas are seamed by their old beds and flooded by their present channels. Immediately to the south the hilly plateau of Chota Nagpur rises above the plain. South of the latter are the highlands of Orissa, and to the extreme south-east the deltaic districts of the Orissa Division lie between them and the Bay of Bengal. Even the alluvial stretches of South Bihar and the Orissa delta have a hilly backbone, and in the whole Province there are only four districts (Saran, Darbhanga, Muzaffarpur and Purnea) in which hills are conspicuous by their absence. The hills generally are either bare rocky excrescences, or bear a thin sterile soil covered with forest.

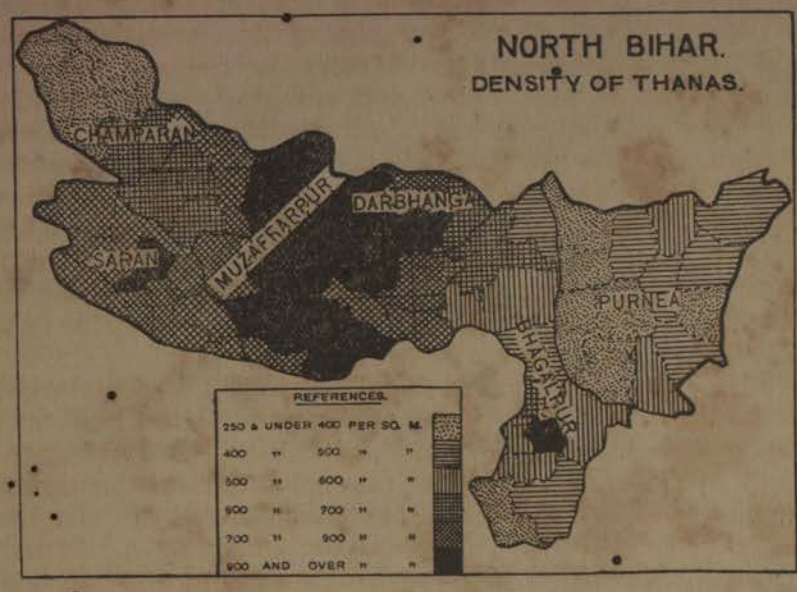
BIHAR AND ORISSA.



34. North Bihar, with 646 persons to the square mile, is the most congested tract in the Province. To the extreme north there is, in some parts, a sub-montane strip, consisting mostly of prairie land and denuded forest, but

NORTH BIHAR.

the remainder is a flat alluvial plain almostentirely under cultivation. It is watered by a number of great rivers which have gradually raised their beds by the deposition of silt and flow on ridges slightly elevated above the general level of the country. Most of them are liable to overflow their banks after heavy rainfall in the Nepal hills, and such inundations have been of increasing frequency and severity in recent years. In the west cultivation is rendered precarious by the oscillations of the Kosi, which, frequently changing its course, spreads over the land a layer of infertile sand that destroys its productive powers. The rainfall is ordinarily ample, the normal annual quantity being 53·36 inches, but it is capricious and its distribution frequently untimely, especially in the north of the Tirhut Division, which has been described as the "blackest of black spots on the famine map." Here the cultivators are practically dependent on one crop, viz., winter rice. The population is dense, wages are low and rents high; when the rains fail, distress ensues among the landless labourers, but is mitigated by their increasing readiness to leave their homes and obtain work and wages elsewhere.



35. The pressure on the soil, especially in Saran, Muzaffarpur and Darbhanga, would be still greater were it not for the area occupied by rivers and marshes. Through the centre of Champaran extends a chain of 43 lakes, with an area of 139 square miles, which mark an old bed of the Gandak and never entirely dry up. In the other districts north of the Ganges there are numerous marshes (*chaurs*) and meres, which represent the deeper portions of old river beds or are formed by the troughlike depressions between the present river banks. They are generally full of water during the rains, when they are filled by the floods of the Ganges and its affluents, and most contain some water even in the cold weather. In those which dry soonest, the ground, which retains abundant moisture, is cultivated with winter rice; in others a precarious crop of early rice is raised before they are again flooded, or a long-stemmed variety is sown broadcast which rises with the water and is reaped from boats. They are often of very large size, one in Saran having a length of 20 miles and attaining a breadth of 2 to 5 miles.

36. These marshes, which confine the limits of human habitation, are mostly a legacy of the rivers debouching from the Himalayas. The vagaries of the same rivers are to-day a barrier to the expansion of cultivation. The most destructive is the Kosi, which is remarkable for the rapidity of its stream, the uncertain nature of its channel and the desolation caused by its floods. Between 1850 and 1875 this river, swinging to the west, cut into and overspread some 20 miles of country, turning fertile fields into wilderness of sand and swamp. Nathpur, a great grain mart in Bhagalpur, was swept away in 1875, and two fertile *parganas* in the north-east of that district have lost a considerable portion of their cultivable area. Similar devastation has taken place along both banks of the river down to its confluence with the Ganges, and, even in tracts not affected by its inroads, the fear of its movements has driven back cultivation, the place of which has been taken by grass jungle.

37. In Muzaffarpur the greatest density is found in the Sitamarhi subdivision to the north, where every thana contains over 1,000 persons per

square mile. Throughout the rest of the district the pressure on the soil is uniformly heavy, all but two thanas having over 900 per square mile. The Samastipur subdivision of Darbhanga, with 938 persons to the square mile, has as dense a population as Muzaffarpur. This subdivision, moreover, contains about 40 square miles of *diara* land, which is practically uninhabited and mainly uncultivated, so that the density on the inhabited and productive area is very little less than 1,000 per square mile. In the Madhubani and Sadar subdivisions the ratio varies from 1,065 to 696 per square mile; these are mainly rice-producing tracts, which cannot support so large a population as Samastipur, where the rich uplands produce more valuable crops. The population is evenly distributed throughout Saran, only two thanas having less than 800 persons per square mile, and only one having a ratio exceeding 900. The Siwan subdivision, with 905 persons to the square mile, is now the most populous part of the district, having changed places with the head-quarters subdivision during the last decade. The most sparsely inhabited is the Gopalganj subdivision, which contains a large area of *diara* land and numerous *chaurs* or rice swamps: notwithstanding this, it has over 800 persons per square mile.

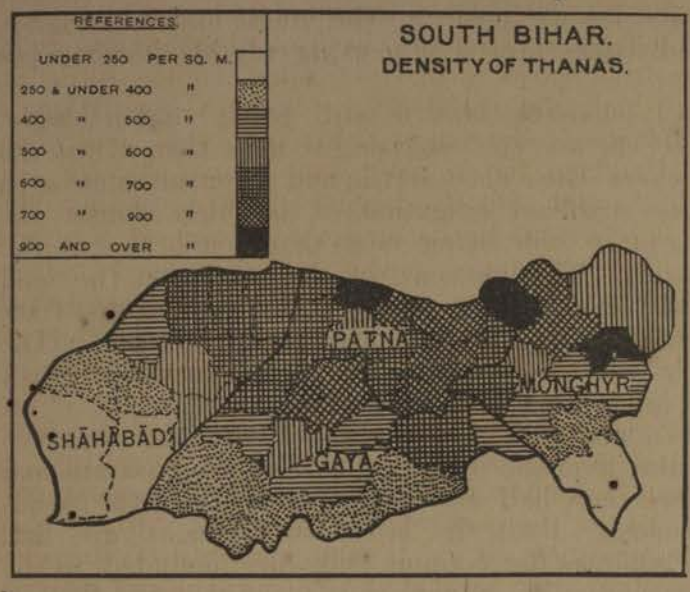
38. Compared with the other districts of the Tirhut Division, Champaran has but a thin population. Not only is the proportion of uncultivable land (17 per cent.) higher than elsewhere, but a large part of the cultivable area still awaits development. Pasturage rather than cultivation has engaged the energies of its inhabitants until fairly recent times. Partly for this reason and partly because of the prevalence of malaria, which saps their energies, the people are but indifferent cultivators, and only two-thirds of the cultivable area has been brought under the plough. There is, moreover, a submontane strip in the north-west, where the Dun and Sumeswar hills cover an area of 364 square miles. These hills and the land at their base are still covered with forest, and further into the plain extend large grassy prairies, which are mainly used for pasturage. The Bettiah subdivision, in which this submontane tract lies, supports only 400 persons per square mile, or a little more than half the number found in the Sadar subdivision. Population is most dense in the eastern thanas of the latter subdivision, which adjoin the Muzaffarpur district and in which similar conditions obtain. These thanas (Adapur, Dhaka and Madhubani) are mainly under rice cultivation, and their average density is over 800. The central thanas, where there is a fair amount of waste land, have from 600 to 700 persons per square mile, while in the northern thanas the ratio is only a little over 300.

39. Bhagalpur is divided in two by the Ganges, and conditions differ considerably in the north and south. The north consists of alluvium as far as the limits of the Kosi floods, after which sandy flats appear. Till 40 years ago the country in the extreme north was covered with jungle or *sal* forest, which has now been cleared away. The land is mainly under rice cultivation and, being traversed by a number of rivers with numerous cross-channels, needs little irrigation. The highest density (641 to the square mile) is found in the Bhagalpur subdivision, which extends along both sides of the Ganges, and has a rich alluvial soil annually renewed by the fertilizing overflow of that river. Further south the surface gradually rises till a hilly tract is reached, in which artificial irrigation is necessary for the cultivation of rice. This broken country corresponds to the Banka subdivision, in which there are only 370 persons to the square mile.

In Purnea the average density of population (398) is less than in any other district in the Province outside Chota Nagpur; it exceeds 500 in only two thanas and falls below 300 in two more. This low density is due mainly to the unhealthiness of the climate and the extent of uncultivated land. The east of the district is a low-lying tract with shallow swamps, stagnant rivers and wide stretches of flooded land, which slowly dry up after the rains. The west is higher in level, and is thickly overlaid with sand deposited by the Kosi river in its gradual westward movement. The greater part is open pasture land, and crops are grown for the most part near the rivers and in irrigated plots close to the villages. Generally speaking, the northern and eastern portions of Purnea district are more thickly populated than the south and west.

40. In South Bihar the rivers do not affect the distribution of population to anything like the same extent as in North Bihar. The great rivers to the north of the Ganges are fed by the Himalayas ; with the exception of the Son those south of the Ganges have by no means so large a drainage area, for they drain only a portion of the Chota Nagpur Plateau. They often come down in freshets after heavy rain, and occasionally sweep over the country in their vicinity, but the area covered by the flood is comparatively small, and they subside as quickly as they rise. Large demands are made on them for irrigation, and within a short time after the rains they run dry. Away from the Ganges, marshes, lakes and water-logged areas are almost unknown. The deficiency rather than the excess of rainfall or river-borne water is the chief difficulty the people have to face. This difficulty is met partly by the Son canal system, from which the greater part of Shahabad and small portions of Patna and Gaya receive an assured supply, partly by an extensive and ingenious system of indigenous irrigation, the cultivators tapping the rivers so long as they have any flow, building artificial reservoirs (*aháirs*) and channels, and taking every advantage of the slope of the country to ensure the conservation of water.

South Bihar.



41. Hills and jungle, so rare in North Bihar, here impose the limits on human habitation. One-fifth of the total area of Shahabad is occupied by the Kaimur Hills, which form a rocky plateau, mostly covered by jungle and incapable of cultivation. Further east, in the south of Gaya and Monghyr, there are a number of ridges and spurs projecting from the plateau of Chota Nagpur, in addition to which there are semi-detached ranges, and isolated peaks which appear to form irregular links between them. Much of this southern tract consists of broken country with a fringe of brushwood jungle. The soil is poor, it has little or no irrigation, and it yields precarious crops. It is thinly peopled, and a large portion of the inhabitants are low semi-Hinduized castes, such as Bhuiyas, Rajwars and Musahars.

The northern portion is an alluvial tract highly cultivated, extensively irrigated and well populated. The difference between this alluvial tract and the southern portion may be realized from the fact that the southern thanas of the division, extending over 4,638 square miles, have an average density of only 268 per square mile, whereas in the thanas bordering on the Ganges the density is more than $2\frac{1}{2}$ times as great, averaging 734 in an area of 2,628 square miles.

42. In Patna, with 778 persons to the square mile, the pressure on the soil is 50 per cent. greater than in other districts of South Bihar. Along the Ganges there are a number of towns and crowded villages almost as populous as towns ; even Bihar, which is the only subdivision away from the Ganges, has a density of 732 per square mile, although a considerable area is covered by the Rajgir Hills. All the other districts of South Bihar have a far scantier population, owing to the hills and jungle in the south. Monghyr is the most populous of them, but the north of the district is an integral part of North Bihar, being a fertile alluvial plain intersected by large rivers and subject to inundation. It has a mean density of 715 per square mile, the maximum of 961 being found in the Teghra thana and the minimum of 562 in Gogri, which is a low-lying tract with a large area of marsh and grass jungle.

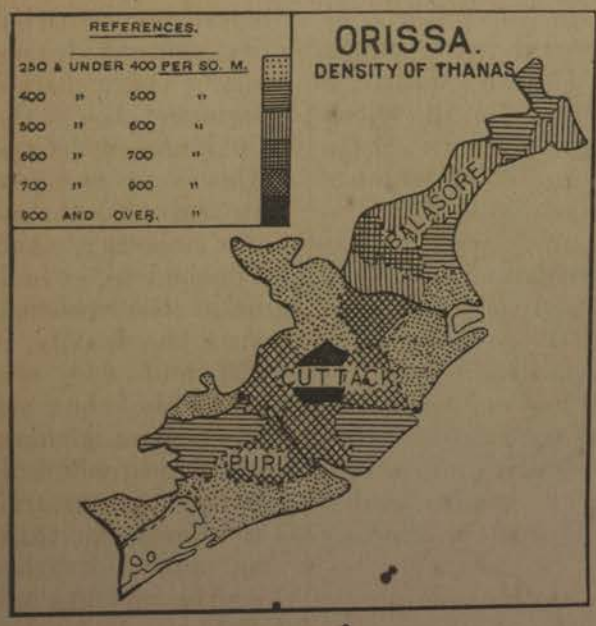
South of the Ganges there are, on the average, only 528 persons to the square mile. Here the most thickly populated tracts are the thanas lying along the Ganges, south of which there is hilly country, bearing a growth of scrub fungle, with a rocky surface unfavourable to cultivation. The Kharagpur thana with 415 persons to the square mile begins to show a marked diminution, which reaches its lowest point in the jungle and waste lands in the extreme south of the district, the soil supporting only 231 persons per square mile in Chakai.

43. In the south of Gaya, where the land is still partly under jungle, the population is very sparse, the average not rising to more than 278 to the square mile. In the north, where the soil is fertile and a considerable area is protected from drought by artificial irrigation, it is fairly dense, the number of persons to the square mile being more than double that in the less favoured tracts to the south. The pressure of the people on the land is greatest in the Jahanabad thana to the north, a highly cultivated tract where the large number of 712 persons to the square mile is found. The minimum (271) is reached in the Barachatti thana to the south, where a considerable area consists of hills, jungle and waste land.

In Shahabad there is an equally marked variation between the northern and southern subdivisions, the average density (305) in the Sasaram and Bhabua subdivisions being less than half of what it is (641) in the head-quarters and Buxar subdivisions. Both the latter subdivisions are flat, fertile and highly cultivated, whereas the Kaimur Hills are included in the two subdivisions first mentioned. The minimum is reached in the Bhabua thana, which, with 187 persons to the square mile, is the most sparsely inhabited tract in South Bihar : a large part of this thana lies on the plateau of the Kaimur Hills, where patches of cultivation are few and far between.

ORISSA.

44. In Orissa the average density is 508 to the square mile, but it varies very considerably in different parts according to their proximity either to the sea or to the hills. All three districts have three zones, the first being an unproductive maritime strip, the second a cultivated central plain formed of rich alluvium, and the third a broken hilly region on the west. The strip along the coast is in many places impregnated with salt, and a great part of it is unfit for cultivation. It is swampy and traversed by sluggish brackish streams, and from its general nature has been described as the Sundarbans on a miniature scale. The central portion, which forms the delta proper, is an alluvial plain with a teeming population and a fertile soil. The third belt consists of a submontane undulating country, in places broken by hills, with a sterile soil which supports a scanty semi-Hinduized population.



45. The greatest density in Cuttack is found in the thanas of Salepur, Jajpur and Kendrapara, where the average is 800 per square mile. The maximum of 963 is reached in Salepur, which lies between two large branches of the Mahanadi and has 31 per cent. of its area irrigated from the Orissa canal system. All the thanas along the sea-coast to the east and in the hilly tract to the west have under 400 to the square mile, the most thinly populated thana being Aul with 308 to the square mile. This thana contains a belt of saline soil, in places 30 miles wide, covered by sand, coarse grass or shrub. Taking the district as a whole, the density per cultivated square mile is

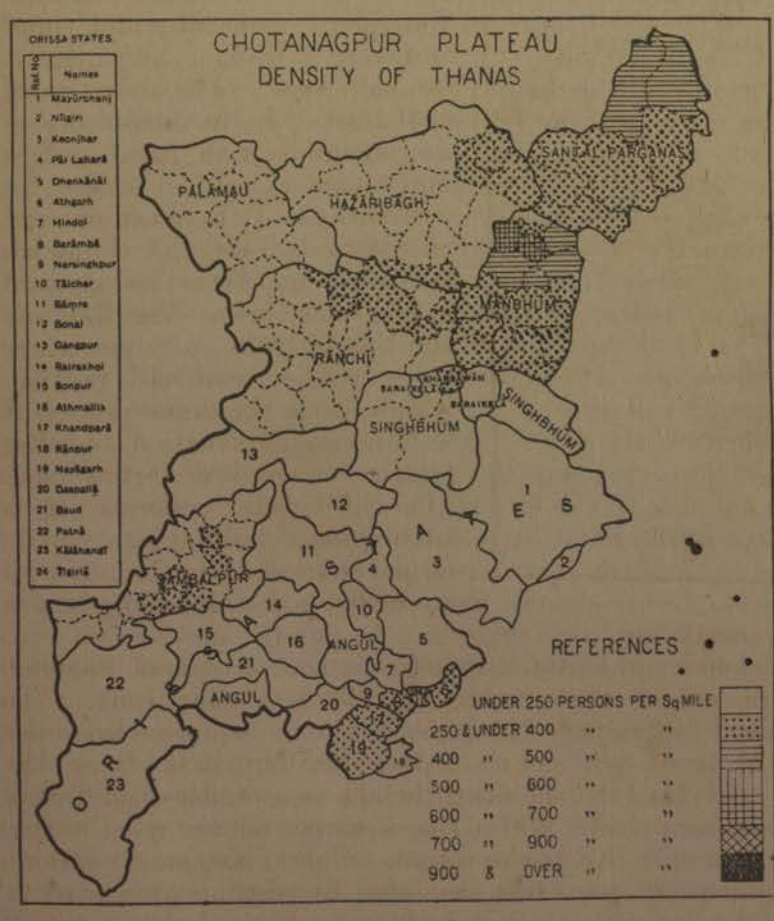
1,099, but it has been calculated that each square mile of unirrigated land would support 1,167 persons, and each square mile of irrigated land 1,515 persons.

46. Balasore at its greatest width is only 40 miles broad from east to west, and every thana contains either saline soil or uncultivated laterite. Density is as low as 288 persons to the square mile in Chandbali, a large part of which is a prairie of high grass merging on the sea-coast in a mangrove forest like that found in the Sundarbans. It is as high as 638 in thana Bhadrakh, which contains only a little hilly country and marches with the fertile central plain of Cuttack. Though it contains the head-quarters of the district, Balasore thana supports only 536 persons to the square mile. the explanation being that part of it is taken up by the maritime saline strip on the east and the undulating tract on the west. The average density (410 persons to the square mile) in Puri is far less than in any other district of Orissa, owing to the area occupied by hills, forests and water. Reserved and protected forests extend over 485 square miles, while the area of the Chilka Lake is about 450 square miles during the rains and 350 square miles for the remainder of the year: altogether 310 square miles of this lake lie in the district. Pipli (including Nimapara) in the centre of the district, which is the only thana without any hills or maritime area, contains as many as 760 persons to the square mile; the average density (349) in the rest of the district is less than half this figure. There are under 300 persons to the square mile in the Puri and Banpur thanas, of which former contains the Chilka Lake, while the latter comprises a large hilly area.

CHOTA NAGPUR PLATEAU.

47. The Chota Nagpur Plateau is an upland tract, with a general elevation of 2,000 feet and over, forming part of the descent from the elevated high lands of Central India. It includes the inner highlands of Orissa, as well as Chota Nagpur proper, and consists of a succession of plateaux, hills and valleys. Cultivation is comparatively sparse; where not under cultivation, the plateaux and their escarpments are thinly covered with forests. The average density is 186 per square mile.

48. The most populous district is Manbhum, which has changed places with the Sonthal Parganas during the last decade owing to the development of the Jheria coal-fields. In the two thanas of Jheria and Topchanchi, which lie in the coal-field area, the density is 667 and 644 respectively, but in the adjoining thana of Tundi, which is purely agricultural, it is only one-third as much. Further south, in Chas and Raghunathpur, where there is more level ground than elsewhere, the facilities for cultivation have attracted permanent settlers, and there are 498 and 442 persons respectively to the square mile. Nirsa, which contains part of the Raniganj coal-field, and Para, in the centre of the



district, are the only other thanas with over 400 to the square mile. Generally speaking, density diminishes from north to south, if the northernmost thana (Tundi) is excluded.

49. The Sonthal Parganas, unlike Manbhum, sends out emigrants and has no large industries to attract labour. Only half the district is under cultivation; a remaining fourth is cultivable and awaits development, but the rate of reclamation is not commensurate with the growth of population, for year by year the Santhals pour out of the district in increasing numbers to do pioneer work elsewhere. The Rajmahal Hills and outlying ranges extend over nearly two-fifths of the district, and there is a long narrow strip of alluvial soil between them and the Ganges on the north and east. It is in this strip and in the portion of the Godda subdivision to the north-west, where the land below the hills is alluvial and fertile, that the population is most dense. In the Damin-i-Koh, which comprises almost the whole of the Rajmahal Hills and consists of hills, plateaux and fertile valleys between the ranges, there are only 284 persons to the square mile, the average for the rest of the district being 360: the marginal statement shows the density according to subdivisions. In the Rajmahal subdivision all the thanas outside the Damin lie in the alluvial belt, one containing the town of Sahebganj. In the Pakaur subdivision the Pakaur and Maheshpur thanas lie partly in this belt and partly in the rolling uplands known as the Sonthali tract of Ambar and Sultanbad, which also contains the whole of Pakuria. In the Godda subdivision density is greatest in the Godda and Mahagama thanas, where the country is more open and mostly under cultivation. The latter thana, which is an alluvial tract, is the most populous in the district next to Sahebganj. Poreya is more hilly, and its density is only a little greater than that of the Godda Damin. In the south and south-west the hills give place to a series of ridges and undulating uplands, with a scanty population. This latter area contains the Deoghar subdivision, which consists chiefly of a high tableland, much of which is of little agricultural value, and the Jamtara subdivision, the least populous of all the subdivisions, in which there is still a good deal of uncleared jungle.

SUBDIVISION.	Damin.	Extra Damin.
Dumka	159	309
Godda	305	500
Deoghar	...	322
Jamtara	...	297
Pakaur	256	488
Rajmahal	332	631

50. The average density in Ranchi and Sambalpur is exactly the same, viz., 195 to the square mile. In the former district it diminishes from the north-east to the south and south-west, the Ranchi subdivision having 256, the Khunti subdivision 226 and the Gumla subdivision only 146 persons to the square mile. In Sambalpur the unsurveyed area (*i.e.*, the Government reserved forests, the zamindari forests and the Mahanadi river) account for about a fifth of the total area: if it is left out of account, the average density is 249 to the square mile. Population is most sparse in the Bargarh subdivision; the Barapahar hills extend over 300 square miles or more than one-eighth of the subdivision, and a considerable area is covered with jungle, whereas in the Sambalpur subdivision there are large expanses of open cultivated country along the Mahanadi. Excluding the unsurveyed tracts, the former subdivision supports 260 and the latter 235 persons per square mile, the difference being due to the fact that in the Bargarh plain the land is under closer tillage than in the Sambalpur subdivision.

51. The population is unevenly distributed in Hazaribagh, the density in the Giridih subdivision, where there are coal-fields, being nearly 50 per cent. greater than in the Sadar subdivision, which is purely agricultural. A slight decrease is found in Singhbhum, owing to the protected and reserved forests, which extend over more than one-fourth of the whole district. If they are left out of account, there are 241 persons to the square mile. In Palamau the population is fairly dense in the valleys and in the north of the district, but gradually decreases to the south, which is a region of hills and jungle. The extreme southern thana, with 64 persons per square mile, is the most thinly populated tract in the whole of the Chota Nagpur Division. Angul supports a smaller population than any other district in the province, being still an undeveloped tract with scattered villages, often in the midst of dense forest. There are 142 persons to the square mile in the Angul

subdivision, where reserved and protected forests extend over 613 square miles, or nearly three-fourths of its area. The average is only 93 in the Khondmals, where five-eighths of the area is under forest.

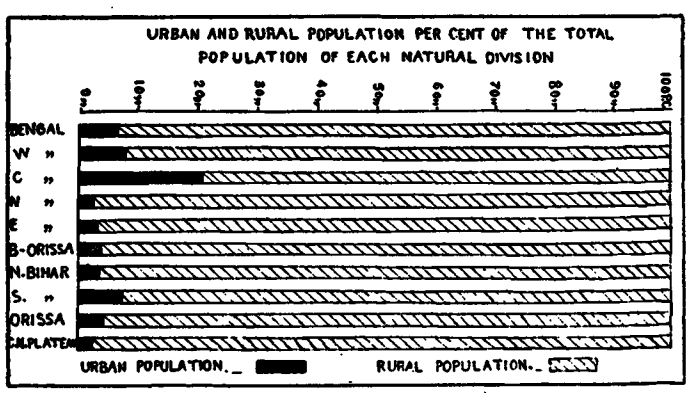
52. The Orissa Feudatory States, as a whole, have a denser population than any district in the Chota Nagpur Plateau, except Manbhum and the Sonthal Parganas. Extraordinary variations, however, are found, the maximum being 505 in Tigiria, which is a fertile tract with easy means of access to Cuttack, while a minimum of 38 is found in Rairakhol, which is still mostly covered with forest. Generally speaking, the States adjoining the sea-board districts have the greatest population, eight of them having over 200 persons to the square mile. The only other State exceeding that ratio is Sonpur, which is traversed by the Mahanadi and includes a large area of alluvial land on either side of its banks.

SIKKIM.

53. At the bottom of the list is Sikkim, with only 31 persons per square mile. This is not to be wondered at, considering the physical configuration of the country. From 15,000 feet and upwards there is a mass of snow-topped peaks and ridges, treeless and uninhabited. From 12,000 to 15,000 feet the ridges are clothed with rhododendron and coniferous forests: occasional grassy plateaux with small lakes are found, to which cattle are driven for pasturage in the summer. The hill slopes from 9,000 to 12,000 feet are very steep and usually covered with virgin forest, but scattered settlements of Bhotia graziers now begin to appear. It is only at the lower levels below 7,000 feet that cultivation is possible. Here the country has been largely denuded of forest, and the slopes utilized for the growth of the staple crops, viz., maize, millets and pulses, the people living in small homesteads surrounded by patches of cultivation.

CITIES AND TOWNS.

54. The places treated as towns consist of three classes, viz., municipalities, cantonments and other places which were treated as towns for census purposes. In deciding the places which were to be included in the last category, their importance as centres of trade, their historic associations, the character of their population and the relative density of dwelling-houses were taken into consideration. Altogether, 22 places in Bihar and Orissa, other than municipalities or cantonments, were treated as towns, of which four had a population over 10,000, twelve contained 5,000 to 10,000 inhabitants, and the remainder had under 5,000 inhabitants. Eight, with an aggregate population of 45,031, were treated as towns for the first time. In Bengal six places were considered sufficiently urban to be treated as towns, one (Kharagpur) having a population of 18,957, another (Saidpur in Rangpur) of 8,287, and the remaining four (all in Rangpur) under 5,000 inhabitants. All but two of these were treated as towns in 1901; the two additions were Kharagpur and Dhulian, the aggregate population of which is 27,255.



55. In neither Province have the people shown any appreciable tendency to desert the villages for the towns. In Bengal the urban population has increased by 13·2 per cent. since 1901, its growth being more rapid than that of the general population, but only 64 out of every thousand persons live in towns. In Bihar and Orissa, on the other hand, an increase in the general population has been accompanied by a decrease of 2·8 per cent. in the number of persons living in towns, who now constitute only 34 per mille of the total population. This decline is, as will be shown later, due mainly to the continued presence of plague in Bihar,

which has not only caused a grievous mortality, but disorganized the industries and trade of the towns it afflicted. *Primâ facie* the Bengalis appear to have a greater predilection for town life than the inhabitants of Bihar and Orissa. Bengal contains 124 towns, whereas Bihar and Orissa contains only 76, *i.e.*, less than two-thirds of the number in Bengal, though its area is one-third greater. The average town population (23,937) of Bengal is more than one-third as great as the average in Bihar and Orissa, and the proportion of townspeople to the total population is nearly double what it is in the latter Province. Calcutta, Howrah and three suburban municipalites, viz., Maniktollah, Cossipur-Chitpur and Garden Reach, contain two-fifths of the urban population. If they are left out of account, the average population of the Bengal town is reduced to 14,672, while the proportion of the urban population to the total population of the Presidency is only 38 per mille, or little more than in Bihar and Orissa.

56. In Bengal there are only three towns containing over 30,000 persons outside Calcutta and the metropolitan districts of Howrah, the 24-Parganas and Hooghly. The first two districts are the most distinctively urban, over one-fifth of their inhabitants being resident in towns, while the proportion is over one-tenth in Hooghly, where there are seven municipalities stretching along the bank of the Hooghly from Tribeni southwards. Of the other districts in the Province, Darjeeling with 93 per mille has the largest urban population, and then *longo intervallo* comes Dacca, where two towns, Dacca and Narayanganj, contain 46 per mille of the district population.

57. In Bihar and Orissa, 13 out of 21 districts have an urban population of over 25 per mille, whereas in North and East Bengal this ratio is reached in only five out of 17 districts. The most distinctively urban tract is South Bihar, where 67 per mille of the population are found in towns; then follows Orissa with 38 per mille, and North Bihar with 28 per mille. The average of the Chota Nagpur Plateau (20 per mille) is largely reduced by the Orissa States; in the Chota Nagpur Division the proportion (26 per mille) is very little less than in North Bihar, which contains an agricultural population and has few towns. Its towns are, however, of considerable size, their average population being 21,145, or very little less than in South Bihar. Even larger towns are found in Orissa their average population amounting to 26,585, a figure higher than in any division of the two Provinces except Central Bengal.

58. The two Provinces dealt with in this report contain eight towns which have at one time or other been imperial or provincial capitals, viz., Bihar, Nadia, Patna, Rajmahal, Monghyr, Cuttack, Dacca, Murshidabad and Calcutta. The oldest of these is Bihar, which gave its name to the Province, and was so called from the great Buddhist monastery (*Vihara*) it contained. It was the headquarters of the Hindu Governors of the Pala Kings, but in 1198-99 the city was sacked, the monastery burnt and the Buddhist monks slain by Bakhtiyar Khilji. A sudden raid was made next year on Nadia (Navadwip), then the Bengal capital of the last of the Sena kings. After this, it is said, the Musalmans, leaving Nadia in desolation, removed the seat of Government to Lakhnauti (Gaur). Bihar appears to have been the headquarters of the Musalman Governors of Bihar until 1541, when Sher Shah rebuilt Patna, which, says the *Tarikh-i-Daudi*, "was then a small town dependent on Bihar, which was the seat of the local Government. From that time Patna became one of the largest cities of the Province." Thenceforward the Mughal Governors of Bihar usually resided at Patna, but the western portion of that Province was under the rule of the Nawabs of Bengal. The latter made Rajmahal their capital from 1592 to 1608, when the seat of Government was transferred to Dacca for strategic reasons, Dacca being a more central position for the defence of Bengal against the raids of the Assamese, Arakanese (Maghs) and Portuguese. In 1639, Shah Shuja again made Rajmahal the capital, but this change did not last for more than 21 years, for in 1660 it was found necessary to re-establish the headquarters at Dacca owing to the continued danger of invasion and also because the Ganges had shifted its channel and receded from Rajmahal. Dacca remained the capital for less than half a century, Murshid Kuli Khan transferring his headquarters to Murshidabad

OLD CAPITALS.

in 1704. The Arakanese had now ceased to be dangerous, and the city along the banks of the Bhagirathi, commanding the trade and military route up the Ganges, was far more central. There the Nawabs of Bengal remained till the downfall of the Mughal power, and the assumption of rule by the British, except for a short interval (1761–1763), when Kasim Ali Khan set up his court at Monghyr. In Orissa, Jajpur, now a small rural town, appears in early times to have been the capital of the north, and Bhubaneswar, now a village surrounded by temples or their ruins, the capital of the south. Cuttack, however, became the capital under the Eastern Ganga kings, and retained its position till the British conquest in 1803. The last capitals to be dethroned are Dacca, the headquarters of the short-lived Government of Eastern Bengal and Assam (1905–1912), and Calcutta, which is no longer the official capital of India owing to the removal of the Government of India to Delhi.

59. With the exception of Cuttack, Dacca and Calcutta, all these towns are decadent. Bihar is now a quiet provincial town far from the main routes of commerce. The old town of Nadia has been swept away by the Bhagirathi, and its modern successor is an unimportant rural municipality. Patna has been decimated by plague and lost much of its trade. Monghyr is a district headquarters, only the remains of its fort and palace (now a jail) attesting its former importance as a provincial capital. Rajmahal is a small country town with 5,357 inhabitants, mostly residing in mud huts, the ruins of the nobler edifices of the old city being buried in jungle.* Murshidabad still contains the palace of the descendant of the Nawabs, but otherwise has few traces of its former grandeur, while its population is but a fraction of what it was. One or other of the causes which operate to bring about the decay of old towns in Bengal have, either singly or together, helped to bring about the downfall of these capitals. They are briefly the action of rivers, the removal of native courts, loss of trade, and the gradual or sudden diminution of population owing to disease.

ACTION OF RIVERS.

60. Nearly all the old towns were built on the banks of rivers, a riparian site being naturally selected, both because it ensured the supply of drinking water and also because the rivers furnished the principal means of transport and communication: there were a few military routes, but otherwise roads were few and far between. Their situation has in many instances proved precarious to the old towns owing to the vagaries of the rivers. Pataliputra, the imperial capital of Asoka, lies buried deep below the modern city of Patna; Tamralipta, the ancient port of Bengal, has been covered by the silt of the Rupnarayan, while Tamluk, which was built over it, is now a riparian village 60 miles from the sea. Satgaon, once the headquarters of a Governor and a city crowded with merchants, sank into insignificance owing to the diversion of the rivers that fed its trade, and at the present day is only represented by a few huts scattered among jungle-covered mounds. Gaur, the capital of Bengal for over seven centuries, was ruined by the Ganges receding westwards, leaving long shallow marshes behind it. Fever followed and depopulated the city, the final epidemic of 1575 being so terrible that the dead could be neither buried nor burnt, after which the few survivors fled from the place. Within half a century the population of 200,000 described by Portuguese travellers* had disappeared. The country was almost a wilderness with few villages, but many buffaloes, swine and deer, and "very many" tigers.† Rajmahal was similarly deserted on account of the Ganges changing its course; in 1640 its current washed the walls of the city, but in 1666 the channel was, according to Tavernier, a good half league away.

61. The ruin of a riparian town may be either sudden or gradual, according as the river quickly or slowly changes its course or gradually dwindles away. The former is a somewhat rare occurrence, but two cases may be mentioned. Nadia, the old capital of the Sena Kings, was swept away by a sudden change in the course of the Bhagirathi in the beginning of the nineteenth century. Its site is now partly *char* land and partly forms the bed of the stream, which passes to the north of the present town. In the

* De Barros, *Da Asia.*
† J. H. Ryley, *Ralph Fitch* (1899).

north of the Bhagalpur district, again, the town of Nathpur was destroyed by a sudden inrush of the Kosi in 1875. More frequently a river gradually changes its course, and an attenuated stream flows down its old channel, the towns along its banks lingering on with a gradual loss of prosperity. Such has been the case with Purnea, which in the eighteenth century was the headquarters of a Military Governor who could put an army of 15,000 men into the field, and which only 100 years ago is said to have covered a space equal to more than half of London.* Formerly the main stream of the Kosi flowed by it, but that river has worked westward and its former channel contains only a small sluggish stream. The town still extends over a large area, but its population is gradually diminishing and is now only 14,784.

In the lower delta it is more common for a river to keep to its old channel, but the silting up of its intake deprives it of a supply of fresh water, and it consequently shrinks in volume and generally deteriorates. It ceases to have a flowing stream, and its bed being choked with vegetation, navigation is rendered impossible, while the drainage of the country is obstructed and malarial fever spreads over the neighbourhood. Such rivers are either dying or dead, and the towns along them have no vitality. Their inhabitants do not desert them, but their fecundity is sapped by sickness, their industries languish, and there is nothing to induce an influx of immigrants. Jessore and Krishnagar may be regarded as typical instances of such towns. Both are headquarters of districts and are situated on the railway, factors which should make for growth, but both stand on dead or dying rivers (the Bhairab and Anjana). Both suffer from persistent endemic fever, and the population of Jessore has been stationary for the last 40 years, while that of Krishnagar is steadily diminishing.

INFLUENCE OF COURTS.

62. In other cases the decline of a town is due to the removal of the Court, to which it owed its prosperity. When the Court is removed, the entourage of nobles, their train of followers, and the industrial classes, which ministered to their luxuries, also leave the place. The traces of old industries may be found, but these industries serve only the demands of a few rich persons. Murshidabad is a type of such a town. On entering it after the victory of Plassey, Clive wrote :—"This city is as extensive, populous, and rich as the city of London, with this difference that there are individuals in the first possessing infinitely greater property than in the last city." After that battle the seat of Government was removed to Calcutta, and in 1799 the civil headquarters of the district were transferred to Berhampore. With the loss of its political importance, the size and population of Murshidabad also declined. At the first census of 1872 its population was 46,182 ; now the inhabitants of Murshidabad and its suburb Azimganj (which was formed into a separate municipality in 1896), taken together, number only 24,996.

LOSS OF TRADE.

63. A third cause tending to check the development of towns in the two Provinces is loss of trade. This may be caused, as already stated, by the removal of a Court or by a river changing its course or silting up; but in modern times it is generally due to trade being rail-borne instead of river-borne. The effect of such a diversion of trade is best illustrated by the case of Patna and Revelganj (in Saran). Patna is marked out by nature as the site of a riparian emporium, for it has a river frontage of 7 or 8 miles in the rains and of 4 miles in the dry season, while its central position near the junction of three great rivers, the Ganges, the Gandak and the Son, gives it natural advantages as a distributing centre. The trade of the city has now diminished owing to the opening out of new lines of railway in the districts north of the Ganges, and to the fact that it is cheaper to book goods direct to Calcutta than (as formerly) to rail them to Patna, and thence send them down by river. The natural advantages of Revelganj as a river-side emporium were equally marked. Commanding as it did the junction of the Gogra and Ganges, it was an important changing station, where boats from Bengal used to tranship their cargoes to boats from Fyzabad and Gorakhpur. Thirty years ago it was the second largest river mart with a Ganges-borne trade in

* Montgomery Martin, *Eastern India.*

Bengal. The railway has now deprived it of most of its trade, and it has also suffered from the river setting towards the opposite bank and from the retreat eastwards of the point of junction of the Ganges and Gogra. Of late years moreover it has suffered from plague, and since 1891 it has lost over a third of its population. In Bengal the history of Kalna and Katwa (in the Burdwan district) is a similar record of decay due to the diversion of trade from the river to the railway.

ENDEMIC AND EPIDEMIC DISEASE.

64. The fourth factor is endemic and epidemic disease. In some parts, notably Central Bengal, the prevalence of malaria is responsible for decline or stagnation; in Bihar plague has been a more deadly scourge. An account of its ravages will be given in the next chapter, and here it will be sufficient to refer the reader to the marginal statement showing the deaths from plague and the total decrease of population during the last decade in 26 towns.* Not only is the number of deaths from plague considerable, but the disorganization of industry, commerce and social life has permanent effects on the prosperity of the towns. In Gaya, for instance, 30 per cent. of the population had left their work and fled from their homes when the census was held. Normal conditions were not re-established till three months later: the result of their absence on the trade of the town even for a short time may easily be imagined. It is scarcely necessary to add that, where plague persists year after year, as has been the case elsewhere, its effects are even more paralyzing.

DISTRICT.	Number of towns.	Population, 1901.	Decrease of population, 1901-11.	Deaths from plague, 1901-10.
Patna ...	5	252,791	26,617	35,309
Gaya ...	3	87,469	9,093	2,033
Shahabad ...	6	118,047	13,634	15,664
Saran ...	3	71,422	12,234	8,243
Muzaffarpur ...	3	78,517	8,210	6,552
Darbhanga ...	4	103,392	6,336	6,873
Bhagalpur ...	2	81,498	4,249	2,852
Total ...	26	793,136	80,373	77,526

65. If deaths from plague are left out of account, the death-rate in towns is smaller than in rural areas, though heavier mortality is caused by epidemics of cholera and small-pox, which spread more rapidly in congested areas. The relatively greater healthiness of towns (excluding epidemics) is due to the fact that it is only there that a proper agency and other facilities exist for carrying out sanitary reforms. These reforms, carried out as far as municipal finances allow, have resulted in a gradual improvement in hygienic conditions, especially in matters connected with surface drainage purity of the drinking water-supply, and conservancy. From the marginal statement† giving the vital statistics in urban and rural areas for five years, it will be seen that in the towns the death-rate for fever is far lower, but that for dysentery, diarrhœa, and respiratory diseases is higher: this may be ascribed to more accurate reporting of the causes of death. On the other hand, the birth-rate in towns is generally lower than in rural areas owing to the operation of one or more of the following causes:—(1) The disproportion in the sexes of the inhabitants, males being in excess of females; (2) the presence of a large floating population; and (3) the custom of sending females to be confined in their parents' homes in the villages.

AVERAGE DEATH-RATE PER MILLE.

	Urban.	Rural.
All causes	37·26	35·41
Fever	12·80	22·85
Plague	6·06	1·32
Cholera	3·37	3·09
Small-pox	0·63	0·35
Dysentery and diarrhœa ...	2·93	0·80
Respiratory diseases	1·60	0·12
Injury	0·58	0·48
Other causes	9·26	6·38

DEVELOPMENT OF MODERN TOWNS.

66. After the somewhat dreary sketch of urban decay, stagnation or decimation by disease given in the preceding paragraphs, it is refreshing to turn to the number of towns, some old, some young and some nascent, which are fast

* In this statement, when a second census was held in 1901 after a plague epidemic had subsided and the people had returned to their homes, the figures of the second census have been taken, as representing the normal population in that year.

† Report of the Sanitary Commissioner for Bengal for 1908.

developing owing to the expansion of trade or industrial enterprise, often introduced and directed by Europeans. The cities will be separately dealt with later, and are therefore excluded from this review of the growth of railway towns, mill towns and commercial emporia. There are three towns in the two Provinces, to which the term railway town applies, and each shows rapid progress. Jamalpur in Monghyr, which contains the large workshops of the East Indian Railway, had an increase of 14 per cent. between 1891-1901, and at this census, in spite of losing 2,000 persons from plague, is almost the only town in Bihar with a substantial increase (8 per cent).* Kharagpur in Midnapore, the headquarters of the Loco., Carriage and Waggon Departments of the Bengal-Nagpur Railway, now has 18,957 inhabitants, whereas in 1901 there were only 3,526 persons resident in railway premises. The population of Saidpur in Rangpur, which is the headquarters of the northern section of the Eastern Bengal State Railway and is also a centre for the trade in jute, has risen by 42 per cent. within the last 10 years.

67. Even more phenomenal increases are recorded by the mill towns. In the 24-Parganas, for instance, Bhatpara, which was formerly famous as a centre of Sanskrit learning, where Brahman pandits studied the Sastras in quiet seclusion, is now the fifth largest town in Bengal and a busy industrial centre. Its population has increased five-fold since 1881, and has been more than doubled during the last decade, rising from 21,540 to 50,414. This increase, however, is merely commensurate with the increase of mill hands. The expansion of other mill towns along the Hooghly is equally remarkable, the aggregate population of seven in the 24-Parganas having risen by 87 per cent. since 1901. Titagarh has nearly trebled its population. Garulia has an addition of 57 per cent. and Bhadreswar on the other side of the Hooghly of 61 per cent. The whole riparian strip along the Hooghly is, in fact, becoming increasingly urban and increasingly populous, and the account given by Mr. Beverley in the Census Report of 1872 no longer holds good. He remarked—"Even in the neighbourhood of Calcutta the so-called townships are mere collections of villages—villages closely studded and densely populated, it is true, but still with small pretensions to be designated towns. The left bank of the Hooghly, like the right, is most thickly inhabited all the way up to Nadia. The villages are grouped together for municipal purposes, and are thus shown in the census tables as towns; but cattle graze, and rice is sown and reaped, in their very midst." Any one who has had occasion to pass through the mill towns will realize how conditions have changed since this account was written.

The latest addition to the list of industrial towns is Sakchi in the Singhbhum district. This is a town which has sprung up owing to its being the headquarters of the Tata Iron and Steel Company: though the latter has only recently started work, it already contains a population of 5,672 persons.

68. The third class of modern towns showing a noticeable development consists of trade centres. Their number is considerable, but two typical examples may be mentioned, viz.; Chittagong, which has an increase of 30 per cent. since 1901, and Narayanganj, the centre of the jute trade, which has been growing by leaps and bounds since 1872: its population had more than doubled in 1901, and since then has increased to 27,876, or by nearly 14 per cent. This is by no means its fullest population, for it is much more crowded during the jute season than at the time of the census (in March), when trade is slack.

69. In both Provinces there is an excess of males living in towns.

PROPORTION OF SEXES.

The preponderance of men is much more marked in Bengal than in Bihar and Orissa; in the former Province there are only 631 females to every 1,000 males, whereas in the latter the proportion of females (932) is only slightly in defect. The reason for the relative excess of males in Bengal is partly that, in the Presidency as a whole, males outnumber females, there being 945 females to every 1,000 males. The reverse is the case in Bihar and Orissa, where there are 1,043 females to every 1,000 males. It is also partly accounted for by the fact

* The area of this town has been very largely extended since 1901; the figure given above is for the town as now constituted.

that there are more industrial centres with a population largely composed of immigrants working in the mills and factories, who leave their wives at home. The larger the town, the greater is the excess of males over females, *e.g.*, in the cities of Calcutta and Howrah there are only 475 and 562 females respectively to every 1,000 males. The proportion of women gradually rises as the towns diminish in size, the maximum (798) being found in towns with 5,000 to 10,000 inhabitants, which are mostly small municipal areas with few urban characteristics. Briefly, in the small towns of Bengal there are 5 males to every 4 females, whereas in the cities there are 2 males to every female.

The proportion in the mill and factory towns is much the same as in the cities. In the mill-towns of the 24-Parganas, such as Bhatpara, Garulia and Titagarh, the males outnumber females by two to one, whereas in non-manufacturing towns, such as Krishnagar, Nadia and Santipur, the sexes are equally represented, or the female element predominates. In the old city of Dacca, which contains a large permanent population—nine-tenths of the present inhabitants were born within the Dacca Division—there are 721 women to every 1,000 men, but 10 miles away in Narayanganj, a comparatively new centre of trade and industry, the proportion of females falls to 488, and in Barisal, the headquarters of the Backergunge district and an important junction of steamer routes, it is only 451.

In Bihar and Orissa the increase in the proportion of females as the towns decrease in size is even more noticeable, for, with the exception of Patna (where there are 922 females to every 1,000 males), the ratio gradually rises from 900 in towns with 50,000 to 100,000 inhabitants to 992 in towns containing under 5,000 inhabitants.

RELIGIONS.

70. In Bengal 67 per cent. of the urban population consists of Hindus and 30 per cent. of Muhammadans, whereas in Bihar and Orissa the proportions are 75 and 22 per cent.. respectively. Considering that in Bengal Musalmans account for over one-half of the total population, the percentage they contribute to the urban population is extremely small. In Bihar and Orissa on the other hand, Musalmans represent only one-tenth of the total population, but their ratio to the urban population is more than double that figure. The larger proportion of Hindus in Bengal may be accounted for by the fact that the population of towns is recruited very largely either from the educated classes, for whom a town opens up avenues of employment, or provides educational facilities not existing in the country, or from well-to-do people who desire the amenities of town-life, or from poor people who are drawn to the towns by the demand for labour. In the Presidency Hindus contribute to these three classes more largely than Muhammadans.

There are altogether 14 towns in Bengal in which Musalmans are in a majority, viz., Garden Reach, Murshidabad and Dhulian in Central Bengal, Pabna, Sirajganj and Nawabganj in North Bengal, and Sherpur (in Mymensingh), Kishoreganj, Netrakona, Jamalpur, Comilla, Patuakhali, Chittagong and Cox's Bazar in East Bengal. Even in Dacca city there are 55 Hindus to 44 followers of the Prophet. In Bihar and Orissa, the most distinctively Muhammadan towns are Sasaram, in which 42 per cent. of the population is Muhammadan, and Bihar, in which the percentage is 36. There are no towns in which Musalmans are in excess, and even in Patna, an old Muhammadan capital, and in modern times the nidus of Musalman religious movements, the Muhammadan inhabitants represent only 27 per cent. of the population.

71. If we consider the ratio of the members of each religion living in towns to the total number professing that religion, the preponderance of Musalmans over Hindus in Bihar and Orissa, and *vice versâ* in Bengal, is equally striking. In Bengal, out of every 1,000 Hindus, 95 live in towns, but in Bihar and Orissa only 31. On the other hand, 80 out of every 1,000 Musalmans are denizens of towns in the latter Province as compared with 36 in Bengal. A similar disproportion is noticeable in the case of Christians, the ratio being 478 per mille in the Presidency and only 72 in Bihar and Orissa. In the latter most of the Christians are aboriginal villagers in Chota Nagpur who cling to their ancestral lands. whereas in Bengal a large proportion of the Christians are resident in Calcutta and other large

towns, where missionary enterprise has been active. Members of other religions are numerically insignificant, and it need merely be mentioned that, as might be expected, most of them Parsis, a foreign race of merchants and traders, are found in towns. The fact that a small proportion of them and a larger proportion of Jains, who are mainly Marwari immigrants engaged in trade, are resident outside towns, is due simply to the circumstance that both Provinces export raw materials, and a certain number have to go to the villages as brokers or set up small agencies there.

72. In Imperial Table V, and in subsidiary Table V to this Chapter, towns are grouped in six classes as shown in the margin. In Bengal there are three towns in class I, viz., Calcutta, Howrah, and Dacca, which contribute two-fifths of the total urban population. More than half the towns are of average size, 28 being in Class III and 40 in Class IV; they account for nearly half of the urban population. The small towns (34 in Class V and 17 in Class VI) contribute only one-tenth of the total number of persons living in towns. In Bihar and Orissa there are only one town in Class I and three in Class II, which between them account for less than one-fourth of the town population. Half the towns are of small size, there being 30 in Class V and 8 in Class VI, which contribute less than one-fifth of the urban population. The remainder are towns of average size (14 in Class III and 20 in Class IV), which contain more than half of the urban population.

CLASSES OF TOWNS.

Class.		Population.
I	...	100,000 and over.
II	...	50,000 to 100,000.
III	...	20,000 to 50,000.
IV	...	10,000 to 20,000.
V	...	5,000 to 10,000.
VI	...	Under 5,000.

Taking each class as a whole, we find that in Bengal all classes of towns have steadily developed during the last two decades. There has been a comparatively small increase in Class I during the last ten years, which is due to the tendency of the people of Calcutta to spread out to the suburban municipalities. In Bihar and Orissa, on the other hand, all but the smaller towns have declined or stood still during the last 20 years, owing, to a large extent, to plague.

73. The population of places that were treated as towns in 1872 has increased by 32 per cent. in Bengal, but by only 8 per cent. in Bihar and Orissa. Including the places subsequently treated as towns, the actual increase in the urban population is 61 per cent. in Bengal as against 21 per cent. in Bihar and Orissa. In Bengal the increase is shared in more or less by all classes of towns except those in Class II, which have a decrease of nearly 50 per cent. This decrease is, however, more apparent than real, being mainly due to the transfer to Class I of Howrah and Dacca, which came under Class II in 1872.

Class.	Number of Towns.			
	Bengal.		Bihar and Orissa.	
	1911.	1872.	1911.	1872.
Under 5,000	17	4	8	7
5,000 to 10,000 ...	34	27	30	14
10,000 to 20,000 ...	40	27	20	17
20,000 to 50,000 ...	28	15	14	9
50,000 to 100,000 ...	2	3	3	4
100,000 and over ...	3	1	1	1
	124	77	76	52

CITIES.

74. There are nine cities in the two Provinces, the population of which is shown in the margin with the percentage of variation since 1901. In considering the recent growth of cities and towns, it will be most convenient to take the cities first and then to group the towns together according to natural divisions forming homogeneous areas.

Before proceeding to discuss the results of the census of cities, one characteristic feature may be referred to. There is a rise in the proportion of foreign-born (*i.e.*, those born outside the district

City.	Population.	Percentage of variation.
Bengal.		
Calcutta	896,067	+ 5·7
Howrah	179,006	+ 13·6
Dacca	108,551	+ 21·0
Manicktollah	53,767	+ 66·0
Cossipur-Chitpur ...	48,178	+ 18·2
Garden Reach... ...	45,295	+ 60·6
Bihar and Orissa.		
Patna	136,153	+ 1·0
Bhagalpur	74,349	− 1·9
Gaya...	49,921	− 30·0

containing the city*), and a fall in the proportion of females to males in every city but Manicktollah and Cossipur-Chitpur, where the proportion of females has slightly increased. As this is the case in cities which have a loss of population, as well as in those which have an increase, it is clear that the cities are tending more and more to attract immigrants from greater distances. It will also be observed that the proportion of foreign-born in all the Bengal cities except Dacca is very much greater than in the Bihar cities, which have as yet not become manufacturing centres.

CITY.	PROPORTION PER MILLE.			
	Females to males.		Foreign born.	
	1901.	1911.	1901.	1911.
Calcutta	507	475	657	714
Howrah	577	562	659	756
Cossipur-Chitpur	556	565	637	661
Manicktollah	692	694	649	750
Garden Reach*	667	637	428	453
Dacca	799	721	164	198
Patna	1,011	922	85	93
Gaya	950	897	79	96
Bhagalpur	907	861	156	169

75. CALCUTTA. The progress of Calcutta and the three suburban municipalities of Cossipur-Chitpur, Manicktollah and Garden Reach is dealt with in a separate report, and here it will be sufficient to state that though each of them is under separate municipal government, they form an integral part of the same city. Howrah may also reasonably be treated as part of the metropolis, for it is only separated from Calcutta by the Hooghly, and, as pointed out in the last Census Report, is really as much a part of that city as Southwark is of London. If this be conceded, the population of the metropolis (1,222,313) is greater than that of any city in the British Empire except London, and among European cities is only surpassed by London, Paris, Vienna, St. Petersburg and Moscow.†

In Calcutta, the rate of increase which was 24·3 per cent. between 1891 and 1901, has now fallen to 5·7 per cent. Its population shows a centrifugal tendency, spreading out to Howrah and to all the suburban municipalities except Cossipur-Chitpur. The number of persons born in Calcutta and resident in that city is now nearly 34,000 less than it was 10 years ago: had it remained the same, an increase of 9·8 per cent. would have been registered. Owing to this exodus from Calcutta and the relatively greater number of immigrants they receive, all four municipalities have grown much more rapidly since 1901 than Calcutta. In Manicktollah and Garden Reach (in the latter of which, however, the growth is partly due to the extension of the municipal boundary) the percentage of increase is over ten times, in Cossipur-Chitpur it is treble, and in Howrah it is double what it is in the area administered by the Calcutta Corporation.

76. HOWRAH. Howrah is a city which owes its development entirely to modern commerce. Originally, it was a small collection of villages, the names of which still survive in the quarters (*paras*) constituting the city. In the 18th century docks were opened along its banks for repairing the wooden vessels plying up the Hooghly, and it also became a kind of suburban retreat in which the wealthier citizens of Calcutta set up villas and laid out gardens. There cannot, however, have been much development in the first half of that century, for, as late as 1750, it is said to have been "a line of mud banks reeking with malaria, corpses in all stages of decomposition floating up and down the stream by the dozen, jungle lining the shore, the abode of the snake and alligator.‡" According to Bishop Heber, it was in 1823 a place "chiefly inhabited by ship-builders," while in 1848 it was referred to as "the Wapping of Calcutta inhabited chiefly by persons connected with the docks and shipping."

77. Howrah began to expand rapidly in the middle of the 19th century, especially after 1850 when it became the terminus of the East Indian Railway. Not only did the docks increase in size and number, but other large industrial concerns were started, such as engineering works, sugar factories, flour mills, cotton mills, jute mills and jute presses. The construction of the bridge over the Hooghly gave a further impetus to its

* It being impossible to distinguish between persons born in Calcutta and Garden Reach, the term district-born in the case of the latter town is taken to include persons in Calcutta.

† A slightly larger population is estimated for Constantinople.

‡ Howrah Past and Present, pp. 18-19.

growth, which in recent years has been stimulated by the Bengal-Nagpur Railway obtaining access to it, by the opening of two light railways, which have linked up the interior of the district with the city, and by the establishment of steamer services along and across the Hooghly. The Calcutta tramway system has been extended to Howrah since 1907, and the ferry steamer service connecting the two cities has been considerably improved and accelerated: consequently, an increasing number of persons who work in the day at Calcutta have their residence in Howrah. The number of persons born in Calcutta and enumerated in the city is double what it was in 1901, while those born in Hooghly have increased from nearly 17,000 to over 29,000. During the 40 years over which the census records extend it has more than doubled its population; the rate of growth was 35 per cent. in 1891-1901 and 13·6 per cent. in the last decade. The decline in the percentage of increase is partly due to the fact that the jute mills were not in full work owing to the dull state of the market, and partly to the fact that the census was taken on a Friday. "Some of the mills," reports the Magistrate, "closed on Friday evening for the week and a considerable number of the mill hands left for their homes, Calcutta and other places."

78. Less than twenty years ago the city was lacking in many urban amenities. In 1889 it was described by the Sanitary Commissioner as being "without exception the dirtiest, most backward and badly managed municipality" he had seen. His successor in 1893 endorsed this verdict and remarked: "Generally speaking the sanitary condition of the town of Howrah is most deplorable. I have never, in fact, seen a town in such a dangerously insanitary condition, and I should be very sorry to live in it myself." Since then much has been done to improve the condition of the town, though it is no easy task on account of its low-lying situation, its rapid growth and past neglect in laying out building sites. Water-works were opened in 1896; a regular system of drainage has been introduced and is gradually being extended; bye-laws have been adopted for regulating the construction of new huts; overcrowded *bastis* are being gradually opened up by new roads, and their sanitation and drainage improved. Density is high, averaging 33 per acre. The outer fringe of the town is thinly peopled, being mostly occupied by fields interspersed with gardens and villas.

Ward.	Number of persons per acre.	
	1911.	1901.
I	39	36
II	16	14
III	70	58
IV	54	60
V	90	83
VI	24	18
VII	76	62
VIII	41	33
IX	9	9
X	9	8

79. The marginal statement showing the districts which contribute over 1,000 to the population sufficiently indicates how heterogeneous the population is. More than two-thirds of the inhabitants were born outside the district, and in the last ten years the number of those born in the city or district has decreased by 10,000, or nearly 20 per cent. The latter now outnumber the immigrants from the United Provinces by less than 6,000. The city is, in fact, ceasing to be a Bengali city. Nearly half the inhabitants were born in the United Provinces or the Province of Bihar and Orissa (which contributes 38,830 to its population), and only 45 per cent. speak Bengali, while 47 per cent. speak Hindi and 3 per cent. Oriya. The railways, mills, factories, docks, iron-works, etc., afford employment to a large number of labourers and artisans, who leave their wives and families at home and huddle together in crowded *bastis*. At the time of the census it was ascertained that jute mills and presses employed no less than 24,000 persons, or over one-seventh of the total population, while 15,000 more worked in other manufacturing concerns, such as cotton mills, rope works, iron foundries, machinery and engineering workshops, etc. There are now only 562 females to every 1,000 males. The proportion of females gradually decreases at each census: since 1872 the number of males has risen by nearly 150 per cent., whereas the females have increased by only 75 per cent.

District, etc.	Number of persons.
Howrah	43,639
United Provinces ...	37,943
Hooghly	29,010
Calcutta	9,847
Shahabad ...	7,575
Saran	6,353
Midnapore ...	6,047
Patna	5,709
24-Parganas ...	4,031
Gaya	3,664
Cuttack	3,446
Monghyr ...	3,150
Muzaffarpur ...	3,035
Burdwan ...	2,247
Balasore	1,655
Dacca	1,365
Darbhanga ...	1,290

DACCA.

80. With a population of 108,551, the city of Dacca contains 58 per cent. more inhabitants than in 1872. Its development has been most rapid during the last ten years, during which it added 21 per cent. to its numbers, while the general ratio of increase in the Dacca district was 12 per cent. Its rapid growth since 1901 is mainly due to its being made the headquarters of the Eastern Bengal and Assam Government in 1905, after which people settled in increasing numbers in the town. The construction of public buildings, moreover, necessitated the employment of a large labour force, and the population was further increased by the staff of the Secretariat and other offices. The extent to which the town owes its accretion to immigrants is shewn by the fact that while the males have increased by 26·5 per cent., the rise in the number of females is only 14 per cent. Out of every thousand persons, 198 are foreign born, their distribution by birthplace being as shewn in the margin. Nearly 4,000 immigrants are natives of Monghyr. Other districts of the United Provinces and Bihar from which immigrants come in large numbers are Muzaffarpur, Darbhanga, Muzaffarnagar, Ghazipur, Ballia, Gorakhpur and Azamgarh. These people work as constables, railway servants, boatmen, general labourers, porters, domestic servants, scavengers, and shoe-makers. The town is far from being a busy manufacturing or industrial centre, for at the industrial census of 1911 only 14 concerns employing 20 hands or more were returned, the aggregate number of their employés being 1,195. Weaving, formerly the glory of Dacca, now supports only 1,029 persons, but shell work and jewellery, for which the city is also famous, are still in a fairly flourishing condition, the former supporting 2,193 persons and the latter 4,032 persons.

Other districts of Eastern Bengal.	66
Assam	12
Bengal excluding Eastern Bengal	24
Bihar and Orissa ...	65
United Provinces ...	24
Other countries	7

81. The density of population in each ward is shewn in the marginal statement, from which also it will be seen that all parts of the town share in the increase except Ward V, which sustained a slight loss owing to the removal of a *basti* from a plot of land acquired by Government. The city consists of three parts—(1) the town proper, which is congested, (2) the fringe area, which is more thinly populated, and (3) the new town, which is very sparsely populated. Wards V and II, which occupy the first and second place respectively in order of density, lie wholly within the town proper. The most congested area in these two wards is Sankaribazar in Ward II, where 2,456 persons were enumerated in 432 *khanas* or census houses. In this quarter the houses, mostly three stories high, are closely packed together with a small frontage along the road, and the unsavoury smell of decomposing shells (in which the Sankharis work) clings to them. Ward I, which comes next, lies partly within the town proper and partly within the fringe area. The next, in order of density, is Ward IV, which covers a portion of the old town proper and also includes a portion of the new town. Wards III and VI lie partly in the town proper and partly in the fringe area. Ward VI lies wholly in the fringe area with one rather congested road. The civil station at Ramna consists of the area acquired by Government for the construction of public buildings and is the least populous part of the city.

No. of ward.	Population.	DENSITY PER ACRE.	
		1901.	1911.
I ...	21,093	33·1	35·4
II ...	14,998	47·1	61·6
III ...	18,536	15·7	19·0
IV ...	19,675	17·8	24·9
V ...	8,765	76·5	76·0
VI ...	11,163	12·5	17·1
VII ...	12,348	21·1	24·4

82. Patna, the capital designate of the new Province of Bihar and Orissa, is a decadent city, as will be seen from the figures given in the margin which show the total population recorded at each census during the last 30 years. Figures are not given for the census of 1872, as it was wanting in accuracy, and an apparent increase of over 11,000 recorded in 1881 is believed to have been due to its incompleteness. In 1901 the census was taken at a time when plague was raging and a considerable proportion of the inhabitants had fled from their homes, the result being a decrease of 18½

Year.	Number.
1881	170,654
1891	165,192
1901	134,785
1911	136,153

per cent. At a second census held in July 1901 the population was found to be 153,739, or nearly 19,000 more, but in spite of this addition, the population was ·7 per cent. less than in 1891. The last census shows an increase of 1,368, as compared with the first census of 1901; but if it be compared with the second enumeration of 1901, there is a decrease of 17,586 persons, or 11·4 per cent. The causes of the decay of the city are twofold, viz., persistent unhealthiness,* repeated epidemics of plague and the loss of trade. Formerly Patna city was an important emporium for rail and river-borne trade, but its trade is declining owing to its diversion to other centres and the disorganization caused by plague, which has been almost an annual visitation since 1900. During the last ten years there have been no less than 17,384 deaths from plague, and the loss of population (17,586) recorded at the present census very nearly corresponds with that figure. The city extends over nine square miles, and for the purposes of municipal government includes the town of Bankipore. There are on the average 23 persons per acre, and 90 per cent. of the inhabitants were born in the city or district.

BHAGALPUR.

83. Bhagalpur grew steadily between 1872 and 1901, adding 10,000 to its population during that period. During the last decade there has been a set back, the number of its inhabitants decreasing by 1,411, or 1·9 per cent. The decrease is mainly the effect of plague, which during the decade caused a mortality of 4,201. Like Patna, the people are nearly all local residents, 83 per cent. being born either in Bhagalpur itself or in the Bhagalpur district, while the sexes are almost equally distributed.

GAYA.

84. When the census of 1911 was taken, plague was raging in Gaya. A large proportion of the people had fled from the town to villages in the interior, and the total population recorded was only 49,921, or 30 per cent. less than in 1901, in which year also the town was suffering from the effects of another epidemic.† Owing to this circumstance, the census gave no idea of the normal population of the town, and a second census was therefore held in June, when the epidemic had subsided. At this second census the population returned was 70,423, or only a little less than in 1901. The comparatively large excess of males, who outnumbered the females by 7,000, may perhaps be taken as an indication that all the females who had been sent away during the plague had not yet returned to their homes, but the same disproportion of the sexes is noticeable in other towns of Bihar. There were no less than 4,780 deaths from plague in the town during the decade ending in 1910. That, in spite of this mortality, the population should be nearly equal to that recorded in 1901 (though in that year also the population had been diminished by plague) may be attributed to the fact that it is a large pilgrim centre, which every year has a considerable floating population, and that it has developed since 1901 owing to its being an important station on the Grand Chord Line.

TOWNS.

TOWNS IN WEST BENGAL.

85. Before 1872 the town of Burdwan suffered severely from the epidemic of fever which took its name from the district, and in the last 40 years has only added 3,600 to its population. Excluding the places treated as cities, however, it is the fourth largest town in the Province. It now contains 35,921 inhabitants, or 899 more than in 1901, but had it not been for an extension of the municipal boundaries, there would probably have been a decrease.

* The *Sair-ul-Mutakharin* gives an account of an epidemic which broke out at Patna in the early part of the 18th century and spread over Northern India :—"At the end of the year (1730-31) there arose, for forty days together, out of the ground such an abominable stench all over the city, that the poor and rich, being equally affected by it, were attacked by an epidemical fever that filled the houses with sick. The shops and markets were shut up, the streets became desert, and the city looked like a place forsaken by its inhabitants. People said that they had never seen or heard of such a calamity. The stench and sickness commenced at Patna and Ilahabad, from whence it proceeded to Akbarabad and Delhi, and continued spreading over Paniput and Sirhind, until it extended to Lahore where it stopped by a favour of divine Providence."

† The coincidence of plague with the census both in 1901 and 1911 led to a belief among the people that there was some connection between the two.

The health of the town has considerably improved since the construction of water-works in 1884-85, but it is still badly drained and suffers from fever. It has, moreover, no large industries such as would attract labour from outside. The most progressive town in the district is Asansol, which is one of the chief centres of the coal industry and an important railway junction. It has developed rapidly of recent years, and has added 50 per cent. to its population during the last ten years : part of this increase must, however, be attributed to the municipal area being extended by 1½ square miles in 1905. The head-quarters of the subdivision were removed to Asansol in 1906, owing to its being a more central and important place than Raniganj, and it has not only supplanted but outgrown the latter town, which now shows a slight decline. Kalna has a slight increase of population, but the other towns of the district, viz., Dainhat and Katwa, are decadent. The three towns last named are situated on the Bhagirathi, and all have suffered from the silting-up or receding of that river and from the diversion of trade to the railway. The prosperity of Katwa should, however, revive on the completion of the Hooghly-Katwa line, which will give its trade an outlet.

86. Suri in Birbhum is a small town, which has recently been opened up by the Ondal-Sainthia line ; its apparent growth since 1901 is probably due to the inclusion of another square mile within the municipal area. The effect of railway communication is marked in the case of the towns in Bankura, to which the Bengal-Nagpur Railway line was extended in 1902. The head-quarters station has increased by 13 per cent. and Bishnupur by 7 per cent., but Sonamukhi, which is not connected with the railway, has a slight decrease. Part of the increase in Bankura must, however, be ascribed to the addition of an area of one square mile to the municipal limits. In Midnapore the only progressive town is Kharagpur, whose rise has already been referred to. In all the other towns the population has diminished, or is practically stationary. Midnapore now has 32,740 inhabitants, or little more than in 1891 : as pointed out in the last Census Report, it has no great industry or trade, and exhibits no tendency to grow. During the last decade it has suffered from fever and cholera, which have also seriously affected the towns of the Ghatal subdivision, viz., Chandrakona, Ghatal, Kharar, Khirpai and Ramjibanpur.

87. Hooghly contains no less than eight towns, of which seven are on the banks of the Hooghly. The most important is Serampore, which has been steadily progressing during the last 40 years. Owing to the proximity of Calcutta and to the industrial concerns started in the town and its neighbourhood, an increase of nearly 12 per cent. at this census has brought its population up to 49,594, or double what it was in 1872. Bhadreswar, a rising factory town, has developed at an extraordinary rate during the last decade, having an addition of 9,203, or over 60 per cent. Another flourishing town is Baidyabati, which has a considerable trade-mart at Sheoraphuli and a large jute mill at Champdani ; it has added 3,342 to its population since 1901. Hooghly-Chinsura, the headquarters of the district, is slowly but steadily declining. It has no industries or trade of any importance ; it is an unhealthy place, the inhabitants of which suffer from dysentery, fever and occasional epidemics of cholera ; the streets or lanes are full of abandoned houses overgrown with jungle, recalling the picture of "ruin and melancholy" sketched by Mrs. Fenton in 1827. The death rate has been higher than in any other town of the district in all but two years of the decade, when it was exceeded in Bansberia. The latter town, which also suffers from defective drainage, was second on the list in the remaining eight years and has also lost population. The other towns, viz., Kotrang, Uttarpara and Arambagh, are small and unimportant. Bally, further down the river, is the only town in the Howrah district besides Howrah ; it adjoins Howrah, and, as it shares in the influx of immigrants, is steadily growing.

88. In no district has there been a greater growth of urban population than in the 24-Parganas, where the average per town has risen by 38 per cent. since 1901. The whole riparian strip along the Hooghly is fast becoming urbanized : owing to their growing density of population the South Suburbs, South

CENTRAL BENGAL.

Barrackpore and Naihati municipalities have been twice subdivided since their creation and formed into nine municipalities. There are now no less than 26 towns in the district, of which five, viz., Cossipore-Chitpur, Manicktollah, Garden Reach, South Suburbs and Tollygunge, adjoin Calcutta and are suburban in character. Taken together, these five towns have added 40 per cent. to their population since 1901, while Calcutta itself has only increased by 5·7 per cent. The second class consists of industrial towns which, with the exception of Budge Budge, stretch northwards from Calcutta along the bank of the Hooghly, viz., Baranagar, the adjoining town of Kamarhati, Naihati, the two contiguous towns of Halisahar and Bhatpara, Titagarh, Budge-Budge and Garulia. The increase in these towns has been phenomenal, averaging no less than 67 per cent. There are three other towns along the Hooghly, viz., South Barrackpore, North Barrackpore and Panihati, which, however, are not industrial centres : of these, only South Barrackpore shows an advance, which is partly accounted for by the increase in the Barrackpore Cantonment. The remaining ten towns are situated inland, and are mostly rural in character : altogether, they have only increased by 6 per cent., the most substantial increases being found in South Dum-Dum, Baruipur and Basirhat. The growth of Baruipur may, however, be partly accounted for by an addition to its area.

89. Nearly all the towns of Nadia are either stationary or decadent. Their aggregate population has had an addition of only 563 since 1901, and has decreased by a little over 17,000 since 1891. The two largest towns, Krishnagar, the headquarters station, and Santipur, once an important weaving centre, have both lost ground : the former has suffered severely from fever, the latter from the decline of its industries. Nadia, the birthplace of Chaitanya and a pilgrim centre, returns 1,600 more persons than in 1901, but this is not altogether a real advance, for a religious festival was approaching, and there were a number of pilgrims present in the town. Of the subdivisional stations, Ranaghat alone, which is a considerable railway junction, has been slowly but steadily growing since 1891 ; in spite of the municipal area being reduced by 200 acres in 1905, its population has increased by 13 per cent. In Murshidabad, Berhampore, the district headquarters, has an increase of 7 per cent., and the two subdivisional headquarters of Kandi and Jangipur have also expanded. Murshidabad and Azimganj (a municipality in its suburbs) both show a continuous and heavy decline since 1891. There are only three towns in Jessore, of which Jessore and Maheshpur are stationary, while Kotchandpur has lost population owing to the falling off in the manufacture and sale of sugar to which it owed its former prosperity.

Towns in North Bengal.

90. In North Bengal there are 19 towns and three cantonments, viz., Buxa, Lebong and Jalapahar, of which, however, the last two are treated as forming part of the town of Darjeeling. Of these 19 towns, only three have over 20,000 and only six over 10,000 inhabitants. The largest is still Sirajganj, a flourishing jute market on the Padma, which has grown slightly since 1901. The only towns that have decreased since that year are Natore and Sherpur (Bogra), the result of malaria and general unhealthiness. All the other towns are progressive except Old Malda, which is stationary : Jalpaiguri has increased by nearly 18 and Bogra by nearly 28 per cent. since 1901. Of the smaller towns, Saidpur, the northern head-quarters of the Eastern Bengal State Railway, has gained 42 per cent. and now contains over 8,000 inhabitants, while Kurigram and Gaibandha, two subdivisional stations in Rangpur, have doubled their population. All the towns in Cooch Behar are small and unimportant, except the capital of the same name, which is not progressive.

Darjeeling.

91. The town of Darjeeling requires special mention on account of its importance as the headquarters of Government for part of the year and also because it is, next to Simla, the most populous hill station in India. In 1872, before the construction of the railway, when the only approach to Darjeeling was by a long tedious march, the population numbered only 3,157, but during the next

nine years it increased by more than 100 per cent. It again doubled itself between 1881 and 1891, after the construction of the Darjeeling-Himalayan Railway had brought the station within a day's journey of Calcutta. The census of 1901 disclosed an increase of 20 per cent., the total population amounting to 16,924, but its progress during the last ten years has been less rapid owing partly to the fact that there is not much room for further expansion, and partly because its expensiveness deters people from taking up a permanent residence in it. In spite of this, the population rose to 19.005, the actual increase according to the census taken in March 1911 being 2,081. or 12 per cent. : altogether 1,576 of the inhabitants were Europeans. Darjeeling, it may be explained, includes not only the town proper, *i.e.*, the area within municipal limits, but also the cantonments of Lebong and Jalapahar including Katapahar. If these two cantonments are excluded and the figures for the town alone are taken, there was an increase of 1,360, or 9 per cent. during the decade.

*Simla ...	...	37,895
Darjeeling	...	24,696
Ootacamund	...	18,829
Naini Tal	...	18,027
Mussorie	...	17,402
Murree	...	16,934

92. A census taken at this time of the year only records the population of the town at the end of winter. It is far greater during the hot weather and rainy months, when it attracts a number of visitors who come to escape the heat of the plains or to recruit in its cool climate. A second census was therefore held in September 1911 in order to ascertain its population at this time of the year. The results of the two censuses are shown in the margin. The population of the whole town was only 3 per cent. more than that recorded at a similar hot weather census held in September 1900, a fact which seems to indicate that the place now attracts very few more people than it did ten years ago. Kurseong, which is a minor hill station, is growing more rapidly, having added 25 per cent. to its population since 1901. It is becoming an educational centre for Europeans and Anglo-Indians, and like Darjeeling attracts visitors, but it is by no means the same trade centre as the headquarters of the district.

	March 1911.	September 1911.
	Total number of persons.	Total number of persons.
Darjeeling Town	17,053	21,553
Lebong ...	1,037	1,569
Jalapahar ...	915	1,574
Total ...	19,005	24,696

93. In East Bengal there has been an increase in the population of all but four towns since 1901. Both Dacca and Narayanganj record large additions, to which reference has already been made. Seven of the eight municipalities of Mymensingh are progressive ; the head-quarters town has grown by 35 per cent., Sherpur by 24 per cent., Netrakona by 20 per cent. and Jamalpur by 18 per cent. The only decadent town in this district is Tangail, the population of which has been steadily falling off since 1881. The two towns of Faridpur show steady increases ; Faridpur is now nearly 13 per cent. and Madaripur, a large jute centre, 9 per cent. larger than in 1901. In Tippera district, Comilla has an increase of 18 per cent. and Brahmanberia of nearly 12 per cent.

TOWNS IN EAST BENGAL.

94. Excluding Dacca, which has already been dealt with, the largest town in the Division is Chittagong, the trade of which has developed very considerably since 1901. It has more than made up the apparent decrease disclosed in 1901, and the population is nearly 30 per cent. greater than that then returned. Trade has grown steadily, and is of sufficient importance for one large steamer line to visit the port. Previously navigation was impracticable owing to the low depth of water in the Karnafuli river, but the channel has been considerably improved by dredging. Jetties and warehouses have sprung up, and the town is now a growing centre of commercial and railway activity. Cox's Bazar in the same district has regained the position it had in 1891. In Backergunge there has been a loss of 15 per cent. in Pirojpur and of 13 per cent. in Nalchiti, but Barisal, the head-quarters, shows an increase of 18 per cent. The other towns are progressive, notably Jhalakati, a large centre of trade, which has increased by 14 per cent. and has more than quadrupled its population since

* The marginal statement shows the season population, according to a special census, of each hill station except Ootacamund, of which no special season census was taken.

1881. In Khulna, the head-quarters station of the same name is advancing rapidly, adding 25 per cent. to its population during the decade. There is also an increase of 30 per cent. in Sátkhira, but a large village extending over three square miles was added to it in 1903. The only other town, Debhatta, has been stationary since 1881.

TOWNS IN SOUTH BIHAR.

95. Nearly every town in South Bihar has suffered from repeated visitations of plague, and has had to sustain a loss of population. The aggregate population of the seven towns of the Patna district has fallen by 15 per cent. Patna city, as already stated, has sustained a loss at each census since 1881, while Bihar shows an even greater relative decrease. Its population numbered 48,968 in 1881, but gradually fell to 45,063 in 1901. It now contains nearly 10,000, or 22 per cent., less than in the latter year, though it has been connected with the main system of the East Indian Railway by a light railway, and its trade should have developed, had conditions been normal. Its area has, however, been reduced by a square mile, and it has suffered severely from disease, no less than 4,082 persons dying from plague; in only two out of ten years ending in 1910 did the birth-rate exceed the death-rate. Dinapore, where there were 3,205 deaths from plague, has decreased by 2,674, and Barh and Khagaul have also sustained heavy losses. Phulwari, which is a collection of villages rather than a town, is practically stationary, and the only town in the district with a substantial increase is Mokameh. This is a rising trade centre and an important railway junction, through which the grain traffic of the north-Gangetic districts passes, and it has added 11 per cent. to its population.

96. In Gaya, the second census of the headquarters station, as already stated, disclosed a population little less than in 1901; and it is still the largest town in the Province next to Patna and Bhagalpur. All the other towns in this district are small in size, none having 10,000 inhabitants. Plague was prevalent in Jahanabad at the time of the census, and many of its inhabitants had consequently deserted their homes. The result was a diminution by 32 per cent. or 2 per cent., more than that shown by the first census of Gaya, where the more urban population has not the same intimate connection with neighbouring villages. Tekari, in which the population was reduced to nearly half in 1901 in consequence of the deaths and desertions caused by plague, is still on the downward grade, and Daudnagar also has a slight falling off. Both these towns were affected by plague when the census was held, and many had left their homes for unaffected areas. Apart, moreover, from this disturbing element, Tekari is 17 miles and Daudnagar 14 miles from the railway, to which trade flows more and more. Aurangabad and Nawada have an increase of 1,000, and Hisua has a slight increase. The last two towns are on the railway, while Aurangabad is close to it, and all three have benefited by the opening of the Grand Chord Line.

97. In the Census Report of 1901 it was remarked that with the solitary exception of Sasaram, all the towns of Shahabad seemed decadent. The result of this census is to confirm this view, for every town has lost population. The loss is insignificant in Sasaram, which has had a mortality of over 1,000 from plague, but has developed owing to the opening of the Grand Chord Line. It is very great in the case of the four northern towns of Arrah, Buxar, Dumraon and Jagdispur, where it averages 17 per cent. Arrah has been especially subject to epidemics of plague since the last census, the mortality from that cause amounting to 8,747; the actual decrease of population according to the census is 7,621. Buxar, which suffered a decrease of 10 per cent. in 1891-1901, has now another decrease of 19 per cent.

98. The town of Monghyr has now 11,033 persons more than were enumerated in March 1901; but at that time plague was raging and the population was abnormally small. A second census taken in July 1901 gave a population of 50,133, so that there has actually been a decrease of 3,220. The number of deaths from plague recorded in the ten years 1901-11 was 9,666, and had it not been for this mortality, there would probably have been a fair increase. Jamalpur was also partially deserted at the census held in March 1901, when the town contained 13,929 persons: a second enumeration taken eight

months later disclosed a population of 16,302. The number has now risen to 20,526 in spite of 2,000 deaths from plague, but this large increase is mainly due to an additional 2½ square miles having been included within municipal limits in 1903. In the town as now constituted the ratio of increase is 8 per cent.: it contains large railway workshops to which labourers and artisans are attracted. There are only two other towns in the district, both of which are local trade centres, viz., Khagaria, north of the Ganges, which has lost slightly, and Sheikhpura, south of the Ganges, which has added considerably to its numbers.

Towns in North Bihar.

99. There has been a general decline of urban population in North Bihar due partly to plague, partly to the increasing volume of emigration and partly to the fact that the towns, as a rule, have no large industries, while their trade, except in agricultural produce, mainly supplies local wants. In Saran the decrease of 1901 has been followed by another serious loss of population, which is shared in by every town. Chapra, which in 1901 recorded a decrease of 20 per cent., has since then had 6,634 deaths from plague, and its population has fallen by 3,528, or 8 per cent. There was a particularly bad epidemic in the early part of 1911, and some of the inhabitants had deserted the town when the census was taken. The rate of decrease is even higher in Revelganj, which has suffered both from loss of trade and from plague mortality: the aggregate number of deaths from plague in the decade 1901—1910 represented 23 per cent. of the population returned in 1901. In Siwan the decrease of 21 per cent. corresponds to the actual loss caused by plague. The two towns of Champaran, Motihari and Bettiah, have been free from this disease, and both have an accretion of population. The municipal area of Bettiah was reduced by half a square mile in 1902, but on the other hand its population was artificially inflated on the day of the census, when the town was visited by the Lieutenant-Governor, and people flocked in from the neighbourhood. These two towns have progressed steadily since 1872, the former having more than doubled, and the latter nearly doubled, its population.

100. The town of Muzaffarpur, which declined by 9 per cent. between 1891 and 1901, has again lost slightly, though there was an addition of a square mile to the municipal area: it now has only 1,200 more inhabitants than it had in 1881. Sitamarhi is the only progressive town in this district; it has been free from plague and has grown steadily since 1881. Hajipur, the only other subdivisional town, is decadent, every census disclosing a further decrease: during the last 10 years plague carried away 17 per cent. of the population. The population returned for Darbhanga in 1901 was unnaturally small, owing to a number of people having left the town for weddings elsewhere; but in spite of this there has been no recovery, but a further small decline (3,616), which may be attributed to plague. Madhubani, on the other hand, was free from plague, but has lost 7 per cent., owing to nearly two square miles being excluded from municipal limits in 1901. Samastipur is stationary, and Rosera has a heavy loss. The two towns of Bhagalpur have also lost ground; the headquarters station, as already stated, has lost slightly, while the population of Colgong has been reduced to under 5,000. In the Purnea district, Katihar, a developing railway junction, is the only town with an appreciable growth. Purnea is decadent, and, though it has added a few hundreds since 1901, is less populous than in 1872. Kishenganj is a centre for the jute trade, but it lies in the fever area and its population is stationary.

Orissa.

101. In Orissa more than half of the urban population is contained within the two towns of Cuttack and Puri. Cuttack with 52,528 inhabitants is the fourth largest town in the Province, and has an increase of 2 per cent. since 1901. In that year it contained a detachment of a Madras regiment, but it has since ceased to be a military station, and there is consequently a slight loss of population on that account.* The other two towns in the Cuttack district are holding their ground. The population of Puri is always an uncertain

* The old cantonment has been absorbed by the municipality. Its population at the time of the census was 3,508.

quantity, owing to the floating population of pilgrims. In 1901 it was returned at 49,334 or nearly 21,000 more than in 1891, but 17,085 pilgrims had come into the town for one of the large annual festivals, and the permanent population was 32,259. At the present census also there was an influx of 5,293 pilgrims on account of the approaching *Dol Jatra* festival; if they are deducted, the resident population of the town numbers 34,393 or 6·3 per cent. more than in 1901. This increase is natural, for Puri has developed considerably during the last ten years, owing to its attractions as a seaside resort. Balasore has grown slightly in consequence of an extension of the municipal boundary, but Bhadrakh is stationary.

TOWNS OF THE CHOTA NAGPUR PLATEAU.

102. The Chota Nagpur Plateau contains 27 towns, but only two (Ranchi and Purulia) have over 20,000 inhabitants. In the Feudatory States there are only six small towns, with an average population of 6,200, scattered over 28,000 square miles. Most of the towns in British territory have not yet been connected by the railway with the outside world; eight, which have obtained railway communication, are making great progress, their average increase in the last ten years being 25 per cent. In the Sonthal Parganas, Sahebganj has now more than made up the loss of population it sustained by plague in 1901, when the town was partially evacuated. Compared with that year, it has nearly doubled its population, but it has only 3,000 more inhabitants than in 1891. This is a town which owes its development to the railway, local produce being received from the districts of Purnea, Malda and Bhagalpur, as well as from the Rajmahal Hills. Of recent years it has been growing in importance as a trade centre, owing to the development of the trade in *sabai* grass, which is here pressed into bales and exported to Calcutta for the manufacture of paper. Deoghar has added no less than 29 per cent. to its population since 1901, but this apparently large increase is due, to a great extent, to an addition of two square miles to its area: it is a favourite place of pilgrimage, but the pilgrims mostly come in January, February and September, and the census figures were but little affected by them. Both this town and Madhupur are attracting an increasing number of Bengali gentlemen, owing to their reputation as health resorts, but the actual growth of population in Madhupur is still very small. Dumka, the headquarters of the district, which has recently been created a municipality, has a population of only 5,629. The old capital of Rajmahal, which is a subdivisional headquarters and a local trade mart, has only a few hundreds less.

103. Hazaribagh, which was described as slightly decadent in 1901, has now increased by nearly 2,000. Access to this town has been facilitated by the opening of the Grand Chord Line, and though it is still 40 miles from the railway, it is already attracting visitors and permanent residents on account of its healthiness and the educational facilities afforded by its college. Giridih in the same district has benefited by the development of the coal-fields, and has increased by 13 per cent.; but part of the increase must be attributed to the municipal area being extended by 1½ square miles. The growth of Ranchi since 1891 has been even greater, for in 1901 it recorded a growth of 28 per cent. and it has now added 7,000 more or 27 per cent. to its population. The area of this town has increased by 1½ square miles since 1901, but it has also developed naturally. At the last census it was 70 miles away from any line of railway, but since then the Bengal-Nagpur Railway has been extended from Purulia, a narrow gauge line being opened at the end of 1907. Its importance both as a sanitarium and as an administrative and commercial centre has increased in an extraordinary degree. New buildings have sprung up, and it has attracted a number of new settlers.

104. Daltonganj, in the district of Palamau, to which the railway has also been extended since the last census, has grown by 23 per cent., while Garhwa, a trading mart, which has been connected with the railway by a good road, has advanced considerably. Purulia, in the Manbhum district, which increased by 42 per cent. between 1891 and 1901, owing to the opening of the Bengal-

Nagpur Railway and the development of the cooly-recruiting business, has registered a further increase of 21 per cent. Chaibasa, the headquarters of Singhbhum, which is still 16 miles away from a railway, has not advanced appreciably, but the opening of the Tata Iron and Steel Works at Sakchi has led to the creation of another town in the district. Sambalpur, the headquarters of the district of that name, returned a population of 14,571 in 1891, but between that year and 1901 some adjoining villages were excluded from the municipal area, and the population consequently fell to 12,870. During the past decade it has advanced very slightly.

VILLAGES.

THE CENSUS VILLAGE.

105. Both in the Presidency of Bengal and the Province of Bihar and Orissa the vast majority of the population live in villages, the proportion per mille of the total population being 936 in the former and 966 in the latter Province. The term village as used in the census records has very different meanings. In the Province of Bengal as constituted at the time of the census the *mauza*, or survey or settlement village, was treated as the census village, except in four districts; in the districts then included in Eastern Bengal and Assam the residential village was taken as the unit. In other words, the census village corresponded to the *mauza* in all the districts of Bihar and Orissa, except Manbhum, Palamau and Hazaribagh, while in Bengal it corresponded to the *mauza* in the Burdwan and Presidency Divisions, to the settlement village in Cooch Bihar, and elsewhere to the residential village, *i.e.*, a group of houses bearing a separate name.

THE MAUZA.

106. The *mauza*, it may be explained, consists of a parcel of land, the boundaries of which were defined either by the revenue survey over half a century ago or by later cadastral surveys. It usually bears the name of the main village or collection of houses found on it when the survey was made, but it does not necessarily correspond with the latter. It may contain only that one village, or it may contain a number of separate villages, or it may be uninhabited. In some cases the portion of the *mauza* which was inhabited at the time of the survey may have disappeared owing to the village or villages being abandoned, or it may be known by a different name, or new villages or groups of houses may have been established. The area and boundaries of the *mauza*, however, remain unchanged. It cannot disappear, except by being diluviated, and it is therefore a constant unit. The adoption of this unit is an innovation in the Bengal census procedure, the residential village being hitherto the unit, except in cadastrally surveyed districts. The term 'village' was, however, elusive and difficult of definition, while its application proved a source of great divergencies. In some parts the only residential village which was locally recognized was the village which gave its name to the *mauza;* this, consequently, was the only village returned, other collections of houses being treated as hamlets (*tolas* or *aras*). Again, groups of houses at a distance from this village, each of which in the general acceptation of the term would be regarded as a separate village, were grouped together, because they bore the name of the parent village. Elsewhere, however, every collection of houses bearing a separate name was treated as a separate village. The general result in 1901 was summarized by Mr. Gait as follows :—

"There is no guarantee that the definition has been rightly or uniformly applied even now, or that a fresh enquiry would not result in many of the so-called hamlets being classed as villages and many of the villages transferred to the category of hamlets. And if it is difficult now to decide precisely what constitutes a residential village, it will be still more so ten years hence to say what was treated as a village at the present census. In the course of ten years many existing villages will have disappeared or changed their names, while new ones will have sprung up; large villages will have absorbed their smaller neighbours, and hamlets will have grown to the status of separate villages. Detailed comparison between the results of the two censuses is thus impossible where the residential village is taken as the unit."

SIZE OF MAUZAS.

107. The size of the *mauza* varies very greatly, and some are surprisingly large and populous. This is due to the fact that, at the time of the revenue survey, large tracts were under jungle and were consequently surveyed in large blocks. Since then the jungle has given way to cultivation, and villages have sprung up in what was waste land. Thus, in the west of Midnapore a large tract of jungle land was delimited in 19 blocks; these blocks, which are known as the Jungle Mahals, contain over 20,000 villages between them. Again, in the Gaya district, one *mauza* Kawakhol, with an area of 60 square miles, was treated as a single *mauza;* it now contains no less than 88 villages or hamlets with 14,608 inhabitants. In Muzaffarpur the average area of a *mauza* is 431 acres or about two-thirds of a square mile; but the individual *mauzas* range from a few acres to three square miles, and in the alluvial formation known as *diaras* extend to 19 square miles: in this district one *mauza*, Sarsand, has a population of 10,120 persons. In Purnea, where the population is not so dense, the average size of a *mauza* is a little under one square mile; here the smallest *mauza* has an area of only 5 acres, while the largest extends over 12,621 acres, or nearly 20 square miles. In Champaran the average is 1·14 square miles, or nearly double that of Muzaffarpur, but one *mauza* (Semra Labedaha) has the enormous area of 40 square miles and a population of 16,135 persons, while another extends over 14 square miles and has 11,540 inhabitants.

108. In Saran, a district where the pressure on the soil is very great, the average size of a *mauza* is only a little over half a square mile. Altogether 218 *mauzas* in this district are over 1,000 acres in area, and 697 between 500 and 1,000 acres; the remainder are all under 500 acres. The smallest has an area of less than 1½ acres; the largest (Shitab Diara) stretches over 16½ square miles, and contains 5,117 inhabitants. The latter is, as the name indicates, a *diara* tract, and its size is not therefore so extraordinary, but its population shows how, in course of time, *diara* lands are taken up for settled habitation and not merely for shifting cultivation. Parsa, an inland *mauza* in the same district, with an area of 6½ square miles and 6,479 inhabitants, is far more densely populated. In Bhagalpur, Khawaspur Milik has a population of 10,452 living in 18 villages, while in Puri the *mauza* of Pratap Sasan includes 27 villages with 5,094 inhabitants. A *milik*, it may be mentioned, was originally a rent-free property granted either for religious and charitable purposes or as a reward for services already rendered or liable to be rendered in the future; a *sasan* was a royal grant of rent-free land to Brahmans, and in Orissa there is still a class of Brahmans designated Sasani as they depend for their subsistence on such grants. The marginal statement shows the number of inhabited *mauzas* and residential villages or hamlets in the districts of the old Province of Bengal which have come under survey and in which the *mauza* was taken as the unit.

Division.	Inhabited *mauzas.*	Inhabited villages.
Burdwan	24,132	29,451
Presidency	13,389	21,233
Patna	13,231	23,566
Tirhut	14,352	29,656
Bhagalpur	19,714	32,301
Orissa	15,875	27,391

POPULATION OF VILLAGES.

109. It would be of little use to discuss the statistics for villages in Bengal owing to the difference between the definitions adopted in those districts which were administered by the two Governments, viz., Bengal and Eastern Bengal and Assam, and to the indeterminate character of the unit adopted by the latter. Suffice it to say that the average number of inhabitants per village is 352, and that nearly four-fifths of the rural population are in villages with a population of under 2,000. The least populous villages, as distinct from *mauzas*, are found in North Bengal, and the most populous in East Bengal, the average population being 261 and 391 respectively. The population of *mauzas* varies from 326 in West Bengal to 574 in Central Bengal. As an instance of the variations which may occur in the number of villages found in a district where the residential village is the unit, it will be sufficient to point to Jalpaiguri, where there were 3,330 villages in 1891, 766 in 1901 and 2,219 at this census.

110. In Bihar and Orissa, however, the *mauza* being a permanent unit, the statistics repay examination, though, owing to the change of the

definition of the census village, it is impossible to institute a comparison with the statistics of last census. In this Province the average population is 318. The mean is, however, reduced by the small villages of the Chota Nagpur Plateau, which contain on the average less than half as many people as the villages of North and South Bihar: in the latter area the most populous villages are found in Darbhanga, where there is a dense cultivating population averaging 900 per village. In the Province as a whole nearly nine-tenths of the rural population reside in villages with under 2,000 inhabitants. Large villages with over 2,000 inhabitants are far more frequent in Bihar than in other parts of the Province, the proportion of the rural population enumerated in them being 255 per mille in North Bihar and 178 per mille in South Bihar, whereas in Orissa the ratio is only 55, and in the Chota Nagpur Plateau 25 per mille.

CHARACTER OF VILLAGES.

111. In the two Provinces dealt with in this report it is not always easy to distinguish between an overgrown village and a small town having few, if any, urban characteristics. The density and numerical strength of the population are in themselves no criteria, for a village may be a compact area with over 5,000 inhabitants, all or nearly all engaged in cultivation. The main points of difference lie in the occupations of the people, for a town is a centre of trade, or at least has shops catering for the wants of its inhabitants and of the surrounding villages, or it is a place where the majority of the residents are engaged in non-agricultural pursuits. In the villages, however, the majority are devoted to agriculture, there being comparatively few of the industrial classes: such as there are have mostly occupations subsidiary to agriculture or are engaged in handicrafts supplying the simple needs of the villagers. As a rule, the village is purely residential, and shops are few and far between, the villagers getting their supplies at the periodical weekly or bi-weekly markets (*hats*) or the fair (*mela*) to which people flock in from the neighbourhood. In this respect rural Bengal is not unlike mediæval England, where nearly all selling and buying took place at weekly markets or annual fairs. The villages are still, to a very large extent, self-contained, the people meeting their wants from their own resources, but as the tentacles of modern commerce are thrown out further and further, this isolation is gradually being broken into. One sign of the change is the creation of a new type of village, which may be most appropriately described as a railway bazar. They are simply collections of houses, godowns, etc., which spring up in the neighbourhood of railway stations and grow by gradual accretion. They are often little more than crowded *bastis*, their inhabitants consisting of traders, coolies and others dependent on the railway and not on the produce of the fields. Another new type of village consists of the cooly lines in the neighbourhood of mills and mines.

112. The character of the residential villages varies considerably in different parts of the two Provinces. In Bengal the village generally consists of small groups of houses scattered through the rice and jute fields: large compact villages, where periodical markets are held, are usually found only on the banks of the rivers. The villagers live, more or less secluded, in detached homesteads, surrounded by a belt of fruit trees or bamboo thickets: the screen of trees and jungle secures that privacy which the Bengali likes for his domestic life. The oldest villages are almost invariably found on the banks of the rivers or in their neighbourhood, where there are ridges of comparatively high land and of considerable extent. The central basins between such ridges are swampy and unhealthy, but as the population increases and the village site becomes more crowded, the people build their houses further away from the river bank on mounds artificially raised in order to keep them above flood-level. A noticeable feature of the Bengal villages as compared with villages up-country is their cleanliness. The difference between them may be illustrated by an account given by a medical officer fresh from the Punjab: "The very first thing that struck me after coming from the Punjab was the cleanliness of the villages as compared with those of that Province, and also the plentiful and comparatively pure supply of drinking-water. In the Punjab there is often but one irregular shallow pond used by man and beast for washing and drinking, the banks

of which are extremely foul with excrement, which is washed into the water by every shower, and as the dry season progresses, makes the water more and more concentrated sewage. There too in every village it is a common thing for men, women and children to go but a very few yards from their houses to defecate. In this district (Murshidabad) I found nothing approaching this state of affairs : the houses were carefully *leeped* every morning, court-yards were swept, and all the lanes were quite free from human excrement ".

113. In Bihar the people are more gregarious. They live in closely packed villages standing on mounds that consist of the *debris* of former habitations ; instead of dispersed homesteads we find clusters of mud-walled houses grouped round a main street with narrow side lanes. In densely populated areas, the establishment of a new village is no easy matter, and the growing population has to find accommodation by over-crowding the existing houses or adding yet another house to the congested village site. Most villages are situated in the open, surrounded by dry cultivation, but in North Bihar many are built on the edges of swampy depressions. Some villages are surrounded by groves of palm trees which furnish liquor—strangely enough, the inhabitants of such villages are often Muhammadans, to whom such indulgence should be taboo—elsewhere they stand compact in the midst of bare treeless plains. The sanitation of the Bihar village leaves much to be desired. There is little or no attempt to secure proper drainage or cleanliness. The wells from which the people get their drinking-water supply are frequently neglected and dirty. Some, moreover, are in the inner court-yards of the houses, surrounded by the house drains, the contents of which gradually soak in and find their way into the water by percolation.

114. In Orissa the villages consist of groups of houses, each with a small compound enclosed by a bamboo fence, and containing a vegetable garden. They are screened by a belt of palm, mango and fig trees ; close by is the village tank, consecrated or married to a god, in the centre of which may be seen a small column or pole sacred to the deity. Most villages contain a small open shed in which the Bhagabat is recited before the assembled villagers, and in Sambalpur there is generally a rest-house for the accommodation of strangers.

115. In Chota Nagpur the villages are generally built on a ridge or near the crest of a slope, above the spot which the first settlers selected for the *bandh* or reservoir from which to irrigate their fields. They consist, as a rule, of a long straggling row of houses or of a single street with houses on each side, but occasionally contain narrow lanes striking off from the main street. When the village is first formed, the houses stand well away from one another, each with a little plot hedged in, but, as it grows, the villagers have to be content with more contracted sites and smaller enclosures. Trees, so conspicuous a feature of the Bengal village, are few in number ; but there is generally a solitary *pipal*, banyan or mango tree near the house of the village head-man, alongside which an open space is usually reserved as the *akhara* or village meeting place. Immediately outside the village, however, there are usually one or more groups of trees (generally *sal* in the villages of aboriginals), or even a single tree representing the grove (*sarna* or *jahira*) sacred to the village deity.* Many of the villages are very dirty, drinking-water being got from unprotected *kutcha* wells, which receive part of the drainage, and their general condition is more like that of Bihar than Bengal. The Santal village is an exception, the Santals keeping their houses and their surroundings exemplarily clean, a fact which partly accounts for the healthiness and vigour of the race.

116. The above account may be taken to apply to most villages in Chota Nagpur, but they vary in character with the race of their occupants. They are not all so bare and treeless as the village described above. The Khond village lies embedded in a leafy grove or at the foot of finely-wooded hills, or crowns some knoll in the valleys. The houses are built in two long rows forming a street ; at the back is a fence enclosing the homesteads

* H. Coupland, Manbhum District Gazetteer.

so that the whole village looks somewhat like a stockade. Both the Ho and Munda villages are distinguished by graveyards with massive slabs, beneath which lie the bones of past generations of the villagers. A collection of these sepulchral monuments invariably marks the site of a Ho or a Munda village; in addition to the slab at the tomb, a massive stone, 5 to 15 feet high, is set up to the memory of the deceased outside the village. The Bhuiya villages in the Orissa hills on the other hand are picturesquely placed at the foot of well-wooded hills by the side of a hill stream. "The village nestles in a fine grove of jack trees, to the fruit of which the Bhuiya is particularly partial. There is one broad street with the houses on either side. The house of the head-man and the village elders is in the centre of the street: on the outskirts live the low castes of Pans and Kols, who perform all the menial tasks of the Bhuiyas. In close proximity to the head-man's house is the *darbar* or *mandap* (drum) house, where the bachelors of the village sleep, and the place in front is used as the village dancing ground. The *darbar* house is also the village guest-house: here are stored the provisions contributed by the villagers and made up into bundles ready for the immediate use of the guest."* The Oraon villages, on the other hand, are generally huddled together without any attempt at a village street; there are no thoroughfares, but only narrow twisting paths—"a most perfect labyrinth leading to an infinite series of *cul-de-sacs*, each one or more puzzling than the last. A European who finds himself in one of these mazes would find it impossible to get out of it without a guide." †

117. The villages of the aboriginals are by no means always permanent. Should a village be attacked by some epidemic disease, the inhabitants believe that the spot is haunted by some evil spirit, demolish their house and move to some more favourable site. Another feature which deserves mention is that in Oraon, Khond, Bhuiya and Sauria Paharia villages there is generally a dormitory for unmarried boys and another for unmarried girls.

118. In Darjeeling and Sikkim there are no villages in the proper sense of the word, but only homesteads nestling on the hill sides or in the valleys. Occasionally five or six houses are grouped together, but generally each homestead stands in its own land near the patches of cleared cultivation. Clusters of houses, which can be dignified by the designation of villages, are only found in a few bazars to which the people go to obtain their weekly supplies of food. In the Tarai the social unit is not, as elsewhere in India, the village, but the *jot* or farm, *i.e.*, the homestead of a substantial farmer or *jotdar* with the houses of his relations, tenants and farm labourers clustered round it. The *jotdar* keeps the little community together and maintains a store-house, elevated on piles, in which his stock of rice is kept and from which he makes loans to his dependants or furnishes them with seed.

DEPRESSED CLASSES.

119. On the outskirts of many villages in the plains may be seen a small cluster of houses in which live the degraded semi-Hinduized castes, the 'untouchables' as they are called by the modern Bengali. These consist of dirty ill-thatched houses, which present a very different appearance to the neat, well-swept and tidy buildings of the better class Hindus. In Orissa these detached hamlets are occupied by such castes as the Pans and Gandas, in Bihar by Musahars and Doms, and in Bengal by the unclean Haris and Bauris.

HOUSES.

DWELLING HOUSES.

120. In Bengal the dwelling house, or, as it may perhaps be more properly called, the homestead, is as a rule composed of four huts, built round and facing a central court-yard, with detached cattle-sheds and out-houses. Two of the huts forming

* L. E. B. Cobden Ramsay, Orissa States Gazetteer, p. 51.

† Rev. P. Dehon, *Religion and Customs of the Oraons*, Memoirs of Asiatic Society of Bengal, Vol. I, 1906.

the house are used as living rooms—one for the male and the other for the female members of the household—another is a cookshed and the third is the *baitakkhana* or sitting room, where visitors are received and the men sit and smoke. They are usually built on raised plinths, and the walls consist of bamboos or reeds plastered with mud, or are built of earth, which, in the lateritic districts of West Bengal, hardens, with exposure to the air, almost of the consistency of stone. The earth required for the plinth and walls is taken from pits dug in the neighbourhood, which in the rains are full of water and afford congenial breeding-grounds for mosquitoes. The roofs are covered with thatch of considerable thickness, and have a curved hog-backed ridge, especially designed for withstanding the heavy rainfall of the delta. Sometimes the roofs are tiled, and those who can afford it are beginning to roof their houses with corrugated iron (in which they adhere to the same immemorial curve), as a protection against arson, which is a favourite form of crime in some parts of Bengal.

121. In Bihar the houses of the cultivators are, as a rule, mud-walled huts, built of earth dug up in the vicinity, with which broken pottery is mixed, so as to impart solidity. The roof is, as a rule, made of thatch—a frequent source of fire; only the well-to-do can afford tiled roofs. There is no provision for ventilation, but this is no great hardship to peasants who regard their houses merely as places for cooking and sleeping in. In riverain tracts, liable to flood from great rivers, the cultivators live in huts with wattled walls and thatched roofs, for the soil is often so sandy that mud for the walls cannot be obtained, while the risk of their houses being washed away makes the use of other materials a piece of useless extravagance.[a] The houses are therefore erected above the level of flood water and are constructed of bamboo framework, thickly plastered over, and thatched with straw. Some of the poorer classes have only huts made of reeds which scarcely support a thatch. These, however, have the advantage of being portable. As a Bengali writer says—"A man like a snail can carry his house anywhere and raise it anew." The richer classes only live in brick-built houses. The ordinary cultivator either cannot get or cannot afford bricks, and, besides this, there is in some places a superstitious belief that brick-built walls attract the evil eye; the well-to-do, however, know that it is harder for a burglar to break through a brick wall than a soft earthen wall. There is also a prejudice in some parts against square houses; houses should be oblong, and the two longer sides should run north and south.

122. In Chota Nagpur the average house consists of three mud-walled and thatched buildings, one of which is the sleeping apartment, one a kitchen and one a cattle-shed. They are arranged on three sides of a quadrangle; on the fourth or open side is a plot of land, on which are grown various crops and vegetables for domestic consumption. Behind one of the three huts is another plot usually enclosed by mud-walls, in which corn is threshed and fodder and manure are stored. The site has to be carefully selected from superstitious motives. A Brahman or Ojha is consulted as to whether the site is a lucky one and what is an auspicious day for commencing building. Some of the wilder tribes place 21 grains of paddy on the spot selected over-night and return in the morning to ascertain the result. If the grain has been disturbed or attacked by white-ants during the night, the spot is abandoned as unlucky; if it is untouched, the building is commenced.

123. Among some of the aboriginal races the houses are of a very primitive kind. "The huts of the Juangs," writes Colonel Dalton, "are about the smallest that human being ever deliberately constructed as dwellings. They measure about 6 feet by 8, and are very low, with doors so small as to preclude the idea of a corpulent householder. Scanty as are the above dimensions for a family dwelling, the interior is divided into two compartments, one of which is the store-room, the other used for all domestic

[a] The name of the headquarters station of the Saran district, viz., Chapra, is believed to be derived from *Chhapar*, meaning a thatched roof, and is evidence of its liability to inundation in early times.

arrangements. The paterfamilias and all his belongings of the female sex huddle together in this one stall, not much larger than a dog-kennel; for the boys there is a separate dormitory." The narrow entrance. into which the owner is obliged to creep on all fours, is characteristic of the rudest huts used by Dravidian races, and it has been suggested that it is a reminiscence of cave-life. This feature is also found in the Oraon houses, which are small and low, most of them consisting of four mud walls, 15 feet long, 7 feet high and 6 feet broad, surmounted by a thatched roof. In the middle of one of these walls there is a hole 4½ feet high, which serves as an entrance, the door consisting of two big planks roughly hewn out of the trunk of a tree. Above is a log which supports the wall and is aptly called the *kanarphora* or forehead-breaker, whilst on both sides there is a raised verandah, under which the household pigs have their sty. Inside, there are three rooms, in the middle one of which the family live and prepare their food. On one side of it is a room in which the bullocks and the goats are kept; on the other is the granary and store-room.

124. The houses of the Khonds call for special mention, as they are made entirely of wood without a single nail being used. The Khond builds his house himself, his only tools being a hatchet and a chisel. With these he hews out thick planks from the log of a tree, and erects grooved posts to form the framework of his house. Planks are slid into the grooves and bound together by cross-stays, which are fixed by wooden pins and keys. The doors are ingeniously made to revolve in grooved blocks fixed to the frame. The roof consists of thin flat rafters with a thatch of straw, and the only repair it requires is the addition of a layer of fresh straw every year. It takes a Khond two years to build a house, and it lasts from 20 to 30 years. The interior generally consists of two small rooms partitioned off by a railing. One is used for cooking and sleeping in, the other serves as a cattle-pen. The younger members of the family and the servants sleep in a separate room. where the stock of grain is also stored. The grown-up girls sleep together in a dormitory in charge of one of the old women, and there is another dormitory for the young men.

125. An entirely different type of house is found among the hill tribes in the Chittagong Hill Tracts. They are raised about six feet above the ground, access being obtained by means of a rough step ladder. In front is a verandah, behind which come the bachelors' quarters. At the back of the latter are the rooms of the married members of the family, separated by mat walls. They are apportioned according to seniority, one being reserved for the eldest married member of the family and his wife, another for the second eldest, and so on.*

126. In whatever part of the country they may be, the houses have one common feature, viz., the absence or rarity of windows by which the interior can be ventilated. This is not a matter of much importance in the houses of the lower classes, for the men live out in the open, and for the women there is a certain amount of ventilation through the thatched or tiled roof, or through the walls where the latter are made of reeds or bamboos: there is, moreover, generally a space between the walls and the roof which allows of the perflation of air. The flimsy nature of the walls is really sanitary, and in the hills of Darjeeling and Sikkim the comparative rarity of consumption may be put down to this account. It is a different matter for *pardah* women living in brick-built house. The rooms are jealously closed, and the windows, if any, are small in size, are near the top of the wall, and are securely latticed, so that these are of little use for ventilation. The insanitary effects of the *purdah* system are accentuated in towns. To quote from a report by Dr. H. M. Crake on the sanitary condition of the northern portion of Calcutta:—"No survey of an oriental city can possibly ignore the potent influence of the *purdah* system on its domestic architecture. Obviously, the house is directly inspired by the necessity of securing absolute privacy for the ladies of the household. To effectually seclude the inner apartments from the vulgar gaze, air and light are shut out and the rooms rendered

* Chittagong Hill Tracts Gazetteer.

unfit for human habitation. It is very common to find the whole of the lower storey of the zanana, even in large and valuable houses, given up to godowns and kitchens, the inmates frankly admitting that none of the rooms are fit to live in. I must confess I am astonished at the average kitchen. It is, in a large number of houses, a gloomy, stuffy den, full of acrid smoke, and yet the ladies of the house have to spend hours in these very unpleasant surroundings. The entire absence of chimneys results in an atmosphere which is almost unbearable when cooking is going on in a particularly ill-ventilated kitchen."

HOUSES AND FAMILIES.

THE CENSUS HOUSE.

127. The house for census purposes is a social and not a structural unit. As at the last census, it was defined "as consisting of the buildings, one or many, inhabited by one family, *i.e.*, by a number of persons living and eating together in one mess, with their resident dependants, such as mother, widowed sisters, younger brothers, etc., and their servants who reside in the house." In other words, the unit is the commensal family, and not the homestead or enclosure. The value of this definition is that it is easily understood and requires very little explanation : it is, in practice, the definition used in the interpretation of the Chaukidari Act, and is no novelty to the people. There were some exceptions to the standard definition, but they were few in number. In the case of Europeans and Anglo-Indians the whole building actually occupied by them was taken as one house. For police lines, jail, dispensaries, lunatic asylums, etc., special arrangements were made. In cooly lines each room with a separate door-way was treated as a separate house, and in the *bastis* of Howrah and Calcutta each hut was numbered as a house.

128. The average number of persons per house as thus defined is 5·3 in Bengal and 5·2 in Bihar and Orissa, which have changed places in this respect during the last decade. The variations between the different divisions are small, the maximum being 5·5 (in Central and East Bengal) and the minimum 4·6 (in West Bengal). In Central Bengal the average is slightly inflated by the figures for Calcutta, where the house was defined as the municipal premise ; in West Bengal the low average is due to the immigrant population of labourers living in huts, *bastis* or cooly lines, where each room was numbered as a house. There are 105 houses per square mile in Bengal, or 38 more than in Bihar and Orissa : for this difference the area of waste, hill and jungle in the Chota Nagpur Plateau is mainly responsible. Their relative density is highest in West Bengal with its numerous towns, and then in North Bihar, which has comparatively few towns but a dense agricultural population. Of individual districts, Howrah has most (433) and the Chittagong Hill Tracts fewest (5) houses per square mile.

129. The figures for cities give very divergent results, owing to the fact that in Calcutta and the suburban municipalities of Cossipur-Chitpur Manicktollah and Garden Reach the unit was the municipal premise. In Howrah city, where the Bengal definition of house was adopted, the average number of persons enumerated in each house is only 2·9 : this low figure is due to the number of *bastis* with a cooly population where each hut or room was treated as a house. The fall of the average since 1901, when it was 3·4, may be regarded as a result of the measures taken to open them out and prevent overcrowding. With this number may be compared the average of Patna city, viz., 4·8. The variation is sufficient proof, if any is needed, of the difference between conditions in a progressive but congested industrial town and a decadent town with no large manufactures. Both in Patna and in Bhagalpur the average number of persons per house has risen since 1901 ; in the former the figure is still below that for the district generally, and in the latter it is exactly the same. In the case of Gaya no such comparison is possible, as the place was half empty at the time of the census.

130. The average number of houses per square mile has increased steadily in each Province and in each Division, except in Central Bengal, where however the apparent decrease is due to the change of definition of house in Calcutta and its suburbs. It is difficult to draw from the statistics of census houses any general conclusions as to the number of families, owing to the joint-family system. A house, as defined for census purposes, does not necessarily imply the existence of one married couple with their children: there are probably also sons and nephews with their wives, and the widows of the husband's brothers or his sons' widows, together with their children. The different conditions prevailing may however be roughly gauged by comparing the number of married females aged 15 and over with the number of houses, as in the marginal statement. It will be seen that the relative number of houses has decreased throughout Bengal, but has increased in Bihar and Orissa except in the Chota Nagpur Plateau.

Province or Division.	Number of houses per 100 married females aged 15 and over.	
	1901.	1911.
Bengal	***104***	***101***
West Bengal ...	118	115
Central ,, ...	109	102
North ,, ...	101	100
East ,, ...	97	95
Bihar and Orissa	***88***	***90***
South Bihar	84	86
North ,, ...	85	87
Orissa	91	96
Chota Nagpur Plateau ...	96	94

Disruption of Joint Families.

131. The increase of houses in the latter Province is partly due to the natural increase of families, but the growing tendency to break up the joint-family is a contributory cause. During the lifetime of the father, the Hindu family, as a rule, lives jointly, *i.e.*, it not only lives together, but its property is also held in common under the managership of the *karta* or head of the family. All who belong to it, *e.g.*, sons, grandsons, nephews, etc., are entitled to maintenance from its funds, and all contribute to the expenses, whether present in the house or absent from it—in the latter case, they are bound to remit their savings home. This system has all the force of a religious institution, being based both on sacred texts and immemorial custom, but there is a general consensus of opinion that the family is now-a-days more frequently broken up when the father dies. In Bihar, it is reported, it was the general practice, within living memory, for families to remain joint for two, three, or even more generations. Now it is estimated that the number in which the joint system is maintained for any considerable time after the father's death—much less for two generations or more—is less than one-fourth. It is a common practice for brothers to set up for themselves either as soon as their father is dead, or, a little later, while their mother is still living. In most cases, when the adult brothers partition the ancestral property in this way, the younger children have to cast in their lot with one or other of them, the mother generally remaining with the youngest of her children. But, though they set up separate establishments, they often continue undivided in legal and other business affecting their property. Where this is the case among the landlord classes, no application for partition of the estate is presented to the Collector; the rent is still collected in a lump sum, but after it is realized, it is divided among the sharers.

132. Among cultivators, holdings sometimes remain joint for a considerable time after the buildings, furniture, etc., have been apportioned among the members of the family, the division of the produce taking place on the *khalihan*, or threshing floor, after it has been reaped. In the majority of cases, however, when the family ceases to live together, a partition is made of the holdings, and their accounts are entered separately in the landlord books. On the whole, the family remains joint among the peasantry for a longer time than among the non-agricultural classes, the reason being simply that the larger the labour force, the easier it is to till the land. In the case of industrial and professional pursuits, where the personal equation is far more important, the individualistic tendency is more pronounced.

133. In Orissa, as in Bihar, the family generally remains joint so long as the father or mother is alive, but is broken up after the death of the parents. The disruption takes place at once if their sons are married and have children, and, if not, later, when they have married and have children of their own. It is estimated that only one-fifth of the families are now joint. In Sambalpur, which is governed by the Mitakshara law, according to which the son has the same rights as the father in the ancestral property, the sons are more

prone to demand or enforce partition during his lifetime. In other parts of Orissa it is practically out of the question for a son to separate unless he has some independent means of livelihood, whereas in Sambalpur the sons are sure of a share in the property. In this latter district, therefore, the family is more likely to break up after one of its members marries and begins to live with his wife. In Orissa generally, however, the longer a family remains joint, the more are its members held in esteem, for the breaking up of the family, though of common occurrence, is looked upon with disapproval. Neither marriage nor the death of the father necessarily causes the sons to leave the ancestral home. On the contrary, they generally continue to live together in the same homestead, but in separate messes.

134. In cities also the tendency is for the family to continue to live under the same roof but in separate rooms. To quote again from the report by Dr. Crake on the sanitary condition of northern Calcutta :—

"The curious system of actually dividing dwelling houses amongst several co-heirs is a very potent factor in the production of insanitary property. It is quite common to come across what was originally a single dwelling split up amongst three or four relatives. Owing partly to the *pardah* system, but very largely to the bad blood engendered by the almost inevitable litigation which the partition involves, each co-sharer erects as lofty a masonry wall as he possibly can, so as to completely shut off his share from the rest. Though carried out with wonderful ingenuity, the result too frequently is that a noble mansion with spacious court-yards is converted into a number of mean little houses with totally inadequate open spaces, and most of the rooms imperfectly lighted and ventilated. Very often one unfortunate heir can only reach his portion through a long tortuous passage."

135. The following are the chief causes to which the disintegration of joint families is ascribed.—(1) Some members of the family take advantage of their position to lead a life of idleness and become simply drones, living on the labour of their brothers. As an instance of this, may be quoted the case of an officer in Government employ, who obtained a large increase of pay but was poorer than before, because his elder brother at once threw up his own post and ceased to contribute to the family income. (2) Misappropriation or misuse of the joint property, *e.g.*, the *karta* may devote part of it to his own personal uses or employ it for purposes which do not benefit the family. (3) The family becomes so large, that there is no longer room for all under the ancestral roof. Some of the joint families are exceptionally large forming small colonies—there is a case on record of a joint family with 500 members.* For the sake of convenience, some of them move out to make homes for themselves. This division is often followed by a partition of the property. (4) Migration. Members of the family leave the home in search of employment, and do not return. Having to live apart from their relations, and finding no chance of returning home, they naturally do not see why they should not obtain a separate share of the property. In Bihar a number of joint families have been broken up on this account, especially among Kayasths who furnish recruits to Government service. Brahmans and Rajputs, who do not so generally find employment in occupations necessitating their absence from home, have, it is reported, a larger number of joint families than the Kayasths. (5) The abolition of the Panchayat system. Formerly disputes between the members of a joint family were settled by the Panchayat quickly and cheaply. Now, they have recourse to *mukhtars* and *vakils*, the result being protracted litigation, embittered feelings, and eventually the impoverishment and dismemberment of the family. (6) Modern tendencies, such as the influence of education, the throwing off of caste-ties, especially in towns, and the consequent weakening of the family bond. These tendencies are confined to the educated classes and mostly come into operation where some members are conservative and others have advanced ideas. The disintegration of the family may be due to their neglect of caste rules or to their wanting to live in a more luxurious or laxer style than their forefathers; in one case a family divided merely because one of them decided to give an English education to his daughters.

* S. C. Bose, *The Hindus as they are* (1883), p. 2.

136. Women are frequently instrumental in producing the dismemberment of families. This is especially the case where the husbands marry girls from some distant village and from families with which they have little or no past connection. Devoted to their husbands' interests, the wives are jealous of their earnings being used by others, particularly by those who do not contribute to the family income. More petty feelings, less disinterested motives, such as the mutual jealousy of the brothers' wives, the quarrels of their children, etc., also contribute to the breaking up of the family. More than one correspondent points out that it is significant that one of the Sanskrit word for wife, viz., *dara*, comes from a root meaning "to tear asunder."

137. Notes on customs regulating inheritance and partition among the aboriginals of the Chota Nagpur Plateau are given in the appendix at the end of this volume. There appear to be signs that these customs, which are the outcome of an earlier state of development, are gradually becoming modified in tracts where aboriginals are brought into contact with more civilized neighbours. Tribal customs are thus gradually breaking down, and this process will, in the nature of things, become more general as the aboriginals adopt the manners and customs of their more advanced neighbours and as their contact with Hindus becomes more frequent. It may be of value, however, to place on record the customs as they still exist.

SUBSIDIARY TABLE I.—Density, Water-supply and Crops of Districts.

District and Natural Division.	Mean density per square mile in 1911.	Percentage of total area.		Percentage to cultivable area of—		Percentage of gross cultivated area which is irrigated.	Normal rainfall.	Percentage of gross cultivated area under—			
		Cultivable.	Net cultivated.	Net Cultivated.	Double-cropped.			Rice.	Jute.	Other food-crops.	Other non-food crops.
1	2	3	4	5	6	7	8	9	10	11	12
BENGAL	**567**	**69·7**	**49·5**	**70·9**	**16·7**	**4·3**	**69·84**	**69·2**	**8·2**	**13·0**	**9·6**
WEST BENGAL	**607**	**79·6**	**53·0**	**66·6**	**5·5**	**20·5**	**56·86**	**79·3**	**2·2**	**10·2**	**8·3**
Burdwan	572	82·6	52·6	63·7	15·4	39·6	56·06	81·9	1·3	11·6	5·0
Birbhum	534	87·8	69·7	79·4	2·5	21·5	55·77	92·8	...	3·2	4·0
Bankura	434	90·0	33·5	37·3	1·6	14·8	55·26	84·3	...	8·6	7·1
Midnapore	544	70·9	56·7	80·0	2·0	10·6	59·45	76·1	0·6	13·3	10·0
Hooghly	918	74·2	52·3	70·5	7·0	31·0	58·69	66·4	11·8	7·0	14·8
Howrah	1,850	83·9	61·7	73·1	13·1	...	56·95	58·3	14·2	11·1	16·4
CENTRAL BENGAL ...	**565**	**65·5**	**41·5**	**63·4**	**13·8**	**3·7**	**58·55**	**75·1**	**4·6**	**12·2**	**8·1**
24-Parganas	502	46·2	33·3	71·8	3·1	0·1	63·10	85·8	7·5	4·3	1·9
Nadia	580	79·0	37·1	46·9	21·2	0·2	57·20	67·6	3·6	15·8	13·0
Murshidabad	640	89·1	42·3	47·6	17·4	18·3	54·04	67·1	3·5	20·6	8·8
Jessore	601	67·3	59·0	80·5	14·3	...	60·72	76·6	3·3	10·4	9·2
NORTH BENGAL ...	**527**	**75·0**	**54·8**	**73·0**	**23·7**	**1·0**	**88·76**	**61·8**	**9·9**	**12·1**	**16·2**
Rajshahi	566	74·3	55·3	75·1	27·0	0·1	59·79	67·6	6·1	11·9	14·4
Dinajpur	428	80·2	50·1	62·4	0·5	...	69·13	79·0	8·8	6·0	6·2
Jalpaiguri	309	70·4	49·5	68·9	15·4	...	130·28	63·3	7·2	4·0	25·0
Darjeeling	223	33·2	23·1	69·6	4·1	16·2	120·33	22·5	2·7	41·6	33·2
Rangpur	686	73·5	65·6	89·3	39·0	3·0	83·80	57·3	13·4	9·9	19·4
Bogra	724	86·2	53·5	62·0	37·7	...	72·79	64·4	16·1	8·0	11·5
Pubna	772	81·3	63·3	77·8	56·6	...	60·63	47·9	14·1	21·8	16·2
Malda	529	86·4	65·0	75·3	16·2	...	54·22	62·3	2·8	19·4	15·5
EAST BENGAL	**573**	**63·9**	**47·9**	**75·0**	**18·9**	...	**82·08**	**69·4**	**10·8**	**6·6**	**14·2**
Khulna	287	38·1	28·6	75·2	5·5	...	65·97	84·3	1·7	3·3	10·7
Dacca	1,066	75·3	67·0	89·0	26·6	...	69·22	65·1	11·0	11·1	12·8
Mymensingh	724	70·4	59·2	84·1	31·9	...	83·93	49·1	21·4	9·3	20·2
Faridpur	524	69·7	61·8	88·7	12·2	...	65·59	70·1	11·3	6·8	11·8
Backergunge	523	79·3	64·9	81·8	12·1	...	84·29	81·6	1·5	4·5	12·4
Tippera	972	82·9	72·0	86·8	22·6	...	83·81	71·3	17·2	2·2	9·3
Noakhali	792	88·8	84·5	95·1	42·4	...	111·92	77·1	2·3	6·5	14·1
Chittagong	605	45·2	37·2	82·2	9·5	...	126·83	90·9	0·03	2·0	7·1
Chittagong Hill Tracts ...	30	51·7	4·6	9·0	...	...	93·03	58·5	...	8·5	33·0

District and Natural Division.	Mean density per square mile in 1911.	Percentage of total area.		Percentage to cultivable area of—		Percentage of gross cultivated area which is irrigated.	Normal rainfall.	Percentage of gross cultivated area under—			
		Cultivable.	Net cultivated.	Net cultivated.	Double-cropped.			Rice.	Maize.	Other cereals and pulses.	Other crops.
1	2	3	4	5	6	7	8	9	10	11	12
BIHAR and ORISSA	**415**	**72·7**	**51·7**	**71·1**	**13·4**	**12·3**	**52·48**	**54·3**	**5·3**	**26·6**	**13·8**
NORTH BIHAR	**646**	**89·6**	**68·2**	**76·1**	**26·3**	**9·8**	**53·36**	**43·2**	**6·4**	**31·6**	**18·6**
Saran	853	91·3	76·9	84·2	29·9	26·1	45·03	24·2	11·2	47·1	17·5
Champaran	540	83·1	55·7	67·0	24·8	6·2	54·09	40·3	6·3	34·3	18·6
Muzaffarpur	937	90·3	74·4	82·4	59·1	1·8	45·88	33·9	7·9	38·8	19·4
Darbhanga	875	91·0	71·9	79·0	32·7	14·9	49·77	41·2	5·1	35·0	18·7
Bhagalpore	506	90·0	66·0	73·3	24·0	12·0	51·27	54·8	7·3	27·8	10·1
Purnea	398	91·4	67·9	74·3	3·4	1·2	72·51	60·4	1·1	11·4	27·1
SOUTH BIHAR	**515**	**75·7**	**59·3**	**78·3**	**14·3**	**22·2**	**45·18**	**44·2**	**6·5**	**37·7**	**11·6**
Patna	778	86·2	84·2	97·6	10·5	29·4	45·26	43·5	9·4	37·2	9·9
Gaya	458	78·2	67·1	85·8	8·4	18·2	42·96	59·7	2·8	27·4	10·1
Shahabad	427	78·6	54·3	69·1	11·3	35·5	43·62	42·7	2·1	42·3	12·9
Monghyr	544	63·7	42·2	66·2	29·9	9·0	49·01	24·1	14·4	48·0	13·5
ORISSA	**505**	**70·3**	**55·1**	**78·3**	**9·2**	**18·6**	**59·10**	**82·49**	**·01**	**7·2**	**10·3**
Cuttack	577	72·8	52·5	72·2	14·3	18·2	60·41	77·7	.....	11·4	10·9
Balasore	506	77·6	69·1	89·2	0·8	12·9	60·79	89·4	.....	1·8	8·8
Puri	410	60·8	47·1	77·4	9·1	25·7	55·66	83·18	0·02	5·9	10·9
CHOTA NAGPUR PLATEAU	**221**	**62·4**	**38·6**	**61·9**	**3·5**	**7·9**	**53·75**	**66·2**	**4·9**	**18·5**	**10·4**
Hazaribagh	184	63·1	39·0	61·7	3·3	7·1	51·29	72·1	8·5	13·6	5·8
Ranchi	195	69·1	40·1	58·0	0·9	0·2	53·54	72·3	0·2	12·7	14·8
Palamau	140	45·6	18·6	40·8	3·1	9·8	48·16	54·0	7·9	25·4	12·7
Manbhum	373	65·2	57·9	88·8	4·3	8·4	52·68	78·1	4·9	10·0	7·0
Singhbhum	178	57·3	34·3	59·9	2·5	8·8	58·32	65·1	3·5	22·5	8·9
Santhal Parganas ...	345	78·3	49·0	66·9	8·6	16·3	53·85	46·9	8·5	33·6	11·0
Angul	119	29·2	22·3	76·6	6·1	5·5	53·37	51·5	1·9	18·8	27·8
Sambalpur	195	72·3	37·2	51·5	0·9	4·8	58·48	74·6	0·1	14·8	10·5

In the calculations for each Province and for each natural division, those areas for which figures are not available have been left out of account.

SUBSIDIARY TABLE II.—DISTRIBUTION OF THE

THANAS WITH A POPULATION

PROVINCE AND NATURAL DIVISION.	Under 150.		150—300.		300—450.		450—
	Area.	Population (000's omitted).	Area.	Population (000's omitted).	Area.	Population (000's omitted).	Area.
1	2	3	4	5	6	7	8
BENGAL, BIHAR AND ORISSA.	*40,515* *20·6*	*3,548* *4·2*	*40,139* *20·5*	*8,608* *10·1*	*32,871* *16·8*	*12,079* *14·2*	*25,862* *13·2*
Bengal	*12,892* *15·3*	*734* *1·6*	*4,110* *4·9*	*976* *2·1*	*13,683* *16·3*	*5,167* *11·1*	*17,017* *20·2*
WEST BENGAL*	...	...	*888* *6·4*	*252* *3·0*	*3,842* *27·5*	*1,428* *16·9*	*4,130* *29·6*
Burdwan	...	...	...	...	532	204	1,273
Birbhum	...	...	...	...	309	139	1,167
Bankura	...	...	...	...	1,589	589	775
Midnapore*	...	...	888	252	1,412	496	752
Hooghly	...	...	...	...	...	...	163
Howrah	...	...	...	...	...	...	...
CENTRAL BENGAL	*3,256* *25·6*	*301* *3·7*	...	...	*619* *4·8*	*267* *3·3*	*3,981* *31·3*
24-Parganas	3,256	301	...	...	...	...	50
Calcutta	...	...	...	...	...	...	...
Nadia	...	...	...	...	316	135	1,596
Murshidabad	...	...	...	...	119	50	1,283
Jessore	...	...	...	...	184	82	1,052
NORTH BENGAL	*412* *2·0*	*49* *0·5*	*2,085* *10·2*	*467* *4·4*	*5,960* *29·0*	*2,254* *21·0*	*5,553* *27·0*
Rajshahi	...	...	...	...	524	176	1,152
Dinajpur	...	...	...	...	2,641	1,018	1,305
Jalpaiguri	...	...	1,361	278	1,286	480	272
Darjeeling	412	49	724	189	...	...	...
Rangpur	...	...	...	...	...	...	974
Bogra	...	...	...	...	...	...	441
Pubna	...	...	...	...	...	...	221
Malda	...	...	...	...	831	306	559
Cooch Behar	...	...	...	...	678	274	629
EAST BENGAL*	*9,224* *25·1*	*384* *2·0*	*1,137* *3·1*	*257* *1·3*	*3,262* *8·8*	*1,218* *6·4*	*3,353* *9·1*
Khulna*	...	...	...	...	263	113	348
Dacca	...	...	...	...	...	...	618
Mymensingh	...	...	...	...	1,110	441	393
Faridpur	...	...	...	...	...	...	190
Backergunge	...	...	624	145	1,133	376	1,326
Tippera	...	...	...	...	...	...	...
Noakhali	...	...	...	...	185	75	258
Chittagong	...	...	513	112	571	213	215
Chittagong Hill Tracts	5,138	154	...	...	...	...	...
Hill Tippera	4,086	230	...	...	...	...	...
Bihar and Orissa	*27,623* *24·7*	*2,814* *7·3*	*36,029* *32·2*	*7,632* *19·9*	*19,188* *17·2*	*6,912* *18·0*	*8,845* *7·9*
NORTH BIHAR	...	...	*1,278* *5·9*	*316* *2·2*	*5,082* *23·3*	*1,862* *13·2*	*3,378* *15·5*
Saran	...	...	...	...	...	...	...
Champaran	...	...	...	...	1,463	446	...
Muzaffarpur	...	...	...	...	...	...	...
Darbhanga	...	...	...	...	...	...	...
Bhagalpur	...	...	534	117	707	301	2,036
Purnea	...	...	744	199	2,912	1,115	1,342
SOUTH BIHAR	...	...	*2,885* *19·1*	*658* *5·5*	*3,856* *25·6*	*1,432* *15·4*	*3,009* *20·0*
Patna	...	...	...	...	...	...	231
Gaya	...	...	693	191	1,983	721	953
Shahabad	...	...	1,604	331	652	236	850
Monghyr	...	...	588	136	1,221	475	975
ORISSA	...	...	*1,433* *17·4*	*390* *9·3*	*2,990* *36·3*	*1,133* *27·1*	*1,594* *19·4*
Cuttack	...	...	...	...	1,669	598	391
Balasore	...	...	213	61	377	159	1,203
Puri	...	...	1,220	329	944	376	...
CHOTA NAGPUR PLATEAU	*27,623* *41·9*	*2,814* *28·7*	*30,433* *45·6*	*6,268* *50·7*	*7,260* *10·9*	*2,485* *20·1*	*864* *1·3*
Hazaribagh	2,742	356	4,279	933	...	...	...
Ranchi	2,477	300	3,853	818	774	270	...
Palamau	2,592	252	2,322	435	...	...	...
Manbhum	...	...	1,238	336	2,317	861	237
Singhbhum	820	54	3,071	640	...	...	...
Santhal Parganas	...	...	1,499	395	3,235	1,031	581
Angul	1,681	199	...	...	...	...	...
Sambalpur	847	87	2,491	469	556	188	...
Orissa Feudatory States	16,464	1,566	11,158	2,098	378	115	46
Chota Nagpur Tributary States	...	...	602	149	...	...	...

* The areas shown for West Bengal and Midnapore exclude 41 square miles of uninhabited river beds
The proportions per cent. which the area and population of each group bear to the total

POPULATION CLASSIFIED ACCORDING TO DENSITY.

PER SQUARE MILE OF—

	600—750.		750—900.		900—1,050.		1,050 and over.	
Population (000's omitted).	Area.	Population (000's omitted).	Area.	Population (000's omitted).	Area.	Population (000's omitted).	Area.	Population (000's omitted).
9	10	11	12	13	14	15	16	17
13,463	***21,263***	***14,243***	***16,294***	***13,287***	***9,427***	***9,119***	***6,821***	***10,394***
15·9	*10·8*	*16·8*	*8·3*	*15·7*	*4·8*	*10·8*	*3·5*	*12·3*
8,833	***13,781***	***9,308***	***8,817***	***7,228***	***5,201***	***4,996***	***5,862***	***9,064***
19·1	*16·4*	*20·1*	*10·5*	*15·6*	*6·2*	*10·8*	*7·0*	*19·6*
2,172	***2,287***	***1,522***	***1,375***	***1,130***	***551***	***513***	***834***	***1,450***
25·7	*16·4*	*18·0*	*9·9*	*13·3*	*4·0*	*6·0*	*6·0*	*17·1*
686	608	404	278	244	...	...	...	...
613	276	183	...	...	...	...	...	...
390	257	170	...	...	...	...	...	...
404	755	499	836	667	373	331	129	152
89	391	266	261	219	178	162	195	354
...	...	...	...	...	...	...	510	944
2,057	***2,456***	***1,665***	***890***	***736***	***358***	***340***	***1,174***	***2,713***
25·5	*19·3*	*20·6*	*7·0*	*9·1*	*2·8*	*4·2*	*9·2*	*33·6*
27	103	74	186	150	157	142	1,092	1,741
...	...	...	...	...	...	...	32	896
828	429	280	449	375	...	...	...	...
691	490	357	...	...	201	198	50	76
511	1,434	954	255	211	...	...	...	...
2,863	***2,831***	***1,860***	***2,735***	***2,239***	***559***	***511***	***40***	***488***
26·7	*15·8*	*17·3*	*13·3*	*20·9*	*2·7*	*4·8*	*2·0*	*4·5*
601	448	289	332	286	142	129	...	...
670	...	...	...	...	...	...	...	...
145	...	...	...	...	...	...	...	...
...	...	...	...	...	28	27	...	...
506	1,188	770	1,122	877	...	...	195	232
226	288	201	241	202	389	355	...	...
108	779	505	639	559	...	...	212	256
259	128	95	381	315	...	...	...	...
319	...	...	...	...	...	...	...	...
1,741	***6,207***	***4,261***	***3,817***	***3,123***	***3,733***	***3,632***	***3,447***	***4,413***
9·1	*16·8*	*29·4*	*10·4*	*16·5*	*10·1*	*19·1*	*9·4*	*23·2*
168	858	587	608	499	...	...	...	...
334	370	265	310	242	218	221	1,261	1,898
315	2,396	1,649	1,180	963	492	493	673	765
113	936	632	262	216	964	915	222	246
673	709	468	466	387	384	380	...	...
...	421	313	551	466	545	541	982	1,110
120	309	195	...	...	644	631	248	281
118	206	152	440	350	486	451	61	113
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
4,630	***7,482***	***4,935***	***7,477***	***6,059***	***4,226***	***4,123***	***959***	***1,330***
12·0	*6·7*	*12·8*	*6·7*	*15·8*	*3·8*	*10·7*	*0·8*	*3·5*
1,745	***2,950***	***1,939***	***4,809***	***3,914***	***3,437***	***3,357***	***888***	***969***
12·4	*13·5*	*13·7*	*22·0*	*27·8*	*15·7*	*23·8*	*4·1*	*6.8*
...	...	...	2,425	1,989	...	...	258	301
...	1,379	896	689	566	...	...	...	...
...	...	...	728	574	2,106	2,060	202	212
...	803	577	967	785	1,150	1,112	428	456
1,070	768	466	...	...	181	185	...	...
675	...	...	...	...	...	...	...	...
1,596	***3,742***	***2,483***	***1,046***	***832***	***486***	***474***	***52***	***293***
20·5	*24·8*	*32·0*	*6·9*	*10·7*	*3·2*	*6·1*	*0·4*	*3·8*
130	1,176	796	535	411	93	89	34	184
480	1,075	717	...	...	...	...	8	50
453	1,261	827	...	...	...	...	6	39
553	230	143	511	421	393	385	4	20
833	***292***	***187***	***1,622***	***1,313***	***303***	***292***	***4***	***40***
15·9	*3·5*	*4·4*	*15·7*	*31·3*	*5·7*	*7·0*	*0·04*	*1·0*
184	...	...	1,291	1,085	303	292	...	...
649	292	187	...	...	...	...	...	...
...	...	...	331	278	...	...	4	40
456	***498***	***326***	...	...	...	...	***15***	***28***
3·7	*0·8*	*2·6*	...	...	...	...	*0·02*	*0·2*
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
118	355	233	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
315	143	93	...	...	...	...	15	28
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
23	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...

in Midnapore. Those for East Bengal and Khulna exclude 2,688 square miles of the Sundarbans in Khulna.
population are given in italics below the absolute figures for each Province and natural division.

SUBSIDIARY TABLE III.—Distribution of Population between Towns and Villages.

District and Natural Division.	Average population per		Number per mille residing in—		Number per mille of urban population residing in towns with a population of—				Number per mille of rural population residing in villages with a population of—			
	Town.	Village.	Towns.	Villages.	20,000 and over.	10,000 to 20,000.	5,000 to 10,000.	Under 5,000.	5,000 and over.	2,000 to 5,000.	500 to 2,000.	Under 500.
1	2	3	4	5	6	7	8	9	10	11	12	13
BENGAL, BIHAR AND ORISSA.	*21,456*	*338*	*51*	*949*	*672*	*199*	*109*	*20*	*19*	*120*	*443*	*418*
Bengal	*23,937*	*352*	*64*	*936*	*702*	*196*	*85*	*17*	*22*	*113*	*453*	*412*
WEST BENGAL	*21,974*	*326*	*73*	*927*	*747*	*97*	*149*	*7*	*5*	*100*	*457*	*438*
Burdwan	15,698	522	61	939	614	165	221	...	4	140	561	295
Birbhum	9,131	418	10	990	...	...	1,000	...	20	102	485	393
Bankura	19,869	233	50	950	768	232	...	...	7	48	338	607
Midnapore	12,732	240	36	964	321	305	329	45	...	50	390	560
Hooghly	18,935	426	139	861	814	...	186	...	6	113	508	373
Howrah	100 700	767	213	787	1,000	...	...	...	...	258	580	162
CENTRAL BENGAL ...	*36,560*	*574*	*204*	*796*	*790*	*145*	*55*	*10*	*34*	*142*	*545*	*279*
24-Parganas	21,097	557	225	775	598	323	79	...	65	146	500	289
Calcutta	896,067	...	1,000	...	1,000	...	...	...	...	...	...	...
Nadia	10,658	643	59	941	523	130	229	118	17	140	616	227
Murshidabad	18914	686	61	939	313	588	99	...	44	228	526	202
Jessore	7.066	462	12	988	...	...	801	199	7	78	544	371
NORTH BENGAL	*9,777*	*261*	*22*	*978*	*305*	*457*	*133*	*105*	*17*	*79*	*355*	*549*
Rajshahi	15,829	217	21	979	739	...	261	...	...	7	297	696
Dinajpur	15,945	173	9	991	...	1.000	...	...	...	27	214	759
Jalpaiguri	5,882	401	13	987	...	975	...	25	25	228	284	463
Darjeeling	12,290	478	93	907	...	773	227	...	45	196	454	305
Rangpur	6.044	316	15	985	...	453	228	319	47	154	402	397
Bogra	6,601	268	13	987	...	...	690	310	...	24	403	573
Pubna	22,026	322	31	969	562	438	...	...	5	49	452	494
Malda	13.796	206	41	959	563	346	...	91	...	24	292	684
Cooch Behar	3.949	482	27	973	...	686	...	314	38	92	587	283
EAST BENGAL	*17,524*	*391*	*25*	*975*	*536*	*370*	*80*	*14*	*29*	*127*	*473*	*371*
Khulna	9,717	685	21	979	...	819	181	...	76	201	516	207
Dacca	68.214	325	46	954	1,000	...	...	...	10	73	472	445
Mymensingh	15,259	363	27	973	173	773	54	...	...	73	426	429
Faridpur	16,102	351	15	985	...	1,000	...	...	9	91	445	455
Backergunge	9.724	475	20	980	462	247	251	40	10	161	536	293
Tippera	19,235	343	24	976	780	220	...	...	...	74	479	447
Noakhali	7.009	475	5	995	...	...	1.000	...	88	306	440	266
Chittagong	16,572	1,066	22	978	868	...	...	132	170	365	363	102
Chittagong Hill Tracts ...	...	384	...	1,000	...	...	...	...	...	...	577	423
Hill Tippera	6,831	96	30	970	...	...	1,000	...	...	...	154	846
Bihar and Orissa ...	*17,407*	*344*	*34*	*966*	*603*	*208*	*163*	*26*	*15*	*128*	*432*	*425*
NORTH BIHAR	*21,145*	*445*	*28*	*972*	*619*	*244*	*126*	*11*	*30*	*225*	*523*	*222*
Saran	17,817	514	31	969	595	175	230	...	7	106	577	310
Champaran	20,335	705	21	979	634	366	...	...	25	281	501	193
Muzaffarpur	20,522	655	29	971	532	357	111	...	39	209	546	206
Darbhanga	24 130	900	33	967	649	171	180	...	30	344	502	124
Bhagalpur	39,323	582	37	963	945	...	...	55	61	254	469	216
Purnea	10,855	484	16	984	...	768	232	...	16	127	538	319
SOUTH BIHAR	*21,554*	*461*	*67*	*933*	*737*	*123*	*124*	*16*	*18*	*160*	*500*	*322*
Patna	33,763	586	147	853	856	65	64	15	22	159	545	274
Gaya	12,759	335	41	959	559	...	388	53	9	65	473	453
Shahabad	17,062	377	55	945	602	257	141	...	7	112	486	395
Monghyr	22,319	808	42	958	755	245	...	...	36	297	508	159
ORISSA	*26,585*	*328*	*38*	*962*	*712*	*288*	...	...	*3*	*52*	*478*	*467*
Cuttack	26,627	360	38	962	658	342	...	...	4	56	518	422
Balasore	19,970	285	28	962	535	465	...	...	...	35	439	526
Puri	39,686	320	39	961	1,000	...	...	...	5	59	437	499
CHOTA NAGPUR PLATEAU	*9,051*	*209*	*20*	*980*	*220*	*274*	*415*	*91*	*1*	*24*	*272*	*703*
Hazaribagh	10,474	148	33	967	...	661	220	119	...	16	176	808
Ranchi	15,572	342	34	966	706	...	294	...	...	29	440	531
Palamau	5,689	218	17	983	...	...	631	369	...	35	219	746
Manbhum	10.229	254	20	980	681	...	193	126	3	34	240	723
Singhbhum	7,341	203	21	979	...	...	1,000	...	...	6	209	785
Santhal Parganas ...	8,832	192	23	977	...	593	407	...	...	20	230	750
Angul	...	136	...	1,000	...	...	...	...	...	11	372	617
Sambalpur	8,835	379	24	976	...	...	735	265	7	52	497	444
Orissa Feudatory States ...	6,200	194	10	990	...	...	889	111	...	21	255	724
Chota Nagpur States ...	...	167	...	1,000	...	...	...	...	...	18	161	821
Sikkim	...	*279*	...	*1,000*	...	...	...	...	...	...	*265*	*735*

SUBSIDIARY TABLE IV.—NUMBER PER MILLE OF THE TOTAL POPULATION AND OF EACH MAIN RELIGION WHO LIVE IN TOWNS.

DISTRICT AND NATURAL DIVISION.	NUMBER PER MILLE WHO LIVE IN TOWNS.					
	Total Population.	Hindu.	Musalman.	Christian.	Jain.	Parsi.
1	2	3	4	5	6	7
BENGAL, BIHAR AND ORISSA.	**51**	**57**	**42**	**204**	**489**	**910**
Bengal	**64**	**95**	**36**	**478**	**572**	**925**
WEST BENGAL	**73**	**71**	**97**	**615**	**292**	**597**
Burdwan	61	58	72	585	402	...
Birbhum	10	10	10	85	...	...
Bankura	50	56	35	341	83	...
Midnapore	36	34	64	647	208	...
Hooghly	139	141	138	702	1,000	1,000
Howrah	213	203	243	811	1,000	878
CENTRAL BENGAL	**204**	**275**	**119**	**687**	**934**	**992**
24-Parganas	225	248	186	283	320	962
Calcutta	1,000	1,000	9,000	1,000	1,000	1,000
Nadia	59	110	25	115	500	500
Murshidabad	61	90	34	869	896	...
Jessore	12	19	8	114	...	...
NORTH BENGAL	**22**	**32**	**15**	**235**	**301**	**605**
Rajshahi	21	55	12	192	337	...
Dinajpur	9	13	7	85	583	...
Jalpaiguri	13	13	18	20	28	1,000
Darjeeling	93	76	132	414	528	1,000
Rangpur	15	27	9	746	239	...
Bogra	13	47	7	130	142	...
Pabna	31	54	23	190	598	...
Maida	41	38	47	19	619	...
Cooch Behar	27	29	19	711	348	...
EAST BENGAL	**25**	**43**	**16**	**117**	**308**	**1,000**
Khulna	21	25	17	139	...	1,000
Dacca	46	71	32	82	600	1,000
Mymensingh	27	51	19	94	299	...
Faridpur	15	25	9	19	...	...
Backergunge	20	40	11	134	...	...
Tippera	24	48	14	605	...	...
Noakhali	5	13	3	61	...	...
Chittagong	22	34	16	676	1,000	1,000
Chittagong Hill Tracts ...	Contains no town.					
Hill Tippera	30	38	13	133	1,000	...
Bihar and Orissa ...	**34**	**31**	**80**	**72**	**369**	**657**
NORTH BIHAR	**28**	**25**	**45**	**554**	**238**	**619**
Saran	31	27	66	393	653	...
Champaran	21	17	42	565	...	...
Muzaffarpur	29	25	53	721	...	1,000
Darbhanga	33	28	66	640	41	...
Bhagalpur	37	30	101	475	207	...
Purnea	16	19	12	390	245	...
SOUTH BIHAR	**67**	**54**	**182**	**767**	**593**	**1,000**
Patna	147	119	370	894	275	1,000
Gaya	41	34	100	880	746	...
Shahabad	55	42	214	467	1,000	...
Monghyr	42	36	93	679	267	...
ORISSA	**38**	**33**	**201**	**547**	**495**	...
Cuttack	38	32	189	828	545	...
Balasore	38	28	345	431	...	...
Puri	39	39	7	151	...	...
CHOTA NAGPUR PLATEAU ...	**20**	**21**	**70**	**35**	**306**	**667**
Hazaribagh	33	29	75	278	397	...
Ranchi	34	49	185	32	492	...
Palamau	17	15	44	10	...	...
Manbhum	20	21	38	166	124	...
Singhbhum	21	29	237	122	...	800
Sonthal Parganas	23	37	44	59	314	...
Angul	Contains no town.					
Samtalpur	24	23	345	62	...	...
Orissa Feudatory States ...	10	11	79	2	132	...
Chota Nagpur States	Contains no town.					
Sikkim	Contains no town.					

SUBSIDIARY TABLE V.—TOWNS CLASSIFIED BY POPULATION.

CLASS OF TOWNS.	Number of towns of each class in 1911.	Proportion to total urban population.	Number of females per 1,000 males.	VARIATION PER CENT. IN THE POPULATION OF TOWNS AS CLASSED AT PREVIOUS CENSUS.				VARIATION PER CENT. IN URBAN POPULATION OF EACH CLASS FROM 1871 TO 1911.	
				1901—1911.	1891—1901.	1881—1891.	1871—1881.	(a) In towns as classed in 1871.	(b) In the total of each class in 1911 as compared with the corresponding total in 1871.
1	2	3	4	5	6	7	8	9	10
BENGAL	***124***	***100***	***631***	***+ 13·2***	***+ 12·2***	***+ 9·9***	***- 2·6***	***+ 31·8***	***+ 60·6***
I.—100,000 and over	3	39·9	507	+ 6·9	+ 25·8	+ 11·4	- 8·3	+ 41·6	+ 87·0
II.— 50,000—100,000	2	3·5	567	+ 21·0	+ 3·9	+ 21·3	+ 5·2	+ 65·2	- 49·6
III.— 20,000— 50,000	24	26·8	710	+ 20·8	+ 2·0	+ 3·3	- 10·2	+ 6·7	+ 85·7
IV.— 10,000— 20,000	40	19·6	742	+ 17·7	+ 7·3	+ 5·0	- 4·1	+ 23·3	+ 60·0
V.— 5,000— 10,000	34	8·5	798	+ 6·8	+ 3·2	+ 13·1	+ 4·0	+ 16.2	+ 24·7
VI.—Under 5,000 ...	17	1·7	774	+ 15·5	+ 10·7	+ 35·5	+ 79·9	+ 179·6	+ 244·9
BIHAR AND ORISSA.	***76***	***100***	***932***	***- 2·8***	***- 6·7***	***+ 6·6***	***+ 13·2***	***+ 7·1***	***+ 21·2***
I.—100,000 and over	1	10·3	922	+ 1·0	- 18·4	- 3·2	+ 7·4	- 14·3	- 14·3
II.— 50,000—100,000	3	14·3	900	- 9·5	- 12·6	+ 6·2	+ 8·3	- 4·8	- 22·8
III.— 20,000— 50,000	14	35·7	923	- 4·4	+ 4·2	+ 8·1	+ 5·2	+ 2·0	+ 48·0
IV.— 10,000— 20,000	20	20·8	947	- 1·9	- 9·7	+ 6·0	+ 18·9	+ 20·2	+ 20·6
V.— 5,000— 10,000	30	16·3	961	+ 3·7	+ 0·4	+ 19·8	+ 29·6	+ 29·5	+ 118·4
VI.—Under 5,000 ...	8	2·6	992	+ 16·2	+ 10·8	+ 37·4	+ 85·5	+ 151·0	+ 42·0

* The percentage in column 5 represents the increase shown by the census of 1911, in the population of the towns included in each class at the census of 1901. The same principle is followed in the case of columns 6 to 9. In column 10 the percentage shows the increase in the population of towns as classed in 1911 over the population of towns as classed in 1872, including the population of towns which have come up from a lower class during the intervening period.

SUBSIDIARY TABLE VI.—CITIES.

CITY.	Population in 1911.	Number of persons per square mile.	Number of females to 1,000 males.	Proportion of foreign born per mille.*	PERCENTAGE OF VARIATION.				
					1901—1911.	1891—1901.	1881—1891.	1871—1881.	1871—1911.
1	2	3	4	5	6	7	8	9	10
Bengal.									
Calcutta	896,067	28,002	475	714	+ 5·7	+ 24·3	+ 11·4	- 3·3	+ 41·6
Howrah	179,006	20,965	562	756	+ 13·6	+ 35·2	+ 28·4	+ 8·0	+ 112·9
Cossipur-Chitpur ...	48,178	14,824	565	661	+ 18·2	+ 29·7	+ 19·5	+ 3·6	+ 89·9
Manicktollah	52,967	15,721	694	750	+ 66·0	+ 15·0	- 41·5	- 10·9	- 0·4
Garden Reach	45,295	13,322	637	453	+ 60·6	+ 1·0	+ 124·2	+ 34·5	+ 389·3
Dacca	108,551	15,917	721	198	+ 21·0	+ 10·0	+ 4·1	+ 14·2	+ 58·2
Bihar and Orissa.									
Patna	136,153	15,128	922	98	+ 1·0	- 18·4	- 3·2	+ 7·4	- 14·3
Gaya	49,921	6,240	897	96	- 30·0	- 11·3	+ 5·2	+ 14·3	- 25·3
Bhagalpur	74,349	8,261	861	169	- 1·9	+ 9·6	+ 1·3	+ 4·4	+ 13·7

* Foreign-born indicates born outside the city or the district containing the city. Calcutta for the purpose of this table is treated as a district. It being impossible to distinguish between those born in Calcutta and Garden Reach, the district-born in the case of the latter town includes persons born in Calcutta.

SUBSIDIARY TABLE VII.—PERSONS PER HOUSE AND HOUSES PER SQUARE MILE.

DISTRICT AND NATURAL DIVISION.	AVERAGE NUMBER OF PERSONS PER HOUSE.				AVERAGE NUMBER OF HOUSES PER SQUARE MILE.			
	1911.	1901.	1891.	1881.	1911.	1901.	1891.	1881.
1	2	3	4	5	6	7	8	9
BENGAL, BIHAR AND ORISSA.	**5·2**	**5·2**	**5·3**	**6·2**	**83**	**78**	**77**	**61**
Bengal	**5·3**	**5·1**	**5·2**	**6·3**	**105**	**100**	**96**	**75**
WEST BENGAL	**4·6**	**4·5**	**4·5**	**5·3**	**133**	**132**	**122**	**102**
Burdwan	4·3	4·2	4·3	4·8	132	135	121	107
Birbhum	4·6	4·3	4·3	4·4	117	120	107	103
Bankura	4·9	4·8	5·0	6·2	88	88	82	64
Midnapore	4·8	4·8	4·9	6·0	112	111	103	82
Hooghly	4·1	4·0	3·8	4·2	221	222	233	195
Howrah	4·3	4·5	4·9	5·6	433	373	307	239
CENTRAL BENGAL	**5·5**	**5·1**	**5·4**	**6·0**	**116**	**119**	**107**	**93**
24-Parganas	5·4	5·4	5·7	5·4	94	79	68	56
Calcutta*	20·3	6·8	10·1	17·7	1,384	3,921	2,110	1,079
Nadia	4·7	4·8	5·0	5·6	122	125	119	106
Murshidabad	4·9	4·7	4·4	4·8	132	132	132	120
Jessore	5·0	4·9	5·4	7·0	121	126	120	100
NORTH BENGAL	**5·4**	**5·3**	**5·5**	**6·0**	**97**	**91**	**83**	**73**
Rajshahi	4·7	5·1	5·3	6·0	119	111	107	95
Dinajpur	5·6	5·4	5·6	5·6	76	73	67	63
Jalpaiguri	5·3	5·0	5·4	6·1	58	53	42	33
Darjeeling	4·1	4·1	4·9	5·3	55	53	39	24
Rangpur	5·9	5·6	5·7	6·2	116	111	105	97
Bogra	5·8	5·9	5·7	7·4	124	106	99	66
Pubna	5·2	5·5	5·5	6·5	148	141	134	109
Malda	5·7	5·4	5·5	5·6	92	86	78	67
Cooch Behar	5·2	5·1	4·9	5·2	87	85	90	89
EAST BENGAL	**5·5**	**5·5**	**5·4**	**7·8**	**94**	**87**	**88**	**57**
Khulna	6·0	5·5	5·7	6·9	48	48	43	35
Dacca	5·4	5·6	5·5	6·9	196	169	157	110
Mymensingh	5·8	5·9	5·6	7·6	124	103	98	64
Faridpur	5·1	5·2	5·3	7·3	162	164	150	99
Backergunge	4·9	4·9	4·9	8·5	107	129	120	61
Tippera	6·0	5·9	5·7	8·5	161	144	125	72
Noakhali	5·7	5·5	5·4	9·4	140	127	114	53
Chittagong	5·0	4·9	5·0	5·4	122	111	101	82
Chittagong Hill Tracts ...	5·8	5·8	5·2	6·8	5	4	4	3
Hill Tippera	5·1	5·6	...	...	11	8	...	...
Bihar and Orissa ...	**5·2**	**5·3**	**5·7**	**6·4**	**67**	**62**	**71**	**61**
NORTH BIHAR	**5·1**	**5·3**	**5·8**	**6·6**	**126**	**120**	**109**	**91**
Saran	4·5	5·2	5·5	7·0	189	175	168	125
Champaran	5·5	5·7	6·2	6·1	98	90	84	80
Muzaffarpur	5·5	5·3	5·9	7·0	169	174	153	123
Darbhanga	4·9	5·0	6·1	7·3	179	178	137	109
Bhagalpur	5·3	5·5	5·8	6·1	96	91	83	75
Purnea	5·4	5·4	5·4	6·1	74	69	72	61
SOUTH BIHAR	**5·2**	**5·3**	**5·8**	**6·6**	**98**	**97**	**92**	**78**
Patna	5·3	5·2	6·1	6·3	146	150	140	134
Gaya	5·1	5·1	5·6	6·1	90	86	81	74
Shahabad	5·0	5·3	5·7	7·1	85	84	83	63
Monghyr	5·5	5·4	5·7	7·0	99	97	91	71
ORISSA	**5·0**	**5·1**	**5·2**	**5·9**	**103**	**99**	**90**	**75**
Cuttack	4·9	5·1	5·1	5·5	117	111	104	90
Balasore	5·0	5·9	5·4	5·9	102	88	89	78
Puri	5·0	4·6	5·3	7·0	82	90	72	52
CHOTA NAGPUR PLATEAU ...	**5·2**	**5·3**	**5·7**	**5·9**	**36**	**31**	**34**	**30**
Hazaribagh	5·7	5·3	6·0	6·0	32	32	28	26
Ranchi	5·6	5·3	5·0	5·6	35	32	32	24
Palamau	5·3	5·4	5·9	5·3	26	23	21	20
Manbhum	5·1	5·3	5·5	5·9	74	59	53	43
Singhbhum	5·0	5·2	5·3	5·3	36	31	27	23
Sonthal Parganas	5·7	5·8	6·1	6·4	61	57	52	45
Angul	4·7	4·8	9·0	5·8	25	24	11	10
Sambalpur	4·5	4·7	...	...	43	36	...	...
Orissa Feudatory States ...	5·1	5·1	...	...	27	22	...	...
Chota Nagpur States	4·9	6·0	...	...	50	39	...	...
SIKKIM	**5·3**	**5·3**	...	...	**6**	**4**	...	...
Cities—								
Calcutta	20·3	6·7			1,384	3,921		
Howrah	2·9	3·4			7,301	5,434		
Cossipur-Chitpur	7·7	6·9			1,931	1,796		
Manicktolla	8·8	5·2	Not available.	Not available.	1,782	1,815	Not available.	Not available.
Garden Reach	8·8	4·2			1,511	1,947		
Dacca	5·5	5·2			2,901	2,530		
Patna	4·8	4·1			3,119	3,653		
Gaya	4·2	5·0			1,494	1,782		
Bhagalpur	5·3	5·2			1,556	1 619		

* The variations in Calcutta are due to changes in the definition of house.

In the calculations for each Province and for each natural division, those areas for which figures are not available have been left out of account.

CHAPTER II.

MOVEMENT OF THE POPULATION.

INTRODUCTORY.

138. The present chapter deals with the variations in the population that have taken place since 1872, when the first census was taken. The changes which occurred between each census up to 1901 will be only briefly referred to, as they have already been dealt with in previous census reports, and the discussion will be mainly devoted to the variations during the last decade. Figures showing the variations in the population of each district and State are contained in Imperial Table II, and similar information for thanas is given in Provincial Table I. These statistics are further illustrated by the subsidiary tables at the end of this chapter, which deal with (1) variations in the population of districts in relation to density since 1872, (2) variations in the natural population of districts between 1901 and 1911, (3) the difference between the vital statistics for each district and the results of the census, and (4) variations by thanas classified according to density.

EARLY ACCOUNTS.

139. There is ample evidence of the scantiness of the population in various parts of the two provinces in the early days of British rule. This was largely the result of the terrible famine of 1770, when, according to the estimate made by Warren Hastings, "at least one-third of the inhabitants perished"; even 18 years later the Governor-General had to report to the Court of Directors that one-third of the Company's territory in Bengal "was a jungle inhabited only by wild beasts." Apart, however, from the effects of famine, the unsettled state of the country had long been instrumental in preventing the expansion of cultivation. Rennell's map shows the lower part of the delta as empty of villages, with the words "Depopulated by the Maghs" written across it. Further south, the oppression of the Marathas had reduced the rich province of Orissa to a pitiable state. Of this we have first-hand evidence by Mr. Motte, who, in order to avoid repetition in his account of the journey which he made through it in 1766, remarked: "In my journey it will be unnecessary to say that any place I came to was once considerable, since all the places which were not so are now depopulated by the Marhattas, and such alone remain as on account of their bulk are longer in decaying." Again: "I passed into the talook of Budruc (Bhadrakh), where I found deep marks of the Marhatta claws on the fine tract of land, formerly well peopled, where a human creature is not now to be seen, except, perhaps, a solitary herdsman, attending a large drove of buffaloes or other horned cattle."* In many parts it took years for the British to establish the settled rule of peace. The border district of Midnapore, for instance, was liable to periodical invasions by the Marathas, while its western portion was covered with jungle and inhabited by predatory tribes. It was perpetually harassed by the inroads of the Marathas, by armed bands of *sannyasis*, who roamed through the country in many thousands strong, by the raids of aboriginal tribes (generally known as Chuars), and by the turbulence of the jungle chiefs. Even in 1800, after nearly forty years of British occupation, the Collector reported that two-thirds of Midnapore consisted of jungle, the greater part of which was uninhabited and inaccessible.

140. In the districts now included in Bihar and Orissa, the reports of the Collectors and the investigations of Buchanan Hamilton show that large areas were waste or very thinly peopled. Herds of wild elephants roamed through the north of Purnea, and some had even made their way to

* *Narrative of a Journey to the Diamond Mines at Sumbhulpoor*, Asiatic Annual Register, 1799.

"the woods in the south." Wild buffaloes were exceedingly destructive, and in the north wolves used to carry off a number of children. "The population seems in some places to be diminishing, for the extreme timidity and listlessness of the people have in some parts prevented them from being able to repel the encroachments of wild beasts."* Only one-fourth of Champaran was under tillage in 1794, and a great part of Darbhanga was uncultivated, partly owing to the famine of 1770, and partly because of the oppression of the farmers of revenue and freebooting zamindars. In 1783 the Collector proposed that cultivators should be recruited from the dominions of the Vizier of Oudh to reclaim "the unpeopled wastes." Thirteen years later one *pargana* was described as "the abode of dreadful beasts of prey," while another was the haunt of wild elephants. Again, Buchanan Hamilton left it on record that part of Shahabad had not recovered from the desolation caused by the wars of Kasim Ali half a century before, and that in some *parganas* a large portion of the land was either overgrown with stunted woods or had lately been deserted. Even as late as the Mutiny the country round Jagdispur was covered with dense jungle in which the mutineers found a retreat, and Government was obliged to have it cleared at a great cost. Similar accounts might be given for other districts, if the limits of space permitted.

EARLY ESTIMATES.

141. In the first half of the 19th century attempts to compute the population, or actually to count it, appear to have been made from time to time. The basis of the calculations varied widely, and some of the figures appear so extraordinary in the light of our present knowledge, that it is surprising that they can have been accepted at all. For instance, a so-called census of the district of Patna was held in 1837, and the total population estimated at 845,790, but 284,132 persons, or nearly one-third of the total, were assigned to the city of Patna.† Another census showed the population of the Tirhut district as 1,660,538, the basis of the calculation being a count of houses and the assumption that each contained six persons. Twelve years later the number had fallen by 150,000, and it was naively explained that the population was "supposed to have increased enormously," but it was now calculated at the rate of $5\frac{1}{2}$ persons a house. Elsewhere, as for instance in Chittagong, the figures were based on an estimate of the area under cultivation, coupled with an assumption that each cultivated acre supported 6 persons. As a rule, however, the estimates were based on the number of houses and the average number of persons supposed to live in each; the average, though generally taken to be 5, was sometimes 4, and in one case as low as $2\frac{1}{2}$. How haphazard these estimates were may be realized from the experience of Sir Henry Thuillier, then a young lieutenant, in charge of the revenue survey of a *varaana* in Sylhet from 1839 to 1842. In 1841 he reported to the Deputy Surveyor-General that he did not know that he was expected to undertake a census, and that it would be difficult to make the count so late in the day. Major Bedford, Deputy Surveyor-General, reprimanded Thuillier, though he candidly admitted that the mistake had probably resulted in economy; if the houses were counted and multiplied by a certain factor, that would suffice. Thuillier then pointed out that this gave no indication of the numbers of the sexes, but even this did not defeat Major Bedford. He seemed astonished that Thuillier's wanderings in Jaintia had not given him a tolerably accurate idea of the relative numbers of the sexes, and eventually the numbers were assigned on Thuillier's visual knowledge of that *varaana*. Afterwards, in 1851-52, Thuillier himself, who had in the meantime been appointed Deputy Surveyor-General, in an annual report to the Board of Revenue, showed the figures for Jaintia as having been obtained by a "census taken of the population."‡

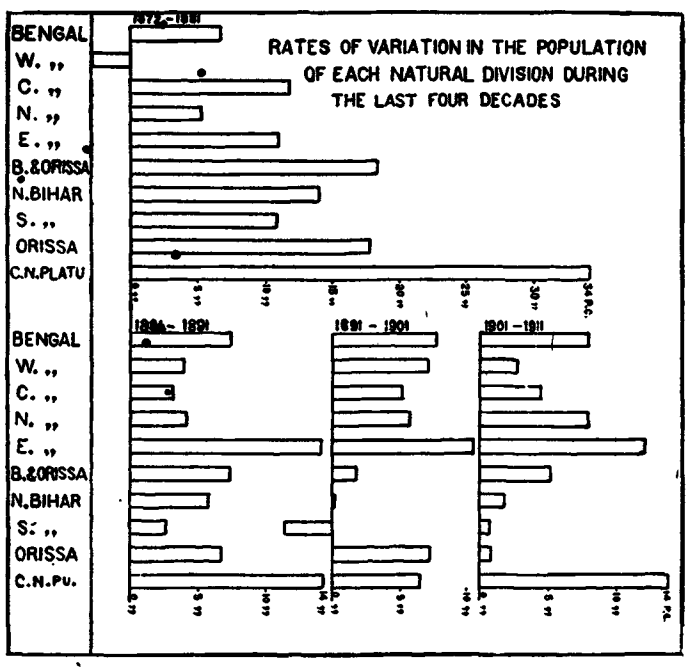
* Montgomery Martin, *Eastern India*. Buchanan Hamilton adds :—"This however is only a local and recent evil, and within the last forty years the population has, I am credibly informed, at least doubled." He also speaks of "the immense population by which the country is overwhelmed." These statements can scarcely be credited, for even in 1788 nearly a quarter of a *pargana* with an area of nearly 1,000 square miles lay waste for want of cultivators (Purnea District Gazetteer, p. 99).

† Bengal and Agra Gazetteer of 1841.

‡ I am indebted for knowledge of the above incident to a note by Captain F. C. Hirst.

142. The marginal table shows the population recorded at each census, and the percentage of increase during each intercensal period. During the 39 years over which the census operations have extended, Bengal has added 11,618,350 persons or 33·5 per cent. to its population. At every census, except that of 1881, the rate of increase has been greatest in East Bengal—a rich and fertile region, which is now more populous by 56·8 per cent. than it was in 1872. Both Central and North Bengal have added to their population at every census but the rates of increase since 1872 are less than half that returned for East Bengal. West Bengal, which has grown only by 11·3 per cent., has been the least progressive, owing partly to the decline between 1872 and 1881, when it suffered from the long continued visitation of Burdwan fever, and partly to the small increase (2·8 per cent.) registered in 1911.

Variations since 1872.			Percentage of increase since previous census.
	Year of census.	Population.	
Bengal	1872	34,687,292	...
	1881	37,014,989	6·7
	1891	39,805,942	7·5
	1901	42,881,776	7·7
	1911	46,305,642	8·0
Bihar and Orissa	1872	28,210,382	...
	1881	33,398,931	18·4
	1891	35,904,575	7·5
	1901	36,557,257	1·8
	1911	38,435,293	5·1



143. The population of Bihar and Orissa has grown by 36·2 per cent. since 1872, but part of the increase is fictitious, *i.e.*, it is the result of improved enumeration and not of natural growth. This has been especially the case in the Chota Nagpur Plateau, where the census was defective, in a major degree in 1872, and in a minor degree at each subsequent census. Even allowing for this element of error, no part of the Province has developed so rapidly as this tract, which is peopled by hardy and prolific races, mostly of aboriginal descent, and in which there is room for expansion, large areas being available for reclamation and calling for cultivators. In North and South Bihar also the census of 1872 was wanting in accuracy and completeness, the result being that an increase of over 10 per cent. was returned for both divisions in 1881. The census of 1891 showed a growth of 5·9 per cent. in North Bihar and of 2·7 per cent. in South Bihar, but since then the former has been almost stationary, while the latter has yet not made good the loss of population which it sustained between 1891 and 1901. Orissa developed rapidly up to 1881, when it was recovering from the effects of the famine of 1866, and it continued to progress until 1901. It has now received a check, its rate of increment during the last decade being under 1 per cent.

144. Up to 1905 both the Provinces dealt with in this report formed part of one Province (Bengal), and when the census of 1911 was held, they were divided between the two Provinces of Bengal and Eastern Bengal and Assam.



The general rate of growth up to the latter year in the united Province of Bengal showed a progressive decline, viz., from 12 per cent. in 1881 to 7½ per cent. in 1891 and to 5 per cent. in 1901: this was undoubtedly due in part to the higher standard of accuracy attained at each successive census. The separation of the figures for the two new Provinces shows that there has been a steady but gradual advance in Bengal, the increase in the percentage of growth being 1·3 per cent. (from 6·7 per cent. in 1881 to 8 per cent. in 1911). In Bihar and Orissa there was a continuous decline in the ratio till 1901, after which there was a sharp rise: in this latter Province the abrupt transition from the increase of 18·4 per cent. recorded in 1881 to that of 7·5 per cent. recorded in 1891 is due to the admitted incompleteness of the first census of 1872. The greater rapidity of growth shown by the present census cannot be connected with any improvement of the census-taking except in the Chota Nagpur Plateau, especially in the Orissa States. Here, there is reason to believe, the better organization effected under the control of the Political Agent resulted in the enumeration of persons who previously escaped the census, and it must therefore be held partly responsible for the high rate of increase recorded (19·6 per cent. as against 9·5 per cent. in 1901). In the case of Sikkim there is no doubt that part of the increase of 49 per cent. recorded at this census is due to improved organization. The census of this State is always a matter of difficulty owing to the nature of the country, its scattered population and the paucity of persons able to read and write. In 1901 an excellent scheme for the census was drawn up by Mr. Gait, then Census Superintendent of Bengal, but, for reasons which need not be discussed here, it was not given effect to. Except in a few bazars, houses were not numbered, and eventually two clerks had to be deputed to conduct the census of the entire population—a task which lasted ten weeks. At the present census, a scheme following the lines laid down in 1901, with some modifications suggested by further experience, was carried out successfully by the Political Officer.

VARIATIONS SINCE 1901—BENGAL.

145. The actual increase of population in Bengal since 1901 has been 3,423,866 or 6·7 per cent. Every natural division contributes to the increase, but in unequal shares, as shown in the margin. Immigration is partly responsible for the accretion. The immigrants from outside provinces outnumber the emigrants who have gone to other parts of India by a little over 1½ millions, the excess having increased considerably during the last decade. The main factor, however, is natural growth, and in different parts of the province this largely depends on the strength of Musalmans, who, as is well known, are more prolific than Hindus.

DIVISION.	Increase per cent.	Excess (+) or deficiency (−) of births over deaths per mille.	Percentage of Musalmans.
Bengal	**6·7**	**+ 4·8**	**52·3**
West Bengal ...	2·8	+ 1·1	13·4
Central ,, ...	4·5	− 0·5	48·1
North ,, ...	8·0	+ 3·95	59·3
East ,, ...	12·1	+ 9·55	67·5

The marginal table sufficiently shows how largely their preponderance affects the proportional growth of the population in each division. The increase is greatest where they are most numerous. viz., in North and East Bengal, and least where they are in a minority, viz., in Central and West Bengal, though the immigrant population is strongest there. It should be added that conditions are somewhat peculiar in Central Bengal. It contains the unhealthy districts of Jessore and Nadia, the only two districts in the province which have sustained a loss, and there is a large body of male immigrants in Calcutta and the 24-Parganas who leave their wives at home: it is on this account that the birth-rate falls below the death-rate.

BIHAR AND ORISSA.

The addition of population in Bihar and Orissa is 1,878,036, or 5·1 per cent. In this province the Musalmans form a small minority, and emigration is active. The number of emigrants to other provinces is in excess of that returned for immigrants by 1½ millions: one-thirtieth of the total population of the province were present in Bengal at the time of the census. The emigrants are mostly adult males, and, though for the most part they leave their homes only for a time, their absence materially affects the birth returns. It is this which mainly accounts for the fact that, while, according to the vital statistics for 1901-10 there was an excess of 1,910,000 births over deaths, the census shows an increase of only 1,240,000 in the areas for which returns of births and deaths were compiled. There has been a substantial increase only in the Chota Nagpur Plateau, where it is due to natural growth among prolific aboriginals. In North Bihar there has been a slight advance, but South Bihar and Orissa are practically stationary. The slow rate of growth in these areas is the combined result of emigration, scarcity and epidemics of disease, notably plague, which has caused a mortality of half a million in Bihar. The conditions prevailing in each district and natural division will be discussed later, and here it may be stated that four districts have sustained a loss of population, viz., Saran, Patna, Shahabad and Monghyr.

DIVISION.	Increase per cent.	Excess of births over deaths per mille.
Bihar and Orissa	**5·1**	**5·2**
North Bihar	1·9	5·1
South ,, ...	0·7	1·5
Orissa	0·9	2·6
Chota Nagpur Plateau ...	14·0	9·1

CONDITIONS IN 1901—1910.

146. In Bengal, where there had been a succession of somewhat unhealthy years at the end of the previous decade, there was an improvement in the general condition of the people up to 1904. The east of the Province was visited by heavy floods in 1905 and by widespread epidemics of cholera in the next two years. Crops were also short, and their partial failure accentuated the tendency to high prices. After 1907, however, there was again a series of healthy years. In Bihar and Orissa the first four years of the decade witnessed a period of fair agricultural prosperity: the number of births increased, while mortality gradually fell. The three years 1905 to 1908, however, were years of distress. The harvests were short and the price of food-grains ruled high: the decline of the birth-rate and the rise of mortality are symptomatic of the unfavourable conditions prevailing. In 1909, however, the outturn of the crops was excellent, prices fell and a marked improvement in the general health was apparent.

It is almost superfluous to add that conditions even in the same year are exceptionally diverse in different parts of the enormous area covered by the two Provinces. The same year may witness drought and excessive rainfall, a failure of the crops on one side and a full harvest on the other, a rapid extension of cultivation in one direction and the lapsing of well cultivated land into jungle in another. These features will be dealt with later in the sections given to each district.

RISE OF PRICES.

147. The most prominent feature of the economic history of the last decade is the rise in prices which took place in 1906 and continued during the two succeeding years. While the average price of food varied little from the normal during the first half of the decade (1901 to 1905), it suddenly rose in 1906 in a marked degree, this rise becoming accentuated in 1907, till in August of that year the average price of rice was 58 per cent., and the maize 70 per cent. above the previous normal. The crops of 1907-08 being also

short, there was no appreciable reduction next year in the price of these cereals, the first of which is the great food-staple of Bengal, while the latter is consumed largely in Bihar. The rise of prices was not confined to these two Provinces, but was more or less general throughout India, and was due to common causes. It is not proposed to trespass into the region of economics by discussing these causes, but some of the contributory factors operating in Bengal, Bihar and Orissa may be mentioned, as illustrating the change of economic conditions in this part of India and the manner in which it has affected the people.

148. In the first place, there was a poor outturn of food-crops, and of rice in particular, for four successive years (1904-07), accompanied by a rapid advance in the price of jute. The area under jute expanded enormously, rice fields being given up to the cultivation of jute in many districts : it has been estimated that 40 crores of rupees were paid for the jute crop of one year (1906), and that of this sum 15½ crores were clear profit. The shortage of the stocks of rice, combined with an increase in the buying powers of a large proportion of the cultivators, led to an unprecedented demand for the grain ; and the high prices naturally induced those who had stores of rice to sell what they could, keeping only the minimum required to carry them on till the next harvest.

The enhancement of prices and the high level at which they stayed were partly also due to the action of dealers and merchants, who, with increased facilities for inter-communication, are now able to control the grain trade to an extent previously unknown. Much of the trade which used to be carried on locally between the actual cultivators and grain-dealers in local bazars, has now come under the control of large firms at the chief commercial centres. Their agents penetrate into rural tracts where they were unheard of 15 or 20 years ago, and buy up surplus stocks of grain before even dealers in the nearest towns can make a bid. Their operations are well described in the following extract :—

"In India the initial stage of 'cornering' is not ordinarily the result of the action of individuals, but of the elements, as represented by a deficient rainfall. This reduces the supply, and prices automatically rise, but the indications are that the prices to which food-stuffs have risen in certain years have not been raised entirely by the natural action of insufficient rain, or even by an inflation of the currency, but have been artificially raised, and this could only have been done by the action of dealers in food-grains. These men do not actually form a ring for cornering purposes, but they belong to the same caste in each of the different parts of the country, and act together apparently by instinct, and hold back their stocks, for the purpose of artificially raising prices, on the smallest encouragement, *i.e.*, the smallest shortage of grain. When normal conditions return, after having reaped a rich harvest, if money is at the same time plentiful, they can afford to hold back their stocks, partly with the object of maintaining high prices, and partly for speculative reasons, gambling for another year of scarcity, and they have been encouraged in this manœuvre by the many years of apparent slight scarcity during the past fourteen years. If successive years of plenty follow, they are forced to sell, and prices fall, but only slowly."[*]

149. Generally speaking, the cultivators, who form the vast majority of the population, benefited by the high range of prices, but it must be remembered that with them an increased income does not always mean increased resources, but rather greater opportunities for unproductive expenditure, *e.g.*, on marriages and other social or religious ceremonies. The savings of years may in this way be spent in a single week. This is especially the case among the less highly developed races, such as the aboriginals of Chota Nagpur, who are improvident and thriftless. Living only for the present, they love to spend all that they get and never trouble to save. In Singhbhum, for instance, the Hos devote their surplus stores of rice to making rice-beer and consume them in drink : it is estimated that in the Kolhan a quarter of the rice crop is converted into liquor. In the Sonthal Parganas, again, the scarcity of 1908 coincided with an abnormal increase in the consumption of country spirit, the cultivators having obtained high prices for their lac.

The high range of prices had a very different effect on the non-agricultural sections of the community, and especially those members of

[*] F. J. Atkinson, *Rupee Prices in India, 1870 to 1908*, Journal of the Royal Statistical Society, September 1909.

the middle classes who are dependent not on agriculture, commerce and industries, but on the fixed salaries which they obtain in clerical and professional employment. To them high prices meant straitened circumstances, if not actual privation. So much was this the case, that the Government instituted a system of "grain compensation allowances" in view of the diminution of their assets which the high prices of food entailed.

150 The landless labourers, formerly the most destitute of all, were not much affected. Those who were ready to travel could find ample employment in the coal-mines, mills, factories, etc., where wages have risen and are far higher than in rural areas. In the case of the coal-mines, for instance, it has been estimated that in the ten years ending in 1903 the wages of all classes of workers had been increased by about 50 per cent. "In the year 1894," writes the Chief Inspector of Mines, "the manager of a large colliery said that he took care that any miner who was willing to do a fair day's work should get not less than 4 annas per day, and that most of his work-people got one hot meal every day, as if these workers were exceptionally well off as compared with other miners. I believe that 4 annas a day was almost above the average wage at that time, but now it is not at all uncommon to be told that the miners earn 5 annas, 7 annas, and more than these amounts per day, that a miner and his wife earn Re. 1 per day between them, and so on. And the wages of other labourers have, consequently, risen accordingly."*

In districts where the wages of field-labour have not risen appreciably during the last ten years, the labourers should *a priori* have suffered from the pinch of high prices. In such districts, however, they are paid not in cash but in kind; and while the quantity of produce they receive remains unchanged, its value has increased. In other parts agricultural labour is paid in cash, but the labourers' wages have risen. In several Bengal districts, indeed, local labour has to be supplemented by the influx of immigrants from Bihar and Orissa, and the demand being in excess of the supply, wages are regulated thereby.

151. There appears to be no doubt that there has been a general rise in the standard of living of the present generation. Many things which were formerly regarded as luxuries are now articles of ordinary use. Not the least significant change is the way in which rice is displacing coarser grains as a daily article of food, *e.g.*, the lowest classes in parts of Bihar and Chota Nagpur, who seldom had a meal of rice, are now able to indulge in it. There has at the same time been an undoubted improvement in the staying powers both of the small cultivators and the landless labourers, especially in North Bihar. In giving an account of the famine of 1873-74, which was most intense in the district of Tirhut (*i.e.*, the present districts of Darbhanga and Muzaffarpur), Sir A. P. (now Lord) Macdonnell stated that the raiyats were so impoverished, and so unable to bear up against the failure of a single season's crop, that one-third of the population was at one period in receipt of relief from the Government. In 1896-97, when the distress was at its highest, more than three-fifths of the persons in the Patna Division who were in receipt of relief belonged to the districts of Muzaffarpur and Darbhanga, but, instead of forming one-third of the population of those districts, they formed less than 3½ per cent. of it. The lapse of ten years has shewn a further improvement, for in the Darbhanga famine of 1906-07 the percentage of persons relieved to the population of the distressed area was only one-third of what it was in the previous famine, while the proportion was still smaller in the famine of 1908-09.

The change which has taken place must be mainly attributed to the greater mobility of labour, which again is the result of the extension of railway communications. When scarcity is felt, a larger proportion of the people leave the district and obtain labour elsewhere, remitting their savings home. The volume of emigration, in fact, corresponds to the state of the crops. If they are good, it diminishes; if there is a failure, it is larger and lasts longer. The one section of the community, which appears to be stationary, consists of the professional middle classes (*bhadralok*) of Bengal who do not engage in commerce or industries. They do not reduce their expenditure on the social and religious ceremonies

* Report of Chief Inspector of Mines for 1903.

incidental to their position, though the expense of maintaining that position has increased. At the same time their ranks are swelled, and competition is rendered keener year after year, by the growing number of recruits from schools and universities.

152. There was, as already stated, a large extension of the area under jute during the first part of the decade, and fears were at one time expressed that the area under food-crops was being reduced below the limit of safety. There appear to be no valid grounds for such fears, for the subsequent lowering of the price of jute resulted in the shrinkage of jute cultivation, and rice has partially regained the area which it lost. Even in the jute-growing districts of Dacca, Mymensingh, Faridpur and Tippera the area under jute is still less than a third of that under rice. It has been proved, moreover, that rice and jute can be raised from the same land, provided that it is fertile enough; but it must be admitted that the ordinary cultivator is averse to such double cropping, as it involves continuous and somewhat exhausting labour.

There is no doubt as to the extension of rice cultivation to tracts where it was formerly unknown. In 1877 Sir William Hunter wrote in the Statistical Account of Bhagalpur:— "Throughout the south of Bihar, all along the hills from Rajmahal to where the Son river enters at the south-west corner of Shahabad district, the people are poor, and the country is barren and only just reclaimed from jungle. Rice has been comparatively recently introduced, and is still too scarce and dear to be the staple food of a people who had long been accustomed to support life on more hardy grains and on jungle produce, such as the fruit of the *mahua* tree." In this area rice is now grown wherever irrigation is possible, and much has been done to extend irrigation. In Monghyr the Kharagpur reservoir alone has so greatly extended the area under cultivation that the rent-roll of the estate benefited by it has risen by 300 per cent. in 30 years.

Formerly the cultivator distributed his capital and labour far more equally between rice and other crops, such as oil-seeds, pulses, etc. Now, the good prices commanded by rice, and the facilities for export afforded by the railway, have led him to concentrate on rice. This is not an unmixed benefit, for rice is often grown on uplands imperfectly irrigated and unsuitable for its growth. Areas which used to produce millets and maize, on which the people subsisted, have been turned into rice-lands of which the outturn is often uncertain and precarious. Such cultivation is, in fact, speculative, the peasants abandoning the grains which form their daily food for the sake of the larger profits which rice yields.

FAMINE AND SCARCITY.

153. Three districts were affected by famine during the decade, viz., Darbhanga, Ranchi and Puri. A number of other districts suffered from scarcity, which in places almost approached famine, but the cultivators, with improved resources, were able to tide over their difficulties with the aid of loans, and it was not necessary to declare famine. Such scarcity was most acute in Bankura, Nadia, Muzaffarpur, Sonthal Parganas, Bhagalpur, Cuttack, Balasore and Angul.

DARBHANGA.

154. In Darbhanga the famine of 1906-07 affected an area of about 1,690 square miles, or one-half of the district, with a population of nearly 1½ million. The famine was primarily due, not as usual to drought, but to floods. In July 1906 the rivers debouching from the Himalayas overflowed their banks, causing considerable damage to the crops. The first flood had scarcely subsided, when it was followed by another of unprecedented height and duration. Almost the whole affected area was submerged for about a fortnight, the *bhadoi* crop being destroyed and the paddy seedlings swept away. By the time the water had subsided, it was too late to plant out fresh seedlings, except in the Madhubani subdivision, where, however, the crop suffered greatly from subsequent drought. In the end the *bhadoi* crop yielded only 12, and the winter rice 27 per cent. of the normal. *Rabi* crops were sown over a larger area than usual, but the prolonged drought which followed the floods, and heavy rainfall at harvest time, reduced their yield to barely half the average. The total outturn of crops for the year is estimated to have been barely one-third of the normal. The harvests of 1905-06, moreover, had been poor, the yield being only two-thirds of the normal. The result was widespread scarcity.

but it was acute enough to necessitate the declaration of famine only in five of the ten thanas of the district, viz., the Darbhanga, Bahera and Rosera thanas in the Sadar subdivision, the Warisnagar thana in the Samastipur subdivision and the Benipati thana in the Madhubani subdivision. Of these, the Rosera and Bahera thanas (in the south-east of the district, where the famine was especially severe) suffered most. The famine continued till August 1907, when a bumper *bhadoi* crop brought it to a close. Except during the first stage of the distress, *i.e.*, immediately after the floods, the percentage of persons relieved to the population of the distressed area was only 2·66 per cent., their average daily number being 38,945.

155. The affected area had not fully recovered from the effects of this famine before it was visited by another, which was brought about by drought. Owing to the failure of the monsoon rains in 1908, the *bhadoi* crop had a very poor yield, and the winter rice crop, which is the mainstay of the people, was an almost total failure. There was scarcity, more or less acute, throughout the district, except in the Dalsinghsarai and Samastipur thanas. The failure of crops was most severe in the Sadar and Madhubani subdivisions, where famine was declared and relief operations had to be undertaken. In the Samastipur subdivision the only area where scarcity existed was the Warisnagar thana. This subdivision is a rich, fertile tract, with uplands suited to the cultivation of *bhadoi* and *rabi* crops, and is not dependent on winter rice like other parts of the district.

Distress was acute from February 1909 up to the end of May 1909, when it was mitigated by the commencement of the rains and the consequent resumption of agricultural operations. During these four months agricultural employment was almost entirely non-existent, except for a short time during the *rabi* harvest. This harvest, however, had little effect in relieving distress, as the crop was very poor for want of moisture. The most severely affected parts were the east portion of the Bahera thana, the Singhia outpost of Rosera, and portions of the Darbhanga, Phulparas and Benipati thanas. The average daily number of persons relieved (53,609) was greater than in 1906-07, but their proportion to the total population of the affected area was less and amounted only to 2·22 per cent.

RANCHI.

156. In 1908 there was famine in Ranchi for the first time since 1900. After that year the crops were more or less normal until 1906-07, when there was a bumper crop, the bulk of which was exported owing to the enhanced demand caused by the failure of crops elsewhere. The famine was due to the early cessation of rain in 1907, and was intensified by the very large exports. The total rainfall was in excess of the normal, but it was very badly distributed. August was abnormally wet; the rainfall in September was quite up to the average in quantity, but the whole of it fell in the first few days, and, except for one or two slight local showers, there was no rain in the district after 9th September. The result was that the early rice suffered from damp, while the winter rice dried up owing to insufficient moisture. The oil-seed crops withered, and the *rabi* was a total failure.

The area in which famine had to be declared consisted of thanas Kurdeg, Kochedega, Chainpur, Bishenpur, Ghagra and Gumla (all in the Gumla subdivision), with an area of 2,261 square miles and a population of 237,238. Relief was also required in Sisai thana and a part of Sonahatu thana, while test-works were opened in Burmu and part of Tamar. The whole affected tract was 3,402 square miles with a population of 447,461. The distress varied from scarcity in Burmu to actual famine in Bishenpur, but in the area in which famine was declared the ratio of persons on relief works to the population affected was only 1·59 per cent. It would undoubtedly have been greater but for the exodus of able-bodied labourers. Instead of the emigration season closing as usual in April, it continued right through the hot weather and even into the rains, when, as a rule, cultivators are very unwilling to leave their fields. The mortality reached a high figure (46·5 per mille) owing mainly to severe epidemics of small-pox, fever and cholera, which were rife throughout the district. There were no deaths directly traceable to starvation or privation, but in consequence of high prices and general distress the people succumbed to disease more readily than would have been the case in an ordinary year.

157. In Puri there was famine, in 1908, in two separate tracts, of which the former consists of the islands and sea-face of the Chilka Lake in the south-west of the district, while the latter, which also borders on the sea, lies at the extreme north-east of the district. The former, which may be described as the Chilka tract, has an area of 143 square miles with a population of 25,038; the latter, which is known as the Marichpur tract, extends over 178 square miles with a population of 74,345. The Chilka tract, which has a sandy, salt-impregnated soil, is not subject to floods to any considerable extent, but it cannot withstand the effects of drought. It has only one crop viz, winter rice, which is very largely dependent on an adequate rainfall in September and October; and in 1907 there was almost an entire failure of this later rain. In the Marichpur tract conditions are somewhat similar, winter rice being the chief, and in some parts the only, crop, but unlike the Chilka tract, it is subject to inundation from the Devi and other rivers running through it. Here floods did great damage to the standing rice crop, and the failure of the September and October rains destroyed much of what the people were able to retransplant.

PURI.

158. Fever is such an important factor in Bengal, Bihar and Orissa, whether considered from a statistical, sociological or economic point of view, that a brief description of its distribution is required, especially as no comprehensive account of the liability of different parts of the two Provinces to fever has hitherto been published. Year by year it is silently and relentlessly at work. Plague slays its thousands, but fever its ten thousands. Not only does it diminish the population by death, but it reduces the vitality of the survivors, saps their vigour and fecundity, and either interrupts the even tenor, or hinders the development, of commerce and industry. "A leading cause of poverty—and of many other disagreeables in a great part of Bengal—is the prevalence of malaria. For a physical explanation of the Bengali lack of energy, malaria would count high."* The present account is brief and sketchy considering the complexity and intrinsic importance of the subject, but, owing to the necessary limitations imposed on a census report; more cannot be attempted.

FEVER.

159. The vast majority of deaths are returned under the generic head of fever, owing to the predilection of chaukidars for fever as the cause of death in any case not palpably due to cholera, small-pox or plague. Inquiries made by competent observers show, however, that the mortality actually due to malarial fever is very much smaller than would appear from the returns. A special investigation (referred to at greater length below), which was conducted for three years in a rural area of Burdwan, where over 70 per cent. of the deaths were ascribed to fever, proved that nearly half were due to other causes, chiefly respiratory diseases. Even worse results were obtained by an investigation into the actual causes of so-called fever deaths in a small Bengal town, where the standard of intelligence and efficiency should *a priori* be higher. In this town a Deputy Sanitary Commissioner went from house to house to verify the recorded deaths, and found that, out of twenty deaths, ascribed to fever, three only were due to malaria, and even these were doubtful. The diagnosis of the cause of death was, in the majority of cases, extraordinary. Three deaths were due to old age, dropsy or bronchitis. One was a case of convulsions, and another of septicoemia. Two deaths could not be traced, and in two other cases living persons were reported as dead.

160. The medical officers deputed to assist the Bengal Drainage Committee, during the special inquiry held in 1906-07, also made investigations into the actual causes of the deaths reported as due to fever. In Nadia they found that 40 per cent. of the cases investigated were due to malaria, acute or chronic, and the remaining 60 per cent. to bronchitis, pneumonia, phthisis, dysentery, diarrhœa, typhoid, Leishman-Donovan infection, and other causes. In Jessore they found that 35 per cent. were due to malaria, while phthisis was responsible for 9 per cent., and dysentery and diarrhœa for 11 per cent. A similar inquiry was held in the Dinajpur district in 1904, when it was found that less than one-third of the deaths classified as due to fever were actually caused by malaria. The general result of these different inquiries is to

* R. E. Vernede *An Ignorant in India*, 1911.

show that approximately one-third of the deaths imputed to fever are the direct result of malaria.

161. In many localities where malaria has long been prevalent and become endemic, it does not cause any exceptional mortality. In parts of the Tarai, in particular, the incidence of malaria is high, but the inhabitants, such as Tharus, Meches and Rajbansis, seem inured to it. Major A. B. Fry, I.M.S., Deputy Sanitary Commissioner, Bengal, writes that, on visiting malarious districts for the first time, he was struck by the well-developed and prosperous appearance of the inhabitants, even in the malarious villages. "Although with spleens large enough to reach the umbilicus, and with malarial parasites in their blood, and having attacks of fever frequently, the children as a whole looked quite fat and healthy, not particularly anæmic, and seemed little affected by the malaria and were playing about and seemed to enjoy life." In fact, though malaria prevents any large increase of population, it is not inconsistent with a small or moderate increase provided that other conditions are favourable.

It must, however, be remembered that malaria is the indirect cause of a large proportion of deaths owing to enfeeblement caused by its repeated attacks. Malaria, and the lowered vitality resulting from it, is a predisposing cause in both phthisis and dysentery, so that it is responsible, in part, for the prevalence of these diseases and for the mortality ascribed to them. There can, in any case, be little doubt as to the prejudicial effect of malaria on the birth-rate, both by causing abortion and still-birth, and also by diminishing the reproductive powers of persons whose systems are weakened by continual attacks. Further, as stated by a statistical authority, "from an economical point of view common sickness is more important than deaths, for it is the amount and duration of sickness rather than the mortality that tell on the prosperity of a community (Dr. Dickson)."*

162. It is only recently that the various types of fever and their prevalence in different localities have been scientifically investigated. It has been shown conclusively that Central Bengal is subject to a bad type of malarial fever, where its prevalence is due to, or is facilitated by, two causes, viz., the water-logged state of the country and the insanitary condition of the villages. The general situation may be summed up in the words used by Captain Stewart and Lieutenant Proctor in their description of one typical district: "The excessive prevalence of malaria as a whole can be attributed directly to the great facilities afforded to the breeding of mosquitoes, chiefly by the presence in and around the villages of jungle, dirty tanks, ditches, marshes and casual water in every direction, and, to a lesser degree, to the *bils* and dead rivers acting in the same way in some cases. In its turn, the presence of so much water in the villages is due, in part, to the carelessness and ignorance of the inhabitants, and in part to the want of natural drainage in the country, owing to its position in a deltaic tract, where the process of land-building is still going on. The rivers are gradually heightening their banks and beds, until the drainage is away from instead of towards them. The subsoil water is unable to drain away rapidly, remains long at high level after the wet season, and prevents the soaking in of rain-water resulting in casual collections of water remaining for long periods in every hollow, natural and artificial. It is the combination of these two factors, the high subsoil water and the jungly and insanitary condition of the villages, that results in so high a malaria rate. The pits, hollows and jungle in the villages† would in themselves be insufficient to account for so great a prevalence of the disease, were they not combined with the lack of natural drainage, which allows the surface collections of water to remain for so long a time; and on the other hand, so far as our present knowledge goes, the high subsoil water has no connection with the disease except in so far as it is a cause of these surface collections of water. The silting up of the rivers is merely one sign of the

* A. Newsholme, *Vital Statistics* (1899), p. 38.

† Major A. B. Fry, I.M.S., to whom I am indebted for assistance in preparing this account of the localities affected by fever, writes:—"Tanks containing enough water to remain full all the year, contain enough fish to destroy all larvæ, provided the latter are unprotected by excess of weeds. Clean tanks are larvæ-free..... The effect of jungle is variable, but I am convinced that the undoubted fact that jungly villages are more malarious is explained by the fact that jungle is a measure of the age of a village; the jungly villages are old villages with broken surface and foul soil. It is quite conceivable that a village situated within a solid unbroken circle of bamboo growth, with only a narrow exit and with tanks and water outside the ring, might be adequately guarded against mosquitoes and fever."

lack of natural drainage, and apart from that is not in itself a cause of malaria to any large extent."*

163. Conditions similar to those described above also exist in North Bengal and the alluvial tract of West Bengal, in both of which malaria is prevalent. In West Bengal it occurs not only in water-logged localities, but also on the dry uplands. The geological formation of the latter is favourable to the retention of water in places where the surface is uneven, for the soil has an impermeable stratum which prevents percolation. Here stagnant pools remain until a dry season comes in. The number of infected mosquitoes goes on increasing, and, *pari passu*, the number of infected people, the one reacting the other. The river districts of East Bengal, such as Dacca, Backergunge and Tippera, are the least malarious. "In these districts the rivers are open, tidal, and clean-banked. When the rivers silt up at the mouth, the district is at once changed in character, and will be found to be amongst the worst in the Province. Dinajpur, Jessore, part of the 24-Parganas, Faridpur and Nadia are examples of this class. In Faridpur these two conditions are seen within thirty miles of each other. The Bhushna thana is as malarious as any part of Bengal, while Palang and the *char* thanas are free from the scourge. Were it not for Bhushna and some of the western thanas, the death-rate from fever in Faridpur would be comparatively low. Jessore and Dinajpur are full of old water courses that have gradually silted up; the natural drainage of the district is upset, the subsoil water is phenomenally high, and the whole place is water-logged."† On the other hand, cholera is nearly always more prevalent in river districts than in drier areas. In the latter, epidemics rage sometimes with extraordinary virulence, but in the river districts cholera is almost an annual visitation. The explanation seems to be the practice of defœcation on the banks of *khals* or rivers and the consequent pollution of the water. The severity of the disease depends on the rainfall and the quantity of water flowing in the rivers. When rainfall is short, the current slow, and the volume of water small, the disease is rife. When rainfall is heavy and there is a good flood, it is rare.

164. In Bihar the condition of most villages is probably more insanitary than in Central and North Bengal. They are mostly free from jungle, but they are congested and badly drained; the drinking-water supply is often neglected, measures being rarely taken to protect the wells and preserve their purity. The earth required for building the houses is dug up in their immediate vicinity, the excavations forming dirty pits, where water remains stagnant for a long time. Moreover, the cowsheds, as in Bengal, are close to the houses, and in them the mosquito finds a resting place undisturbed by smoke.‡ Most villages are surrounded by rice cultivation, this wet crop being grown almost up to the doors of the houses. In North Bihar many villages stand on the edge of large marshes. Extensive areas are swampy and liable to inundation from the rivers, which leave water lying over the country till November. The drainage, as in North Bengal, is obstructed by deserted or silted-up river channels.§ Conditions, however, vary even in the same district. The south of Champaran, a dry area, is practically free from malaria, but in the submontane swampy area to the north the people are fever-sodden. In the north of Bhagalpur malaria is rife; on the south bank of the Ganges in this district, and also in Patna and Monghyr, it is absent.

165. In the upland plateau of Chota Nagpur, where there is good natural drainage, where the soil is dry and porous, and where wet crops are not grown to such an extent as in Bengal and Bihar, malaria is far less common, but it is distinctly prevalent in the valleys. "In some of the shut-in valleys in this part of the Province it is possible to find places that are perfect death-traps. These are usually valleys with rich marshy soil and a slow stream wandering through them. Malarial infection is so rapid and so deadly, that inhabitants of the districts usually avoid these spots choosing

* Report of the Drainage Committee, Bengal, 1909.

† Report of the Sanitary Commissioner, Bengal, for 1904.

‡ The proximity of cowsheds to houses is an important factor in the propagation of malaria.

§ In Saran the country is so closely cultivated, that in some places the natural drainage channels have been brought under tillage.

the more healthy parts for locating their villages."* The villages at the foot of the *ghats* below the plateau or on the escarpments, and those actually on the edge of plateau, are also very malarious.

166. In Orissa most of the big villages have a main street, with houses in a row, which is kept clear of trees and jungle, but behind the houses, and on the outskirts of the village, conditions are similar to those obtaining in a typical Bengal village. The country is deltaic, rice cultivation is seen everywhere, and Cuttack is largely under irrigation from the canals. "One can," remarks Major A. B. Fry, I.M.S., "only suppose that the reason why the irrigated portions of this district are not heavily malarious is that the amount of existing infection has not reached a numerical value high enough to cause widespread epidemics. The supply of anophelines has certainly in many places reached the numerical value, and I shall not be surprised to find a heavy epidemic occurring in the near future."

PLAGUE.

167. Plague first appeared in Bengal in 1898, when there were two outbreaks, one in Calcutta and the other in Backergunge.† In the early part of 1899 it again visited Calcutta, and there were also outbreaks in ten rural districts. In the cold weather of 1900-01 the disease spread over a larger area. Since the last census it has established itself firmly in Bihar, coming and going with the seasons with wonderful regularity. It is most prevalent in the winter, practically disappears or remains dormant throughout the hot and rainy seasons, and recrudesces with the advent of the cold weather, attaining its greatest virulence in the first three months of the year. At first, the epidemic was confined to those parts where easy communication and grain markets existed, *e.g.*, in Patna from 1900 to 1904 the tract along the East Indian Railway and the surroundings of Bihar were attacked every year, while the south-west of the district remained immune. The disease thrives in congested areas, and the people have recognized this by evacuating their houses and encamping in the open. This so far is practically the only measure they take to avoid attack, and inoculation has found little favour. The only district in which it has been resorted to on any extensive scale is Gaya, where 23,000 persons were inoculated by their own free will during the epidemic of 1900-1901. The success of inoculation in this district was due to the popularity of the Collector and the Civil Surgeon and to the confidence they inspired : in no other district has the same result been obtained.

1901	...	70,388
1902	...	25,369
1903	...	56,972
1904	...	70,450
1905	...	116,769
1906	...	56,708
1907	...	79,867
1908	...	14,105
1909	...	9,613
1910	...	45,209
Total	...	545,450

168. The marginal table shows the actual number of deaths recorded as due to plague during the ten years in the Province of Bihar and Orissa. The vast majority occurred in Bihar, for Chota Nagpur and Orissa have been almost immune from this scourge. The districts of Patna, Saran and Shahabad have suffered particularly severely, the ratio of plague deaths during the decade to the population of 1901 being 90, 80 and 30 per mille respectively.

The trading classes appear to have lost most heavily from plague. "It is not the trade itself but its environment that is responsible for the increased death-rate or for the immunity of those who engage in it. All shop-keepers, especially grain-dealers and Halwais or sweetmeat vendors, show a very great mortality from plague. It is almost always the village shop-keepers who are first attacked with plague : they usually introduce the disease, and they always suffer the most. These men have dark, rat-infested store-godowns. In Calcutta, Barh and many other towns it is the bania's quarter from which most deaths are

* Report of the Sanitary Commissioner, Bengal, for 1904.

† As the people of Bengal, Bihar and Orissa seem to think that plague is a new visitation, and that its causes are mysterious, I may be permitted to mention that a Muhammadan historian (Mu'tamad Khan) gives an account of an outbreak of plague in the Deccan in 1619, which clearly indicates the presence of the plague rat and also that the evacuation of houses was then, as now, the chief means of escape from attacks. "When it was about to break out, a mouse would rush out of its hole, and, striking itself against the door and the walls of the house, would expire. If, immediately after this signal, the occupants left the house and went away to the jungle, their lives were saved ; if otherwise, the inhabitants of the whole village would be swept away by the hand of death. If any person touched the dead, or even the clothes of a dead man, he also could not survive the fatal contact."

returned. Unfortunately the raiyat's house is his grain-store as well as his abode ; hence in the agricultural villages the conditions under which he and his family live are not very different from those of the city bania. Consequently, though not a shop-keeper, his family usually suffers severely. The men who have no such houses, such as Nats (gipsies) fishermen, herdsmen, almost entirely escape. Further, in rural areas, the death-rate is much greater amongst women and children than amongst men, the reason being that they are more at home."*

169. Epidemics of plague have been conspicuous by their absence in nearly all parts of Bengal, Calcutta being the only place in which there has been any considerable mortality. The immunity of Bengal is remarkable, because there is a constant influx of labourers coming from infected areas to seek employment in the fields, or on the railways, or in the mills and factories. The causes of this immunity were first pointed to, in 1906, by Major Clemesha, I.M.S., who remarked :—"A house so constructed as not to be suitable for rats to live in, and not containing any food to attract rodents, would probably remain non-infected unless a case of pneumonic plague was placed in it. There is the greatest difference between the Bihar and purely Bengali villages. In Bihar the mud houses are closely packed together, so as to utilise every available inch of ground. There are no streets, narrow passages between the walls only remaining. The village is compact ; it may consist of several *tolas* or hamlets separated by a considerable distance, but each such *tola* is a compact, overcrowded unit. In Bengal exactly the opposite tendency prevails. Villages are long straggling lines of houses built on the highest part of the land to be above flood level. Usually each house is buried in a thicket of bamboos and rank vegetation, having its own compound, and the individual houses being often some distance apart. Undoubtedly, the Bihar village is the ideal type for plague to flourish in. It is certainly infested with rats. Conversely, it would appear that the Bengali village is not a suitable location for the virus. Plague has fairly frequently been introduced into such a village, but it has not spread to more than the members of the household. To-day, practically all Eastern Bengal is free from the disease. I am not able to say definitely as to whether this type of village contains less rats than the Bihar variety, but I am inclined to think that this is the case. It is, however, certain that there is comparatively little overcrowding : the houses are much better ventilated and lighted, and are made of a material which allows rapid exchange of air. I am inclined to the view that these conditions and the possible scarcity of rodents have a considerable influence in rendering these districts unsuitable as a habitat for the plague virus. With most of these districts there is considerable communication with Calcutta, and yet it is only in Bihar and in cities that plague has obtained a foothold.†"

170. Further inquiries have established the fact that the immunity of Bengal, and particularly of Eastern Bengal, is due to the scarcity of plague rats, which again is a consequence of the structure of the houses and the habits of the people. The results of these inquiries are summarized as follows :—

(1) Eastern Bengal has suffered very little from bubonic plague ; a few epidemics only of pneumonic plague have occurred. (2) The physical features of the country protect it, to some extent, from the importation of infection and would tend to limit the opportunities for spreading the disease if it once broke out. (3) The freedom from plague can chiefly be attributed to the scarcity of rats in the houses as compared with other parts of India. (4) *Mus rattus* is comparatively rare in Bengali houses, because of the habits of the people, in respect of their greater regard for neatness and tidiness, both in and around their dwellings, which diminishes the food supply of the rodents. (5) The structure and design of the Bengali home, whether it be of the solid masonry type on the one hand or of the flimsy matting or grass type on the other, afford little shelter for rats. (6) The presence of natural enemies of *Mus rattus*, such as the musk rat, may assist in maintaining a low rat infestation of the houses.‡

Commenting on these conclusions, the Editor of the *Indian Medical Gazette* remarks :—"The writer gives a very favourable account of the neatness

* Major W. W. Clemesha, I.M.S., *An Account of Plague in Bengal*, Indian Medical Gazette, Volume XLI, 1906.

† The Journal of Hygiene, Plague Supplement I, p. 192 (Dec 1911).

‡ *An Account of Plague in Bengal*, Indian Medical Gazette, Volume XLI, 1905.

and tidiness of the houses of Eastern Bengalis as compared with those of the inhabitants of other parts of India, and while these habits of tidiness help to keep away the rats, the construction of the houses does even more. When the houses are *mucka*, *i.e.*, built of brick and mortar, the walls naturally afford little harbourage for rats, and the *kutcha* houses of thin bamboo matting or wattle, with a roof of corrugated iron, split bamboo or thin thatch, gives even less shelter to these rodents. The country-tiled roofs, which are so infested with rats up-country, are not seen in Eastern Bengal. In fact, the rat is a domestic animal in the thick mud-walled houses necessary to protect against heat and cold in Upper India, while in Eastern Bengal he finds but few places to live in and, in fact, is not a domestic animal.*"

CANALS.

171. The principal irrigation works in the two Provinces are the Son, Orissa and Midnapore canals. The average area (in square miles) irrigated in each of the decades ending in 1890, 1900 and 1910 was as shown in the margin. During the last decade the Dhaka canal, a small work in the Champaran district, was completed, being first used for irrigation in 1905. The Tribeni canal in the same district is under construction and has been partially completed. The Son Canals irrigate the greater part of the district of Shahabad and small portions of Gaya and Patna. After their introduction in Shahabad a large area of waste land was brought under the plough, and the cultivation both of rice and sugarcane rapidly increased: the area under rice is now 50 per cent. more than it was before the construction of the canals. In Gaya the canals, which irrigate the two thanas of Daudnagar and Arwal, have turned neglected waste into fertile fields. Speaking of Daudnagar in 1812, Buchanan Hamilton remarked :—"Some of the best land even is neglected, and is chiefly occupied by poor-looking woods of the *palas* (*Butea frondosa*)." His description of the country round Arwal is equally depressing, for he observed :—"A great portion is neglected, and, where the soil is poor, is chiefly over-grown with thorns of the stunted *jujub*. Where the waste land is rich, it is over-grown with harsh long grass." The appearance of this tract is now very different, as it includes some of the best rice-growing land in the district. Since 1872, Daudnagar has added nearly 45 per cent. and Arwal 33 per cent. to its numbers.

CANALS.	DECADE ENDING IN—		
	1890.	1900.	1910.
Son	447	645	849
Orissa	472	256	378
Midnapore ...	128	119	119

172. In Shahabad also there was a large increase of population in the irrigated areas up to 1891, but the census statistics of the last 20 years do not show any general correlation between growth of population and the benefits of canal irrigation. In fact, throughout the northern thanas (numbered 1—4 in the marginal statement) there has been a serious decline since 1891, whereas in the central thanas (numbered 5—7) the pressure on the soil has been increasing. There is a general belief locally that since the construction of the canals malarial fever has spread and increased in intensity. That this belief is well founded may be gathered from the finding of the Committee appointed to enquire into the administration of the Son Canals. "It is," they remarked, "a matter of notoriety that Shahabad was formerly one of the healthiest districts in Bengal. It was not free from fever by any means, but it was less malarious than other districts. There can be no doubt that it does not now enjoy this comparative exemption from malaria. Fever now appears to be more common in all the Bihar districts than it formerly was, but nowhere is the increase greater and more marked than in Shahabad. This change is attributed partly to the dampness of the subsoil occasioned by irrigation, and partly to the obstruction of drainage occasioned by the canal embankments. It is an obvious conclusion to connect increased malaria with increased dampness. The change, moreover, cannot be attributed to any other cause with any show of reason.

THANA.	Percentage of area under irrigation in 1910.	DENSITY PER SQUARE MILE.			
		1911.	1901.	1891.	1881.
(1) Shahpur	22	655	739	789	777
(2) Piro	37	634	661	643	581
(3) Buxar	25	473	497	528	528
(4) Dumraon	25	661	735	769	749
(5) Bikramganj... ...	34	525	508	497	490
(6) Dehri	31	546	537	496	466
(7) Karghar	14	376	365	391	393

* *An Account of Plague in Bengal*, Indian Medical Gazette, Volume XLI, 1906.

We think that, in the districts irrigated by the Son, the complaints of injury to health are well founded, and that the tracts so irrigated suffer now more severely than other tracts which are not commanded by canal water.*"

On the other hand, the census returns for the last twenty years show that, of the three thanas where most land is under canal irrigation, Bikramganj and Dehri have a substantial increase, while in the third (Piro) the decrease is very small. The explanation of the results of these twenty years must be sought elsewhere. It appears to be simply that the thanas in which there has been the greatest loss (all of which lie along the banks of the Ganges and are traversed by the railway) have suffered severely from plague and that this scourge has not affected those further inland and away from the railway.

173. In Cuttack, which is served by the Orissa canal system, the inquiries made during the settlement show that the increase of cultivation in irrigated lands is no greater than in non-irrigated lands. "All the inquiries made have failed to elicit any evidence of a substantial extension of cultivation to lands which but for the canal water were not likely to have been reclaimed."† The canals have, however, given a large area immunity both from famine and flood. The canal embankments protect nearly 550,000 acres, where every year the inhabitants used to be kept on the alert, for two or three days at a time, waiting for a signal to fly to the highest ground available, and were obliged to see their houses washed down on all sides without having any power to save them. As regards famine, it will be sufficient to refer to the terrible famine of 1866, when the mortality was estimated at one-fifth to one-fourth of the population, and to quote the conclusion of the Indian Irrigation Commission that now few parts of India are more secure from such visitations. The marginal statement shows the increase in density since 1881 in the thanas where over 5 per cent. of the area is under canal irrigation.

THANA.	Percentage of area under irrigation in 1910.	DENSITY PER SQUARE MILE.			
		1911.	1901.	1891.	1881.
Jagatsinghpur	21	770	764	725	685
Tirtol...	9	470	461	432	438
Salepur	31	963	899	893	874
Kendrapara	19	832	771	714	650

174. Three large schemes, designed for the drainage and consequent reclamation of swampy areas, have been carried out in Bengal. Two of these schemes, viz., the Howrah and Rajapur schemes, which drain an area of 50 and 270 square miles respectively, benefit the district of Howrah: the former was completed in 1885 and the latter in 1894-95. The third, which was brought into operation in 1873, consists of the Dankuni drainage works in the Hooghly district. These schemes have proved very successful in reclaiming useless swamps and improving other lands. In years of heavy rainfall the surplus water is drained away by means of channels and sluices; in years of drought water from the Hooghly is let in for the purposes of irrigation. The people are thus assured of good harvests both in years of drought and in years of heavy rainfall. Two thanas have been especially benefited by these schemes, viz., Dumjor, where the population has increased 33 per cent. since 1891, and Jagatballabhpur, where the increase amounts to 17 per cent. In Amta, where there is waterlogging similar to that which used to prevail in these two thanas, the increase has been only 9 per cent., though it has been opened up by the Howrah-Amta Light Railway.

DRAINAGE SCHEMES.

175. The most important drainage scheme in progress during the decade was the Magra Hat scheme, which is designed for the drainage of a marshy tract extending over 290 square miles in the south of the 24-Parganas. The conditions which formerly existed in this tract may be realized from a description written 30 years ago. Fever was constantly present in every village; other diseases found a congenial home; the productiveness of the land was only a fraction of what it should be. The inhabitants, it was said, might be regarded as "inured to a semi-amphibious life by a long course of preparation resulting in the survival of the fittest." This state of affairs is becoming a thing of the past, even though the scheme has not yet been fully

* Report, 1888, pp. 27-28.

† S. L. Maddox, Orissa Settlement Report

completed. Its effect is already shewn in the census returns for villages situated in the affected tract. The population of these villages was 272,734 in 1901, but has now risen to 352,702, the rate of increase being 29 per cent., which is far in advance of the general rate of growth for the district, in spite of the fact that this tract contains an agricultural population, whereas others are industrial centres. In the same district the construction of a sluice in the neighbourhood of Diamond Harbour has already had remarkable effects, though it was only completed in 1909. Prior to its construction there were 100 square miles of swampy or waste land: now this area is covered with rice cultivation, the annual value of which is nearly 38½ lakhs of rupees, while the value to the tenantry of one year's crops only is estimated as approximately twice the actual cost of the scheme.

RAILWAYS.

176. There have been large extensions of railway communications in both Provinces during the decade. The Eastern Bengal State Railway has completed a line from Kaunia to Dhubri and Gauhati, and thus brought the Assam Valley within easy reach of Calcutta and Bihar. The extension of the Bengal and North-Western Railway from Hajipur to Katihar and the linking up of Katihar with Godagari by the Eastern Bengal State Railway have furnished a through route across the North Gangetic districts, stimulating emigration from Bihar to Bengal and Assam. The Assam-Bengal Railway has completed communication through the hill section of the line between Chandpur and Dibrugarh, thus opening up a new route to the tea-gardens of the Assam and Surma Valleys, which is much quicker and easier than the old river routes. Lines have also been made to Noakhali and Ashuganj on the Meghna, and from Kaunia to Bogra on the Eastern Bengal State Railway. The increase in the mileage of railways in this part of Bengal has not resulted in a decrease in steamer traffic: on the contrary, the facilities afforded by the steamers have been increased considerably, and they now penetrate the recesses of the delta more extensively than they used to do. In West Bengal the East Indian Railway has constructed a line from Ondal to Sainthia, which passes through the centre of Birbhum and connects the Sadar station (Suri) with the Chord Line at Ondal on one side and with the Loop Line at Sainthia on the other. In Central Bengal the Murshidabad-Ranaghat branch has been added to the Eastern Bengal State Railway system: it takes off from the main line at Ranaghat in Nadia and runs through that district to Lalgola Ghat on the Ganges in the extreme north of Murshidabad.

177. In Bihar and Orissa the Grand Chord line, traversing the districts of Manbhum, Hazaribagh, Gaya and Shahabad, was opened in 1906, and the Purulia-Ranchi line, a light railway connecting Ranchi with Purulia on the Bengal-Nagpur Railway, in 1907. In the latter year also, the Midnapore-Bhojudih-Gomoh section of the Bengal-Nagpur Railway in the districts of Manbhum, Bankura and Midnapore was opened to traffic. This line completed the through connection, east and west, of the Manbhum colliery area, and also opened up the district of Bankura, which till then was not served by a railway. Further west the Barun-Daltonganj branch of the East Indian Railway has given Palamau connection with the main line and afforded its coalfields access to the north-west. North of the Ganges the most important new line is the Hajipur-Katihar extension of the Bengal and North-Western Railway along the left bank of the Ganges. Its value to North Bihar can hardly be exaggerated, as it traverses the districts of Purnea, Bhagalpur, Monghyr, Darbhanga and Saran, and has branches to the Ganges opposite Bhagalpur and Monghyr. Another extension of the same railway is the Mansi-Bhaptiahi line, which runs from Mansi near the Ganges to Bhaptiahi near the Nepal frontier. This line connects Bhagalpur city with the Supaul and Madhipura subdivisions, and in Monghyr provides direct communication across a tract seamed with rivers and channels, where traffic by road used to be most difficult.

Three light railways have also been opened. In Patna a light railway was built from Baktiyarpur to Bihar, which was extended to Silao in 1909. In 1905 a line from Barasat to Basirhat in the 24-Parganas was opened, and in the same year the Mayurbhanj light railway, which connects Baripada, the headquarters of the Mayurbhanj State, with Rupsa on the Bengal-Nagpur Railway.

178. The value of new lines of railway to the tracts which they traverse, in affording an outlet for their produce and also to their surplus labour, needs no explanation; but, as pointed out in the Census Report of 1901—"It does not follow that this benefit will be reflected by an immediate growth of the population. Very frequently the tendency is the other way. The cultivator benefits by the rise in prices, but there is no waste land available for new settlers, while on the other hand the landless labourer is enabled to move more easily and for shorter periods to places where there is a greater demand for his services." These remarks are very fully confirmed by the results of this census, which show that the effect of a new line of railway is to relieve congestion in populous districts rather than to attract permanent settlers to undeveloped tracts. In the densely populated district of Cuttack, in which railway facilities have recently been introduced, the number of immigrants has not increased, whereas the number of emigrants has increased by over 50 per cent. In Ranchi, again, where the railway is only a few years old, the number of emigrants has already so far increased that they represent 20 per cent. of the number born in the district.

179. Sometimes the construction of railways appears to be, actually or potentially, prejudicial to the health of the people, owing to the facilities which they afford for the introduction or dissemination of disease. Their function is that of carriers, and they carry disease as well as goods. This seems to be especially the case with plague, a contributory cause, no doubt, being that grain godowns, infested by plague rats, are naturally most common in the towns or stations situated along the railway. In twelve towns, for instance, situated on the railway in Bihar (which are also along or within a few miles of the Ganges), we find that since 1901 there have been 60,000 deaths from plague, representing one-eighth of the population of that year.* On the other hand, the districts of Orissa and West Bengal (such as Bankura and Midnapore) have benefited by the railway taking pilgrims, who previously plodded wearily on foot to and from the shrine of Jagannath at Puri, and spread cholera in all directions along their line of march. There is also a potential source of danger in the form of 'borrow pits,' from which earth is excavated for railway embankments. They are not continuous but separated from each other by intervening banks of earth. In the rains they form stagnant ponds, from which water cannot drain away and in which the fever mosquito breeds and multiplies.† So far there is ground for the common, though vague, belief of the people that fever is connected with the railway, but, on the other hand, where such embankments exist, fever is as prevalent in tracts far removed from the railway as it is near the line.

180. Railway embankments may also obstruct the drainage of the country. The Indian Railway Act (section 11) requires railway administrations to provide waterways sufficient to enable the water to drain off the land near or affected by the railway as rapidly as before its construction, but it is open to question whether it is physically possible to do so, and there is no doubt that in areas liable to inundation, the embankment does frequently alter the drainage of the country. On one side the floods are deeper and last longer than before, and the soil becomes waterlogged; on

* *Cf.* the following remarks in the Journal of the Royal Statistical Society for February 1912 (page 348)—"Two further Reports by Mr. M. Greenwood, Jun., on the Statistical Investigation of Plague in the Punjab, are published in a supplementary number of the *Journal of Hygiene* (December, 1911). The first deals with the relation between the proximity of villages to railways and the relative frequency of epidemics. Comparing villages in the whole of the Amritsar district with those within two miles of the railways, it was evident that the latter were the more frequently infected. But villages near the railway are, on the whole, the larger villages, and a large village is, *ipso facto*, the more likely to be infected. Correction being made for size, it appears that proximity to the railway still remains as an effective factor, leading to an increase of plague owing to the increased personal communication between villages and centres of constant infection."

† The following remarks recorded by the Sanitary Commissioner in his Annual Report for 1904 apply *mutatis mutandis* to railway embankments. "The question of constructing roads is in certain districts a matter of the greatest importance and one that is constantly neglected. In districts, like Backergunge, where all roads have to be raised, the greatest possible care should be taken in obtaining earth for this purpose. All borrow-pits should be made in the form of a ditch and should open into a *khal*. All earth should be taken from one side of the road only. If this were done, the result would be a channel useful as a waterway for small boats. It would be scoured out at each tide, and thus could not possibly be a breeding place of mosquitoes, while it would be very beneficial to the drainage of the country. A series of stagnant pools covered with weeds, as one only too frequently sees, is a standing danger to health."

the other, the land does not receive the same amount of moisture or the same fertilizing deposit of silt. The resultant advantages and disadvantages may, however, counterbalance one another. In the former area the cultivators may merely lose 'catch crops'; in the latter they may be benefited by protection from floods.*

INDUSTRIAL DEVELOPMENT.

181. Since the last decade there has been a revival of the small industries of Bengal owing to the Swadeshi movement, *i.e.*, a movement aiming at the resuscitation of dead or dying indigenous industries, the development of such as have maintained their vitality, and the initiation of new forms of industrial enterprise, directed and managed by Indians and employing Indian labour. Its effect has been principally to enable weavers to regain some of the ground which they had lost owing to the produce of their looms being driven out of the market by cheaper machine-made goods. Interest in the movement has fallen off lately, but for some years it had a stimulating effect, as may be realized from the remarks recorded in 1907 by the Magistrate of Hooghly: "It appears that while formerly the weavers had to take advances from the middlemen and were always more or less indebted to the latter, they are now very much better off, and if anything, the middlemen are sometimes indebted to them. I was told the other day by the President of the Dwarhatta Union that a young widow of the weaver caste, who would formerly have in all possibility suffered great privation, was now earning Rs. 16 or 17 a month and maintaining herself and her younger brother and sister in some comfort. In Dhaniakhali I was told that a weaver earns about Rs. 20 a month, and the Subdivisional Officer of Serampore reported that a weaver there earns Rs. 25 a month. A large dealer in Dhaniakhali was complaining that he was doing less business now than before, because now dealers from Chandernagore and elsewhere are coming to the villages, whereas formerly he and a few others had a sort of monopoly." The Swadeshi movement has also been instrumental in the starting of a number of small factories in the metropolitan districts for the manufacture of such articles as soap, ink, pencils, tin boxes, steel trunks, combs, buttons etc., but it has not made much headway as regards large manufactures employing mechanical power. Joint-stock companies have been started, but few have had any real vitality, and nearly all the important industrial concerns are still chiefly under European supervision and supported by European capital. There is one notable exception in the case of the Tata Iron and Steel Works, recently established at Sakchi in Singhbhum, which owe their creation to the enterprise of Messrs. Tata, but in this case also the management consists of Europeans and Americans.

COAL MINING.

182. The decade has witnessed a most remarkable development of coal mining, as may be realized from the marginal statement. Coal mining in Bengal is now nearly a century old, but in spite of the natural advantages conferred by the geographical position of the coal-fields and easy mining conditions, its true development has only taken place during the last 20 years, progress being most rapid in the last ten. The output of coal in 1910 was 95 per cent. of the total production of India, and nearly all of it was raised from the Raniganj coal-field in the districts of Burdwan and Manbhum and the Jheria coal-field in Manbhum. The latter has developed most rapidly and produces more coal than any other field: the labour force in Manbhum has grown accordingly, for whereas there were 157 mines with 32,194 workers in 1901, the number of the former rose in 1910 to 232 and of the latter to 56,179. The only other coal-fields of any importance are the Giridih field in Hazaribagh, with 10 collieries and an output of 674,000 tons (in 1910), and the Daltonganj coal-field in Palamau with one mine from which 85,000 tons were raised. The history of the industry during the decade was somewhat chequered owing to a boom and a subsequent slump. The boom culminated in 1908, when the output was over 11½ million tons, the maximum ever reached; next year the trade received a decided set-back, the demand being no longer equal to the supply. The value

YEAR.	NUMBER OF—		Output in tons.
	Mines.	Workers.	
1901 ...	292	79,652	5,703,876
1910 ...	418	98,268	10,777,306

* See Monghyr District Gazetteer, pages 116, 117, 119-121, Bhagalpur District Gazetteer, pages 132, 133, and Purnea District Gazetteer, page 105.

of coal properties shrunk enormously, but the Chief Inspector of Mines pointed out—"In spite of the slump in values, the industry to-day is probably in a healthier condition than it was two years ago; the fever has departed; many weak members, which should never have been allowed to grow, have been excised, and, with normal conditions, a steady and profitable future is bound to supervene."* At present the mining is easy, for most of the coal is raised from inclines driven into the outcrops of the seams, and the majority of the mines are at depths varying from a few feet to 350 feet.

OTHER MINES.

183. Mica mining has also developed, though not so rapidly, the number of mines rising from 18 in 1901 to 168 in 1910 and the number of workers from 6,254 to 10,581. Slate is quarried on a small scale in Monghyr, and copper is raised in Singhbhum. There are also iron workings in the latter district, from which 17,646 tons were raised in 1910.

MANUFACTURES.

184. The development of other large industries, which was so marked a feature of the economic history of the last decade, has continued. The marginal table will sufficiently illustrate the expansion of jute manufacture, and it need merely be added that the capital invested in the mills is £13,000,000, that the wages paid to the 200,000 employés are 4 crores per annum and the annual value of the products is 25 to 30 crores. The number of cotton mills has risen in the decade from 10 to 15, and the average number of operatives from 8,000 to over 11,500, while in 1910-11 the number of other factories in Bengal as then constituted was 184, with an average daily labour force of 84,657.† Of the latter 21,914 persons were employed in 10 railway workshops, 9,850 in 36 jute presses, 8,288 in 9 dock-yards, 7,232 in 4 arms and ammunition factories, and 5,310 in engineering workshops.

Year.	No. of mills.	Average daily number of operatives.
1880-1	19	33,994
1890-1	25	61,563
1900-1	34	110,057
1910-1	58	199,670

185. Nowhere has there been a greater outburst of manufacturing and industrial activity than in the 24-Parganas, where the number of factories rose from 74 to 124 between 1901 and 1911 and the number of operatives from 94,186 to 169,310. The mill towns along the banks of the Hooghly show a most extraordinary growth of population, which is accounted for by the influx of labourers, as illustrated in the marginal table. The character of the population has changed so greatly owing to this influx, that some mill towns are now practically foreign towns planted in the midst of Bengal. In Bhatpara four persons speak Hindi to each person speaking Bengali: in Titagarh 75 per cent speak Hindi, 8 per cent. Telugu, and 4 per cent Oriya, while 11 per cent. only speak Bengali.‡

Towns.	Increase of Population, 1901-1911.		Increase of Operatives, 1901-1910.	
	Actual.	Per cent.	Actual.	Per cent.
Bhatpara	36,763	81	30,835	215
Naihati				
Halisahar				
Titagarh	29,106	181	21,649	162

VITAL STATISTICS.

186. The present system of reporting births and deaths and compiling vital statistics from the returns is of recent origin, having been introduced in 1892. Births and deaths in towns had, it is true, been registered since 1873, but in rural areas deaths alone were registered. Under the present system, compulsory registration is in force in the towns, *i.e.*, parents, guardians or the persons directly concerned are required to report births and deaths to the town police. In rural areas each *chaukidar* or village watchman is provided with a pocket

* Report of Chief Inspector of Mines in India for 1909.

† There was 1 cotton mill in Eastern Bengal and Assam with 285 operatives, and 115 other factories employing, on the average, 11,875 operatives daily.

‡ The existence of a large new colony of Madrasis in Titagarh came to light in a curious way in the course of compilation. In the tickets for Table XIII for that town there were over 3,000 entries of Doanguli, *i.e.*, two-fingered. This seemed at first inexplicable, but one of the tickets gave a clue, as Madrasi was entered after Doanguli. The tickets for Tables X and XI were then compared. The former contained over 3,000 entries of Telugu and the latter a corresponding number of entries of Ganjam and Vizagapatam as the birthplace. Investigation showed that the caste was Devangula or Devanga, a common Madras weaving caste. It may be added that in 1901 the number of persons born in Madras and enumerated in the whole district (24-Parganas) was only 618, and the number of Telugu speakers 294.

book, in which he is required to have all births and deaths that may occur within his jurisdiction recorded by himself or the village *panchayat*; these are reported on parade days at the police stations and outposts, which are the registering centres. The statistics thus obtained are compiled by the police, and submitted monthly to the Civil Surgeon, who prepares returns for the whole district for inclusion in the annual report of the Sanitary Commissioner. The statistics are checked from time to time by Inspectors and Sub-Inspectors of Vaccination, and the *chaukidars* punished, if necessary, for neglecting to report properly. In the towns, the higher level of intelligence and the fear of legal penalties tend to make registration and the classification of diseases more accurate than in the rural tracts. In the latter the reporting *chaukidar* is generally illiterate, and vital registration is less correct, the chief defects being the registration of still births and the omission of actual births in outlying villages and among the lower castes.

187. In addition to the periodical checks above mentioned, a special inquiry was held between August 1906 and July 1909 in order to ascertain, by way of test experiment, how far births and deaths in rural areas are correctly registered. The area selected was a portion of thana Galsi in the district of Burdwan, containing a population of about 53,000 persons. This area was divided into three sections, each of which was under the charge of a medical officer. The reports of births and deaths as registered by the *chaukidars* were obtained every week. Inquiries were then made in the villages to verify their reports, and also to ascertain locally if any births and deaths remained unregistered. A special local inquiry was also made in each case of death, either recorded or unrecorded in the thana register, to determine its probable cause. At the commencement of each year's operation a rough census of the population under observation was taken with the object of checking the work done. During the three years over which the enquiry extended, the number of births and deaths that actually occurred was 4,670 and 6,910, respectively, as against 4,690 and 6,917 entered in the thana register. There was thus an excess of 20 births, which was due to the inclusion in the birth returns of 2 cases of abortion and 26 cases of still-births, to the double registration of 3 births, and to failure to report 11 actual births. In the case of deaths there was an excess of 7 only, due to the erroneous inclusion of 3 cases of abortion, 23 cases of still-birth and one death that occurred before the enquiry began, and to the omission of 20 deaths that actually occurred.

188. This enquiry shows that the vital statistics as at present collected and compiled are vitiated mainly by errors regarding still-births and, in a minor degree, by the omission of births and deaths, but that the net difference between the number of vital occurrences and the number registered is very small: the latter is in excess by 4 per mille in the case of births and one per mille in the case of deaths. The number of births that escaped registration was only 2 per mille of the total number, the corresponding ratio for deaths being 3 per mille. There was considerable variation in the manner of registering still-born infants. Altogether, 53 cases of still-birth were reported, of which 23 were recorded as deaths and 26 as births, while the remaining four were correctly recorded as still-births. The medical officers, however, found that there were 34 cases of still-births which were not recorded at all, 22 being males and 12 females. The effect of still-born infants being erroneously included in the birth returns is to produce an infinitesimal excess of males, for the proportion of males among still-births is always high:—it varies according to the figures given by Darwin from 135 to 150 males per 100 females[o] and in the 34 cases reported by the medical officers the ratio per 100 females was as high as 183. The proportion of still-births actually registered as births to the total number of births that occurred was, however, under 6 per mille, and it is doubtful therefore whether the excess of males due to this error is more than 4 per mille.

189. If the results of this enquiry could be regarded as typical, the vital statistics could be accepted with some confidence, but it may be taken for granted that, while the inquiry lasted, the *chaukidars* felt that they were on their trial and took trouble to report every birth or death of which they

[o] Descent of Man, 2nd edition, p. 243.

had cognizance. Further, the testing conducted annually by the Inspectors and Sub-Inspectors of Vaccination shows that the reporting is still incomplete in some districts. The maximum in the case of deaths (21 per cent.) was recorded by Shahabad* in 1904 and in the case of births (12) by Rajshahi in 1903; the worst district record in a single year for births and deaths was 9 per cent. Such bad results, however, are exceptional. The returns (given in the margin) showing the percentage of unreported births and deaths in Bengal that were detected by the vaccination staff (out of several millions inquired into) are proof that there has been a gradual improvement in accuracy, especially since 1905, after which the Eastern Bengal districts are excluded as they were transferred in that year to the newly formed province of Eastern Bengal and Assam. In these districts the same standard of efficiency has not been reached as elsewhere, and it is noticeable that the census results do not correspond with those afforded by the vital statistics so closely as in Bihar and Orissa and other districts of Bengal.

YEAR.	PERCENTAGE OF OMISSIONS.	
	Births.	Deaths.
1901	1·23	0·99
1902	0·85	0·66
1903	0·80	0·62
1904	0·91	0·81
1905	0·87	0·72
1906	0·66	0·61
1907	0·68	0·57
1908	0·42	0·38
1909	0·28	0·30
1910	0·42	0·29

On the whole, the number of omissions is comparatively small, and the general conclusion seems to be (1) that though the vital statistics of the whole Province approximate to the truth (in a major degree for Bihar and Orissa and in a minor degree for Bengal), those for individual districts, especially in Eastern Bengal, are not always reliable; and (2) that the tendency to omission is greater in the case of births than deaths.

CAUSES OF DEATH.

190. The returns of deaths under different heads are notoriously inaccurate. The reported number of deaths from fever is invariably in excess of the actual number owing to the fact that the *chaukidars*, who are primarily responsible for their registration, group under this head nearly all the deaths which are not due to the well known diseases of plague, cholera or small-pox. Even deaths from cholera are often returned as deaths from fever: a year in which there is a cholera epidemic in a district will usually show a large rise in fever mortality, because the *chaukidar* goes on returning cholera deaths as due to fever until the epidemic is so bad that he is forced to recognize it. Detailed investigation of the returns yields the most extraordinary results. In one district, for instance, out of a total of 83 deaths registered as due to dysentery or diarrhœa, only 32 were actually found to be due to those diseases. Fever accounted for 23, cholera for 18, teething for 2; one was really a case of still-birth and another of death during labour; the remainder were caused by old age, obstruction of the spleen, phthisis, want of milk and actually snake-bite. Other instances of the extraordinary nature of the diagnosis have already been given. The explanation is that the *chaukidar* is ignorant and careless, that he rarely sees the corpse, and that in any case he nearly always takes the word of the relatives as to the cause of death: the latter are generally equally ignorant, and sometimes they deliberately conceal the actual cause.

VITAL STATISTICS AND ESTIMATES OF BIRTH AND DEATH RATES.

191. Taking the figures given in the marginal table above as typical of each Province, and making necessary allowance for omissions, we get the results given in Subsidiary Table III A. In this table the effects of migration have had to be ignored, as there are no statistics to show how many were immigrants and how many were emigrants in each year. They do not, however, affect the results very materially, for in Bengal, Bihar and Orissa, as a whole, the emigrants exceeded the immigrants by only 1 per mille in 1901 and by 3 per mille in 1911: in Bengal the excess of immigrants over emigrants is now 6 per mille more than in 1901, while in Bihar and Orissa the excess of emigrants over immigrants is greater by 13 per mille. From this subsidiary table it will be seen that the average annual birth rate and death rate during 1901-10 are 37·9 and 32·9 in Bengal, 42·1 and 36·1 in Bihar and Orissa, and 39·7 and 34·3 respectively in Bengal,

* The actual number was 104 out of 484 deaths.

Bihar and Orissa as a whole. Mr. G. F. Hardy, F.I.A., F.S.S., the Actuary who was retained by the Census Commissioners of 1881, 1891 and 1901 to deal with the age statistics of those years, estimated the birth and death rates for 1891-1901 in Bengal, Bihar and Orissa as a whole at a little over 43·9 and 38·9 per mille, respectively.

192. As regards the net increase due to vital occurrences, the subsidiary table shows the actual annual excess of births over deaths to be 5·3 per mille in Bengal, Bihar and Orissa as a whole, and this may be taken as a fair approximation, for as shown above, the net excess of emigrants over immigrants in the two Provinces taken together was very small. There is, however, a considerable difference if the two Provinces are considered separately. The balance of migration is against Bihar and Orissa, the emigrants from which add to the death roll of other countries and thus lower the death rate of their own Province: they outnumbered the immigrants by 36 per mille in 1901 and by 49 per mille in 1911. In Bengal, on the other hand, there is a gain from migration, for the number of immigrants is far greater than that of emigrants. It contains a large floating population from Bihar and the United Provinces, who add to its mortality returns, but very little to the number of births, as the immigrants live mostly without their families. These immigrants outnumbered the emigrants by 26 per mille in 1901 and by 32 per mille in 1911. Thus, as compared with the birth rate, the real death rate is lower in Bengal, but higher in Bihar and Orissa, than would appear from the returns of vital occurrences. The average rate of increase, therefore, may be taken as a little above 5 per mille in Bengal and a little below 6 per mille in Bihar and Orissa. The rate for the two Provinces together (5·4 per mille) corresponds closely with Mr. Hardy's estimate of 5 per mille as the actual rate during the previous decade.

193. It also appears on a detailed examination of the vital statistics that each Province would have shown a higher rate (about 7 per mille) as the average annual rate of increase, had there been no lean years like 1905-08, when the rates of increase were abnormally low. Mr. Hardy's estimate of 7 per mille as the normal rate of increase in Bengal, Bihar and Orissa as a whole (deduced from the estimated actual rates of 8·0, 7·7 and 5·0 per mille in 1872-81, 1881-91 and 1891-1901, respectively) is therefore corroborated by the vital statistics for 1901-10, during which the registration of vital occurrence appears to have reached a higher standard of accuracy. The table in the margin shows how much more closely the rates according to the vital statistics now correspond to the estimated rates of births and deaths, the difference being reduced from 16 and 13 per mille in 1901 to only 4·2 and 4·6 per mille in 1911.

	1891-1901.			1901-1911.		
	Births	Deaths.	Increase.	Births.	Deaths.	Increase.
Rates according to Mr. Hardy's estimate for the previous decade ...	51·8	44·8	7·0	43·9	38·9	5·0
Rates according to vital statistics ...	35·8	31·8	4·0	39·7	34·3	5·4
Difference ...	-16·0	-13·0	-3·0	-4·2	-4·6	-4

WEST BENGAL.

BURDWAN.

194. The district of Burdwan consists of two distinct portions, the eastern half being a low-lying alluvial tract subject to inundations from the Damodar, Ajay, Khari, Kunur and Bhagirathi rivers. The effects of floods are accentuated by the obstruction of drainage caused by dams and weirs erected across rivers and creeks for the purposes of irrigation Large tracts are consequently water-logged for a considerable portion of the year, especially in the south-east of the Sadar subdivision. The western portion of the district, which corresponds to the Asansol subdivision, consists of rolling uplands, with a dry laterite soil, and is far healthier. Between 1872 and 1891 the population of the district decreased by 94,535, or nearly 6½ per cent. This diminution

was due to a virulent type of fever known as Burdwan fever,* which raged between 1862 and 1874, and, in 12 years, is estimated to have carried off three-quarters of a million people. After 1874 the health of the district 'improved,' but the after effects of the disease persisted for some time. A decided recovery was witnessed in the decade 1891-1901, and at the census taken in the latter year an advance of 10 per cent. was recorded.

BURDWAN.	Population, 1911.	PERCENTAGE OF VARIATION. 1901—1911.
DISTRICT TOTAL	**1,538,371**	**+ 0·37**
Sadar Subdivision	**663,101**	**- 2·81**
Burdwan	66,572	- 4·65
Sahibganj	83,964	+ 1·32
Khandaghosh	60,362	- 2·24
Raina	101,801	- 3·38
Satgachia	105,115	- 3·30
Jamalpur	69,853	- 4·24
Galsi	90,271	- 3·18
Ausgram	85,163	- 2·74
Asansol Subdivision	**388,582**	**+ 4·74**
Asansol	177,312	+ 5·61
Raniganj	175,783	+ 5·95
Kajora	35,487	- 4·56
Katwa Subdivision	**261,463**	**+ 3·32**
Kengaon	89,673	+ 5·14
Katwa	94,955	+ 2·43
Mangalkot	76,835	+ 2·33
Kalna Subdivision	**225,225**	**- 0·53**
Kalna	89,411	- 0·91
Purbasthali	73,250	- 3·13
Maureshwar	62,564	+ 3·29

195. During the next decade conditions were generally unfavourable, and the number of deaths exceeded the number of births by 20,000. There were repeated epidemics of cholera, those of 1907 and 1908 being especially virulent and accounting for over 20,000 deaths, while malaria continued to levy its annual toll. Some scarcity was felt in 1904, and, in the Katwa subdivision, in 1908; but owing to the demand for labour in the coal fields, mills, factories, etc., the poorer classes are able to get ample employment and are not so much affected by the failure of crops and the pinch of high prices as in purely agricultural districts. The cultivators have benefited by the higher price which they get for their crops, and labourers by the rise in wages: the rate of interest, it is reported, was formerly 30 to 75 per cent. but has fallen to 12 per cent. Sons of middle-class cultivators now think it beneath their dignity to work in the fields like their fathers. The supply of agricultural labourers is unequal to the demand in the sowing and harvesting seasons, and wages have consequently gone up.

196. The population has only increased by 5,655 or less than one-half per cent., and there would have been a decrease, had it not been for the influx of labourers attracted by the good wages obtainable in the coal-fields. The causes of this check to development are the ravages of disease and the greater exodus of persons born in the district, whose absence materially affected the returns. Their number has risen by 32,000 since 1901, but is still only about two-thirds of that returned for immigrants, who have increased by 21,000. The increase would have been greater had it not been for the dull state of the coal market, which resulted in the number of workers in coal mines falling from 39,383 in 1901 to 37,665 at the time of the census. Owing

BURDWAN.	1911.		1901.	
	Male	Female.	Male.	Female.
Actual population ...	770,324	768,047	764,842	767,874
Immigrants	92,390	87,200	84,208	74,139
Emigrants	57,978	68,552	43,537	50,877
Natural population ...	735,912	749,399	724,171	744,612

* As I shall have frequently occasion to refer to "the epidemic of Burdwan fever," it may be explained that this was an endemic fever which became generally epidemic, and that the name by no means implies that it was peculiar to Burdwan. It appears first to have attracted notice in the Jessore district about 1825, and began to affect the Nadia district about 1832; it came across the Bhagirathi or Hooghly river into the Hooghly district in 1857-59 and it affected other districts of the Burdwan Division at a later period. During the period of its greatest virulence, a number of medical officers made inquiries into its origin and character and into the condition of the affected tracts. The general consensus of opinion was that the disease was a malarial fever of an intensely aggravated type, attended by an unprecedented mortality. The causes most generally assigned were over-population and obstruction of drainage caused by the silting up of rivers; but it cannot be said that any completely satisfactory reason has been put forward, which accounts for the outbreak of the fever, its gradual spread from east to west, and its disappearance. The fever was called by the natives *jwar bokar* (literally, fever without sense) *i.e.*, fever with delirium, a term which in recent years has also been applied to cases of plague. [*See* Hooghly District Gazetteer, pp. 127-128.]

mainly to this cause, the Asansol subdivision, which grew by nearly 20 per cent. between 1891 and 1901, has now added only 4·7 per cent. to its population. Immigrants represent over one-fifth of the total population of the subdivision and nearly one-half of the total number of immigrants in the whole district. All the thanas in the subdivision have added to their population except Kaksa, which is an agricultural tract that suffers from endemic fever. The only other subdivision in which there has been any growth of population is Katwa, where it is independent of immigration. In the Sadar and Kalna subdivisions the loss is shared in by every thana except Manteswar and Sahebganj.

BIRBHUM.

197. Like Burdwan, the district of Birbhum was decadent until 1891 owing to the ravages of fever. The population decreased by 7 per cent. between 1872 and 1881, and in the succeeding decade increased by less than 1 per cent. In the latter year there was a falling off of nearly 4 per cent. in the Sadar subdivision, which suffered severely from fever, but an accretion of 10 per cent. in the Rampur Hat subdivision to the north. In the next 10 years conditions were generally favourable, and the district added 13 per cent. to its population, the two subdivisions sharing nearly equally in the increase.

BIRBHUM.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	***935,473***	***+ 3·68***	***+ 13·03***
Sadar Subdivision	***544,472***	***+ 1·59***	***+ 13·97***
Suri	138,806	- 0·87	+ 10·95
Dubrajpur	135,511	- 1·82	+ 15·58
Bolpur	119,176	+ 2·01	+ 17·28
Sakulipur	82,516	+ 6·14	+ 14·08
Labpur	68,463	+ 6·50	+ 11·55
Rampur Hat Subdivision ...	***391,001***	***+ 6·73***	***+ 11·68***
Rampur Hat	110,469	+ 7·45	+ 10·03
Mayureshwar	97,107	+ 3·48	+ 8·57
Nalhati	90,154	+ 7·94	+ 11·93
Murarai	93,271	+ 8·23	+ 17·18

198. Birbhum is almost entirely dependent on agriculture. The crops had a fair outturn up to 1905, but from 1906 to 1908 they were short, and there was some scarcity in 1908-09. The poorer middle classes, small cultivators and landless labourers were embarrassed by the high range of prices. The Santals and other semi-aboriginal tribes, who could not get sufficient employment locally, went further afield, but the good crops and revived prosperity of the next two years brought them back to their homes. A serious flood occurred in 1902, when heavy rain caused the Brahmani in thana Murarai, the Bansloi in Nalhati and the Mor in thana Suri to overflow their banks and inundate the surrounding country. Great loss was caused to cultivators in the four thanas of the Rampur Hat subdivision, and more especially Nalhati and Murarai. The health of the district was generally good, except in 1906-1908, when it suffered from a wave of fever and epidemics of cholera ; in these three years deaths exceeded births by 41,000.

199. In the decade as a whole, there was an excess of 18,689 births, and the census shows an increase of 33,193 or 3·68 per cent. The number of immigrants has risen very little and is nearly counterbalanced by the exodus of the native-born. The growth of population is shared in by all thanas except Suri and Dubrajpur in the west of the Sadar subdivision. Here the soil is not so fertile as it is to the east, and heavy mortality was caused by cholera and small-pox in 1908. Owing to the loss in these two thanas, the average increase in the Sadar subdivision is only 1·59 per cent., whereas it is 6·73 per cent. in the Rampur Hat subdivision. The latter is one of the most progressive tracts in West Bengal, and is now more populous by 29 per cent. than it was in 1881, whereas the population of

BIRBHUM	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ...	463,838	471,635	444,689	457,591
Immigrants	28,897	35,182	27,194	33,412
Emigrants	21,681	28,521	20,876	26,485
Natural population ...	456,622	464,974	438,371	450,664

the Sadar subdivision has increased by only 11 per cent. As in 1901, the Murarai thana in the extreme north, which benefits from the immigration of Santals, has grown most rapidly, but it is closely followed by Rampur Hat and Nalhati, which are prosperous agricultural areas traversed by the railway.

BANKURA.

200. The district of Bankura has had a sustained growth of population since the first census was taken in 1872. The census of 1881 showed an increase of 7·6 per cent. and, in spite of the prevalence of disease in the east of the district, there was a further advance of 2·7 per cent. in 1891. The succeeding decade was, on the whole, a healthy one, the endemic or epidemic fever known as Burdwan fever having died out. The census of 1901 showed a total population of 1,116,411, representing an increase of 4·37 per cent.

BANKURA.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901
DISTRICT TOTAL ...	*1,138,670*	*+ 1·99*	*+ 4·37*
Sadar Subdivision ...	*746,964*	*+ 4·90*	*+ 2·85*
Bankura	158,373	+ 6·39	+ 5·88
Onda	120,591	- 1·89	+ 0·44
Gangajalghati	189,440	+ 2·17	+ 4·06
Khatra	129,996	+ 12·73	+ 3·00
Raipur	110,078	+ 8·32	- 0·79
Simlapal	38,486	+ 0·99	+ 9·37
Vishnupur Subdivision	*391,706*	*- 3.13*	*+ 7·17*
Vishnupur	139,735	- 2·20	+ 5·07
Sonamukhi	82,042	- 0·54	+ 9·27
Kotalpur	91,989	- 4·82	+ 5·91
Indas	77,940	- 5·35	+ 10·39

201. On the whole, conditions between 1901 and 1910 were favourable to a further expansion. The railway was introduced in the early part of the decade, and the public health was good during the first 7 years, the number of births exceeding the registered deaths by 41,000. At the close of 1907, however, the crops failed over a large area, the parts most affected being thanas Raipur, Onda and Khatra. Distress was felt from January till September 1908, and relief was given to an aggregate of 414,798 persons. The effect of the scarcity is seen in the vital statistics. In 1908 the births fell by nearly 8,000 and deaths increased by 4,000, but in the next 2 years there was a rapid rise in the birth rate, which culminated in 1910, when the number of births exceeded that recorded in 1908 by 46 per cent.

202. In the whole decade the excess of births over deaths was 58,178, while the increase of population now brought to light is only 22,259 or 2 per cent. The difference is due, almost entirely, to the increasing volume of emigration, to which the construction of the railway has given an impetus. It has long been the practice for the aboriginal tribes or semi-Hinduized castes of aboriginal descent, such as Santals, Bauris and Bagdis, to emigrate to other districts in the cold weather when their harvests are reaped, and to return at the beginning of the hot weather or later. Many, moreover, work in the coal mines in the Burdwan and Manbhum districts, as well as in the mills along the river Hooghly. This exodus has been facilitated by the railway, and since 1901 the number of those who have left their homes for employment elsewhere has increased by 20 per cent., the actual number of male emigrants rising by 20,000.

BANKURA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ...	562,585	576,085	549,484	566,927
Immigrants	13,191	32,991	11,740	17,794
Emigrants	95,757	79,627	75,269	71,249
Natural population ...	645,151	622,721	613 013	620,382

203. While the Sadar subdivision has added 4·9 per cent. to its population, the Vishnupur subdivision has suffered a loss of 3·13 per cent. At first sight, this result appears somewhat surprising, for the headquarters subdivision is an undulating tract of rocky, often barren soil, whereas the Vishnupur subdivision is a fertile alluvial plain. On the other hand, the undulating uplands are well-drained and the people suffer little from malaria, which is prevalent in the lowlands of the Vishnupur subdivision. The difference in climatic conditions has resulted in a very different rate of progress, for between 1872 and 1891 the population of the headquarters subdivision

increased by 24 per cent., while that of the Vishnupur subdivision declined by more than 8 per cent. Only once, viz., in 1901, has the latter had any increase of population since the census operations began. The inhabitants of the headquarters subdivision are, moreover, to a large extent aboriginals or semi-aboriginals, who do not suffer from disease as much as the better castes of Hindus, owing probably to the healthier lives they live, to their residence in the dry uplands, and to their more nourishing diet. They are prolific races, in whose villages swarms of children may be seen. Had it not been for the fact that they supply most of the emigrants from the district, the increase of population would have been even greater. In spite of emigration, the Khatra and Raipur thanas, where they are most numerous, have developed most rapidly.

MIDNAPORE.

204. The population of Midnapore declined by 1 per cent. between 1872 and 1881 owing to the ravages of Burdwan fever, which first appeared in 1871 and raged till 1877, causing an estimated mortality of a quarter of a million. During the next twenty years there was a steady growth, representing 4·6 per cent. in 1891 and 5·99 per cent. in 1901. The increase was rapid along the sea-coast and the estuary of the Hooghly. There was a fair natural development in the healthy, but barren and sparsely inhabited up-lands in the west of the district. Stagnation or decline prevailed in the ill-drained depression that intervenes between these two extremes.

MIDNAPORE.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL ...	**2,821,201**	**+ 1·15**	**+ 5·99**
Sadar Subdivision ...	**1,300,080**	**+ 1·75**	...
Midnapore	78,171	- 0·19	+ 9·07
Kharagpur	132,875	+ 19·69	+ 10·51
Jhargaon	80,351	+ 5·59	+ 7·10
Binpur	112,270	+ 6·94	+ 2·41
Salbani	59,578	- 2·09	+ 9·70
Debra	66,799	- 1·58	- 1·26
Sabang	147,590	- 4·04	...
Narayangarh	123,328	+ 4·13	+ 0·85
Garhbeta	126,353	- 5·08	+ 5·29
Keshpur	83,357	- 3·72	+ 3·33
Dantan	126,236	+ 2·18	+ 2·28
Gopiballabhpur	163,172	+ 0·01	+ 7·36
Ghatal Subdivision ...	**301,396**	**- 7·26**	**- 0·90**
Ghatal	83,341	- 9·49	- 0·48
Daspur	123,157	- 5·75	- 2·37
Chandrakona	94,898	- 7·19	+ 0·68
Tamluk Subdivision	**601,502**	**+ 3·13**	...
Tamluk	151,872	+ 2·67	...
Maslandpur	102,323	+ 5·53	+ 8·27
Sutahata	74,246	+ 5·10	+ 11·30
Panskura	143,991	+ 0·44	+ 4·07
Nandigram	129,070	+ 3·80	+ 12·57
Contai Subdivision	**618,223**	**+ 2·50**	**+ 10·59**
Khajri	57,366	- 0·35	+ 14·74
Contai	177,708	+ 2·93	...
Ramnagar	76,007	+ 1·32	...
Bhagwanpur	126,731	+ 4·97	+ 8·59
Egra	78,329	+ 0·57	+ 5·49
Pataspur	102,082	+ 2·81	+ 6·16

205. Conditions during the decade 1901-1910 were not favourable to a further advance. In 5 years only did the death rate exceed the birth rate, the only really healthy years being 1903, 1904 and 1908-10. This was the result mainly of epidemics of cholera, which were particularly widespread in 1901, 1902, 1906 and 1907; in 1902 there was also an epidemic of smallpox, which was responsible for 17,000 deaths. Fever, the most important factor in the health of the district, was rife in the water-logged areas, and the outturn of the crops was poor for several years.

206. The census discloses an increase of 32,087 or 1·15 per cent. The volume both of emigration and immigration has increased owing to the extension of the railway, the main line of the Bengal-Nagpur Railway being opened in 1901 and the Jherria extension in 1903. The emigrants, however, outnumber the immigrants by 96,000. All the subdivisions have a growth of population except Ghatal, which was also the only subdivison that sustained a loss in 1901. The reasons for its decadence are not far to seek. It is an alluvial tract with a fertile soil, but it is liable to floods, and the people suffer from constant malaria and periodical epidemics of cholera. The inundations affect the public health in two ways. On the one hand, the flood water scours out holes and ditches, and carries off surface filth and rotting vegetation, depositing a protective layer of silt. On the other hand, the stagnant water, slowly drying up, affords a congenial breeding ground for malaria-bearing mosquitoes. There are thus two divergent effects. At first, the flood water cleanses the country and cholera disappears. After the floods are over,

there are large collections of stagnant water, and fever becomes rife. In bad seasons, moreover, the people migrate freely to Calcutta and the Sundarbans, and such a bad season occurred in 1909, when no less than 548 villages were flooded and the crops over 85,000 *bighas* destroyed by unusually high floods.

MIDNAPORE.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,410,714	1,410,487	1,390,233	1,394,881
Immigrants	41,089	34,536	27,754	22,107
Emigrants	92,188	79,584	72,129	62,116
Natural population	1,461,813	1,455,535	1,434,608	1,433,890

207. The Contai and Tamluk subdivisions, which lie along the sea-coast and the estuary of the Hooghly, are comparatively free from malaria, and both have increased at about the same rate. Little progress has been made in the Midnapore subdivision, where six thanas have added to their population but as many more have to record a loss. In one thana, viz., Kharagpur, the advance is very nearly 20 per cent., but this is almost entirely due to the expansion of Kharagpur town, an important railway centre containing the workshops of the Bengal-Nagpur Railway. In other thanas the growth of population must be ascribed to their natural advantages. They all lie to the west where the lateritic uplands descend to meet the plains. The climate is dry, there is a good system of natural drainage, and a large proportion of the population consists of hardy aboriginals. The thanas with a decrease consist of two blocks, viz., (1) an upland block to the north consisting of the Garbheta, Salbani and Keshpur thanas, all lying to the north of the Kasai, and (2) an alluvial block to the south-east consisting of Midnapore, which lies mainly to the south of the Kasai, and Debra and Sabang, which are entirely to the south of that river. In the former of these areas cholera and fever have been prevalent and the harvests poor: the latter block is a low-lying depression mostly under canal irrigation, where the country is fertile, but swampy and malarious.

HOOGHLY.

208. Hooghly, like other districts of the Burdwan Division, suffered severely from the ravages of Burdwan fever between 1857 and 1877. Though the fever persisted for 20 years, it must not be supposed that it afflicted all parts of the district throughout that period. It spread from place to place, its average duration in any one locality being from three to seven years. The mortality was enormous, being estimated by various observers at from one-third of the population up to nine-tenths in certain very severely affected places. In 1881 the Census Superintendent estimated the loss of population due to it at no less than 650,000; for apart from actual mortality, the fever reduced the vitality of the survivors, thus diminishing the birth-rate, and also forced a number of the inhabitants to leave the district for healthier areas. "It is true," he remarked, "that this terrible epidemic did not claim so many victims in the decade which has elapsed since the census of 1872 as in that which preceded it, but the ravages of the disease have not yet been repaired, the ruined villages have not yet been rebuilt, jungle still flourishes where populous hamlets once stood, and while many of those who fled before the fever have not returned, the impaired powers of the survivors have not sufficed to fill the smiling land with a new population." The census of 1881 showed a decline of 13 per cent.; but the disappearance of the fever was followed by a recovery, the result being an increase of 6 per cent. in 1891. In the next decade there was but little advance owing to the drain caused by deaths from fever; the increase of 1·4 per cent. recorded in 1901 was mainly due to a rise in the number of immigrants.

HOOGHLY.	Population, 1911.	PERCENTAGE OF VARIATION. 1901—1911.
DISTRICT TOTAL	***1,090,097***	***+ 3·91***
Sadar Subdivision	***312,042***	***+ 1·16***
Balagarh	45,553	+ 2·56
Pandua	71,040	+ 3·47
Dhaniakhali	97,226	+ 1·71
Polba	43,451	- 0·03
Hooghly	54,772	- 2·80
Serampore Subdivision ...	***461,296***	***+ 11·65***
Haripal	103,704	+ 7·01
Jangipara	58,561	+ 1·50
Serampore	112,154	+ 19·82
Singur	75,083	+ 14·60
Chanditola	111,794	+ 12·42
Arambagh Subdivision ...	***316,759***	***- 3·23***
Arambagh	119,405	- 1·60
Goghat	98,170	- 8·07
Khanakul	99,184	- 0·07

209. Since 1901, there has been a gain of 41,056 or 3·91 per cent., part of which is due to the influx of immigrants rather than to natural growth. There was an excess of recorded deaths over births during the decade amounting to nearly 36,000, and the birth-rate surpassed the death-rate in only 3 years, viz., 1904. 1909 and 1910. Fever is rife and is a natural consequence of the natural configuration of the country. It is for the greater part a semi-aquatic rice plain traversed by large and small rivers, with low-lying depressions between them ; many of the rivers have more or less silted up, and no longer drain the land, which remains swampy and water-logged. Malarial fever," writes Lieut. Col. D. G. Crawford, I. M. S., formerly Civil Surgeon of Hooghly, " is still the prevailing disease of the Hooghly district, though fortunately it is no longer the scourge that it was 50 to 30 years ago. Something has been done since that time to alleviate its ravages, particularly the flushing of some of the 'dead' rivers of the district since the construction of the Dankuni drainage channel in 1873 and the opening of the Eden canal in 1881. Still, however, the physical conditions of the district remain much as they were half a century ago ; and thus they must always remain, for no human agency can alter them. The district is little above sea-level, it has a heavy rainfall, it is traversed by numerous 'dead' or silting up rivers, and it is chiefly devoted to the growth of rice, a crop which requires the ground to be a swamp during several months of the year for its cultivation. These conditions necessarily lead to its being water-logged in the rains. Practically, every house built in the district necessitates the excavation of a small tank or pit (*doba*) to get the earth, which forms a plinth, to raise the house above flood-level. Efficient drainage is an impossibility, as there is not sufficient fall. The tanks which abound in the towns—in the Hooghly-Chinsura Municipality alone there are 700—the drains, with their inefficient fall, forming chains of stagnant pools instead of running streams, and the vast expanses of rice cultivation, all supply ample breeding grounds for the mosquito by which malarial fever is spread."*

The population is still nearly 30,000 less than it was in 1872, and if, as observed by Mr. Gait in the Census Report of 1901, it is very doubtful whether the district will ever fully recover its losses until the drainage problem is solved, that consummation appears a very remote contingency.

210. Emigration is fostered not merely by the industrial, but also by the agricultural demands of the district. The mills, factories and brick-fields along the Hooghly create a constant, and the needs of cultivation a periodic, demand for labour, which is supplied by outside districts, mainly the districts of Orissa, Chota Nagpur and Bihar. The larger industries have developed steadily—the average daily number of operatives in registered factories rose by 10,000 between 1901 and 1911—and the increasing wealth of the cultivators enables them to employ outside labour instead of tilling their fields themselves. Agricultural labourers from the Sonthal Parganas and Chota Nagpur have consequently settled in the villages, and some have become cultivators on their own account. The number of persons from outside districts who were enumerated in Hooghly is now 46,000 more than in 1901. Part of the increase is due to an extra number of coolies being at work on the construction of the Hooghly-Katwa Railway at the time of the census, but even so the number exceeds that returned in 1891 by 86 per cent. On the other hand, the number of those born in the district who were enumerated elsewhere has increased by 27,000 since 1901—the number in Howrah alone has risen by 13,000. So much, however, of the migration, whether inwards or outwards, is of a temporary character, that its effect on the population cannot be gauged with any precision. A large proportion of the immigrants only come for the agricultural season, or work in the mills and brickfields during the cold weather, after which they return to their homes. The emigrants, again, consist, to a large extent, of clerks and others who have to work in Calcutta

HOOGHLY.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	555,823	534,274	528,179	520,862
Immigrants	108,087	78,222	81,823	57,591
Emigrants	88,423	61,085	70,290	52,551
Natural population	536,159	517,137	516,646	515,522

* Hooghly District Gazetteer.

and elsewhere, but keep up homes in this district, to which they return at intervals. They number 37,000 less than the immigrants enumerated in the district, but the figure returned for them is still 7,000 less than in 1891.

211. The only progressive part of the district is the Serampore subdivision, where the increase (48,000) exceeds the total gain of the district. This subdivision has grown steadily since 1881, and owes its prosperity to its marshes having been reclaimed by the Dankuni and Rajapur drainage schemes, to the establishment of jute mills along the river bank, to its connection with the main system of the East Indian Railway by a branch from Tarakeswar to Seoraphuli, and to the opening of the Howrah-Shiakhala Railway in 1897, which has linked it up with Howrah. The population of the headquarters subdivision is stationary. The two inland thanas of Pandua and Dhaniakhali have a small increase, which is ascribed to the influx of agricultural labourers: the other three thanas have been declining since 1891. The Arambagh subdivision has sustained a loss of population in every thana, the decrease being greatest in Goghat, a tract difficult of access, which lost population between 1891 and 1901. It has suffered from malaria, which is also common in the other two thanas, where the land is swept by the annual floods of the Damodar and Dwarakeswar. Their drainage is defective and the flood water remains stagnant in depressions choked with weeds, which foster the propagation of fever-bearing mosquitoes.

HOWRAH.

212. The history of the last 40 years in Howrah has been one of continued progress. It suffered severely from Burdwan fever, which, it was estimated, had caused 50,000 deaths by 1881, but the census of that year brought to light an increase of 6 per cent., the result partly of defective enumeration in 1872 and partly of immigration. The proportional growth rose to 13 per cent. at the next census, and another increase of 11·38 per cent. was recorded in 1901. The rate of progress has been slower during the last decade, the census just concluded bringing to light a gain of 92,988 persons or 10·93 per cent.

HOWRAH.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	***943,502***	***+ 10·93***	***+ 11·38***
Sadar Subdivision ...	***492,225***	***+ 14·14***	***+ 17·73***
Bally	22,394	+ 69·28	...
Liluah	22,635	+ 4·11	...
Howrah, Sibpur and Golabari ...	179,389	+ 13·83	+ 35·15
Dumjor	164,063	+ 10·79	...
Jagatballabhpur	99,724	+ 7·73	+ 5·60
Uluberia Subdivision ...	***451,277***	***+ 7·64***	***+ 5·52***
Amta	177,192	+ 6·15	+ 2·39
Bagnan	77,602	+ 7·13	+ 5·24
Uluberia	107,971	+ 13·87	+ 8·78
Syampur	88,512	+ 7·76	+ 8·66

213. Agricultural conditions were, on the whole, favourable, and the number of births exceeded the deaths by a little under 17,000. These criteria, however, are not of such importance as in other districts. No district in West Bengal is so little dependent on the outturn of its crops—the special industrial census shows that one-seventh of the total male population is at work in manufacturing or industrial concerns employing 20 persons or more. The ratio of births is, moreover, unusually small, because a large proportion of the people consist of male immigrants and of persons living in the towns of Howrah and Bally. The former leave their wives at home and those townspeople who have wives with them, send them home to their villages before confinement: in many cases these villages lie in other districts, and the births are consequently excluded from the returns for Howrah. Immigration has been stimulated during the past decade by the development of commerce and manufactures. The dull state of the jute trade led to a partial stoppage of work in some jute mills, but at the time of the census they had a labour force of over 44,000. The list of other concerns is a long one, *e.g.*, cotton mills, jute presses, iron foundries, machinery and engineering works, brick-fields, railway workshops, oil mills, flour mills, etc. The extension of the tramway from Calcutta and the improvement of the service of ferry steamers between the two cities have further induced persons who formerly resided in Calcutta to cross to this side of the river: the number of persons born in Calcutta but enumerated in Howrah has increased by over 7,000 since 1901. The result of these combined influences is an increase in the number of immigrants by 45,000. This, however, only accounts for less than a half of the total gain, and there is a loss of 17,000

by emigration to be set off against it. A considerable part of the increase must therefore be attributed to natural growth. The immigrants congregate in the neighbourhood of the mills along the river bank from Bally on the north to Uluberia on the south. The inland area in the north of the district has been opened up by the Howrah-Amta and Howrah-Shiakhala Light Railways, while the execution of the Howrah, Barajol and Rajapur drainage schemes has done much to reduce water-logged and uncultivable areas in thanas Dumjor and Jagatballabhpur. The south of the Uluberia subdivision is not so water-logged and is generally healthier. The city of Howrah accounts for nearly one-fourth of the total increase, but all parts of the district are progressive. The rate of growth is twice as rapid in the Sadar subdivision, which is the centre of industrial activity, as in the Uluberia subdivision, where the population is mainly agricultural. Of the rural thanas, Uluberia has the highest ratio of increase (13·9 per cent.), while the percentage is least in Amta, where much of the land is water-logged and fever is prevalent.

HOWRAH.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	493,641	444,861	439 525	410,989
Immigrants	125,215	64,769	98,641	45,979
Emigrants	23,435	16,957	14,884	8,512
Natural population	396,861	397,049	355,768	373,522

CENTRAL BENGAL.

24-PARGANAS.

214. Though the 24-Parganas, as a whole, has steadily added to its population since 1872, the growth has not been uniform. In 1881 there was a net increase of 7 per cent., but the Barrackpore subdivision declined by 9 per cent. owing to the prevalence of malaria. At the next census a proportional growth of 11·9 per cent. was registered, and this was followed by another increase of 9·9 per cent. In the last of these decades (1891-1901) the riparian population grew by 12 per cent. owing to the development of the industrial towns along the Hooghly, but the rate of growth was twice as fast in the southern thanas, where the progress of reclamation in the Sundarbans attracted numerous settlers. On the other hand, the central and northern thanas remained stationary or lost population.

24-PARGANAS.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911	1891—1901.
DISTRICT TOTAL	**2,434,104**	**+ 17·12**	**+ 9·89**
Sadar Subdivision ...	**756,348**	**+ 12·67**	**+ 11·83**
Barnipur	100,309	+ 10·49	+ 4·63
Matia	92,637	+ 15·49	+ 41·87
Jaynagar	123,866	+ 14·72	+ 18·62
Bhangar	87,470	+ 14·36	+ 12·58
Behala	50,482	+ 0·93	...
Tollygunge	66,487	+ 19·42	...
Sonarpur	47,355	+ 8·17	...
Vishnupur	91,768	+ 9·26	+ 6·21
Budge Budge	95,974	+ 16·34	+ 13·93
Diamond Harbour Subdivision.	**515,725**	**+ 11·93**	**+ 14·36**
Magra Hat	140,019	+ 7·36	+ 4·40
Falta	57,301	+ 9·35	+ 5·20
Diamond Harbour	78,535	+ 11·11	+ 11·44
Kulpi	152,224	+ 12·68	+ 26·80
Mathurapur	87,646	+ 21·50	+ 23·02
Barrackpore Subdivision	**292,524**	**+ 41·69**	...
Baranagar	43,910	+ 13·62	+ 12·75
Barrackpore	72 259	+ 102·80	+ 28·12
Noapara	25,236	+ 48·59	- 7·48
Khardaha	22,465	- 13·66	+ 7·3
Dum-Dum	26,887	+ 13·98	...
Naihati	101,767	+ 55·18	...
Barasat Subdivision ...	**292,791**	**+ 10·84**	...
Habra	74,320	+ 5·21	- 0·90
Deganga	62,232	+ 12·72	1·02
Barasat	156,239	+ 12·96	...
Basirhat Subdivision ...	**429,476**	**+ 15·39**	**+ 7·22**
Baduria	141,940	+ 7·34	+ 4·56
Basirhat	98,720	+ 13·57	+ 10·66
Haroa	68,433	+ 11·18	- 1·96
Hasnabad	120,383	+ 31·61	+ 15·29
Suburbs of Calcutta ...	**147,240**	**+ 45·28**	**+ 15·82**
Cossipore-Chitpur	48,178	+ 18·23	+ 29·68
Manicktollah	53,767	+ 66·01	+ 15·01
Garden Reach	45,295	+ 60·56	+ 1·03

215. Conditions between 1901 and 1911 were in favour of a further growth of population. The public health was good, the births exceeding the deaths by 100,000. The outturn of the crops was well up to the average during the first half of the decade. In 1905 they were short owing to heavy but unevenly distributed rainfall, while the rainfall next year was deficient and the outturn was again poor. Consequently, in 1907 there was distress, to meet which agricultural loans and

other relief measures were necessary. On the other hand, there was a continued and increasing activity in manufacturing and industrial centres, which led to an addition of 50 registered factories (*i.e.*, factories employing 50 hands or more) and of 75,000 employés. Considerable progress was also made in the reclamation of the Sundarbans, where agricultural colonies are growing rapidly in spite of the absence of facilities of communication. The north of the district has no such drawbacks, for areas which were without railway communication are now served by the Barasat-Basirhat Light Railway, which was opened to traffic in 1905 and extended to Hasnabad in 1909; another line from Beliaghata Bridge to Patipukur was opened in 1910. The suburban traffic between Calcutta and stations in this district has also developed rapidly: in 1910 the number of season tickets issued to and from Sealdah was 31,766. The extension of the Calcutta Electric Tramway to Alipore, Tollygunge and Behala has assisted in the development of those places, while the Port Commissioners' steamer service has popularized the riverain municipalities in the Barrackpore subdivision. The result is that a growing number of clerks employed in Calcutta offices live outside the city and are daily passengers on the trains, trams or ferry steamers.

216. The total increase of population since 1901 is 355,745 or 17 per cent., nearly half of which may be ascribed to the increased number of immigrants, most of whom are attracted by the good wages offered in the mills along the Hooghly or for agricultural labour in the interior. Their number has risen by 176,000 since 1901, and they now constitute one-sixth of the total population. On the other hand, there has been a loss of 20,000 by emigration, and more than half of the increment of population must be attributed to natural growth. The Barrackpore subdivision has a phenomenal increase, representing 42 per cent., which is nearly entirely due to the influx of mill-hands—the proportion of males to females in the whole subdivision is 5 to 3. In none of the other subdivisions, whether industrial or agricultural, is the rate of growth under 10 per cent. There is no sign of a drain of the population to Calcutta; on the contrary, the development of suburban railways and river steamer services points to the fact that an increasing proportion of the workers in Calcutta prefer to have their homes outside the city.

24-PARGANAS.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,305,700	1,128,404	1,092,916	985,443
Immigrants	275,479	126,144	154,953	70,414
Emigrants	72,097	67,148	63,017	55,943
Natural population	1,102,318	1,069,408	1,000,980	970,972

217. A special inquiry, made by the Bengal Drainage Committee in 1906-07, showed that the district, as a whole, is not abnormally unhealthy nor specially malarious, though some portions of it return high rates of mortality. The noticeably malarious thanas were proved to be those of Dum-Dum, Khardaha, Barrackpore, Noapara, Naihati, Deganga and Habra; the least malarious areas are Bhangar, Matla, Diamond Harbour and Budge Budge. In the healthy thanas the rate of increase has varied from 11 to 17 per cent.; in the unhealthy thanas the natural loss of population by death or lowered vitality is counterbalanced by immigration. Five of the seven unhealthy thanas lie along the Hooghly in the Barrackpore subdivision, where mill-towns cluster closely together, and the effect of malaria is obscured by the shifting of population to industrial centres. Habra has an increase of only 5 per cent., a rate a little below that in the adjoining thana of Baduria. The two thanas last mentioned lie in the extreme north-east of the district, and have advanced at a relatively slow pace compared with the thanas immediately to the south of them, viz., Barasat, Deganga and Basirhat, which have all benefited by the opening of the light railway and have grown at a uniform rate of 13 per cent. There has been even more rapid progress in the Sundarbans thanas to the south and south-east, where cultivation is rapidly spreading. Hasnabad, which has also been opened up by the railway, has a gain of 32 per cent., and Mathurapur of 21 per cent.

218. The census of Calcutta forms the subject of a separate report, and has also already been referred to; its results will therefore be only briefly mentioned here.

CALCUTTA.

The increase in the population of the city since 1901 (48,271 or 5·69 per cent.) is entirely due to immigration, the excess of deaths over births in the decade being no less than 145,534. The large excess of deaths is due to the relative deficiency of females in the population, to the practice of married women being sent away to their native villages for their confinement, which results in births being entered in the returns for outside districts, to the considerable proportion of prostitutes among the female population, and last, but not least, to high infantile mortality. The number of those born in Calcutta represents only 29 per cent. of the total population, and the males outnumber the females by more than two to one. The number of immigrants has risen by 82,000 since 1901, but there has also been an increasing exodus of the native born, particularly to the suburban municipalities and to Howrah.

CALCUTTA.	Population, 1911.	PERCENTAGE OF VARIATION. 1901–1911.	1891–1901.
TOTAL	**896,067**	**+ 5·69**	**+ 24·25**
Shampukur Ward No. 1 ...	53,036	+ 13·14	+ 28·74
Kumartoli ,, 2 ...	33,073	+ 9·68	+ 13·30
Bartala ,, 3 ...	54,610	+ 5·75	+ 37·84
Sukea's Street ,, 4 ...	48,112	+ 14·46	+ 20·67
Jorabagan ,, 5 ...	52,114	+ 6·21	+ 25·24
Jorasanko ,, 6 ...	59,541	+ 12·37	+ 27·20
Barrabazar ,, 7 ...	30,495	- 3·42	+ 52·93
Kalutola ,, 8 ...	57,094	- 9·62	+ 24·39
Muchipara ,, 9 ...	63,362	- 1·17	+ 29·60
Bowbazar ,, 10 ...	25,014	- 7·33	+ 19·34
Paddapukur ,, 11 ...	29,966	+ 6·79	+ 35·16
Waterloo Street ,, 12 ...	6,284	- 0·29	+ 6·24
Fenwick Bazar ,, 13 ...	28,436	- 5·88	+ 10·02
Taltola ,, 14 ...	32,112	- 0·33	+ 10·37
Collinga ,, 15 ...	11,395	- 32·15	+ 26·95
Park Street ,, 16 ...	5,294	- 13·35	+ 32·35
Bam n Bustee (Victoria Terrace). ,, 17 ...	3,125	- 42·70	+ 18·62
Hastings ,, 18 ...	5,550	- 6·25	+ 22·80
Entally ,, 19 ...	45,072	+ 16·69	+ 13·97
Beniapukur ,, 20 ...	37,881	+ 34·32	+ 22·51
Ballygunge-Tollygunge ,, 21 ...	39,952	+ 46·84	+ 19·17
Bhowanipore ,, 22 ...	54,569	+ 9·93	+ 16·55
Alipore ,, 23 ...	19,749	+ 11·46	+ 19·68
Ekbalpore ,, 24 ...	21,869	+ 1·21	+ 40·86
Watgan] ,, 25 ...	43,506	+ 15·53	+ 41·31
Fort William	4,411	+ 32·46	- 7·90
Port	26,890	- 9·67	+ 11·95
Canals	3,265	- 26·58	+ 111·56

POPULATION.	1911. Male.	1911. Female.	1901. Male.	1901. Female.
Actual population	607,674	288,393	562,596	285,200
Immigrants	471,125	168,223	401,602	155,537
Emigrants	47,944	40,204	17,922	18,398
Natural population	184,493	160,374	178,916	148,061

NADIA.

219. The records of the district of Nadia for the last 50 years are a depressing chronicle of disease, either endemic or epidemic. Between 1857 and 1864 it was swept by the fever which was later known as Burdwan fever, but which at first was given the designation of Nadia fever. An inquiry into its causes and prevalence was made in 1862 by a special officer, Mr. J. Elliot, who described the disease as differing only in its intensity from the ordinary form of malarious fever, and attributed its virulence to villages being undrained, tanks uncleaned and overgrown with noxious weeds, and houses unventilated and surrounded by a tangled growth of jungle and rank vegetation. The mortality from this epidemic in some villages amounted to 60 per cent. of the population, and those who had escaped lingered on in a state of apathy and despair. The causes of the epidemic were further inquired into by a committee, usually referred to as the Epidemic Commission. The

NADIA.	Population, 1911.	PERCENTAGE OF VARIATION. 1901–1911.	1891–1901.
DISTRICT TOTAL	**1,617,846**	**- 2·44**	...
Sadar Subdivision ...	**368,614**	**+ 2·01**	**+ 3·53**
Kaliganj	53,122	+ 0·63	+ 12·70
Chapra	70,440	+ 7·58	+ 3·14
Nakasipara	56,163	- 0·96	+ 0·29
Kissengunge	35,752	- 5·12	+ 15·38
Krishnagar	108,196	+ 2·18	- 5·69
Hanskhali	44,941	+ 5·06	+ 14·49
Kushtia Subdivision ...	**456,885**	**- 4·25**	...
Kushtia	109,304	- 3·69	+ 7·19
Mirpur	138,663	- 3·66	+ 2·17
Daulatpur	81,760	- 2·73	- 0·12
Kumarkhali	127,158	- 6·27	...
Meherpur Subdivision ...	**329,563**	**- 5·33**	**+ 3·39**
Karimpur	94,350	- 6·53	- 5·38
Gangani	69,361	- 12·72	+ 8·52
Meherpur	73,582	- 0·39	+ 6·26
Tehatta	92,270	- 1·68	+ 7·72
Chuadanga Subdivision	**242,032**	**- 4·93**	**+ 3·74**
Chuadanga	51,865	- 9·25	+ 1·84
Damurhuda	69,042	- 0·66	+ 9·65
Alamdanga	87,012	- 4·62	+ 1·75
Jibannagar	34,113	- 7·06	+ 1·54
Ranaghat Subdivision ...	**220,752**	**+ 1·69**	...
Santipur	48,947	- 1·23	- 8·16
Ranaghat	80,438	+ 5·64	- 1·20
Chakdaha	91,367	- 0·01	...

Commission ascribed the epidemic to obstructed drainage and waterlogging of the soil, which had been intensified by a gradual filling-up of the *bils* by the deposit brought in from the rivers : this again had been supplemented by a gradual but continuous rise in the level of the river beds, thus causing a general derangement of levels so serious as to affect the natural drainage of the country. There was another serious epidemic which lasted from 1880 to 1885 ; this was inquired into in 1881-82 by the Nadia Fever Commission, which came to the conclusion that the epidemic was due to the insanitary condition of the villages and the silting up of its rivers. A further investigation was made by the Bengal Drainage Committee in 1906-07, whose conclusions may be stated at some length as throwing light on the unhealthiness of different parts of the district and the results of the present census.

220. "It is impossible to differentiate between the physical features of the different portions of the Nadia district. The whole area consists of an alluvial plain, which still receives a fair share of the Gangetic flood through the channels of the Jalangi, Matabhanga and Garai, but it is subject to general inundation in years of high flood only. Back-waters, minor streams and swamps intersect it in all directions. A low-lying tract of black clay soil, known as the Kalanter, stretches from the adjoining district of Murshidabad through the Kaliganj and Tehatta thanas on the west, but these areas do not present any special features from the point of view of health. Taking the average annual district death-rate from fever for 1901-05 (33·3 per mille) it may be said that those thanas which have a corresponding rate of 35 and over are specially unhealthy, and those with a rate of 30 and under are compatatively healthy. On the basis the most unhealthy thanas in Nadia are those of Gangni and Karimpur adjoining one another on the north-west, and Jibannagar, Kumarkhali and Naopara* in the east. The more healthy thanas comprise those of Krishnagar, Chapra and Meherpur, forming a little strip from north to south in the centre of the district, and Chakdaha in the extreme south. It is difficult to connect the figures showing the variations in population, in the three censuses of 1881, 1891 and 1901, with a theory of the progressive deterioration of health in thanas which now show the highest rates of mortality from fever, but the outbreaks of epidemic fever in the district between 1861 and 1864, and again between 1880 and 1886, have complicated the conclusions as to normal health which may be deduced from the various fluctuations. * * * Looking to the available evidence touching the medical history of the district, we arrive at the following conlcusions :—(*a*) the whole district is very unhealthy ; (*b*) similarly, the whole district is feverish ; (*c*) investigation upon a small scale has demonstrated the fact that some of the fever is probably due to Leishman-Donovan infection, but that the greater part is malarial ; (*d*) the most malarious thanas are those of Gangni, Karimpur, Jibannagar, Kumarkhali and Naopara ; (*e*) the least malarious areas are the Krishnagar, Chapra, Chakdaha and Meherpur thanas."

221. It may easily be imagined that in such a district as this the population is not likely to be progressive, for, apart from actual mortality, the constitution of the survivors is enfeebled, and their reproductive powers reduced by constant sickness. In 1881, it is true, a gain of 11 per cent. was recorded, but this was due merely to the deficiencies of the first enumeration, for from 1872 onwards it suffered terribly from fever and in one year lost no less than 40 per mille of its inhabitants. In the next decade there was a decrease of 1·2 per cent., and in 1901 there was an advance of only 1·5 per cent. Conditions between 1901 and 1910 were very unfavourable, the only healthy years being 1904, 1909 and 1910. Not only was fever generally prevalent, but there were constant epidemics of cholera from 1902 to 1908, the death-rate from cholera in these seven years averaging over 4 per mille and being as high as 7 per mille in 1907. There was, moreover, scarcity in 1908, when conditions approached those recognized as famine conditions, though it was not found necessary to declare famine under the Famine Code. The rainfall of 1905, though in excess of the normal, was badly distributed. Next year it was below the average and was again unequally distributed, while in 1907 it was still more deficient, there being an almost complete failure of the monsoon in September and October. The distress which ensued in 1908

* The Naopara thana is now called Murpur.

affected about 800 square miles, comprising the whole of the Meherpur subdivision and parts of the Sadar and Chuadanga subdivisions.

222. It may be added that the land appears to be of low fertility. "The soil," reports the Collector, "is universally a light sandy loam possessing but little fertilizing power. The light manuring which is applied is generally insufficient, and there is no doubt that the soil is getting less and less fertile. The average yield of crops is low, as will be seen from the fact that the average yield of winter rice is 805 lbs. per acre, and of autumn rice 835 lbs., while in Jessore it is 1,145 and 870, and in 24-Parganas it is 943 and 1,014 respectively. Another noticeable feature is that it is becoming necessary to allow the land to lie fallow for longer periods between croppings. During the five years from 1904-05 to 1908-09 only about 40 per cent. of the total cultivable area was actually cropped. The corresponding percentages in the two sister districts of Khulna and Jessore were about 74 and 89 respectively. It can safely be concluded that the soil in Nadia is not sufficiently fertile to enable the same percentage of the population to depend upon agriculture as in other districts."

223. A light railway connecting Ranaghat and Krishnagar was opened in 1898, and the Murshidabad branch of the Eastern Bengal State Railway in 1904-05. All the subdivisions, except Meherpur, are traversed by at least one branch of the Eastern Bengal State Railway. The Meherpur subdivision is cut off from the railway and has no water communications, for the only river which traverses it, the Bhairab, is silted up. Emigration is more active than in any other district of the division, and the number of persons leaving it in order to escape its unhealthy climate or for the sake of employment elsewhere, has risen by 11,500 since 1901. On the other hand, the number of immigrants has increased by 14,000, owing largely to the number of workmen required for the construction of the Lower Ganges Bridge at Damukdia. There is therefore a net gain from migration of 2,500.

224. Only five thanas, viz., Kaliganj, Chapra, Krishnagar and Hanskhali in the Sadar subdivision, and Ranaghat in the subdivision of the same name, have an increase of population. Krishnagar and Chapra are among the thanas classed by the Drainage Committee as among the least malarious in the district. The other two placed under this category, viz., Chakdaha and Meherpur, are so far fortunate, that they have not lost population but are stationary. The increase of the Ranaghat thana is due to the fact that it contains the town of Ranaghat, an important railway junction where labour is attracted by work on the line and in the brick-fields. The malarious thanas mentioned by the Drainage Committee have all lost population at a rate varying from 4 to 13 per cent. The Karimpur thana, in particular, shows a gradual decrease of about 6,000 during each of the two decades since 1891. Apart from malaria, the decrease is partly due to the fact that a portion of the north of the thana has been subject to diluvion by the river Padma, necessitating migration to the northern districts of Pabna and Rajshahi.

NADIA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	812,580	805,366	822,992	835,289
Immigrants	37,610	35,543	30,226	28,784
Emigrants	74,266	61,013	71,160	52,577
Natural population	849,236	830,736	863,925	859,082

Every subdivision has lost ground, but the loss in the Kushtia subdivision is greater than the figures indicate, because at the time of the census, a labour force of 3,447 was employed on the construction of the Lower Ganges Bridge, many of whom were immigrants from outside. The falling off in the Chuadanga and Meherpur subdivisions, on the other hand, must be discounted by the fact that, at the time when the census was taken, a number of labourers had migrated temporarily to the adjacent districts of Jessore, Khulna, etc., where labour is required for harvesting winter rice. The main crop of Nadia is *aus* or early rice, whereas in these districts winter rice predominates. When therefore the *aus* crop has been harvested, a large number of agricultural labourers find employment in reaping the winter crop of other districts.

MURSHIDABAD.

225. Owing to the ravages of Burdwan fever, the increase of population between 1872 and 1891 was very small (barely 3 per cent.), but a good recovery was made between 1891

and 1901, the proportional growth recorded in the latter year being 6·6 per cent. The increase was far from uniform, for in the low-lying water-logged tracts to the east it was only 3 per cent., whereas in the higher and better drained thanas, to the west it averaged 12·9 per cent. It was as high as 26 per cent. in Sagardighi and Nabagram, two sparsely populated thanas which attract immigrants from Birbhum and the Sonthal Parganas.

MURSHIDABAD.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**1,372,274**	+ **2·93**	+ **6·57**
Sadar Subdivision ...	**517,723**	- **0·65**	...
Gorabazar	35,074	- 13·41	...
Berhampore	26,143	+ 7·16	+ 3·75
Beldanga	118,938	+ 13·85	...
Noada	50,779	- 2·00	+ 6·23
Hariharpara	53,351	- 6·70	+ 3·66
Daulatbazar	35,543	4·82	- 4·25
Ranjnagar	85,553	- 3·81	+ 3·88
Domkal	112,342	- 3·54	+ 3·22
Lalbagh Subdivision ...	**195,128**	+ **2·73**	...
Bhagwangola	59,254	+ 5·64	+ 4·67
Shahanagar	21,902	- 13·43	- 18·94
Manullabazar	11,169	- 8·65	- 18·64
Asanpur	12,546	- 3·80	...
Sagardighi	40,424	+ 9·02	+ 26·01
Nobogram	49,833	+ 7·84	...
Jangipur Subdivision ...	**357,930**	+ **7·10**	+ **5·36**
Lalgola	66,517	+ 7·91	+ 1·53
Raghunathganj	73,559	+ 0·07	+ 10·74
Mirzapur	47,597	+ 4·01	+ 9·30
Suti	67,292	+ 4·44	- 6·40
Shamshirganj	102,965	+ 15·89	+ 11·93
Kandi Subdivision ...	**301,493**	+ **9·75**	...
Bharatpur	96,821	+ 0·63	...
Khargaon	68,549	+ 7·49	+ 13·92
Kandi	34,798	+ 9·00	+ 12·51
Barwan	74,378	+ 6·55	+ 9·78
Gokaran	26,947	+ 2·60	...

226. The decade 1901-1910 was one of chequered prosperity. In 1904 there were severe floods by which a considerable area was submerged. Next year there was an epidemic of cholera, which caused over 8,000 deaths, and this was followed by an epidemic of small-pox in 1907. In that year too the Lalitakuri embankment in the Lalbagh subdivision gave away, and there was a partial failure of the winter rice crop. Taking the average of the whole decade, however, crops were almost normal, and a demand for labour was created by the construction of three new railway lines, viz., the Ranaghat-Murshidabad branch of the Eastern Bengal State Railway, and the Barharwa-Azimganj-Katwa and Ondal-Sainthia branches of the East Indian Railway. The effect of these lines being opened is already apparent in an increasing exodus of labourers during the cold weather. The number of persons enumerated outside the district is now nearly 29,000 more than in 1901, and the district has lost nearly 25,000 by migration more than it has gained.

227. The west and east of the district are clearly differentiated by their physical configuration. The portion lying to the west of the Bhagirathi is a continuation of the Chota Nagpur plateau: its general level is higher than that of the rest of the district, the surface is undulating and the climate is comparatively dry. The eastern portion is a deltaic tract in which the land is still being gradually raised by the deposition of silt from the rivers which traverse it. Here, however, the action of nature has been interfered with by the marginal embankment along the left bank of the Bhagirathi, which prevents the inundation which would otherwise occur.

MURSHIDABAD.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	678,292	693,982	653,346	679,838
Immigrants	39,002	36,569	35,848	35,248
Emigrants	51,958	55,325	38,469	40,227
Natural population	691,248	712,738	655,967	684,817

228. The healthiness and unhealthiness of different parts are described as follows by the Drainage Committee :—"Taking the average annual district death-rate from fever (1901-1905) as 29·7 per mille, it may be concluded that the thanas which exhibit a rate of 25 and under are fairly healthy and those in excess of 40 noticeably the reverse. The healthy thanas of Khargaon, Kandi, Barwan, Gokaran and Bharatpur (constituting the Kandi subdivision) occupy a compact block in the south-west corner, while Raghunathganj and Mirzapur, along with Shamshirganj, Suti and Sagardighi (the rates for which are only slightly higher), lie to the north-west—all (except Raghunathganj) to the west of the Bhagirathi. The only other thanas on that side of the river are Nobogram, which is neither specially healthy nor the reverse, and Asanpur, which falls within the area of high rates. The feverish thanas group themselves in a well-defined tract, running north to south along the east bank of the Bhagirathi from Bhagwangola, through Manullabazar,

Shahanagar, Daulatbazar, Asanpur (west of the Bhagirathi), Sujaganj and Gorabazar. The average annual mortality from fever (1901—1905) in these thanas runs higher than any rates in either Jessore or Nadia, with the one exception of the Gangni thana in the latter, which exceeds the rate of Bhagwangola only. Hariharpara adjoining to the east is only slightly less unhealthy.. The eastern portion of the district, Gowas, Jalangi and Noada, stands midway in point of health between the two areas above defined. The variations in the total population recorded in the three censuses, especially during the decade 1891—1901 (the figures for the earlier ten years were affected by epidemic fever), corroborate these conclusions in a marked manner. The district as a whole showed a fair increase in population at the last census (+6·6 per cent.), but all the noticeably unhealthy thanas, with the exception of Bhagwangola and Hariharpara, and Gorabazar and Asanpur, which were practically stationary, showed a falling off, although it is to be remarked that the declining prosperity of the trade which used to centre around Murshidabad and Cossimbazar has also contributed to this result. The only other decrease, in Suti, which is a healthy area, was due to emigration, not illness. The facts then may be summarised as follows :—(*a*) Some portions of the district are extremely unhealthy, more so than Jessore or Nadia ; other portions are comparatively healthy. (*b*) Fever is similarly distributed. (*c*) Local investigation has shown that the fever is malarial, but the presence or absence of Leishman-Donovan infection is an open question. (*d*) The most malarious thanas are Bhagwangola, Manullabazar, Shahanagar, Daulatbazar, Sujaganj, Hariharpara, Asanpur and perhaps Jalangi. (*e*) The least malarious areas are comprised in the whole of the Kandi subdivision, and the thanas of Shamshirganj, Suti, Raghunathganj, Mirzapur and Sagardighi.'

The conclusions of the Drainage Committee are confirmed by the results of the present census. The Kandi subdivision has developed at the rate of 9·75 per cent., and all the other thanas mentioned as least malarious have an increase except Raghunathganj, where the population is stationary. Shamshirganj, which made the greatest progress between 1872 and 1901, is still growing rapidly and has added another 16 per cent. to its population. Of the eight thanas mentioned as most malarious, six are distinctly decadent.

JESSORE.

229. Jessore, like Nadia, is a land of moribund rivers and obstructed drainage, and its history during the last half century is also one of recurring epidemics of disease and declining population. It forms a flat alluvial plain intersected by several rivers debouching from the Ganges, and by numerous lesser channels and *khals*, many of which have lost their connection with feeder streams, and have become merely channels for local drainage. Some rivers, such as the Garai and Madhumati and a portion of the Chitra in the Narail subdivision, though they have degenerated considerably, still have flowing streams. Others have practically no current for the greater part of the year. Except in the rains, when they maintain a languid vitality, these so-called rivers are merely a chain of long stagnant pools overgrown with weeds. In the south, however, the lower reaches of the rivers are affected by the tide. After the rainy season a portion of the country is under water, either from the over-flowing of neighbouring rivers or from the local rainfall. When the land dries up, extensive *bils* are left, some of which remain stagnant throughout the year.

JESSORE.	Population, 1911.	PERCENTAGE OF VARIATION. 1901—1911.
DISTRICT TOTAL	**1,758,264**	- **3·03**
Sadar Subdivision ...	**462,305**	- **3·82**
Bagherpara	50 303	- 9·40
Jessore	126,984	- 2·09
Jhikergacha	76,926	+ 0·44
Manirampur	135,685	- 4·23
Keshabpur	72,407	- 6·00
Jhenida Subdivision ...	**362,518**	- **6·13**
Sailkopa	166,626	- 0·89
Jhenida	88,229	- 7·74
Kaliganj	71,303	- 13·20
Kotchandpur	36,360	- 9·75
Magura Subdivision ...	**265,948**	- **4·12**
Magura	143,048	- 4·96
Muhammadpur	82,627	+ 0·26
Salika I. O. P.	40,273	- 9·42
Narail Subdivision ...	**360,509**	+ **2·54**
Lohagara	120,664	+ 7·15
Narail	149,289	- 2·71
Barkalia	90,556	+ 5·91
Bangaon Subdivision ...	**306,984**	- **3·27**
Maheshpur	92,125	- 6·37
Sarsha	68,683	+ 0·71
Bangaon	104,136	- 3·14
Gaighata	42,040	- 2·80

230. In 1817 the district suffered from a virulent epidemic of cholera,* and in 1836 there was the first outbreak in epidemic form of that malignant type of fever which was first known as Nadia and then as Burdwan fever.† It lasted for seven years, and seemed to disappear in 1843. In 1846, however, it broke out again, and in the next two years spread over the whole district. After a temporary cessation the fever re-appeared in 1854-56, and about this time began to spread westward to Nadia and the 24-Parganas. This epidemic ceased in 1864, but between 1880 and 1885 there was another serious epidemic, for which the Nadia Fever Commission of 1881-82 could discover no specific cause. It held, however, that it had its source in the silting up of the main rivers and the general insanitary condition of the villages. Since 1891 fever of a less virulent type has been prevalent and formed the subject of a special inquiry by the Drainage Committee in 1906-07. The conclusions at which the Committee arrived were—"(1) The whole district is extremely unhealthy; (2) malarial fever prevails extensively everywhere; (3) by a rather arbitrary comparison, the thanas of Jhenida, Gaighata, Salikha and Bagharpara are the most malarious; (4) the least malarious are thanas Barkalia, Lohagara, Kotchandpur and Gadkhali.‡"

231. The only census at which there has been any increase of population is that of 1881, but the apparent increase then recorded must be attributed to incomplete enumeration at the preceding census. In 1891 there was a decline of 2·6 per cent., and this was followed by a further decrease of 4 per cent. in 1901. In the latter year the south-eastern corner was the only tract which showed even a nominal improvement, and the loss of population was greatest in the country running west and south-west from the Muhammadpur thana on the eastern boundary. The unhealthiness of the district was no less conspicuous during the decade 1901—1910, in which the total number of deaths exceeded the births by 70,000, while the death-rate was above the birth-rate in all but three years (1901, 1909 and 1910). This drain on the population is not counterbalanced by an influx from outside. The immigrants, it is true, are more numerous by nearly 13,000 than in 1901, but still fall short by 13,500 of the number of those born in Jessore who, owing either to necessity or choice, were resident elsewhere at the time of the census. All but five thanas have lost population. Three of these fortunate thanas, viz., Barkalia, Lohagara and Jhikargachha, are among those mentioned by the Drainage Committee as the least malarious in the district; Barkalia and Lohagara form part of the Narail subdivision, which is the only tract which has made any progress.

Jessore.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	901,092	857,172	914,025	899,130
Immigrants	29,774	29,999	24,446	22,551
Emigrants	39,434	33,874	43,775	27,359
Natural population	910,756	861,047	933,315	903,938

NORTH BENGAL.

Rajshahi.

232. Rajshahi is covered with recent alluvium, except in the north-west, where the elevated tract known as the Barind has a quasi-laterite soil. With the exception of the Padma, which forms the southern boundary of the district, and

* It is commonly believed that 1817 was the first year and Jessore the first place in which cholera broke out in a virulent epidemic form, and that it had appeared before only in a mild endemic form. This belief does not appear to be justified, for there are historical references to earlier cholera epidemics. Not to multiply instances, Hicky's Bengal Gazette of 22nd April 1781 refers to an outbreak in Calcutta as follows:—"The plague has now broken out in Bengal and rages with great virulence; it has swept away already about 4,000 persons. Two hundred or upwards have been buried in the different Portuguese churches within the last few days." *See* also Jessore Gazetteer, p. 61.

† It was reported on in 1863 by Dr. Elliott, who traced it back as far as 1824 and noted that "a peculiar type of fever was prevalent in Jessore for many years previous to its first appearance in the district of Nadia."

‡ Now the Jhikargachha thana.

of the Mahananda, which runs for a short distance along its western border, the river system consists of a network of moribund streams and water-courses. The district slopes slightly from west to east, and its drainage, instead of being carried off by rivers, flows into a chain of marshes and swamps, the neighbourhood of which is malarious and unhealthy. Rajshahi is, in fact, one of the most malarious districts in North Bengal, and had a higher death rate during the past decade than any district in the division except Jalpaiguri and Dinajpur. Nator is the most malarious, and Naogaon the least malarious subdivision in the district, the Sadar subdivision occupying an intermediate place between the two.

RAJSHAHI.	POPULATION.	PERCENTAGE OF VARIATION.	
	1911.	1901—1911.	1891—1901.
DISTRICT TOTAL	**1,480,587**	**+ 1·37**	**+ 1·58**
Sadar Subdivision ...	**564,224**	**+ 1·41**	**- 1·27**
Boalia	98,628	+ 3·49	- 3·87
Tanor	92,398	+ 4·37	+ 1·77
Godagari	55,097	+ 17·06	+ 1·28
Puthia	86,362	- 6·46	- 3·57
Charghat	102,612	- 6·26	- 5·82
Bagmara	129,127	+ 4·38	+ 3·95
Naogaon Subdivision ...	**517,405**	**+ 8·91**	**+ 12·13**
Naogaon	187,291	+ 10·52	+ 14·23
Manda	135,738	+ 7·66	+ 10·81
Panchupur	99,915	+ 4·22	+ 8·67
Mahadebpur	94,461	+ 12·90	+ 14·10
Nator Subdivision ...	**398,958**	**- 7·02**	**- 4·76**
Nator	118,160	- 12·53	- 9·84
Singra	120,508	+ 0·48	+ 1·61
Baraigram (Burigaon)	98,963	- 7·80	- 6·77
Lalpur	61,327	- 8·12	- 1·28

233. Between 1872 and 1891 the population of the district, as a whole, was almost stationary, and in the next ten years there was a small increase of 1·6 per cent. During these 30 years the south and centre of the district were decadent, the thanas concerned sustaining a loss of 12·8 per cent.; on the other hand, the population increased by 25·6 per cent. in the Barind, and by 59·3 per cent. in the rich *ganja*-growing thanas of Naogaon and Panchupur in the Naogaon subdivision. The Barind has developed owing to the immigration of Santals, Mundas and Oraons, who find congenial employment in clearing it for cultivation, while the Naogaon subdivision has benefited by the movement of the inhabitants of unhealthy water-logged areas to the healthier and more prosperous thanas included in it.

234. The history of the last decade has been uneventful. Fever has continued unabated, and the reported births exceeded the deaths by only 31,266, representing an increase of 2·14 per cent. on the population of 1901. The actual increase of the population according to the census is 20,003 or 1·4 per cent. only, a result which is practically unaffected by the variations in the number of immigrants and emigrants since 1901. Both are now more numerous than they were in that year, but the increase of the former (15,888) is nearly counterbalanced by the increase of the latter (15,114). The rate of growth in the natural population indeed closely corresponds with that of the actual population. Small as the increase is, it is mainly attributable to the greater fecundity of the Muhammadans. While they have increased in numbers, the Hindus have decreased by 2·79 per cent.

RAJSHAHI.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	755,206	725,381	740,764	719,820
Immigrants	57,040	38,027	46,815	32,364
Emigrants	20,097	18,629	11,941	11,671
Natural population	718,263	705,983	705,896	699,127

235. The deterioration of the Nator subdivision, which is the chief centre of malaria, has continued, and a further loss of 7 per cent. is now recorded. Singra is, as in 1901, the only thana that has gained population, but the gain is under one-half per cent. The Sadar subdivision has a slight growth of 1½ per cent. The most progressive thana in this subdivision, or indeed in the whole district, is Godagari, which lies within the Barind and has added 17 per cent. to its numbers. Elsewhere, there has been a small increase of 3½ to 4½ per cent., but Puthia and Charghat have again decreased. Naogaon subdivision, where two thanas (Manda and Mahadebpur) lie in the Barind and two (Naogaon and Panchupur) in the *ganja* area, is by far the most

progressive part of the district, all its thanas contributing to the increase. It is now more populous by 22 per cent. than it was in 1891, whereas the Sadar subdivision has remained practically stationary, the increase at this census not having made up entirely for the loss in 1901. The Nator subdivision is steadily declining and has lost 11 per cent. of its population the last 20 years.

DINAJPUR.

236. The district of Dinajpur is a flat alluvial plain broken in the south by the Barind, which rises in low undulating ridges, and in the north-west by low hills along the Kulik river. It is well drained, and the rainfall is abundant and well-distributed; but these advantages are minimized by the prevalence of fever of a malignant type. A comparatively thinly peopled district, its population has been steadily growing, the ratio of increase, though never very great, rising at each successive census. Conditions between 1901 and 1910 were generally favourable to a further increase. The death rate was the highest in North Bengal, but was surpassed by the birth rate (43·7 per mille), the result being an excess of 45,626 of births over deaths. There was some scarcity during 1908 and 1909 in the west of the district and in the south-east corner in Ghoraghat, which was met by the distribution of loans and relief works. All distress disappeared with the bumper crops reaped next year, and it does not seem to have affected the growth of population, for during these two years there were 17,000 more births and 11,500 less deaths than in the two preceding years.

DINAJPUR.	POPULATION.	PERCENTAGE OF VARIATION.	
	1911.	1901—1911.	1891—1901.
DISTRICT TOTAL	**1,687,863**	**+ 7·72**	**+ 5·70**
Sadar Subdivision	**694,954**	**+ 9·04**	**+ 4·04**
Dinajpur (Kotwali)	217,776	+ 4·08	+ 0·31
Kaliaganj	116,560	+ 10·66	+ 4·66
Raiganj	88,520	+ 3·08	+ 1·56
Banshihari	99,692	+ 15·31	+ 8·81
Parbatipur	84,044	+ 12·09	+ 8·41
Nawabganj	88,362	+ 17·08	+ 13·71
Balurghat Subdivision ...	**447,343**	**+ 15·77**	**+ 14·21**
Patnitala	110,181	+ 17·36	+ 19·05
Para	63,605	+ 15·88	+ 8·39
Balurghat (Patiram)	106,501	+ 14·22	+ 23·95
Gangarampur	95,686	+ 14·80	+ 4·92
Phulbari (Chintamani)	71,370	+ 16·95	+ 12·67
Thakurgaon Subdivision	**545,566**	**+ 0·46**	**+ 2·20**
Thakurgaon	223,671	- 1·65	+ 1·25
Ranisankail	72,981	- 0·99	- 0·45
Pirganj	104,503	+ 6·41	+ 3·66
Birganj	144,411	+ 0·47	+ 4·16

237. The increase disclosed by the census is far greater than is apparent from the vital statistics, amounting to 121,018 or 7·7 per cent. The difference is due to the largely enhanced number of immigrants, of whom there are 65,000 more than in 1901, while emigrants have only increased by 6,000. The immigrants are mainly found in the Barind, where they are fast reclaiming the waste: they are nearly all aboriginals, such as Santals, Mundas and Oraons, who have increased from 90,345 to 131,668. Owing mainly to their presence, all the thanas in the Barind, viz., Kaliganj, Banshihari, Parbatipur and Nawabganj of the Sadar subdivision and the whole of the Balurghat subdivision, have increases of over 10 per cent. The Thakurgaon subdivision, which grew by 2 per cent. between 1891 and 1901, is now practically stationary, one thana only (Pirganj) having gained population. This subdivision suffers severely from malaria, and the loss it thereby sustains is not compensated for by immigration, as in the Sadar and Balurghat subdivisions.

DINAJPUR.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	889,876	797,987	823,852	742,993
Immigrants	119,502	77,541	80,789	51,214
Emigrants	11,056	12,424	8,420	8,773
Natural population	781,430	732,870	751,483	700,552

JALPAIGURI.

238. Between 1872 and 1891 there was a continuous growth of population in Jalpaiguri due to the development of the tea industry (which was introduced in 1874) and to the influx of labourers and cultivators. The increase was, however, almost

entirely confined to the Western Duars, *i.e.*, the country lying along the foot of the Himalayas between the Tista and Sankos rivers, which comprises the Alipur subdivision (thanas Alipur and Falakata) and two thanas in the Sadar subdivision, viz., Mainaguri (with the Dhupguri outpost) and Damdim. In this area there was a remarkable increase, the tea gardens having attracted labour, while new settlers came in from other districts and the Cooch Behar State, as well as from the west of the district. With regard to the condition of the district during the past decade, the Deputy Commissioner (Mr. Lees) writes:—"The chief feature of the history of the past ten years is the great expansion of cultivation, especially in the Alipur subdivision. There are now few jungle blocks in the district outside the reserved forests. The rapid development of the tea industry between 1881 and 1891, when the area under tea rose from 35,683 acres to 76,158 acres, suffered some check owing to the depression of the tea industry, but the area under tea has now risen to 88,000 acres. Almost all the available land which is suited for tea cultivation in this district has now been taken up, and further expansion of tea cultivation cannot be very great. The decade was a period of prosperity without any serious flood or other calamity." There were, it is true, severe floods in 1902, 1906 and 1910, but comparatively little damage was done to the crops. Epidemics of cholera broke out in the two last years, and also in 1908, a year of deficient rainfall, carrying off altogether 5,000 persons. Fever, which is endemic in this Tarai district, continued unabated. Of all the districts in North Bengal only Dinajpur had a higher death-rate, and the excess of births over deaths was only 4,646.

JALPAIGURI.	POPULATION.	PERCENTAGE OF VARIATION.	
	1911.	1901–1911.	1891–1901.
DISTRICT TOTAL	**902,660**	**+ 14·79**	**+ 15·70**
Sadar Subdivision ...	**699,959**	**+ 4·95**	**+ 9·85**
Jalpaiguri	85,783	+ 4·20	- 0·70
Rajganj	73,480	+ 4·06	- 3·31
Boda	163,401	- 0·97	- 0·82
Pathgram	58,624	+ 4·93	- 5·28
Mainaguri	200,636	+ 9·23	+ 31·17
Damdim	116,035	+ 7·87	+ 26·26
Alipore Subdivision ...	**202,701**	**+ 69·83**	**+ 64·75**
Alipore	126,980	+ 81·49	+ 70·01
Falakata...	76,721	+ 53·45	+ 57·87

239. The census shows an addition to the population of 116,334, or 14·8 per cent., for which immigration is mainly responsible, the natural population having increased by only 7 per cent. There are now no less than 265,268 immigrants in the district, their number having risen by 77,045 or 41 per cent. since 1901 : the emigrants have also increased, but only by 3,573, leaving a balance of 73,472 in favour of the district. The population in the Western Duars has continued to grow rapidly, the percentage of increase being 26 per cent.: the marginal table sufficiently shows the effect its development has had on the expansion of the district. While, however, the Alipur subdivision to the east has grown by no less than 70 per cent., the percentage of increase in Mainaguri and Damdim, in the west, is only 9 and 8 per cent., respectively, whereas it was 31 and 28 per cent. in 1901. These two thanas, lying nearest to the old settled tracts, were naturally the first to attract new cultivators. They filled up rapidly, and there is now little waste land available for new settlers. In the Alipur subdivision, however, cultivation is extending in every direction, and there is a constant stream of immigrants attracted by the fertility of the land and the lowness of the rents. In the rest of the district there has been a

JALPAIGURI.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	490,228	412,432	422,350	363,976
Immigrants	153,309	111,959	104,914	83,309
Emigrants	9,677	11,322	7,585	9,871
Natural population	346,596	311,795	324,991	290,538

CENSUS.	INCREASE IN—	
	District.	Western Duars.
1891	100,250	103,661
1901	106,703	114,258
1911	116,334	108,766

small natural growth of population, varying only from 4 to 5 per cent., except in the Boda thana, which records a slight loss : this thana is an old settled tract, with a fairly high density of population, which offers no prospects to immigrants, and is decadent, having lost population at each census since 1881.

DARJEELING.

240. No district in either Province exhibits more clearly the effect of settled peace than Darjeeling. Writing in 1854, Sir Joseph Hooker said that there were not a hundred inhabitants under British protection when Darjeeling was first transferred, but that, during the two years in which he witnessed its development, its progress resembled that of an Australian colony not only in the amount of building, but also in the accession of native families from the surrounding countries.* The influx of immigrants has continued almost unabated up to the present time, and the population has been nearly trebled during the 39 years over which the census returns extend. The census of 1881 disclosed an addition of 64 per cent., which was partly due, however, to the incompleteness of the first enumeration, and this was followed by an increase of 43·5 per cent. in 1891. During the next decade the rate of progress was reduced to 11·55 per cent., owing largely to the depression of the tea industry. The rate of progress in these ten years was very different in the hills and the Tarai at their base. The former is a healthy tract, where the development of the tea industry has attracted labour, while new settlers have come in to bring waste lands under the plough. The latter is an unhealthy tract the inhabitants of which are sodden with fever. It sustained a loss of 3·5 per cent. between 1891 and 1901, whereas there was a growth of nearly 15 per cent. in the hills.

DARJEELING.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**265,550**	**+ 6·65**	**+ 11·55**
Sadar Subdivision	**152,697**	**+ 13·96**	**+ 26·83**
Darjeeling	75,367	+ 8·54	+ 13·76
Jorbhinga ow	27,210	+ 20·76	+ 24·58
Kalimpong	49,520	+ 19·29	+ 55·87
Kurseong Subdivision ...	**41,207**	**- 8·81**	**+ 1·39**
Kurseong	41,207	- 8·81	+ 1·39
Siliguri Subdivision ...	**72,246**	**+ 2·53**	**- 3·47**
Siliguri	72,246	+ 2·53	- 3·47

241. The present census shows a decline in the rate of increase, the actual addition of population being 16,433, or 6·65 per cent. The explanation is that there is only a limited area in which there is room for an increase of population. Over one-third of the district is covered by reserved forests, while the tea gardens extend over about one-seventh of its area. While they were being opened out and developed, labour poured in and a phenomenal growth of population resulted. Now all the land suitable for tea cultivation, within the area reserved for it, has been taken up. On the tea gardens therefore no considerable increase of population can be expected. As it is, tea occupies a third of the cropped area, and the tea gardens employ a labour force of 53,000, or one-fifth of the total population of the district. As regards ordinary cultivation, only one-third of the district is cultivable, and it cannot therefore hope to support a teeming agricultural population. Even in Kalimpong, where nearly half the land is reserved for native cultivation, it is recognized that it has reached the limit of safety in some parts, and in such localities it has been found necessary to prohibit further extension.

The net result is a progressive decline in the rate of increase and a shrinkage of the volume of immigration. The immigrant population decreased between 1891 and 1901, and there is now a further falling off of 7,000. The immigrants still, however, account for two-fifths per cent. of the total population, and, as pointed out in the last Census Report, the reduction in their numbers merely means that the flow of fresh immigrants is growing less and not that it is closing. The earlier foreign-born immigrants are dying out, and their place is being taken by their children born in Darjeeling.

* *Himalayan Journals.* Sir Joseph Hooker refers to the hills portion of the district excluding Kalimpong, which was then part of Bhutan.

242. To turn to the variations of population in different parts of the district, the most progressive thanas are Jorbungalow and Kalimpong. The former grew by 24·5 per cent. between 1891 and 1901 and now records a further increase of 20·9 per cent. In the latter the rate of growth has fallen from 55·9 per cent. to 19·3 per cent., a result due to the check imposed on immigration by the small proportion of cultivable land that is left. As stated in the District Gazetteer (1907) :— "Whether the volume of immigration will be so great in the future is very doubtful. The best lands have been taken up, those now being developed consist of the poorer and more remote lands, which have been the last to attract settlers, though they are largely taken up by those who can get no good land elsewhere, and there are only 30 square miles of reclaimable jungle left." The growth in the Darjeeling thana is normal and calls for no remarks. The Kurseong subdivision, which was almost stationary in 1901, has now sustained a loss. In that year it was shown that the tea garden population was slightly reduced, but this loss was more than counterbalanced by the access of new settlers for ordinary cultivation, and to some extent by the development of the town of Kurseong. The result of this census seems to show that the limit which the land can support has been reached, and it appears probable that some of the cultivators have moved to Kalimpong and Jorbungalow. The Tarai (Siliguri subdivision) has a slight increase, but it has not yet entirely made good the loss that occurred between 1891 and 1901, and the number of its inhabitants is still slightly less than it was 20 years ago.

DARJEELING.	1911.		1901	
	Male.	Female.	Male.	Female.
Actual population	142,094	123,456	133,005	116,112
Immigrants	67,102	50,056	71,289	53,102
Emigrants	5,863	4,553	8,128	3,566
Natural population	80,855	77,953	64,844	65,579

243. The population of Rangpur declined at each census until 1901, when an increase of 4·3 per cent. brought back the number of its inhabitants to the same figure as in 1872. The improvement in 1901 was due in great measure to immigration; if this were left out of account, the increment would barely have exceeded 1 per cent. Immigration was stimulated by the extension of railway communications, for the Gaibandha subdivision was rendered more accessible by the line running from Santahar to Fulchari on the Brahmaputra, the Cooch Behar Railway was opened to traffic, and the Tista river was bridged. The earthquake of 1897, moreover, appears to have had beneficial effects on the public health; for by changing the levels of a great part of the district, it facilitated its drainage, and consequently malaria seems to have been less prevalent.

RANGPUR.

RANGPUR.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**2,385,330**	**+ 10·73**	**+ 4·30**
Sadar Subdivision	**683,127**	**+ 3·77**	**+ 1·84**
Rangpur (Kotwali)	147,292	+ 3·70	+ 7·53
Mahiganj	102,027	- 4·09	- 3·41
Kaliganj	159,161	+ 4·98	+ 1·69
Badarganj	98,248	+ 3·74	+ 7·96
Mithapukhur	96,813	+ 5·32	- 5·73
Pirganj	77,586	+ 10·98	+ 2·79
Nilphamari Subdivision ...	**491,860**	**+ 6·38**	**+ 3·01**
Nilphamari	168,796	+ 5·13	+ 9·57
Dimla	139,692	+ 3·98	- 2·52
Jaldhaka	183,372	+ 9·30	+ 1·78
Kurigram Subdivision ...	**589,520**	**+ 15·29**	**+ 1·31**
Kurigram	65,534	+ 14·13	- 4·62
Lalmanirhat (Barabari)	131,580	+ 4·80	+ 4·27
Nageshwar	155,157	+ 10·18	+ 0·65
Ulipur	237,249	+ 26·48	+ 1·83
Gaibandha Subdivision ...	**620,823**	**+ 18·91**	**+ 12·22**
Gaibandha	231,731	+ 19·24	+ 14·68
Gobindganj	206,512	+ 17·49	+ 15·98
Sundarganj	106,089	+ 24·49	- 1·60
Shaghatta	76,491	+ 14·58	+ 15·85

244. The improvement in the general condition of the people continued between 1901 and 1910. The crops were good and the people prosperous except in 1908-09, when there was scarcity in parts of the Sadar and Gaibandha subdivisions, which was relieved by means of loans and does not seem to have affected the growth of population to any appreciable extent. The Kaunia-Bonarpara line has been opened, and, as it touches the town of Gaibandha, has done much to open out that subdivision, while the extension

of the Eastern Bengal State Railway, first to Dhubri and then to Gauhati, has established direct communication between Assam and Calcutta.

The ravages of malaria diminished, and, though there was some recrudescence after 1905, the number of births exceeded the reported deaths by no less than 133,512 or 6·2 per cent. The increase of population returned at the census is nearly 100,000 more, amounting to 231,215 or 10·7 per cent. To this increase immigrants have contributed largely, for they are more numerous by 69,000 than in 1901, while the district has only lost 7,000 by the exodus of the native born. Owing partly to immigration and partly to natural causes, the Muhammadans, who preponderate in the district, have increased by 14·54 per cent., while the Hindus have increased only by 3·46 per cent. The greatest growth of population has taken place in the Kurigram (15·3 per cent.) and Gaibandha (18·9 per cent.) subdivisions, where there has been a great influx of Musalmans from Pabna and Mymensingh, who have settled on the *chars* of the Brahmaputra. In the Nilphamari and Sadar subdivisions the rate of progress has been influenced more by natural causes than by immigration. Both these subdivisions suffer from fever and epidemics of cholera: the former has gained 6·4 per cent. and the latter only 3·8 per cent. All thanas, however, are progressive except Mahiganj, which has been decadent since 1881.

RANGPUR.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,254,717	1,180,613	1,124,991	1,029,124
Immigrants	125,851	52,639	80,420	28,996
Emigrants	32,249	29,286	26,692	27,470
Natural population	1,161,115	1,107,260	1,071,263	1,027,596

245. The Karatoya river traverses Bogra from north to south and divides it into two unequal portions with distinct characteristics. The eastern portion is covered with the alluvium of the lower Brahmaputra valley, while the western and larger portion forms part of the Barind. Numerous marshes have been formed by the silting up of the old river beds, one of which, locally known as the Bara Bil, is connected with the great Chalan Bil in Rajshahi. The district suffers from malaria, but it is the healthiest district in the Rajshahi Division, its mortality from fevers during the past decade being the lowest returned. Since the first census was taken, there has been a sustained and fairly rapid growth of population, the increment recorded both in 1891 and 1901 exceeding 11 per cent. Conditions during the decennium ending in 1910 were favourable to a further advance. The resources of the district were developed by the railway, running from west to east through Bogra, which was completed in 1900-01. Good harvests were reaped throughout the first half of the decade, and though there was some scarcity in 1908-09 and 1909-10, it was short-lived and the cultivators were able to tide over their difficulties by means of loans. The public health was, on the whole, good, and the births exceeded the deaths by 100,932, representing an increase of 11·8 per cent. on the population of 1901.

BOGRA.

BOGRA.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911	1891—1901.
DISTRICT TOTAL	***983,567***	***+15·24***	***+11·78***
Bogra	270,002	...[a]	...
Adamdighi	108,752	...[a]	...
Dupchanchia	48,878	...[a]	...
Shariakandi	152,553	+ 15·29	+ 16·18
Shibganj	92,170	+ 20·78	+ 13·45
Panchabibi	107,983	+ 21·30	+ 22·06
Khetlal	63,483	+ 23·32	+ 11·24
Sherpur	54,448	+ 2·43	+ 7·09
Dhunot	85,298	+ 5·70	+ 18·89

[a] Figures showing the percentage of variation in Bogra, Adamdighi and Dupchanchia are not available owing to changes of jurisdiction.

246. The increase of 130,063 or 15·2 per cent. brought to light by the census is mainly due to natural growth. Compared with 1901 there is a net gain of only 15,431 from increased migration, immigrants having risen from 37,897 to 63,148 and emigrants from 15,756 to 25,576. All parts of the district contribute to the

BOGRA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	502,526	481,041	436,871	416,633
Immigrants	37,780	25,368	22,358	15,539
Emigrants	12,959	12,617	8,137	7,619
Natural population	477,705	468,290	422,650	408,713

increase. The least progress has been made by the Sherpur thana, an unhealthy tract lying along the moribund Karatoya river: here the deaths during the decade exceeded the births, and there would have been a loss of population but for the immigration of aboriginals and others, who are bringing the jungle under cultivation. The higher ratio of increase (5·7 per cent.) in thana Dhunot, a fertile area which is the most densely populated part of the district, is due to natural growth and not to immigration. In the other thanas, which are mostly healthy and dry, the increase is due to the combined effect of natural causes and immigration. One noticeable feature of the increase is that the Hindus, who were practically stationary between 1891 and 1901, have now increased by 8·24 per cent. The Muhammadans, who form the bulk of the population, show still greater improvement, the ratio of increase among them having risen from 5·6 to 16 per cent.

PABNA.

247. Between 1872 and 1881 the population of Pabna increased by 8·3 per cent., but part of the increase is attributable to the incompleteness of the first census. In 1891 there was a gain of 3·85 per cent., while in 1901 there was a further addition of 4·3 per cent. Between 1901 and 1910 the growth of population was retarded. Crops were poor in the first four years and the jute trade slack in the last three. High floods occurred in 1903, 1906 and 1910, of which that of 1906 did considerable damage to the crops, while the inundation of 1910 caused some temporary distress. In four of the ten years the reported deaths exceeded the births, and in the decade as a whole there was a loss, the deaths outnumbering the births by 11,238. "Malaria," writes the District Magistrate "is a permanent scourge and has its strongest hold in thanas Sainthia (formerly Dulai) and Chatmohar of the Sadar subdivision and in Raiganj and Ullapara in the Sirajganj subdivision. Thana Pabna in the Sadar subdivision is not free from its ravages. This is due mostly to the existence of a number of *bils* of various sizes in the interior, in most of which the water lies stagnant almost all the year round except in years of excessive flood."

PABNA.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**1,428,586**	**+ 0·51**	**+ 4·34**
Sadar Subdivision... ...	**599,266**	**+ 2·04**	**+ 2·10**
Sainthia (Dulai)	169,950	+ 2·08	- 7·20
Pabna	204,084	+ 2·30	+ 6·21
Mathura	94,636	+ 7·20	- 6·33
Chatmohar	130,596	- 1·35	- 3·89
Serajganj Subdivision ...	**829,320**	**- 0·57**	**+ 9·42**
Sirajganj	270,168	+ 2·27	+ 10·93
Shahzadpur	256,336	- 2·31	+ 8·76
Raiganj	104,370	- 1·96	+ 11·56
Ullapara	194,446	- 1·29	+ 7·17

248. There has, according to the recent census, been an increase of 7,191, or only one-half per cent. No less than 7,154 persons however were employed on the Lower Ganges bridge works at Sara, of whom the majority were newcomers from outside districts: if they are left out of account, the increase is entirely wiped out. The returns for birth place show that since 1901 immigrants have increased from 49,040 to 64,900 and emigrants from 64,305 to 102,667. The loss by emigration thus considerably exceeds the gain by immigration, the net loss being 22,502, which added to the excess of deaths over births, should have caused a considerable decrease of population: it is therefore open to question whether the vital statistics are correct. There was an increase of 12,515 or 2 per cent. in the Sadar subdivision, of which more than half was due to the employés enumerated in the Lower Ganges bridge works: without them the Pabna thana would have substained a loss. In Chatmohar, a malarious area, there is a slight decrease, and in Sainthia a slight increase, while Mathura, which has a better climate and more fertile soil, has a substantial growth. The population declined in all the thanas of Sirajganj subdivision, except the Sirajganj thana, where there was a very small

PABNA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	723,591	704,995	709,848	711,547
Immigrants	44,710	20,190	34,158	14,882
Emigrants	58,842	43,825	39,917	24,388
Natural population	737,723	728,630	715,607	721,053

increase. The loss in Raiganj and Ullapara is due to the ravages of malaria. As regards the decrease in this subdivision, the Subdivisional Officer writes : " It is difficult to account for the fall in population shown by this census, but I am of opinion that the ravages of malaria have had a serious effect. The jute trade appears also to have reached its high water-mark as regards the number of persons employed. No attempt has been made to revive or rebuild the jute mill at Sirajganj, which fell in the earthquake of 1897 and used to employ about 2,000 hands. The immigration of Bunas into Raiganj thana has gone on steadily, but this has been more than counterbalanced by the emigration of cultivators from the riparian tracts to the districts of Rangpur and Goalpara, where they find more land for settlement." But for emigration the subdivision as a whole would have shown an increase instead of a decrease.

MALDA.

249. Malda consists of two distinct tracts separated by the river Mahananda. The western portion is composed of recent alluvium, a part of it being still subject to the fluvial action of the Ganges. The eastern portion lies in the Barind, which has an undulating surface and stiff laterite soil. The population has been growing steadily in spite of the prevalence of malaria. An increase of 14·5 per cent. was recorded in 1891, and another increase of 8·5 per cent. in 1901. Between 1891 and 1901 the greatest progress was made by the Gajol and Old Malda thanas in the Barind, two sparsely populated areas where Santals settled in large numbers. There were also large increases in Kaliachak and Sibganj to the south-west, where new alluvial formations attracted a number of Musalman cultivators from Murshidabad on the other side of the Ganges, as well as from English Bazar and Nawabganj in this district : the two latter thanas consequently sustained a loss.

MALDA.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	*1,004,159*	*+13·88*	*+ 8·49*
Gomastapur	67,414	+ 11·77	+ 6·61
English Bazar	94,617	+ 4·61	- 6·23
Nawabganj	71,274	- 0·34	- 1·63
Kaliachak	164,801	+ 11·20	+ 14·30
Sibganj	150,871	+ 15·79	+ 11·01
Kharba	82,399	+ 16·37	+ 10·41
Ratua	135,093	+ 22·16	+ 4·43
Tulsihatta	37,205	+ 19·90	+ 0·35
Old Malda	84,222	+ 16·41	+ 22·90
Gajol	96,763	+ 21·05	+ 25·00

250. Since 1901 the district has made steady progress. The births exceeded the deaths in every year but 1907, the result being a net excess of 98,484 or 11·2 per cent. on the population of 1901. The crops were good except in 1908-09, when short rainfall led to a failure of the winter rice crop—the principal crop in the Barind area—and some distress was experienced, which was remedied by the issue of loans and the opening of relief works. In the rest of the district, however, good *bhadoi* and *rabi* crops were obtained and, owing to the high prices of food-grains, the condition of the people was, if anything, more prosperous than in other years. The scarcity had no deterrent effect on the growth of population : on the contrary, the birth-rate in 1909 (52 per mille) was higher and the death-rate (29·4 per mille) lower than in any other year of the decade. The most important feature in the economic history of the decade was the opening of the Katihar-Godagari Railway, which traverses the district from north to south. The railway has already done much to develop the district. The Collector reports—" At every railway station a bazar has sprung up, and the cultivator has profited largely by the competition of traders in jute, rice and other country produce. Growers of the mango fruit are now able to reach markets at a greater distance and command better prices. Another result has been to facilitate immigration into the thinly populated areas in the east of the district."

251. The outcome of the above conditions is a gain of 122,425 or 13·9 per cent. This increase is to be attributed to natural causes, for though there has been a greater influx of immigrants, the outward movement of the people has resulted in a corresponding increase in the number of emigrants. The immigrants consist chiefly of Santals, who are reclaiming waste lands in the Barind, and of Musalmans who cultivate the new alluvial formations

in the *diara* tract, besides Bihari settlers who come into the northern thanas of the district and into Manikchak. The Santals alone have increased by 14,394 during the past decade. Population increased in every thana except Nawabganj, in which there was a small decrease owing to the diluviation of their lands having compelled a certain number of the people to move to other thanas in the district. English Bazar, in which there was a decrease of 6·23 per cent. in 1901 (attributed to declining trade, unhealthiness and migration to adjoining thanas) now shows an increase of 4·61 per cent. The proportional growth in all the remaining thanas exceeds 10 per cent. As in other districts of North and East Bengal, the Muhammadans have a larger share in the increase than the Hindus, owing to their numerical superiority as well as to their greater procreative capacity.

Malda.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ...	498,547	505,612	436,497	443,237
Immigrants ...	64,678	34,193	55,070	42,817
Emigrants	21,368	20,140	13,220	13,544
Natural population ...	455,237	471,559	394,647	413,964

252. From 1881 to 1901 Cooch Behar suffered from persistent unhealthiness, and both the census of 1891 and that of 1901 showed a declining population, the decrease recorded being 3·9 per cent. and 2 per cent., respectively. Conditions during the decade ending in 1910 were more favourable. Fever, which is the scourge of this low-lying water-logged country, showed some abatement, while cholera, from severe epidemics of which the State suffers periodically, was not so prevalent: the number of deaths due to it was 9,000 as compared with 17,000 in the previous decade. There was a serious inundation in 1906, from which the north of the Matabhanga subdivision suffered most, but otherwise there were no seasonal calamities. Cultivation expanded, the cultivators benefited by the rise in the price of agricultural produce, and there was an increasing demand for labour, which was met by the import of coolies from up-country. The Gauhati extension of the Eastern Bengal State Railway (from Gitaldaha to Gauhati) was built during the decade and has already helped to open out the south-east of the State.

Cooch Behar.

Cooch Behar.	Population, 1911.	Percentage of variation.	
		1901—1911.	1891—1901.
TOTAL	**592,952**	**+ 4·58**	**- 2·05**
Kotwali	130,602	+ 7·40	- 4·4
Tufanganj	80,147	+ 9·14	- 1·3
Dinhata	148,529	+ 3·22	- 2·0
Matabhanga	142,604	+ 1·73	- 3·5
Mekliganj	51,345	+ 2·87	- 1·0
Haldibari	39,525	+ 5·01	+ 9·9

253. The result of the census is to show an increase of 25,978 or 4·6 per cent., part of which is due to immigration, the immigrant population being greater by over 10,000 than in 1901, while the number of emigrants has risen by a little under 4,000. These immigrants are mostly temporary, and the males outnumber the females by 2 to 1. All parts of the district share in the increase, but the most progressive is Tufanganj, a sparsely-inhabited area, where there has been a rapid extension of cultivation. It is closely followed by the Cooch Behar (Kotwali) thana, which has gained by immigration from Mekliganj. Haldibari, a thriving jute-centre, has a normal increase of 5 per cent. The least progress has been made in Matabhanga, which suffered from the floods of 1906, and is reported to be the most unhealthy part of the State. The small increase in Mekliganj is due to the fact that it lost to the Cooch Behar thana, while Dinhata was visited by a severe epidemic of cholera in 1909.

Cooch Behar.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ...	316,548	276,404	301,382	265,592
Immigrants	36,293	18,849	27,884	16,959
Emigrants	19,121	17,115	16,761	15,782
Natural population ...	299,376	274,670	290,259	264,415

EAST BENGAL.

254. The Khulna district was created in 1881, when the census showed a small increase of 3·2 per cent. This was followed by a further increase of 9 per cent. in

Khulna.

1891, but the percentage of growth was reduced to 6·4 per cent. in 1901. This latter census disclosed considerable variations in different parts. While the Sadar subdivision in the centre of the district had an increase of 17·7 per cent., the ratio was only 6·6 per cent. in the Bagherhat subdivision to the east, while there was a falling off of 1·5 per cent. in the Satkhira subdivision to the west, where, however, the decrease was practically confined to the two unhealthy thanas of Kalaroa and Asasuni.

KHULNA.	Population, 1911.	PERCENTAGE OF VARIATION. 1901—1911.
DISTRICT TOTAL	**1,366,766**	+ **9·08**
Sadar Subdivision ...	**441,245**	+ **10·07**
Khulna	129,870	+ 3·20
Dumuria	84,624	+ 5·22
Baitaghata	43,360	+ 7·15
Paikgachha	183,191	+ 22·68
Satkhira Subdivision ...	**547,380**	+ **11·91**
Kalaroa	73,661	+ 2·67
Magura	95,241	+ 2·37
Satkhira	111,749	+ 8·90
Kaliganj	204,367	+ 21·57
Asasuni	62,362	+ 16·24
Bagerhat Subdivision ...	**378,141**	+ **4·16**
Mollahat	61,337	+ 0·01
Bagerhat	164,130	+ 3·22
Rampal	69,179	+ 6·14
Morelganj	83,495	+ 7·70

255. The condition of the people was generally prosperous in the decade succeeding the census of 1901. The east of the district benefited by the construction of a light railway from Barasat to Basirhat, which was opened in 1905-06, while the deepening and widening of the Madaripur Khal, so as make it navigable throughout the year, were of immense advantage to the jute trade. The health of the people was on the whole good, the births during the decade exceeding the deaths by 93,000. The crops were fair, and though the poorest classes felt the pinch of high prices from 1906 to 1909, the cultivators benefited by the enhanced value of their crops. There was a severe cyclone in October 1909, accompanied by a tidal wave which carried all before it along the waterways near the sea-face. Cattle were destroyed in thousands (the estimated loss was 80,000), trees blown down on all sides, *kutcha* buildings levelled to the ground, and the rivers swept clear of country boats. Fortunately, however, there was very little loss of human life, and only slight damage was done to the rice crop. The rainfall in the previous part of the year had moreover been copious and well distributed, and next year bumper crops were reaped.

256. The increase of population disclosed by the census is 113,723 or 9 per cent.—a result due to natural growth, rather than to any large accretion from outside districts. Though the immigrant outnumbers the emigrant population, the excess is only 15,000, whereas in 1901 it was nearly 40,000, the change being due to emigrants increasing by 14,000 and immigrants falling off by 11,000. All parts of the district have progressed in a major or minor degree. The variations seem to depend on two factors, viz., the relative healthiness and unhealthiness of different parts, and the reclamation of cultivable land on the fringe of the Sundarbans. On the former point the conclusions of the Bengal Drainage Committee were—"(*a*) The district is not conspicuously unhealthy in any part; (*b*) malaria is prevalent, but not especially so; (*c*) the most malarious thanas are Kalaroa, Mollahat, Magura and perhaps Satkhira; and (*d*) the least malarious are Kaliganj, Paikgacha, Asasuni and Rampal."

KHULNA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ...	709,379	657,187	653,470	599,573
Immigrants	30,914	25,816	43,697	22,020
Emigrants	20,115	19,732	14,068	11,815
Natural population ...	698,780	653,103	623,841	589,368

257. On the whole, the most unhealthy part is the northern tract adjoining Jessore, and the central portion of the district is not so insalubrious, as it is more open and the jungle less dense. In the Sundarbans, to the south of this cultivated belt, there are few permanent inhabitants, the land being low and subject to inundation, and fresh water scarcely obtainable. It is on the northern edge of this latter tract that cultivation is advancing most rapidly. To this cause and to their comparative healthiness must be ascribed the uniformly large increase (16 to 23 per cent.) in Kaliganj, Paikgachha and Asasuni, which all lie along the central line of the Sundarbans. The three thanas, which were classified by the Drainage Committee as malarious, viz., Kalaroa and Magura on the north-west and Mollahat on the north-east, have advanced very slightly, the ratio of increase in the two

former being under 3 per cent.. while Mollahat is stationary. Satkhira, about the unhealthiness of which there was some doubt, has added 8·9 per cent. to its population, owing mainly to its development by the light railway: at the previous census it decreased by 1·75 per cent., in consequence of bad health and the loss of trade caused by the diversion of its boat traffic.

Considering the variations in the population of each subdivision, the rate of advance has been rapid in the Satkhira and Sadar subdivisions, but comparatively slow in the Bagerhat subdivision. It was noticed in the last census report that the clearance of jungle in the Sundarbans was proceeding at a relatively slow rate in this last subdivision, and the same feature is now even more marked. This is partly due to the cyclone of 1909, which killed off the deer, and consequently led to an increase in the number of man-killing tigers in this part of the Sundarbans: how serious an obstacle these brutes are to the expansion of cultivation is familiar to all having a knowledge of the Sundarbans.

DACCA.

258. Hemmed in on three sides by the Jamuna, Padma and Meghna, Dacca is subject to all the vicissitudes of alluvion and diluvion, as well as to the periodic inundation and silt fertilization characteristic of Eastern Bengal. These great rivers, as well as numerous smaller streams which intersect the district, annually flush and fertilize the land, except in an elevated tract to the north, which includes the Madhupur Jungle. Malaria, which checks the growth of population in so many districts, has no strong hold over this district, except in a small area on the western side. Blest with a fertile soil, a good water-supply, a healthy climate and a comparatively high standard of material prosperity, the people in this favoured tract are multiplying with great rapidity, though most of the district is very densely populated. The northern part towards Mymensingh, which is higher and undulating, is somewhat sparsely populated, but is now being rapidly opened out, on account of the pressure on the soil in other parts of the district. Between 1872 and 1901 there was an increase of over 10 per cent. at each census, and in the year last named the population was greater by 45 per cent. than it was 29 years before.

DACCA.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**2,960,402**	+ **11·95**	+ **10·61**
Sadar Subdivision	**1,013,619**	+ **14·99**	+ **11·45**
Dacca (Kotwali)	113,086	+ 21·33	+ 12·02
Keraniganj	242,047	+ 16·89	+ 10·76
Kapasia	220,008	+ 26·13	+ 22·33
Nawabganj	173,803	+ 1·73	+ 0·09
Sodhar	264,675	+ 12·17	+ 13·72
Narayanganj Subdivision	**779,564**	+ **17·99**	+ **15·00**
Narayanganj	187,761	+ 18·84	+ 14·32
Raipura	329,094	+ 18·88	+ 15·70
Rupganj	262,709	+ 16·30	+ 14·63
Munshiganj Subdivision	**692,407**	+ **9·34**	+ **9·86**
Munshiganj	315,221	+ 6·67	+ 10·25
Srinagar	377,186	+ 11·67	+ 9·52
Manikganj Subdivision	**474,812**	+ **1·25**	+ **4·46**
Manikganj	221,377	+ 6·35	+ 3·87
Seabo (Gheor)	168,046	— 1·17	+ 7·51
Harirampore	95,389	- 5·79	+ 1·11

259. During the past decade there has been nothing to retard the further growth of the people. It was a period of general prosperity and good health, the only year in which there was any interruption to the even tenor of development being 1906, when unusually high floods led to a failure of crops in part of the district; the distress, however, was temporary and not sufficient to call for relief measures. The census shows an increase of 315,967 or 12 per cent., which is due entirely to natural growth, for, compared with 1901, the balance of migration is against the district, causing a loss of 20,000. Both emigration and immigration are more active than in 1901, but while the number of immigrants has risen by 29,322, the increase of emigrants is 49,416. The latter number 177,903, but the excess of males over females shows that 45 per cent. are temporary absentees.

DACCA.	1911.		1901.	
	Male.	Female.	Male.	Female.
tual population	1,477,690	1,482,712	1,309,733	1,334,702
mmigrants	74,640	39,981	56,767	28,532
Emigrants	124,545	53,358	94,842	33,645
Natural population	1,527,595	1,496,089	1,347,808	1,339,815

260. The Sadar subdivision, where cultivation is spreading fast in the Madhupur Jungle, has an increase of 15 per cent. Both the Dacca and Kapasia thanas have grown by over 20 per cent., owing to the development of Dacca city in the former and the expansion of cultivation in the latter. The increase in thana Nawabganj is very small, although there has been an accelerated rate of progress compared with the previous decade The density in this thana is extremely high, and there is a considerable exodus of its male inhabitants; in consequence of this the females exceed the males by one-eighth. The increase of 18 per cent. in the Narayanganj subdivision is evenly distributed between the thanas comprised in it. It is a fertile healthy tract, to which up-country immigrants are attracted in increasing numbers by the demands of the jute industry, and from which there is very little emigration. In the Munshiganj subdivision the increase amounts to 9·3 per cent., which is very little less than at the previous census. The density in this subdivision is the highest in Eastern Bengal, and there is, in consequence, a flow of emigration from it. The rate of progress has diminished in thana Munshiganj, but increased in thana Srinagar in spite of its higher density. The least progressive part of the district is the Manikganj subdivision. Situated between the Dhaleswari and Padma, it is liable to constant changes by alluvion and diluvion. In recent years it has lost by diluvion, the Padma having cut away a considerable area along the western boundary in thanas Sealo and Harirampur. The rate of increase has now fallen from 4·5 to 1·25 per cent. owing mainly to this outward movement.

MYMENSINGH.

261. The Madhupur Jungle divides Mymensingh into two unequal and ethnically dissimilar portions. The western and smaller portion, which is watered and drained by the Jamuna, resembles in its general character the alluvial tract on the other side of that river. The eastern portion is fertilized by the Surma (called the Meghna in its lower reaches), by the old Brahmaputra and numerous other streams. Towards the east, there are extensive *haors* or marshes, where people gather in the cold weather to catch fish, grow *boro* rice, and graze their cattle. Along the border of the Garo Hills, the country is hilly and sparsely populated. The greater part of the Tangail subdivision is malarious like the tract on the other side of the Jamuna, but the remainder of the district is practically free from this scourge. Taken as a whole, Mymensingh must be regarded as one of the healthiest districts in Bengal, its death-rate being lower than that of every other district in North and East Bengal except Tippera. The soil is fertile and admirably suited to the cultivation of jute, which is extensively grown throughout the district. The people are so prosperous that they can afford to look down upon menial work and leave most of it to immigrants from Bihar and the United Provinces, who serve as earth-diggers, *palki*-bearers, domestic servants, boatmen and general labourers. The Musalman form nearly three-fourths of the inhabitants of the district. Owing to the preponderance of this prolific and hardy community and to the constant influx of immigrants, the population

MYMENSINGH.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**4,526,422**	**+ 15·53**	**+ 12·75**
Sadar Subdivision ...	**1,185,330**	**+ 21·26**	**+ 14·59**
Mymensingh (Kotwali)	223,337	+ 16·91	+ 14·40
Phulbaria	130,631	+ 18·38	+ 18·99
Gaforgaon	189,769	+ 16·81	+ 15·54
Nandail	132,271	+ 14·25	+ 9·22
Iswarganj	195,786	+ 21·94	+ 12·10
Phulpur	215,129	+ 31·51	+ 17·67
Muktagacha	96,407	+ 33·88	+ 14·40
Netrakona Subdivision ...	**655,295**	**+ 14·01**	**+ 7·12**
Netrakona	189,857	+ 18·42	+ 8·61
Kendua	175,868	+ 14·17	+ 10·58
Durgapore	127,035	+ 11·13	- 1·22
Borhatta	122,772	+ 10·89	+ 8·61
Kholiajuri	39,763	+ 12·39	+ 10·58
Jamalpore Subdivision ...	**813,306**	**+ 20·23**	**+ 16·10**
Jamalpore	255,859	+ 20·37	+ 15·94
Nalitabari	124,553	+ 25·37	+ 16·01
Dewanganj	177,553	+ 19·89	+ 11·72
Sherpore	179,658	+ 22·63	+ 21·19
Madarganj	75,683	+ 8·25	+ 15·95
Tangail Subdivision ...	**1,049,772**	**+ 8·20**	**+ 12·89**
Tangail	268,305	+ 2·84	+ 10·59
Kalihati	160,982	+ 10·20	+ 10·41
Gopalpur	218,282	+ 13·38	+ 19·44
Nagarpur	118,371	+ 5·90	+ 10·59
Mirzapore	108,781	+ 14·43	+ 10·59
Ghatail	95,196	+ 12·33	+ 10·41
Sharihbari (formerly Jagannathganj.)	79,905	+ 0·87	+ 19·44
Kisorganj Subdivision ...	**822,719**	**+ 14·40**	**+ 11·78**
Kisorganj	190,604	+ 12·06	+ 10·10
Kathiadi	178,430	+ 15·57	+ 11·04
Bajitpur	155,645	+ 11·50	+ 14·16
Badla	146,630	+ 15·19	+ 10·10
Bhairabbazar	53,513	+ 21·19	+ 14·16
Astrogram	97,897	+ 17·00	+ 14·16

has been rapidly growing. the ratio of increase exceeding 12 per cent. at every census. Between 1901 and 1910 there was nothing to cause a check to the development of the district. The public health was good, and even in 1905, when there was a virulent epidemic of cholera causing a mortality of 10½ per cent., the deaths were outnumbered by the births. The agricultural seasons were also good. Some distress was caused by severe floods and high prices in 1906, but their effects did not last long. Both emigration and immigration have been stimulated by the extension of the railway to Fulchari.

262. The addition of population now returned is 608,320 or 15½ per cent. It would have been even greater but for emigration, for though immigrants have increased by 46,000 since 1901, there has been even greater rise in the number of the district-born who were enumerated outside the district. The latter are now more numerous by 76.000, and their aggregate (156,993) is very little less than that of immigrants (161,395). There would be a net loss of 30,000 if the whole of the migration were permanent, but the excess of the males shows that the majority of the emigrants are temporary absentees; immigrants are, to an even greater extent, only temporarily resident in the district.

MYMENSINGH.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ..	2,339,603	2,186,819	2,016,393	1,901,709
Immigrants	111,176	50,219	82,760	32,250
Emigrants	85,957	71,036	45,971	34,594
Natural population ..	2,314,384	2,207,636	1,979,604	1,904,053

In the Sadar subdivision, which includes extensive cultivable wastes, the population has increased by 21·3 per cent.: in no thana is the increase less than 14 per cent. The increase in all the other subdivisions also exceeds 14 per cent., except in Tangail. This sub-division suffers from malaria and lost heavily from the cholera epidemic of 1905: this was so virulent that in many places the villagers, being unable to dispose of their dead by ordinary means, either threw the bodies into the rivers or left them to rot on the ground. It has, moreover, lost from the emigration of Musalmans to the *chars* of the Brahmaputra in the Goalpara district, where they can obtain land on easy terms. Owing to these causes, the percentage of increase has fallen to 2·8 and 5·9 per cent. respectively in thanas Tangail and Nagarpur. Both these thanas are liable to malaria, both are very densely populated, and it is from them that the Musalmans have chiefly emigrated to the higher reaches of the Brahmaputra. All the remaining thanas in the subdivision have a gain of over 10 per cent. In spite of emigration there is an increase of 18·8 per cent. among the Musalmans, or 12 per cent. more than among the Hindus, though the latter are reinforced by immigrants from upcountry.

263. Faridpur is bounded on three sides by the Meghna, the Padma and the Garai, and is intersected by numerous smaller streams. In the north and east the land is comparatively well-raised, but the level sinks towards the south, and on the confines of Backergunge the whole country is practically a marsh intersected by strips of high land, the remains of rivers that have at various times flowed through this tract. The district is malarious, the mortality from fevers during the past decade being exceeded in Eastern Bengal only by that of Chittagong. Since 1872 the population has made steady progress, there being a total increase of 37·65 per cent., which is almost evenly distributed between the four decades ending in 1911.

FARIDPUR.

FARIDPUR.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL ...	**2,121,914**	+ **8·71**	+ **6·20**
Sadar Subdivision ...	**546,499**	+ **5·20**	+ **6·14**
Faridpur	113,194	+ 6·81	+ 9·57
Bhanga	232,632	+ 8·83	+ 13·93
Nagarkanda	110,488	+ 6·62	+ 1·29
Bhushna	90,185	- 6·17	- 6·45
Goalundo Subdivision ...	**328,963**	- **0·003**	- **9·05**
Goalundo	124,073	+ 2·54	- 5·21
Baliakandi	81,902	- 7·43	- 9·47
Pangsha	122,988	+ 2·92	- 12·35
Madaripur Subdivision ...	**816,203**	+ **15·04**	+ **11·82**
Madaripur	245,563	+ 12·00	+ 12·85
Palong	355,332	+ 16·60	- 0·62
Sibchar	215,308	+ 16·09	+ 38·91
Gopalganj Subdivision ...	**430,249**	+ **9·20**	+ **11·82**
Gopalganj	121,698	+ 10·95	+ 14·52
Kotalipara	102,345	+ 10·72	+ 15·28
Maksudpur	112,277	+ 7·46	+ 8·77
Kashiani	93,929		

264. In the last decade there has been an increase of 169,971 persons or 8·71 per cent. Births exceeded deaths by 101,560, but there would have been a greater excess had it not been for epidemics of cholera in five years, which carried off nearly 37,000 persons. The number of immigrants has increased from 73,483 to 96,333 and of emigrants from 75,810 to 81,469, there being therefore a balance of about 17,000 in favour of the district. Much of the migration is temporary, as is apparent from the proportion of the sexes, for males exceed females by 17,345 among immigrants and 23,769 among emigrants.

FARIDPUR.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,074,959	1,046,955	977,385	974,578
Immigrants...	56,889	39,494	46,673	26,810
Emigrants	52,619	28,850	54,190	21,620
Natural population	1,070,889	1,036,311	984,882	969,388

In the Sadar subdivision there has been an increase of 5·2 per cent., which is shared in by all the thanas, except Bhushna, where there is a decrease of 6·17 per cent.: this thana is very malarious owing to the silting up of old drainage channels. The high percentage of increase (15 per cent.) in the Madaripur subdivision is due partly to natural causes and partly to immigration, for the fertile alluvial formations in that subdivision attract Musalman settlers from the adjoining districts. In the Gopalganj subdivision there is a fair increase of 9·2 per cent., but the Goalundo subdivision is stationary: the latter is notoriously unhealthy, and the deaths during the decade exceeded the births considerably.

BACKERGUNGE.

265. Backergunge is a part of the alluvial delta formed by the river systems of Eastern Bengal, and consists partly of mainland and partly of islands in the estuary of the Meghna. The mainland forms an unbroken plain, intersected by a network of tidal rivers and channels. Along the coast lie the Sundarbans, a semi-aquatic area of forests, swamps and tidal creeks, in which cultivation is gradually extending. The soil is extremely fertile, being annually enriched by the silt brought down by the rivers. Owing to its low level, and the peculiar character of its river system, the district is practically protected against drought by natural irrigation, but it is exposed to the devastation of cyclones and tidal waves. The more fertile tracts in the lower levels, which are exposed to tidal waves and infested by wild animals, offer few attractions to the people of other districts. They cannot stand the climate of the Sundarbans as well as the natives of Backergunge, and the new settlers are mainly people from other parts of the district, who have either lost their old homes by diluvion are attracted by low rentals. Even the Maghs, who take up lands in this area, are chiefly colonists from other localities in the district, from which they have been ousted by Bengalis. The people generally are in easy circumstances. "There is no doubt," writes the Collector, "that the average Backergunge cultivator could, if he would take the trouble, be a wealthy man. In all but the most exceptional years his lands give him an abundant crop of rice with the minimum of exertion, and, in addition, his cocoanut and betelnut trees can nearly always be depended upon to give him a plentiful and valuable crop of nuts without any labour on his part beyond that of plucking them."

BACKERGUNGE.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**2,428,911**	**+ 5·98**	**+ 6·40**
Sadar Subdivision ...	**985,184**	**+ 4·21**	**+ 7·53**
Barisal	153,397	+ 6·84	+ 7·66
Gaurnadi...	263,596	+ 10·32	+ 14·79
Mehendiganj	174,154	+ 5·12	+ 6·19
Jhalokati...	167,305	- 3·95	+ 6·48
Nalchiti	82,335	+ 2·40	+ 2·55
Backergunge	144,397	+ 1·26	+ 2·10
Pirozpur Subdivision ...	**550,418**	**- 0·56**	**+ 6·52**
Pirozpur	122,935	- 6·04	+ 5·33
Sarupkati	204,669	- 0·70	+ 13·71
Matbaria	123,005	+ 7·71	- 4·32
Bhandaria	99,809	- 2·48	+ 7·99
Patuakhali Subdivision ...	**574,972**	**+ 10·01**	**+ 5·22**
Patuakhali	208,470	+ 5·67	+ 1·45
Ranphal	120,255	+ 8·75	- 0·41
Amtoli	144,952	+ 16·25	+ 11·28
Galachipa	101,295	+ 12·42	+ 13·79
Dakshin Shahbazpur Subdivision.	**318,337**	**+ 17·80**	**+ 4·56**
Bhola	167,092	+ 11·04	+ 4·24
Baranadi...	151,245	+ 26·30	+ 4·96

266. Between 1872 and 1881 the development of the district was checked by the disastrous cyclone of 1876, but in the next decade it recovered from its effects and grew rapidly, an increase of 13·3 per cent. being recorded in 1891. A further advance of 6½ per cent. was made in the decade ending 1901, the greatest growth of population taking place in the areas where reclamation was most active, viz., in the northern thanas of Gaurnadi (14·8 per cent.) and Swarupkati (13·7 per cent.) and in two of the Sundarban thanas, Amtali and Galachipa (11·3 and 13·8 per cent.). The first half of the decade ending in 1910 was a prosperous period, but in 1905 a partial failure of the early paddy, followed by a general failure of the winter crop, resulted in scarcity. Relief operations had to be started and help given in the shape of loans. Some suffering was also felt in 1909, when a cyclone, accompanied by floods, swept over the country. All parts of the district suffered more or less, but the southern portion of the Dakshin Shahabazpur, Patuakhali and Pirozpur subdivisions were especially affected. The effect of these conditions is seen in the returns of vital occurrences. In the first half of the decade the births exceeded the deaths by nearly 114,000; in the second half there was a reduced birth rate and an enhanced death rate, resulting in the excess of births being only 18,000. In the decade, as a whole, the excess amounted to 132,788, an increase of 5·8 per cent. on the population of 1901.

267. The actual increase shown by the census corresponds very closely with this figure, being 137,159 or 6 per cent. This increase is the result of natural growth, for migration has increased very little since 1901, and the immigrants outnumber the emigrants by only 16,000. The greatest development has taken place in the Dakshin Shahabazpur subdivision (17·8 per cent.), where the large increase is due to the settlement of immigrants on newly formed *chars*. The rate of growth has been slower in Patuakhali (10 per cent.), but the two Sundarbans thanas of Amtali and Galachipa have large gains owing to the expansion of cultivation and colonization. The same cause accounts for the increase in the Matbaria thana of the Pirojpur subdivision. All the other thanas in the latter subdivision have lost ground. It appears to have become more malarious: complaints are made that the water in the tidal creeks has become more brackish, and in the last five years of the decade deaths exceeded births. Conditions were more favourable in the Sadar subdivision, where the population increased by 4 per cent. Here the most progressive tract is the swampy Gaurnadi thana, where reclamation is steadily going on as fresh deposits of silt gradually replace the water and extend the cultivable area.

BACKERGUNGE.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,244,835	1,184,076	1,175,903	1,115,849
Immigrants	46,915	14,497	51,085	6,900
Emigrants	28,629	16,440	24,535	14,477
Natural population	2,226,549	1,186,019	1,149,353	1,121,426

268. Tippera consists of a flat alluvial plain broken only by the Lalmai Hills, which cover a comparatively small area. It is fertilized by the Meghna, which flows along its western border, and by a number of smaller streams that bring down silt from the hills to the east. In the tract watered by the Meghna the soil is admirably suited to the cultivation of jute, while the remainder of the district forms one of the most important rice producing tracts in Bengal. In point of climate Tippera occupies the first place in North and East Bengal, its death rate from fever being the lowest in these two divisions. The population increased rapidly between 1872 and 1901, the rate of growth increasing at each successive

TIPPERA.

TIPPERA.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**2,430,138**	**+ 14·74**	**+ 18·79**
Sadar Subdivision ...	**1,099,050**	**+ 14·76**	**+ 16·61**
Comilla (Kotwali)	198,058	+ 11·61	+ 12·38
Daudkandi	243,487	+ 17·96	+ 19·27
Muradnagar	234,622	+ 14·61	+ 15·72
Chandina	130,961	+ 18·30	+ 18·64
Chanddagram	146,625	+ 9·78	+ 14·46
Laksam	145,297	+ 16·38	+ 20·74
Brahmanbaria Subdivision.	**757,283**	**+ 11·84**	**+ 14·74**
Brahmanbaria	335,490	+ 11·48	+ 14·43
Kashba	159,463	+ 11·02	+ 12·82
Nabinagar	262,340	+ 12·83	+ 16·35
Chandpur Subdivision ...	**573,805**	**+ 18·75**	**+ 30·05**
Chandpur	247,036	+ 18·23	+ 36·76
Hajiganj	167,300	+ 20·70	+ 26·81
Matlabganj	159,469	+ 17·56	+ 23·93

census and reaching 18·8 per cent. in 1901. The next decennium opened in a cycle of general prosperity. The jute industry was thriving, and trade of all sorts expanded with the advent of the railway, for the Assam-Bengal Railway was opened in 1895, while a branch line running from Laksam to Noakhali was added in 1903. This period culminated in 1904-05, when exceptional prices were realized for a fine harvest of jute, and rice was so cheap that it sold at 14 to 15 seers per rupee. In 1906, however, heavy floods caused extensive loss of both the rice and jute crops, and the distress was enhanced by a fall in the price of jute. Two lean years followed, but there were good harvests in 1908-09. In 1910 the floods were abnormally high and prolonged and did much damage, especially in the north of the district.

269. The result of adverse conditions during the latter half of the decade was to reduce the rate of increase to 14·7 per cent., the actual increase being 312,147. It would, however, have been greater had not the balance of migration been against the district, the number of emigrants rising from 55,529 to 95,757, while immigrants only increased from 56,752 to 60,360. Emigration is especially active in the Sadar subdivision, from which people move freely to Hill Tippera, where they can obtain land on easy terms. The increase is shared by all the subdivisions and by all the thanas within them, the range of variation being comparatively small. The greatest advance has been made by the Chandpur subdivision to the south and the least by the Brahmanbaria subdivision to the north, the Sadar subdivision, which lies between them, having an intermediate position. In Tippera, as elsewhere, the Musalmans are multiplying more rapidly than the Hindus, the rate of increase of the latter (8 per cent.) being less than half what it is among the former.

TIPPERA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,243,081	1,187,057	1,085,989	1,032,002
Immigrants	34,363	21,997	37,095	19,657
Emigrants	58,737	37,020	30,367	25,162
Natural population	1,263,455	1,202,080	1,079,261	1,037,507

270. Noakhali consists of a tract of mainland together with a number of islands in the mouth of the Meghna, the largest of which are Sandip and Hatia. The mainland is a flat alluvial plain broken only by a hilly tract in the extreme north-east corner. The soil is fertile throughout the district, the lands subject to the direct fluvial action of the Meghna, being especially rich, although subject to sudden changes, as that river cuts away the land in one place, and builds it up in another. The district is one of the most densely populated in Bengal, and since 1901 the area under cultivation has increased by 15 per cent.: the limits within which further extension is possible have almost been reached. Rice covers about three-fourths of the district and the annual yield is estimated at ten million maunds, of which about six millions are required for local consumption, leaving a balance of four millions for export. The area under jute has risen from 7,000 acres to over 30,000 acres since 1901, and the profits derived from its sale are considerable : it is reported that in 1911 about 400,000 maunds of jute were exported, and that the cultivators obtained at least 24 lakhs of rupees from its sale. Besides this, betelnut and cocoanut plantations cover a large area and yield a handsome profit, the value of their produce being estimated at 25 to 30 lakhs of rupees. These resources place the peasantry above the margin of want, and they are, in fact, in easy circumstances.

NOAKHALI.

NOAKHALI.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	***1,302,090***	***+14·05***	***+13·08***
Sadar Subdivision ...	***961,527***	***+16·85***	***+14·42***
Sudharam (Noakhali)	194,695	+ 10·55	+ 3·90
Lakhipur	204,398	+ 22·74	+ 18·95
Begamganj	224,046	+ 17·88	+ 18·83
Ramganj	143,181	+ 19·60	+ 20·44
Sandip	119,652	+ 3·93	+ 13·98
Hatia	75,605	+ 36·50	+ 12·25
Feni Subdivision	***340,563***	***+ 6·81***	***+ 9·74***
Chhagalnaia	138,292	+ 2·20	+ 8·87
Feni	202,271	+ 10·22	+ 10·39

271. When the census of 1881 was taken, the population of the district had been reduced by the disastrous cyclone of 1876, when the island of Hatia lost a quarter and Sandip a sixth of its inhabitants. This cyclone was followed by a terrible epidemic of cholera, and the mortality from both causes was estimated at 100,000. The result was a decrease of 2·3 per cent., but since 1881 the population has been growing very rapidly, an increase of 23 per cent. being recorded in 1891 and of 13 per cent. in 1901. Since 1901 there has been a series of prosperous years, except 1906, when there was a failure of the crops and relief operations had to be undertaken. Some damage to the crops was also caused by heavy floods in 1909, when fever broke out in a virulent form, causing a mortality of 33,817 as against 26,670, the average for the decade. In the whole decade the reported births exceeded the deaths by 165,754, representing an increase of 14·5 per cent. on the population of 1901.

272. The census shows that the population is greater by 160,362, or 14 per cent., than it was at the last census, in spite of a slight loss by emigration. Emigrants are now more numerous by 15,000 than in 1901, but nearly half are temporary absentees, males being in excess of females by 23,242. The Sadar subdivision, which is less fully developed than the Feni subdivision, has grown more than twice as fast as the latter. There are, however, some remarkable variations in the rate of growth, for while Sandip has an addition of less than 4 per cent., the abnormal figure of 36½ per cent. is reached in Hatia. Such variations, extraordinary as they may appear, are due to the changes wrought by the constructive and destructive powers of the great rivers and the consequent shifting of the population. Chhagalnaia, a densely populated thana, which in 1901 had 1,033 persons per square mile, has only a slight increase (2 per cent.), but the pressure on the soil has been no check to growth in Begamganj or Ramganj, of which the former had 864 and the latter 1,023 persons to the square mile in 1901 : they now have a gain of 18 and 20 per cent. respectively.

NOAKHALI.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	645,898	656,192	568,777	572,951
Immigrants	12,777	10,067	10,775	8,568
Emigrants	35,915	12,673	26,139	7,444
Natural population	669,036	658,798	584,141	571,827

273. Chittagong consists of a long narrow strip of coast, valleys and low ranges of hills lying between the Bay of Bengal and the Chittagong Hill Tracts; its average breadth is only about 15 miles, while its length is 165 miles. The soil is fairly fertile, but malaria is more prevalent than elsewhere in East Bengal: between 1901 and 1910 it had the highest death rate from fever of all the districts in the Division. It is also exposed to cyclones, and its census history is one of fluctuation caused by disastrous cyclones and epidemics of cholera consequent on the pollution of the water supply. The first cyclone occurred in 1876, and was accompanied by a storm-wave, which swept the sea board : 12,000 persons were drowned, and 15,000 perished in the cholera epidemic which followed. The loss of population caused by this calamity, by the ravages of disease in other years, and by emigration, reduced the growth of population to less than one-half per cent. in

CHITTAGONG.

CHITTAGONG.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	*1,508,433*	*+11·47*	*+ 4·89*
Sadar Subdivision	*1,259,842*	*+ 9·26*	*+ 4·62*
Chittagong (Kotwali)	113,137	+ 13·89	+ 10·79
Mirsarai	112,918	+ 10·42	+ 9·31
Sitakund	76,344	+ 3·10	+ 9·37
Hathazari	98,788	+ 4·46	+ 5·82
Phatikchari	117,441	+ 4·57	+ 2·44
Raozan	177,214	+ 7·73	+ 2·91
Pattya	238,682	+ 7·87	+ 7·12
Satkania	172,976	+ 10·45	- 1·09
Banskhali	152,342	+ 18·86	+ 0·29
Cox's Bazar Subdivision	*248,591*	*+24·19*	*+ 6·47*
Cox's Bazar	88,360	+ 16·32	+ 9·91
Moiskhali	47,971	+ 37·37	- 7·35
Chakaria	61,314	+ 29·66	+ 1·51
Teknaf	50,946	+ 21·30	+ 21·29

1881. The next ten years, however, were healthy and prosperous, and a considerable increase of population was recorded in 1891, the number of inhabitants being 13·9 per cent. more than in 1881. In the next decade again the district suffered from a destructive cyclone, which burst in October 1897. A series of storm-waves swept over the island of Kutubdia and the villages on the mainland, drowning many thousands of men and cattle, sweeping away homesteads and destroying standing crops: the loss of life by drowning alone was estimated at 14,000 souls. Cholera broke out in a severe form, and in Kutubdia alone it was estimated that more than one-tenth of the population died during the epidemic. The result was that in 1901 an increase of only 4·9 per cent. was recorded, which is about half what it would have been but for the cyclone and its after effects. The greatest growth occurred in the thanas along the coast which escaped the brunt of the cyclone, viz., Teknaf and Cox's Bazar in the south, and Chittagong, Sitakund and Mirsarai in the north.

274. Since 1901 the district has been free from any such calamity. There was, however, a partial failure of the rice crop over the lowlying tracts of the district, which were affected by heavy floods in 1906. "The people generally," writes the Collector, "are prosperous. In normal years, they get two crops of paddy from their fertile soil, and can obtain house materials and fuel from the neighbouring hills at a nominal price. Those who have no lands of their own and are not so well off go down in thousands in December to cut paddy in Aracan, where they earn a rupee a day, and return in February and March with large sums of money." The effect of these favourable conditions is seen in the addition of 155,183 persons or 11½ per cent., a result which is not due to the accretion of population from outside, for the number of immigrants is only 18,701 or about 7,000 more than in 1901. There has also been a small gain from a slightly diminished exodus of the district-born, the number of whom has fallen from 106,037 to 99,627. In 1901 the figures were inflated by the outward movement of the people, after the cyclone, and also by a poor rice crop in some parts. The emigration is moreover of a temporary character, and does not represent a permanent loss, most of the emigrants being labourers engaged in cutting rice crops in Burma, or lascars on steamers engaged in the eastern trade, who leave their wives at home: the excess of males over females in the emigration population is no less than 81,733.

CHITTAGONG.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	722,837	785,596	641,392	711,858
Immigrants	12,618	6,083	7,029	4,310
Emigrants	90 680	8.947	85,804	20,233
Natural popu'ation	800,899	788,460	720,167	727,781

275. The local variations exhibit very clearly how completely the tracts affected by the cyclone have recovered from its effects. Maiskhal, which in 1901 had a decrease of 7 per cent., has now gained 37 per cent. Chakaria and Banskhali, which were stationary, have an increase of 30 and 19 per cent. respectively. Satkania, where the loss was partly due to the cyclone and partly to enhanced emigration, has advanced by 10½ per cent. Of the two subdivisions, Cox's Bazar has made most progress, this being a sparsely populated area, which is fast developing as cultivation expands. Here the growth has been greatest in the two thanas Maiskhal and Chakaria, which suffered from the cyclone, but Cox's Bazar and Teknaf have also large increases. In the Sadar subdivision the most progressive thanas are those that have recovered from the cyclone and the two northern thanas of Chittagong and Mirsarai. Chittagong owes its expansion largely to the development of Chittagong town, while Mirsarai has a more fertile soil than the inland thanas and has benefited from railway communication. The remaining thanas have a more or less uniform rate of growth, varying from 3 to 8 per cent.

CHITTAGONG HILL TRACTS.

276. The greater part of the Chittagong Hill Tracts consists of hills and ravines covered with dense jungle. The aboriginal tribes, who constitute over 94 per cent. of the population, live chiefly by *jhuming*; and plough cultivation has not made much progress. The area so cultivated is not more than 1 per cent. of the

total area, while the net cropped area only slightly exceeds 5 per cent. Reserved forests alone extend over 1,020 square miles or one-fifth of the district. The district being sparsely inhabited, and the people consisting of hardy aboriginals, they have been steadily adding to their numbers. A phenomenal increase of 46 per cent. was recorded in 1881, but this is merely a proof of the incompleteness of the first census. It is, in fact, known to have been vitiated by the fact that, the Chief's revenue being based on capitation tax, it was to his interest to return a small population. With improved enumeration, the rate of increase was reduced to 5·6 per cent. in 1891, but rose again to 16·3 per cent. in 1901. Since 1901 the history of the district has been uneventful. The health of the people was good, and the only year in which there was any shortage of the crops was 1906, when there was some scarcity necessitating the grant of loans.

CHITTAGONG HILL TRACTS.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	***153,830***	***+ 23·30***	***+ 16·29***
Chakma	77,155	+ 58·14	+ 7·1
Mong	27,613	— 13·43	+ 40·4
Bohmong	49,062	+ 11·81	+ 12·9

277. The rate of increase according to the present census is 23·3 per cent., the actual increment being 29,068. The returns of birthplace are not altogether reliable: such as they are, they show very little variation compared with 1901, and the increase must therefore be attributed to natural growth. The local variations are extraordinary. Chakma, which both in 1891 and 1901 increased by 7 per cent., has now a sudden rise of 58 per cent. Mong, which declined by 40 per cent. between 1891 and 1901 and increased at the same rate in 1901, has now a loss of 13 per cent. Bohmong has an increase of 11 per cent., which is very nearly the same as was recorded in 1901. These variations are ascribed to the nomadic habits of the people, who move from place to place as they *jhum*: it is reported that that a large number migrated from the Mong circle to the Mioni valley in the Chakma circle. It is further reported that there has been immigration from Hill Tippera, though this is not apparent in the returns of birthplace. It is possible that the changes may be due, in part at least, to differences in the dividing line between the circles as drawn at different censuses.

CHITTAGONG HILL TRACTS.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ...	82,695	71,135	68,238	56,524
Immigrants	5,050	1,245	6,337	527
Emigrants	680	591	1,109	762
Natural population ...	78,325	70,478	63,010	56,759

HILL TIPPERA.

278. The State of Hill Tippera consists of several ranges of hills, running north and south with an average interval of 12 miles, and increasing in height towards the east. The hills are clothed for the most part by bamboo jungle, while the low ground is covered with tree jungle, cane bush and thatching grass. The nomadic cultivation known as *jhuming* is almost universal in the hills, plough cultivation being confined to the plains, and in particular to the narrow strip adjoining British territory. The first census of the State was admittedly incomplete, and that of 1881 was also probably inaccurate, so that the abnormal increase of 171 per cent., recorded in 1881 and the very high rate of 44 per cent. returned in 1891 must be discounted. The first reliable census was that of 1901, according to which the number of inhabitants was 26 per cent. more than ten years before.

HILL TIPPERA.	Population, 1911.	PERCENTAGE OF VARIATION. 1901—1911.
DISTRICT TOTAL	***229,613***	***+32·46***
Sadar Subdivision ...	***74,672***	***+13·80***
Kailashahar	31,609	+52·90
Khowai	21,398	+107·85
Dharmanagar	19,056	+87·37
Sonamura	18,062	+48·30
Udaypur	40,114	
Belonia	19,167	-9·66
Subram	5,515	

279. Since 1901 uniform and steady progress has been made. The revenue has been doubled, communications improved, and the reclamation of cultivable waste has proceeded rapidly, attracting numerous new settlers. There were no epidemics of disease, and crops were on the whole good. The census recently concluded shows that the population has increased by 56.288 per cent. or 32½ per cent. Large as this increment is, it is accounted for partly by the natural growth of the people, but mainly by the influx of immigrants, who are more numerous by 37,769 than they were 10 years ago: over one-third of the population consists of persons born outside the State. They include a large number of Muhammadans from the adjoining British districts of Tippera and Sylhet: owing to this addition to their numbers, the Muhammadans have increased by 43·3 per cent., a ratio double that of the previous decade. The increase in population is shared by all parts except Bilonia and Sabrum, the decrease in which is attributed to the movement of the people to Udaypur and across the border into the Chittagong Hill Tracts.

Hill Tippera.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ...	121,820	107,793	92,495	80,830
Immigrants	50,761	30,902	24,828	19,066
Emigrants	703	664	88	64
Natural population ...	71,767	77,555	67,755	61,828

SOUTH BIHAR.

Patna.

280. The census of 1881 disclosed an increase of 12·6 per cent., but this apparently large increase was due, in part at least, to the incompleteness of the first census held in 1872. In 1891 the population was found to be stationary, the recorded increase being barely 1 per cent., while in 1901 there was a decrease of 8·4 per cent. This decrease was directly due to the direct and indirect losses caused by plague, viz., mortality, the flight of the district-born, the absence of those who had homes in other districts and the disorganization of the census staff, which led to a defective enumeration. Plague appeared in epidemic form in January 1900, broke out again in the next cold weather and was at its height at the time the census was taken.

Patna.	Population, 1911.	Percentage of variation.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**1,609,631**	**— 0·93**	**— 8·4**
Patna City Subdivision* ...	**140,063**	**— 0·65**	**— 21·76**
PATNA CITY*	**136,153**	**+ 1·01**	**— 18·41**
Pirbahor Ward 1	33,435	...	...
Alamganj 2	20,334	...	...
Khwaja Kalan 3	33,308	...	...
Chauk Kalan 4	19,880	...	...
Mulsalami 5	14,6?4	...	...
Rural area	18,422	...	...
Sadar Subdivision	**201,193**	**— 0·29**	**— 10·45**
Phulwari	89,056	— 1·46	— 9·94
[Bakarganj]*	[14,512]	...	...
Masaurhi—Buzurg	112,137	+ 0·67	— 11·37
Dinapore Subdivision ...	**314,644**	**— 0·33**	**— 10·36**
Maner	80,248	— 7·87	— 11·57
Dinapore	62,328	— 4·01	— 11·19
Bikram	172,068	+ 5·17	— 9·36
Barh Subdivision	**378,621**	**+ 3·64**	**— 10·52**
Fatua	74,757	+ 2·96	— 18·66
Barh	197,814	+ 1·01	— 4·41
Mokameh	106,050	+ 9·45	— 15·10
Bihar Subdivision	**575,110**	**— 4·30**	**— 0·95**
Bihar	335,939	— 4·81	+ 0·09
Hilsa	161,564	— 4·03	— 2·39
Islampur	77,607	— 2·65	— 2·36

281. Conditions during the ten years which have since elapsed were not such as to give any hope of recovery. Plague continued to levy a heavy toll year after year: in three years only was the mortality ascribed to it under 3,000, and in four years (1901, 1904, 1905 and 1907) it was over 20,000. Altogether, over 140,000 deaths have been caused by this

* For municipal purposes the Bakarganj police-station forms part of Patna City, but for general administrative purposes it forms part not of the Patna City subdivision but of the Sadar subdivision. The figures relating to it are therefore included in those for Patna City, but are excluded from those for the Patna City subdivision.

scourge, representing a death-rate of 8·7 per cent. on the population of 1901: it is probable that the actual number was even greater owing to deaths from plague being returned under the comprehensive head of fever. Epidemics of cholera have also been frequent, that of 1910, which caused nearly 14,000 deaths, being specially virulent: the aggregate number of deaths due to this disease during the decade was over 50,000. Altogether, the number of deaths from all causes exceeded the number of reported births by no less than 111,632. The birth-rate has risen, the average being a little under 41 per mille as compared with 38 per mille in the preceding five years. But it has failed to keep pace with the death-rate, the average for which is 47·50 per mille, the highest ratio in either Province: only twice, and then only slightly, have the births in any one year outnumbered the births. There has, moreover, been no commercial or industrial development which would attract population from outside. The Bihar-Bakhtiyarpur Railway has, it is true, helped to open up the south-west of the district, but, on the other hand, the towns are decadent and the river-borne trade, on which they largely depended, is dwindling. There were partial failures of crops in four years, and in 1901, 1905 and 1909 there were floods. That of 1901, which inundated the country all along the Ganges, was the greatest within living memory, but it subsided rapidly and did very little damage. The inundation of 1909, however, had serious effects on the *bhadôi* crops of the Masaurhi thana.

282. The census discloses a decrease of 15,111, or a little under 1 per cent. That the decrease should be so small in view of the adverse circumstances sketched above is partly due to the fact that at the time of the census there was a large influx of labourers from other districts, who came to cut *rabi* crops in the low-lying lands in the Mokameh thana. Owing largely to this accession, the number of immigrants is 10,000 more than in 1901, but it is still 43,000 below what it was in 1891. The volume of emigration has increased to a far greater extent, the number of persons born in the district who were enumerated outside it rising in the last ten years by 31,000. There was a widespread epidemic of plague from December until after the census was taken—8,000 deaths from plague were reported in January and February 1911—and there was a certain amount of desertion on that account. It did not, however, interfere with the completeness of the census, for those enumerators or supervisors who fled from their villages either provided substitutes—the well-to-do paid for their services—or returned to assist both in the preliminary enumeration and the final census. There was, in fact, only one case in which a break-down of the census organization seemed imminent. Patna city had been free from plague till the first week in March, when it broke out in two wards. Some of the enumerators in one of these wards absconded a few days before the final census, but their schedules were recovered and a special staff deputed to fill their places. The census organization stood the strain put upon it, and no part of the decrease can be ascribed to a failure in the enumeration.

PATNA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	809,778	799,853	804,450	820,292
Immigrants	38,241	53,865	32,262	50,175
Emigrants	95,459	78,456	73,982	68,334
Natural population	866,996	824,444	846,170	838,448

283. All parts of the district share in the decrease or are stationary, except the Barh subdivision and the Masaurhi thana in the south of the Sadar subdivision. In the former, however, the increment is mainly due to the influx of labourers already referred to, as a result of which the Mokameh thana has an increase of over 9,000, or over three-fourths of the increase of the whole subdivision. The Masaurhi thana is a fertile tract traversed by the railway, and it is noticeable that the adjoining thana of Jahanabad is the most progressive area in the Gaya district. Taking the district as a whole, the population is now 8 per cent. less than it was in 1881, and there seems, under present conditions, but little hope of recovery.

GAYA.

284. Between 1881 and 1891 the population of Gaya was practically stationary, the increase recorded in 1891 being only 0·6 per cent. The reason for this slow

growth appears to be that the district suffered from the ravages of fever and that emigration increased greatly, while immigration fell off. The result of the census of 1901 was even more unsatisfactory, for the population decreased by over 78,000 or 3·7 per cent. Not only was the decade 1891—1901 an unhealthy period, but there were two years of scarcity. A virulent epidemic of plague also broke out towards the end of 1900, causing heavy mortality and a general panic, which drove large numbers from their homes. In one thana alone (Tekari) it was found that 11,000 people had fled outside its boundaries between the preliminary enumeration and the final census.

GAYA.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**2,159,498**	+ **4·83**	- **3·67**
Sadar Subdivision ...	***785,334***	+ ***4·74***	- ***9·68***
Gaya Town	49,921	- 29·97	- 11·31
Gaya	191,484	+ 6·60	- 6·55
Atari	94,375	+ 3·36	- 14·88
Tekari	165,610	+ 11·80	- 19·78
Barachati	113,968	+ 5·61	- 2·35
Sherghati	169,976	+ 12·18	- 1·43
Jahanabad Subdivision	***422,287***	+ ***9·33***	- ***1·85***
Jahanabad	289,196	+ 7·27	- 3·95
Arwal	133,091	+ 14·06	+ 3·25
Aurangabad Subdivision	***490,881***	+ ***4·52***	- ***1·02***
Daudnagar	122,394	+ 6·28	+ 6·06
Nabinagar	101,421	- 1·63	- 0·50
Aurangabad	267,066	+ 6·23	- 4·15
Nawada Subdivision ...	***460,996***	+ ***1·49***	+ ***3·25***
Nawada	295,063	- 0·94	+ 4·63
Rajauli	77,290	+ 4·87	+ 0·46
Pakribarawan	88,643	+ 7·24	+ 1·04

285. The plague epidemic of 1901 caused over 10,000 deaths, and since then there have been severe epidemics in the four years 1904—1908, the worst year on record being 1905, when there was a death-roll of over 16,000. In the other five years of the decade the district has been almost free from the pestilence, the aggregate mortality being under 1,400. Gaya has thus suffered far less than Patna, the total number of plague deaths being only a little over 41,000, or less than one quarter of those recorded in the latter district. There have, moreover, been no serious epidemics of cholera: the number of cholera deaths in each year has been under 1,000. The total number of recorded births has exceeded the deaths by 70,000. Except in the three years 1906-08, when deficient or unevenly distributed rainfall resulted in poor harvests, the outturn of the crops was fair. There has been no noticeable development of industries, but the interior of the district has been opened out and emigration stimulated by two new lines of railway, viz., the Grand Chord and Barun-Daltonganj lines.

286. The present census discloses an increase of 99,565 or 4·83 per cent., the result of natural growth and a return to normal conditions. Plague, it is true, prevailed in the early part of 1911, causing 1,000 deaths before the census. When it was taken, the disease raged in the towns of Gaya and Jahanabad, and also in several villages in the interior. Nearly all the inhabitants of Jahanabad encamped in huts outside the town. In Gaya town the people, who had good cause to remember the terrible epidemic of 1900-01, were panic-stricken, and a large number left the town. The result was a loss of over 20,000, which was made good at a second census held three months later. Apart from this disturbing influence, the general increase would have been greater, had it not been for the persons who left the district for employment elsewhere. The number of the district-born who were enumerated elsewhere has risen by 31,000 since 1901, and, even after allowing for an increase of 15,000 in the number of immigrants, there is a balance of 16,000 against the district. All but two thanas share in the increase. The effects of recovery from the effects of plague are specially noticeable in the Tekari thana, which now records a growth of 12 per cent. in place of a decline of 20 per cent. The expansion of the Arwal thana in the north-west has been even more rapid, but, as stated

GAYA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,061,291	1,098,207	1,011,271	1,048,663
Immigrants	22,104	39,290	18,875	27,239
Emigrants	116,911	87,653	96,530	76,939
Natural population	1,156,098	1,146,570	1,088,926	1,098,362

elsewhere, this is a fertile canal-irrigated area, where the population is steadily growing. Arwal and the adjoining thana of Daudnagar were the only thanas outside the Nawada subdivision (which was then free from plague) which had any increment in 1901. The Jahanabad subdivision, in which the Arwal thana is included, is the most progressive part of the district. The least progressive is the Nawada subdivision, which was the only subdivision with any increase of population in 1901. Here there is a slight decline in the Nawada thana, which may be ascribed to the fact that in 1901 its population was temporarily swelled by plague refugees. The only other thana with a falling off is Nabinagar in the south-west, an infertile tract, where the soil is poor and there is practically no irrigation, so that the crops are scanty at the best of times.

SHAHABAD.

287. Between 1872 and 1891 the population of Shahabad grew steadily, owing largely to the development of cultivation and the influx of immigrants caused by the opening of the Son Canals. If, however, immigrants are excluded, the district actually lost as a whole rather than gained in the decade ending in 1891, owing to persistent fever which was never absent since the year 1879. During the next decade the district was visited by famine, while fever continued to cause great mortality. The census of 1901 disclosed a decrease in the population amounting to 97,883 or 4·8 per cent., which was partly the result of the adverse conditions of the preceding years, and partly due to the loss of temporary immigrants. An epidemic of plague broke out in the north-east of the district shortly before the census, and the fact that the number of foreign-born males enumerated in the district decreased by over 45,000 must, in part at least, be ascribed to the panic it caused. The falling off was most marked in the Bhabhua subdivision, where it was due to the unhealthiness of the climate and to the migration of the people to more favoured parts of the district, especially during the famine of 1896-97. Elsewhere the decrease was greatest in the Arrah thana, where it was ascribed to the exodus on account of the plague. The only thanas that gained ground were Sasaram, Bikramganj and Dehri, three of the four thanas forming the Sasaram subdivision. The area under irrigation is greater here than in other parts of the district, and the construction of the Mughalsarai-Gaya Railway caused an influx of labour. On the other hand, Piru thana in the headquarters subdivision, which borders on this tract and which also has a plentiful supply of canal water, was practically stationary.

SHAHABAD.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**1,865,660**	**- 4·94**	**- 4·8**
Sadar Subdivision	**631,226**	**- 9·82**	**- 5·87**
Arrah Town	38,549	- 16·51	- 9·40
Arrah	235,564	- 11·98	
Shahpur	161,804	- 11·34	- 5·43
Piro	195,309	- 4·05	+ 2·83
Buxar Subdivision	**382,971**	**- 8·10**	**- 5·02**
Buxar	149,840	- 4·35	- 6·02
Dumraon	234,131	- 10·05	- 4·42
Sasaram Subdivision	**544,374**	**+ 0·88**	**+ 1·18**
Bikramganj	192,531	+ 3·24	+ 2·27
Khargahar	99,561	+ 2·92	- 6·59
Sasaram	160,583	- 3·41	+ 1·33
Dehri	91,699	+ 1·70	+ 8·14
Bhabhua Subdivision	**307,089**	**+ 0·22**	**- 11·16**
Mohania	136,567	+ 2·32	- 13·66
Bhabhua	170,522	+ 2·89	- 8·92

288. The decade 1901-1910 was a very unhealthy period in Shahabad. Plague was present throughout the ten years, carrying off 68,000 persons, and there were also virulent epidemics of cholera in no less than six years, the mortality being 18,000 in 1910 alone and over 60,000 in the whole decennium. Fever prevailed and was especially virulent in 1905, when the death-rate returned as due to it was no less than 37·8 per mille: the total death-rate for this year reached the appalling figure of 58·65 per mille. Owing to these epidemics, the natural growth of the population was retarded, and the returns show that the net excess of births over deaths was less than 20,000. The agricultural classes, moreover, had to contend against a succession of bad harvests. The decade opened unfavourably, the outturn of crops being less than half the normal, and the strain on the cultivators' resources was all the greater because this was their second bad season in succession.

Good or fair harvests were reaped during the next four years, but in 1906-1907 they were again short, and in the two years 1907-1909 the failure was serious, the outturn of rice being only half the normal. The good rainfall of 1910 however improved the situation and averted a threatened scarcity.

289. The effect of the unfavourable conditions which prevailed is seen in the results of the census, which has brought to light a decrease of 97,036 or nearly 5 per cent. This diminution of population is due largely to the stimulus given to emigration by the stress of bad years, as well as to the continued check of immigration into the district. The number of male emigrants has increased by over 17,000 since 1901, while the immigrant population has declined by over 24,000; the number of foreign-born males enumerated in the district is now only a quarter of what it was in 1901. The loss has been greatest in the Sadar and Buxar subdivisions to the north, where every thana has receded, the rate of decrease varying from 4 per cent. to 12 per cent. The Sasaram and Bhabhua subdivisions to the south are practically stationary, but every thana in them has progressed slightly, except Sasaram and Mohania, both sparsely populated thanas with a large area of hill and jungle.

SHAHABAD.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	904,950	960,710	988,544	1,026,152
Immigrants	18,642	41,965	26,877	57,036
Emigrants	99,631	76,615	82,579	77,134
Natural population	985,989	995,360	992,246	1,046,250

290. Between 1872 and 1881 there was a growth of population, which was sustained during the next decade, the census of 1891 showing an advance of 3·4 per cent. The greatest gain was in the Begusarai subdivision, but in the Jamui subdivision there was practically no increase and in the north-east of that subdivision there was a heavy loss due to the ravages of fever. At the census of 1901 it was found that the population had only increased by 1·6 per cent., a result attributed to a severe epidemic of plague in 1900-01 and, in a minor degree, to emigration from the district. Monghyr was, however, the only district in South Bihar which escaped a loss of population, a result due to the fact that to the north of the Ganges there is a rich alluvial tract, which attracts immigrants and was free from plague in 1901. The portion south of that river sustained a small loss, an increase in four thanas having been more than obliterated by a heavy loss in the areas where plague had appeared, viz., the town of Monghyr and its environs, and two thanas in the west, to which the epidemic spread from the Patna district.

MONGHYR.

MONGHYR.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL ...	**2,132,893**	+ **3·10**	+ **1·61**
Sadar Subdivision ...	**1,094,563**	+ **4·16**	+ **0·44**
Gogri	422,859	+ 6·01	+ 2·55
Monghyr	154,447	+ 18·78	- 9·51
Jamalpur	20,526	+ 8·27	- 14·41
Surajgarh	74,580	- 12·83	Not available.
Lakhisarai	129,733	+ 9·31	
Kharagpur	149,092	- 5·52	+ 6·84
Sheikhpura	143,326	+ 1·72	- 7·30
Begusarai Subdivision	**651,765**	+ **1·37**	+ **5·17**
Teghra	230,522	- 1·36	+ 7·76
Begusarai	421,243	+ 2·93	+ 3·74
Jamui Subdivision ...	**386,565**	+ **3·08**	- **0·49**
Sikandra	107,418	- 0·66	- 1·60
Jamui	143,295	+ 20·17	+ 6·62
Chakai	135,852	- 7·97	+ 4·76

291. The result of the census of 1911 is an increase of 64,089 or 3·10 per cent. The preceding decade was, on the whole, healthy. The births outnumbered the deaths in every year except 1907, the net excess being 137,000, or double that returned for any other district in South

MONGHYR.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,043,477	1,089,416	1,011,580	1,057,224
Immigrants	36,341	56,798	37,524	44,545
Emigrants	129,650	110,054	96,544	87,575
Natural population	1,136,786	1,142,672	1,070,600	1,096,254

Bihar. Plague was present throughout these ten years and affected nearly all parts except some tracts north of the Ganges, but the mortality never reached such a high figure as in other districts of South Bihar. The maximum number of plague deaths was 11,000 in 1905, and the aggregate for the decade was 43,000, or 4,000 less than the number of deaths caused by cholera. Plague also prevailed in the early part of 1911, causing 3,300 deaths in January and February, and was raging in parts of the Sadar and Begusarai subdivisions (particularly in the Sheikhpura, Surajgarh, Gogri, Lakhisarai, Begusarai and Teghra thanas) at the time of the census; but in spite of the great difficulties it caused, the final census was carried out smoothly, and no loss was caused by omissions from the returns. The people have now become accustomed to this disease, and it no longer inspires the same wild terror. They move out of their villages and encamp near the affected area, but do not fly far and wide as they did when it first appeared. The harvests were, on the whole, fair, though the south of the district, particularly the Jamui subdivision, suffered from drought in some years. Though the northern portion of the district is not, as a rule, liable to suffer much from short and unseasonable fall, it is exposed to floods from the overflowing of the Ganges and its affluents. Severe floods occurred in 1904 and 1906, but fortunately that of 1904 took place when most of the *bhadoi* crops had been harvested, and in 1906, when considerable damage was caused in the Gogri and Khagaria thanas, it was not found necessary to institute relief measures, the people being enabled to tide over their losses by means of agricultural loans. The north of the district benefited by the opening of the Mansi-Bhaptiahi line in 1908, and in the same year the Peninsular Tobacco Company opened a factory at Monghyr for the manufacture of cigarettes. This factory employs over 1,000 hands and is the only new industrial concern of any importance.

292. The increase now recorded must be attributed to natural growth. The population has received a comparatively small accretion from outside. On the other hand emigration has developed to a remarkable extent: the number of the district-born who were temporarily or permanently resident outside its limits has risen by nearly 55,000 or 30 per cent. since 1901. At the time of the census there was a large influx of temporary labourers engaged in cutting crops in the *tals* of the Lakhisarai thana. These are low-lying areas which are covered with water during the rains, but are brought under cultivation in the winter season when the water dries up. The crops are ready for harvest about the end of February or beginning of March, when crowds of labourers come and reap them, encamping out in the open until the harvest is complete. Over 8,000 such temporary labourers were enumerated in the Lakhisarai thana, where their presence mainly accounts for the increase of 9 per cent.

293. The Sadar subdivision has a net increase of 4·16 per cent., but its development is by no means uniform. The Kharagpur thana has declined by over 5 per cent., but this decline is more apparent than real, for in 1901 it harboured a considerable number of plague refugees from Monghyr thana, which is now more populous by nearly 19 per cent. than it was in that year. On account of the partial depletion of the latter thana and the temporary addition to the population of the former at the last census, the figures of 1901 are misleading, and it is necessary to go back to 1891 to see how far they have advanced or stood still. Such a comparison shows that Monghyr has grown by 8 per cent., while Kharagpur is practically stationary. Surajgarh to the east of Monghyr shows a falling off of a little under 13 per cent., which is partly due to the incidence of plague and partly to the migration of labourers to the *tals* in Sheikhpura. In the Jamui subdivision to the south the Jamui thana has expanded rapidly owing to the spread of cultivation, but Chakai has lost population, for which emigration is partly responsible, while the Sikandra thana, which was decadent from 1881 to 1901, is stationary. The Begusarai subdivision exhibits only a slight advance, for though the Begusarai thana has developed, Teghra, which has suffered from plague, has lost ground.

NORTH BIHAR.

294. The population of Saran grew steadily until 1891, there being an increase of 10½ per cent. in 1881 and a further advance of 7·4 per cent. in 1891. The next decade witnessed a set-back, for in 1901 the population declined by 2·3 per cent., owing to four causes, viz., famine, a consequent reduction of the birth-rate, plague and emigration. There was famine in 1897, which reduced the vitality of the people and lowered the birth-rate, though it was not directly responsible for any great mortality. By the time of the census the people had recovered from the results of the famine, but unfortunately plague broke out in epidemic form during the winter of 1899-1900 and re-appeared with renewed intensity in the succeeding cold weather. The decline of population was general in the Sadar subdivision and greatest in those thanas where plague was worst. The Darauli thana in Siwan also declined, but the rest of the district had the same population as in 1891. The conclusion arrived at by Mr. Gait was that, while the general want of progress was due to the adverse balance of migration and to the fact that Saran was unable to support a much greater population than it already possessed in 1891, plague was to blame for the greater part of the decrease that had occurred in the southern part of the district.

SARAN.

SARAN.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891-1901.
DISTRICT TOTAL ...	**2,289,778**	**-4·94**	**- 2·3**
Sadar Subdivision ...	**894,248**	**-6·64**	**-5·53**
Manjhi...	121,247	-12·57	- 2·42
Chapra	290,096	- 2·45	- 7·30
Chapra Town	42,373	-23·86	
Parsa	213,081	- 4·56	- 4·89
Mashrak	133,630	- 6·00	- 1·56
Sonpur	95,821	- 6·79	-10·34
Siwan Subdivision ...	**758,699**	**-7·15**	**+0·13**
Siwan	287,259	-14·99	+ 1·26
Darauli	213,243	- 3·43	+ 4·11
Basantapur	258,197	+ 6·25	+ 1·11
Gopalganj Subdivision	**636,831**	**+0·28**	**+0·07**
Mirganj	370,857	+ 3·22	- 0·08
Gopalganj	265,974	- 3·54	+ 0·26

295. The most noticeable feature of the last ten years in Saran has been the great increase in mortality, the average death-rate being 40·50 per mille, whereas in the preceding nine years it was under 30 per mille. For this increase plague is partly responsible. This disease is now an annual visitation and never leaves the district altogether except for a short period in the rains. It increases all through the cold weather and reaches its height in March: deaths are numerous in April, but there is usually a decrease in May and a further considerable decrease in June. During the ten years the aggregate number of deaths returned as caused by plague was 166,000, or nearly a fifth of the total mortality. Plague alone, however, will not account for the enhanced mortality, unless, as is quite possible, a large number of deaths from plague were returned as deaths from fever and from other causes. The returns under the head of fever account for two-fifths of the total mortality, and a virulent type of fever was prevalent in 1903, 1905, 1909 and 1910: in each of these years it carried off over 40,000 persons. The birth-rate has also been high, averaging 40·13 per mille, but it has failed to keep pace with the death-rate, and the net result is that the deaths during the decade exceeded the births by nearly 9,000. "The general opinion," writes the Collector, "that the district has become more unhealthy is unquestionably justified. . . . There are no large rivers inside the district, and there are many low-lying *chaurs* and swamps; the pressure of population and extension of cultivation has probably obstructed the existing drainage channels, and in part of the district the closing of the Saran canals has perhaps contributed to the unhealthiness of the tracts they previously drained."

296. The outturn of crops was on the whole good except in 1906, when the district suffered from floods, and in 1907 and 1908, when it was affected by drought. The harvests were average during the years 1901, 1904 and 1905, and were up to or above the average in the remaining four years, viz., 1902, 1903, 1909 and 1910: especially good harvests were reaped in the last year. The Collector (Mr. B. Foley) remarks :—"It is doubtful if the material condition of the people was much affected even during the three bad years. In 1906 and 1908 an attempt was made to open test relief works, but no one attended them. Takavi loans were found sufficient for the people's need. Emigration, which has increased largely during the ten years, now renders the people more or less independent of the agricultural condition of the district . . . The years of highest mortality, namely, 1903, 1905 and 1910, were years of prosperity, while in the three years 1906 to 1908, during which the crops were bad, there was no remarkable mortality. In a dry year the general health is better, while the poor classes do not depend on their crops in the same way as the people of other districts. There is always an enormous emigration from the district, and though this seems to be unaffected by moderate variations in the conditions of agriculture, anything approaching scarcity drives large numbers away from the district to procure high wages elsewhere, a large proportion of which they remit to their homes."

297. There is no sign of real industrial development, and agriculture still almost entirely monopolizes the energies of the people. The indigo industry has continued to decline, and its decline has helped to swell the number of emigrant labourers. The acreage under this crop decreased steadily from 37,600 acres in 1901 to 9,205 in 1910; six factories were closed entirely, another ceased to grow indigo, and other concerns reduced their establishments. Sugarcane has not yet taken the place of indigo, and its cultivation so far shows no tendency to expand. The cultivation of opium has been steadily reduced, and shortly after the census was discontinued altogether, but the manufacture of saltpetre, which was a declining industry during the preceding ten years, is said to have shown some improvement, though the actual number of refineries decreased from 118 in 1901 to 92 in 1910. There has been a further extension of railway communications within the district. A railway line, 17 miles long, has been constructed between Siwan on the main line of the Bengal and North-Western Railway and Thawe in the northern portion of the district. Another short branch line has connected Daronda on the main line with the grain mart of Maharajganj, while the Chapra-Masrakh Railway line, 26 miles long, has established direct communication between Masrakh and Chapra.

298. In addition to losses by death, the district has lost heavily by migration, the number of the district-born who were enumerated outside the district being 39,000 more than it was in 1901, while the number of immigrants has fallen by nearly 10,000. It must not be imagined that the adverse balance of migration represents any permanent loss, since most of the emigrants leave the district only for a time. The labourers generally go away in November and December, and return in the hot weather, though others return only at intervals of one, two or three years. The annual exodus of the able-bodied is now an established feature. As the Collector remarks :—"The system of annual emigration no doubt had its origin in the inability of the soil to support its population. Having, however, once acquired the habit of emigrating for wages, and having found that it is easy to save money in this way, the people now emigrate yearly as a matter of habit to supplement their incomes, whether agricultural conditions are prosperous or the reverse." At the time of the census the presence of plague (which caused over 8,000 deaths in the two preceding months) most probably helped to swell the numbers of the

SARAN.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,064,333	1,225,445	1,095,117	1,313,697
Immigrants...	12,299	34,635	18,452	37,972
Emigrants	203,876	80,325	175,498	69,436
Natural population	1,255,910	1,271,135	1,252,163	1,345,161

emigrants, and also to prevent them returning as early as they might otherwise have done. The net result of these combined influences, viz., an increased mortality, which has not been altogether counterbalanced by an enhanced birth-rate, a greater exodus from the district, and the check of immigration, is a decrease of 119,036 or a little under 5 per cent.

299. In reviewing the results of the census of 1901, Mr. J. H. Kerr remarked in the Saran Settlement Report :—"On the whole, leaving out of account any great economic revolution, of which there are at present no signs, the situation may be summed up thus. Gopalganj is capable of supporting a slight increase of population. Siwan has probably reached its utmost limits, while in the Sadar, and particularly in the densely populated thanas of Manjhi Chapra and Sonpur, with their high rent-rates, decline has already set in. Nor is this to be deplored. On the contrary, it is a matter for congratulation that emigration should be growing in favour in a district where it is so greatly needed, and it is a further relief to the situation that North Bihar contains no body of peasantry more alert or more industrious, or more ready, within limits, to take advantage of improvements calculated to increase the productive powers of the soil." The result of the census is to confirm these conclusions. Gopalganj is the only subdivision with any increase, and even here the Gopalganj thana has lost population. Mirganj, on the other hand, has an increase of over 3 per cent., which is ascribed to its general healthiness and to the comparative absence of plague in the western portion of the thana, from which moreover there is less migration than elsewhere. It is probable also that the opening of the railway from Siwan to Thawe and the consequent development of trade contributed to the increase. Both the Sadar and Siwan subdivisions exhibit a general decline, the only thana which has gained ground being Basantpur, where the increase is due partly to natural growth and partly to the opening of the railway from Daronda to Maharajganj. The greatest decrease has taken place in thanas Manjhi and Siwan, representing 12½ and 15 per cent., respectively, as against the district average of 4·9 per cent.

CHAMPARAN.

300. The district of Champaran is still a sparsely peopled district, with ample scope for the extension of cultivation. It developed steadily between 1872 and 1891, owing mainly to immigration. Between 1891 and 1901, however, the district suffered from a series of lean years, as well as from persistent unhealthiness, while immigration received a sharp check. Scarcity culminated in famine in 1897, when practically the whole district was affected. Though no deaths occurred from starvation, the vitality of the people was lowered, and the sequel of the famine was a reduced birth-rate. These three factors—the unhealthiness of the decade, the diminished fecundity of the people consequent on a series of bad years, and the stoppage of immigration—led to a loss of population in every thana in the district except Bettiah and Adapur; the latter is a thana with an exceptionally fertile soil, in which a full rice crop was secured even in 1896, when there was a disastrous failure elsewhere.

CHAMPARAN.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**1,908,385**	**+ 6·59**	**- 3·72**
Sadar Subdivision ...	**1,101,498**	**+ 5·85**	**- 5·37**
Motihari	185,155	+ 6·73	- 8·80
Adapur	175,676	+ 4·67	+ 0·25
Dhaka Ramchandra ...	286,906	+ 10·82	- 4·01
Kesaria	185,351	+ 4·18	- 3·13
Madhuban	103,920	- 0·35	- 4·61
Gobindganj	164,488	+ 4·13	- 11·76
Bettiah Subdivision ...	**806,887**	**+ 7·60**	**- 1·32**
Bettiah	360,914	+ 5·38	+ 2·52
Bagaha	208,468	+ 5·93	- 0·94
Shikarpur	237,505	+ 12·79	- 1·26

301. The prosperity of the people appears to have been restored by the end of 1900, and with this recuperation the birth-rate increased rapidly. In the six years ending in 1906 the average birth-rate was 41·85 per mille, as

compared with 34·31 per mille in the preceding nine years, and there was an excess of 85,708 births, whereas the number of deaths in 1892—1900 exceeded the births by 1,059. In the second half of the decade there were three years of agricultural depression (1906—1908). In 1906 the north of the district was visited by floods, and later in the year considerable damage was done to the rice crops, while in 1907 and 1908 the outturn was poor owing to scanty and unevenly distributed rainfall. The rice crop was again injured by floods in 1910, but otherwise the harvests were good. Four new railway branches were opened in 1906 and 1907, viz.—(1) from Bettiah to Narkatiaganj, (2) from the latter place to Bhiknathori, (3) from Bairagnia to Shikarpur and (4) from Narkatiaganj to Bagaha. The construction of these lines, besides facilitating trade and helping to develop the resources of the district, afforded employment to a large number of labourers, and a further demand for labour was created by the construction of the Tribeni canal. Taking the decade as a whole, its most conspicuous feature was a low death-rate, viz., 32·66 per cent. (the lowest in Bihar), and a high birth-rate, viz., 42·85 per mille (the highest in North Bihar), the result being an excess of births over deaths amounting to 182,000.

302. The census shows that the loss recorded in 1901 has been more than made good, the population having been augmented by 117,922 or 6·6 per cent. This gain is to be ascribed mainly to the natural increase of population, in a district where the pressure on the soil is light, during a series of years which were on the whole favourable to development. The gain would have been greater but for the continued shrinkage of the volume of immigration. There are now 11,000 less immigrants than in 1901, while their number is 40 per cent. less than it was in 1891. Not only are fewer settlers attracted from outside, but an increasing number leave the district for employment elsewhere: the number of persons born in the district but enumerated outside it has risen by nearly 12,000 since 1901. This is somewhat surprising, for the district is but imperfectly developed and requires all the available labour for its own needs, but the north of the district, where there is most room for expansion, has an evil reputation for fever. The wages of field labour, moreover, have remained stationary, and the labourers have come to realize that they can get better terms elsewhere. The immigrants are, however, still twice as numerous as the emigrants, the actual excess being nearly 48,000.

CHAMPARAN.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	942,012	966,378	885,607	904,856
Immigrants	42,012	53,506	55,047	51,734
Emigrants	29,409	18,423	16,442	19,635
Natural population	929,409	931,290	847,002	872,757

303. The growth has been general throughout the district, except for the Madhuban thana in the south-east, which is stationary. In 1901 this was the most densely populated thana, and it suffered most from the floods of 1906 and 1910. The Magistrate is disposed to suspect that rack-renting may have something to do with the result, and, whether that is a predisposing cause or not, the figures for the sexes show that the thana has lost by emigration, the number of males falling off while the females have added to their number. The rate of increase is fairly uniform in the other thanas of the Sadar subdivision, except in Dhaka, which has grown by nearly 11 per cent. This is a fertile rice-growing tract, which has suffered less than others from flood and drought. Both the Motihari and Gobindganj thanas suffered severely in the famine and lost heavily between 1891 and 1901. At this census both show a substantial increase, but neither has quite recovered its former position. In the Bettiah subdivision there is a fair growth in thanas Bettiah and Bagaha, though the percentage of increase in the latter is surprisingly low considering the extensive tracts available for cultivation, while there has been a rapid expansion in the Shikarpur thana, where reclamation is steadily progressing.

304. At each of the censuses held since 1872 there has been an increase of population, but the rate of increase shows a steady falling off, which is not altogether

MUZAFFARPUR.

to be wondered at considering the density of population. In 1881 there was an advance of 15 per cent., which was partly due to the deficiencies of the first enumeration. In 1891 the percentage of growth fell to 5 per cent., while in 1901 it still further decreased to 1·5 per cent. The slowness of growth in the decade ending in 1901 is due to the fact that the district suffered from flood, famine and disease. The whole district except the south of the Hajipur subdivision was affected by famine in 1897, next year there were heavy floods, and epidemics of cholera broke out in four years carrying off over 76,000 persons. In other respects, however, the public health was fairly good; the floods caused no permanent injury; and, thanks to a succession of good harvests, the effects of the famine were not long felt. By the end of the decade the people had entirely recovered their normal condition; every thana in the great rice-growing tract north of the Baghmati, where the stress of famine was greatest, showed a marked increase, while every thana south of that river lost population. In the former tract the increase was due mainly to immigration; in the latter emigration was active.

MUZAFFARPUR.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**2,845,514**	**+3·24**	**+1·5**
Sadar Subdivision ...	**1,081,475**	**+2·86**	**-2·27**
Paru	316,706	+ 5·02	- 2·85
Muzaffarpur	547,279	+ 0·97	- 5·62
Katra	217,490	+ 4·66	+ 8·74
Sitamarhi Subdivision	**1,053,739**	**+6·81**	**+6·73**
Shiuhar	212,331	+ 8·38	+ 7·22
Sitamarhi	369,784	+ 6·63	+ 9·02
Belsand	214,363	+ 7·57	+ 4·09
Pupri	257,211	+ 5·18	+ 5·36
Hajipur Subdivision ...	**710,300**	**-1·10**	**+0·57**
Mahuwa	299,560	+ 1·86	- 0·38
Lalganj	154,083	+ 0·61	- 1·05
Hajipur	256,657	- 5·27	+ 2·59

305. Since 1901 there has been no famine, but no little distress has been caused by floods. The Sitamarhi subdivision was swept by a high flood in 1902, when great damage was done to the *bhadoi* crop. Except in a small part of the subdivision, however, the winter rice did not suffer appreciably, and there was ample time for replanting even where it had been destroyed. In other parts, the heavy alluvial deposits benefited the flooded lands, and, in spite of the temporary suffering caused by the flood, no relief was either asked for or required. In 1906 there was a much more serious and widespread inundation, which affected about half the entire district: the thanas which suffered most were Paro, Muzaffarpur, Mahua, Katra and Belsand. For some little time there was acute distress, which was aggravated by the high prices prevailing, but in the end the cultivators were able to weather through with the help of loans, and no famine relief had to be provided. The next two years were years of deficient and badly distributed rainfall, which led to partial failures of the crops, but the ample and reasonable fall of 1909 resulted in good harvests, which to a great extent removed the distress caused by the two preceding years. In 1910, however, there was another inundation which did considerable damage to crops in the Sitamarhi subdivision as well as in part of the Sadar subdivision. Loans had to be granted to the distressed cultivators, and the subsidence of the flood was followed by a virulent epidemic of cholera.

306. Speaking generally, the first half of the decade was a period of prosperity, and the latter half was one of depression owing to repeated crop failures. The health of the people was, however, good, the birth-rate being in excess of the death-rate in every year except 1910. As in Saran, the famine of 1897 was followed by greater productiveness, especially during the first five years of the decade, when the ratio of births was consistently above 43 per mille and ranged between that figure and 48 per mille. The result, for the decade as a whole, has been an excess of births over deaths amounting to 200,000. The actual increase disclosed by the census is 89,384 or 3·24 per cent. The difference between the two figures is due to the movements of the people, immigrants from outside falling off by 14,500 and emigrants increasing by no less than 41,000. Every year many thousands of able-bodied labourers leave the district at the beginning of the cold weather in search of work on the roads, railways and fields

in other districts, returning to their homes at the end of the hot weather, in time for the agricultural operations which commence with the bursting of the monsoon. This exodus is now an annual affair, and its volume is increasing steadily with the opening of through connection with North Bengal.

MUZAFFARPUR.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,360,200	1,485,314	1,319,110	1,437,020
Immigrants	24,0 9	49,197	32,200	55,502
Emigrants	113,753	81,061	86,507	67,030
Natural population... ...	1,449,864	1,517,178	1,373,417	1,448,548

307. Both the Sadar and Sitamarhi subdivisions, which suffered from famine in 1897, have added to their population, and the Hajipur subdivision, the south of which was not affected by it, has lost population. The loss in the latter is due to the decline in the Hajipur thana, which has suffered severely from plague. The Sitamarhi subdivision, which is a great rice-growing tract, is the most progressive part of the district, the rate of increase being the same as in 1901. This subdivision attracts settlers not only from Nepal, but also from the south of the district, owing to the fertility of its soil as well to the fact that it is not affected by plague: here the increase has been greatest in Shiuhar, Belsand and Sitamarhi, which march with the Nepal frontier. In the Sadar subdivision the Muzaffarpur thana is stationary on account of the emigration of men to Bengal in search of work, a cause which also has operated very largely in the decrease so noticeable in Hajipur.

DARBHANGA.

308. The first census of Darbhanga was proved to have been wanting in accuracy by a careful census of the head-quarters and Madhubani subdivisions taken in 1874 and 1876 respectively; the increase of 23 per cent. recorded in 1881 must therefore be largely discounted. The rate of progress during the next ten years was 6·5 per cent., but fell to 3·9 per cent. in 1901. This diminution of growth was due to the unfavourable conditions prevailing after 1891. The decade was not very healthy, and in 1892 there was scarcity, necessitating relief measures in the Sadar and Madhubani subdivisions, while in 1897 there was famine, which affected the whole district except the Samastipur and Dalsinghsarai thanas in the Samastipur subdivision. The result of the census of 1901 was to show that, while there was only a slight increase (1·9 per cent.) in the Samastipur subdivision, which was least affected by the famine, the growth was greatest (7·8 per cent.) in Madhubani, where distress was most acute both in 1892 and 1897. It was only 1·6 per cent. in the head-quarters subdivision, where the Darbhanga thana showed a slight decrease. This, however, was due to an unusually large number of persons having been present in Darbhanga town at the time of the previous census; if the urban area is excluded, the thana had a slight increase. The only other thana in the district in which there was any decrease was Warisnagar, where the population declined by 1·13 per cent

DARBHANGA.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**2,929,682**	**+ 0·59**	**+ 3·95**
Sadar Subdivision ...	**1,063,849**	**- 0·16**	**+ 1·60**
Darbhanga	455,°39	+ 4·68	- 1·36
Baheršh	347,524	- 2·11	+ 2·74
Roserah	260,486	- 5·32	+ 5·09
Madhubani Subdivision...	**1 135,771**	**+ 3·78**	**+ 7·85**
Banipati	265,123	+ 8·27	+ 8·83
Khajauli	271,362	+ 9·72	+ 8·01
Madhubani	282,700	+ 2·72	+ 3·86
Phulparas	316,586	- 3·17	+ 10·21
Samastipur Subdivision	**730,062**	**- 3·00**	**+ 1·92**
Warisnagar	166,227	- 1·67	- 1·13
Samastipur	303,409	- 2·54	+ 2·00
Dalsinghsarai	260,426	- 4·35	+ 3·81

309. The first half of the decade 1901-1910 was, on the whole, a prosperous period for the people of Darbhanga. Crops were fair, and the recovery from the effects of famine was marked by a rapid increase in the birth-rate which averaged 43·35 per mille in the quinquennium ending in 1905. The death-rate was comparatively low (34·9 per mille), and the result was that in these five years the number of births exceeded that of deaths by 153,000 the excess being 87,000 more than that recorded in the preceding nine years After 1905 the district had a series of bad years. Heavy floods in July and

August 1906 were followed by a drought in September and October, and their effect on the crops was disastrous. The distress caused by their destruction was accentuated by the poor outturn of the preceding year, and famine ensued. The rainfall in 1907 was about normal, well distributed and timely, but the year 1908 was a year of protracted drought, which injured almost all the crops of the year, and famine had again to be declared. The last year of the quinquennium was the only one in which the people were free from the pressure of scarcity, the outturn of almost all crops being good and prices almost down to the level of 1905-06. The effect of the famines on the birth-rate and the rapidity with which the people recovered are sufficiently apparent from the marginal figures, which also show the number of deaths in each year of the quinquennium and in 1911. The period of greatest distress was in the latter half of 1906, when the first famine broke out: the births next year fell by over 17,000, but quickly rose in 1908 and 1909 to a figure exceeding that attained in 1906. The second famine lasted till the 31st July 1909, distress being most acute in the first five months of the year, and another drop of 19,000 occurred, but next year there was an extraordinary rise of over 31,000. No deaths were caused by starvation, and it is noticeable that from 1906 to 1908 the deaths steadily decreased, the number recorded in 1908 being the minimum of the decade. On the other hand, the maximum was reached in 1909, when the people had to bear the brunt of the second famine.

Year.	Births.	Deaths.
1906 ...	108,213	96,723
1907 ...	90,882	90,938
1908 ...	113,651	83,395
1909 ...	118,676	122,788
1910 ...	99,716	100,549
1911 ...	131,147	94,694

310. The net excess of births over deaths during the decennium was 190,000, but the increase disclosed by the census is less than one-tenth of this figure, being only 17,071 or a little over one-half per cent. The difference between the two figures is due to migration, and in particular to the greatly enhanced number of those born in Darbhanga who were enumerated elsewhere. The exodus of the native-born has now attained large dimensions, rising by more than 65,000 since 1901; the number of male emigrants alone has been nearly doubled. Owing both to the outward movement of the people, and also to their movements from thana to thana inside the district, it is difficult to gauge the actual effect of the two famines of the decade on the population of the affected areas. The whole of the south and west of the district has a diminution of population, while throughout the north-east and north (except in Phulparas in the extreme north-west) there is an increase. The Darbhanga and Benipati thanas, which suffered in a major degree during the first famine and also, though less severely, during the second famine, have substantial increases. The thanas of Bahera and Roserah, in the Sadar subdivision, where famine prevailed in both years, have a decline, but so also have Samastipur and Dalsinghsarai in the Samastipur subdivision, which remained immune. Owing mainly to epidemics of plague and the loss of population caused by emigration, the decline in the two latter thanas is greater than in Warisnagar, the only thana in the Samastipur subdivision in which famine prevailed.

DARBHANGA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,412,669	1,517,013	1,416,474	1,496,137
Immigrants	27,160	59,588	33,473	57,107
Emigrants	102,675	73,391	58,443	53,069
Natural population	1,488,184	1,530,816	1,441,444	1,492,099

311. No definite conclusions can be drawn as to the effects of famine in different parts, but some broad and general inferences can be made on a survey of the variations during the last 30 years, as shown in the margin. At each of the three censuses the Madhubani subdivision to the north has gained population, but in a diminishing degree: the Sadar subdivision in the centre had become stationary by 1901, and is now decadent. In the Samastipur subdivision in the south the census of 1901 showed that the loss of population had been temporarily checked, but the downward tendency is now again pronounced. The density of population in these three tracts has much to do with the variations in the

SUBDIVISION.	VARIATION PER CENT.		
	1881—1891.	1891—1901.	1901—1911.
Madhubani	+ 12·0	+ 7·8	+ 3·78
Sadar	+ 8·0	+ 1·6	- 0·16
Samastipur	- 2·3	+ 1·9	- 3·0

rate of progress. The Samastipur subdivision is the most fertile but densely populated part of the district, and its surplus population are seeking relief by emigration. In the Sadar subdivision, which, unlike Samastipur, is practically dependent on one crop (winter rice), the pressure on the soil has become so great that it does not appear capable of supporting a greater population, while Madhubani is approaching the same condition.

BHAGALPUR.

312. The progressive decline of the rate of increase, which has been observed in the case of Darbhanga, is equally noticeable in the case of Bhagalpur. In 1881 there was an addition of 7·8 per cent., but the rate of progress was reduced to 3·3 per cent. in 1891 and to 2·8 per cent. in 1901. The increase in the latter year was mainly the result of the development of the Supaul and Sadar subdivisions, for there was only a small advance in the Banka subdivision and the Madhipura subdivision lost ground. The result of the census of 1911 is to show that the district, though still continuing to grow, is developing even less rapidly than in the preceding three decades, the net increment of population being only 50,365 or 2·4 per cent.

BHAGALPUR.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**2,139,318**	**+ 2·41**	**+ 2·77**
Sadar Subdivision ...	**598,377**	**+ 2·24**	**+ 5·97**
Sultanganj	95,198	+ 0·09	+ 3·08
Bhagalpur	185,178	- 2·53	+ 8·54
Colgong	207,964	+ 1·90	+ 6·59
Bihpur	110,037	+ 14·55	+ 2·73
Supaul Subdivision ...	**493,671**	**- 3·37**	**+ 6·09**
Supaul	356,034	- 0·97	+ 3·65
Pratapganj	137,617	- 9·08	+ 12·37
Madhipura Subdivision	**609,610**	**+ 8·99**	**- 2·81**
Bonagaon	153,705	- 7·58	+ 2·03
Madhipura	292,909	+ 5·95	- 3·78
Kishenganj	162,996	+ 39·55	- 6·89
Banka Subdivision ...	**437,660**	**+ 0·96**	**+ 2·40**
Amarpur	154,895	+ 1·27	+ 5·46
Katauria	117,481	- 1·23	+ 3·56
Banka	165,284	+ 2·28	- 1·14

313. The decade 1901 to 1910 was fairly healthy, for the birth-rate exceeded the death-rate in all but three years, viz., 1905, 1906 and 1907, and there was a net excess of births over deaths amounting to 103,000. In the three years above mentioned the death-roll was swollen by epidemics of cholera, small-pox and plague, and especially by cholera, which carried off 17,000 persons in 1906. Its ravages were most severe in the Madhipura and Pratapganj thanas, which lost over 15 per cent. of their population from cholera mortality; here the Kosi river has caused extensive water-logging of the soil, and, wells being scarce, the people resort to the numerous old channels of the Kosi for their drinking water. In the first part of the decade good harvests were reaped and a fair degree of prosperity prevailed, except in 1902 when there was scanty and ill-distributed rainfall. In 1906 floods swept the country round Supaul and Madhipura, and in 1908 the failure of the monsoon brought about one of the worst years of scarcity known in this district. The distress was relieved by the liberal grant of loans—over 20 lakhs were advanced—and by the institution of village relief works, while the readiness of the people to leave their homes for tracts where there was a demand for labour did much to mitigate their difficulties. The scarcity was acutely felt in the western part of the Madhipura and Supaul subdivisions, where famine conditions prevailed and relief works were started. There was also a severe drought in the country south of the Ganges where, however, the harvests of the previous year had been good and the people migrated freely from the affected areas. "The landless labourers cleared out in thousands to look for employment elsewhere. The luxuriant crops in the east of Supaul and Madhipura subdivisions north of the river attracted many: large numbers went into the lands on the Purnea side of the Kosi, and many others went eastwards to get employment in the rice-lands of Bengal proper. It was computed that in August and September about 70,000 people crossed the Ganges from south to north in search of work."* Next year the rainfall was ample and well distributed, the result being excellent crops, but it is reported that the people generally had not fully recovered from the effects of the scarcity by the time the census was taken.

* Bhagalpur District Gazetteer, p. 104.

314. The movements of the people do not appear to have affected the results very materially. As in other districts of North Bihar, the people are availing themselves more freely of the facilities for travel, and the number of emigrants now exceeds that returned in 1901 by over 43,000. On the other hand, settlers from outside have also come into the district in increasing numbers : there are 28,000 more than 10 years previously, so that the net loss to the district is only 15,000. The Supaul subdivision, which was the most rogressive part of the district between 1881 and 1901, is the only part of the district which has sustained a loss. Both the thanas included in this subdivision suffered from the scarcity of 1908-09, and as already stated Pratapganj, in which the loss is greatest, had a heavy mortality from cholera in 1906. Both, moreover, suffer from a severe type of malaria ; low-lying areas are inundated in the rains, and the subsidence of the floods is followed by outbreaks of fever and cholera. In the Madhipura subdivision, immediately to the south, there is a decline in the Bongaon thana, which adjoins the decadent portion of Darbhanga, but there is a fair growth in Madhipura, which is partly the sequel of its being opened out by the railway, and Kishenganj to the west had a remarkable increase. This thana borders on the water-logged area, but owing to the fact that the river Kosi has for some years past not shown any marked tendency to shift its channels, there has been a rapid expansion of cultivation in its south-eastern portion, which was till recently a desolate tract of swamp and jungle. Apart, however, from this, the number of persons enumerated in this thana was swollen at the time of the census by the influx of *dohatwars*, *i.e.*, non-resident cultivators, who came with their labourers to cut the crops. Owing to their presence the male population outnumbered the females by over 7,000, this being the only thana in either the Supaul or Madhipura subdivision in which males were in excess. In the Sadar subdivision there is only a slight development, but thana Bihpur has increased by nearly 15 per cent. owing chiefly to the spread of cultivation in an area which previously lay waste. There is a small increase in Colgong, but Bhagalpur has decreased in consequence of plague and the dislocation of trade it has caused, while Sultanganj is stationary. The Banka subdivision has advanced slightly, but the Katoria thana, a sparsely populated tract in the extreme south, has lost ground.

Bhagalpur.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,057,876	1,081,442	1,027,535	1,062,418
Immigrants	65,978	69,929	58,006	49,532
Emigrants	84,925	82,972	66,491	57,814
Natural population	1,076,823	1,094,485	1,036,020	1,069,700

315. The population of Purnea, a sparsely peopled district in which the soil is fertile and in which there are still large areas awaiting development, grew steadily between 1872 and 1891, an increase of 7·8 per cent. being recorded in 1881 and a further increase of 5·2 per cent. in 1891. The next decade witnessed a set-back, the result of the unhealthy conditions prevailing and of virulent epidemics of disease. In 1901 it was found that the population had diminished by 3·6 per cent., the only thanas in which there was any progress being Katihar and Damdaha.

Purnea.

Purnea.	Population, 1911.	Percentage of variation.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**1,989,637**	**+ 5·98**	**- 3·6**
Sadar Subdivision	**942,716**	**+ 13·30**	**- 2·65**
Purnea	159,764	+ 2·44	- 0·81
Kasba Amur	126,202	+ 5·93	- 2·73
Damdaha	176,362	+ 19·81	+ 1·68
Korah	118,082	+ 7·97	- 5·51
Gopalpur	80,810	+ 3·37	- 22·30
Kadwa	149,855	+ 12·82	- 7·94
Katihar	131,641	+ 33·94	+ 2·63
Kishanganj Subdivision	**606,688**	**- 1·90**	**- 4·85**
Kishanganj	145,625	- 2·90	- 5·01
Bahadurganj	206,261	+ 1·47	- 5·45
Islampur	254,802	- 3·91	- 4·29
Araria Subdivision	**440,233**	**+ 5·58**	**- 3·57**
Raniganj	89,509	+ 2·82	- 2·92
Forbesganj	139,032	+ 7·47	- 2·36
Araria	211,672	+ 5·29	- 4·60

316. In the past decade conditions have been far more favourable, and the number of births exceeded that of deaths in every year except 1905,

1906 and 1907, the aggregate excess being 34,000. From 1905 to 1907 heavy mortality was caused both by fever and cholera, the latter causing 43,000 deaths between October 1905 and June 1907. Next year (1908) the premature cessation of the monsoon led to a partial failure of the crops, and some distress ensued among the small cultivators and field labourers, but, as the Collector remarked :—"The regularity and abundance of the rainfall, the natural moisture of the soil, and, consequent on these conditions, the comparative ease with which the soil is prepared and the crops grown, have all tended to make the typical cultivator of this district a most fortunate being. Well irrigation and the unceasing toil of the typical Central Bihar cultivator are not necessary for him. With such antecedents it is clear that the economic condition of the cultivators in the affected area before the scarcity could not be anything but satisfactory in spite of high prices prevailing." The population generally has reached a high standard of prosperity as compared with other districts of Bihar. Rents are low, for much of the land has been recently brought under cultivation and cultivators are in demand. The soil is fertile, being inundated and enriched by silt nearly every year. The cultivation of jute has expanded steadily, to the profit both of cultivators, middlemen and merchants, and the district generally is being developed by the extension of railway communications. Two new lines have been opened since 1901, one from Forbesganj to Jogbani on the Nepal border and the other from Katihar to Godagari, the latter being an important line which gives direct communication with Lower Bengal.

317. The census shows that since 1901 there has been an addition to the population of 112,308 persons or 6 per cent. A large part of this increase is due to the influx of new settlers. The actual number of immigrants has now risen to 200,000. the increase since 1901 being no less than 84 per cent. That most of the new immigrants have come to the district for permanent settlement will be apparent from the close correspondence between the figures for male and females, the increase in the number of the former being a little under 49,000, while in the case of females it is 42,000. The influx of immigrants has led to very large accretions of population in some thanas, notably Katihar and Damdaha. Katihar, which owes its growth very largely to the development of the railway junction of Katihar, has added 34 per cent. to its population since 1901 and is now more populous by 72 per cent. than in 1891. Damdaha thana in the south-west has grown less rapidly, but has an increase of 20 per cent., the result of cultivation being rendered possible now that the Kosi has swung to the west and the thana is no longer swept by its floods. Here waste land is fast disappearing: what used to be a vast jungle area is now a fertile and well cultivated tract. There is a general increase throughout the west and south of the district, and the only decadent portion is the Kishanganj subdivision in the north-east. This a notoriously unhealthy Tarai tract, which has been steadily losing population since 1891, but even here the Bahadurganj thana, which is an important jute-growing centre, has advanced slightly since 1901.

PURNEA.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,016,421	973,216	959,723	917,606
Immigrants	117,690	81,179	69,032	33,973
Emigrants	20,505	17,943	20,615	16,803
Natural population	919,236	909,960	911,306	895,436

ORISSA.

CUTTACK.

318. As in many other districts, the census of 1881 showed a very large addition to the population of Cuttack (16·2 per cent.), but it has been estimated that at least 100,000 persons escaped enumeration in 1872. and that the real rate of

growth was 13·6 per cent. A further increase of 7·9 per cent. was registered in 1891, which was followed by an advance of 6·5 per cent. in 1901. Throughout these 29 years the district developed steadily, its growth being most rapid in the first decade, when it was recovering from the effects of the famine of 1866. Its development was retarded to some extent in the next decade by the terrible cyclone of 1885, but between 1891 and 1901 there was steady progress, throughout the district.

CUTTACK.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901-1911.	1891-1901.
DISTRICT TOTAL	**2,109,139**	**+ 2·37**	**+ 6·46**
Sadar Subdivision ...	**1,068,772**	**+ 3·24**	**+ 5·43**
Cuttack	228,193	+ 2·03	+ 4·97
Banki	78,604	+ 5·26	+ 7·36
Salepur	291,852	+ 7·19	+ 4·48
Tirtol	183,739	+ 1·90	+ 6·78
Jagatsinghpur	286,384	+ 0·71	+ 5·40
Kendrapara Subdivision	**485,918**	**+ 4·37**	**+ 8·33**
Kendrapara	250,406	+ 7·97	+ 7·09
Patamundi	120,352	- 1·15	+ 10·66
Aul	115,160	+ 2·90	+ 9·83
Jajpur Subdivision ...	**554,449**	**- 0·89**	**+ 6·38**
Jajpur	270,080	+ 0·60	+ 4·26
Dharmsala	284,369	- 2·27	+ 8·77

319. There would probably have been a similar growth of population during the decade ending in 1910, had it not been for a succession of bad agricultural seasons leading to a large increase of emigration. Owing to floods in 1900 and to drought in 1901, the two opening years of the decade were years of short crops. In the next two years excellent harvests were reaped, but the course of the seasons in 1904 was a repetition of 1901, and the outturn was again poor. In 1907 again there were severe floods early in the rains followed by a drought towards their close. The floods caused great distress in the Jajpur and Kendrapara subdivisions, and, when the water receded, a severe outbreak of cholera supervened. The damage to the crops caused by the floods was completed by subsequent drought, and it became necessary to institute relief measures. In the following year scarcity continued, and floods, followed by drought, again reduced the outturn of the crops. The last two years of the decade, however, were exceptionally prosperous, and by the time of the census the people had completely recovered from their distress. The effect of the failure of the harvests on vital occurrences is very marked. In the first two years of the decade the birth-rate was below 38 per mille : it then remained above 42 per mille for five years, but suddenly dropped as the result of scarcity, falling to 36 per mille in 1908 and reaching the minimum (32 per mille) in 1909. In the last year of the decade, with reviving prosperity, it as suddenly rose to the high ratio of 46 per mille. Except in the three years 1901, 1907 and 1908, the birth-rate was always in excess of the death-rate, though there were epidemics of cholera in all but two years, which carried off altogether 100,000 persons : the mortality in 1907 and 1908 alone, when there were epidemics synchronizing with floods, drought and scarcity, accounted for very nearly half the number.

320. Altogether, the births outnumbered the deaths by 96,610, but the census shows that the addition to the population is only about half that number, viz., 48,826 or 2·4 per cent. The difference is due to the exodus of the district born, the number seeking employment elsewhere having risen by no less than 56,000, or 48 per cent., since 1901. The growth of the volume of emigration is due partly to the facilities of communication with the outside world, which the railway has afforded within the past decade, and partly to the stress caused by repeated crop failures. The scarcity of 1907 and 1908 gave a further stimulus to migration. In 1908 the Subdivisional Officer of Jajpur took a rough census and found that 50,000 persons from that subdivision alone had left their homes under the pressure

CUTTACK.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	1,001,175	1,107,964	994,166	1,066,147
Immigrants	10,349	22,905	11,606	21,338
Emigrants	118,377	54,397	81,283	35,476
Natural population	1,109,103	1,139,456	1,063,843	1,080,885

of scarcity and the pinch of high prices.* The emigrants are mostly able-bodied men, and the males are more than twice as numerous as the females. One indirect result is that while the female population has grown by 4 per cent. since 1901, the increase among males is under 1 per cent.

321. There has been a slight loss of population in two thanas, viz., Patamundi in the east and Dharmsala in the north-west, where the loss is due to emigration. In both thanas there has been an increase of the female population, but the males in Patamundi are less by 3,000 than in 1901, and in Dharmsala, which is traversed by the railway, the decline in their number amounts to 7,000. The same feature is noticeable in the neighbouring thana of Jajpur, and it is on this account that the Jajpur subdivision is stationary. In the south-east of the district also there is little or no progress. Here, too, emigration is responsible, for both the thanas concerned (Tirtol and Jagatsinghpur) have lost part of their male population, while the female population has increased. The greatest advance has been made by the Salepur and Kendrapara thanas, which have enjoyed a considerable amount of prosperity on account of the steady expansion of jute cultivation and the facilities for irrigation, which make them fairly independent of seasonal variations. The increase is all the more noticeable in Salepur, because in 1901 it was already the most densely populated thana in the district. After them, comes Banki, a somewhat thinly populated thana, which has been steadily developing since 1891.

BALASORE.

322. When the first census was taken, the people of Balasore were recovering from the famine of 1866, which had decimated their numbers. In 1881 an increase of 28 per cent. was recorded, which was due partly to improved enumeration and partly to a recovery from the losses caused by the famine. The next decade witnessed a greatly diminished rate of progress, the total number of persons recorded in 1891 being only 5 per cent. more than in 1881. The increase was very evenly distributed all over the district, except the Jellasore and Chandbali thanas. In the former thana, where malarial fever of a malignant type had prevailed, the population was practically stationary; in Chandbali there was an increase of 11 per cent. due to the development of trade and the reclamation of waste land. During the next decade the public health was good and the condition of the people prosperous, owing to a succession of good harvests. The result was a further increase of 7·7 per cent., in which all parts of the district shared except the Dhamnagar thana, where the destruction of crops by floods shortly before the census had driven a number of the males to seek employment elsewhere. The Chandbali and Basudebpur thana had the greatest expansion; in other parts of the district the rate of growth was remarkably uniform.

BALASORE.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901-1911.	1891-1901.
DISTRICT TOTAL	**1,055,568**	**- 1·68**	**+ 7·69**
Sadar Subdivision ...	**594,936**	**+ 0·40**	**+ 8·35**
Jellasore	61,866	+ 3·39	+ 7·91
Baliapal	104,522	+ 3·65	+ 8·67
Basta	80,298	+ 0·21	+ 8·00
Balasore	121,636	- 0·78	+ 8·90
Soro	226,614	- 1·16	+ 8·14
Bhadrakh Subdivision ...	**460,632**	**- 4·25**	**+ 7·44**
Bhadrakh	186,353	- 3·65	+ 8·96
Basudebpur	74,222	- 5·90	+ 12·0
Dhamnagar	134,623	- 2·91	+ 0·26
Chandbali	61,434	- 6·77	+ 14·08

323. During the first half of the decade ending in 1910 the crops were below the average except in 1903-04, while there was a serious failure in 1901-02. Though the births exceeded the deaths by 24,000, the quinquennium witnessed a falling off in the birth-rate as compared with the previous five years, the ratio of births declining from 40 to 39 per mille, while the death-rate rose from 31 to 34 per mille. After 1905 the district entered on a series of unhealthy years. In three years, viz., 1906, 1907 and 1908, the deaths exceeded the births, especially in 1908 when the number of deaths was actually double the number of births. Two epidemics of cholera

* The crop failure of the preceding years led to 1,087 persons being recruited under the Emigration Acts in 1903 and 2,657 in 1908: the total for the remaining eight years was only 1,494.

occurred in these three years, the first breaking out in 1907 and being responsible for 11,000 deaths, while the second, which broke out next year, was a terrible visitation which carried off 28,000 people. These epidemics were the result partly of scarcity and high prices, which compelled the poorer classes to resort to dietary of a coarser kind than usual, and partly of the failure of winter and spring rains, which greatly reduced the water-supply.

324. The resources of the people were also severely tried by a succession of bad harvests. The rainfall was scanty in 1905 and 1906, and the crops were consequently short. In 1907 a high flood destroyed the rice crop in the greater part of the Bhadrakh subdivision, and the inundation was followed by a prolonged drought. The combined effect of flood and drought was scarcity, which necessitated the opening of relief works and the distribution of gratuitous relief. The year 1908 also was not a prosperous one. The winter rice over a large part of thanas Dhamnagar and Chandbali was again damaged by floods, while scanty rainfall in October, when the crop was in the ear, seriously affected the outturn over a great part of the district. The damage caused by alternating floods and droughts in these successive years was intensified in its effects by the fact that the district is practically dependent on one crop, viz., winter rice. The distress of the poorer classes is, however, relieved by emigration, which has been facilitated by the opening of the Bengal-Nagpur Railway in 1899 and of the Mayurbhanj State Railway in 1904 : a third railway connecting Balasore with the capital of the Nilgiri State has also been opened recently, but there is as yet practically no passenger traffic.

325. Taken as a whole, the decade was unfavourable to any growth of population. The births outnumbered the deaths by only 2,000; only in one year were the harvests up to the average, and in four years there were serious crop failures. The result of the census is to show that the population has diminished by 18,074 or 1·7 per cent. For this decrease three factors are responsible, viz., emigration, the prevalence of epidemic disease, and failure of crops resulting in scarcity. The two latter factors have already been referred to, and it will suffice to say that the greater activity of emigration is evidenced by the fact that the number of persons who sought employment outside the district has risen by 21,000 since 1901, and that whereas in that year the excess of females over males in the district was 36,000, it is now 46,000.

BALASORE.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	504,615	550,953	518,786	554,856
Immigrants	12,257	20,172	13,235	16,232
Emigrants	43,319	29,157	29,024	22,739
Natural population	535,677	559,938	534,575	561,363

326. The only thanas which show any increase are Jellasore and Baliapal in the north, a result which is at first sight somewhat surprising, as they are reputed to be the home of malaria and the most unhealthy part of the district. On the other hand, this area is one of the most prosperous parts of the district. It is for the most part a permanently-settled tract, and the land revenue demand is very light. The agricultural classes are more prosperous than in the centre and south of the district, and not only was the pinch of scarcity less felt, but the ravages of cholera were not so widespread. Even more important, however, is the fact that there is very little emigration from these thanas. On the contrary, they attract settlers. Labourers come to Baliapal from the adjoining subdivision of Contai in the Midnapore district to reclaim waste land and jungle, while Santals and other aboriginals come to Jellasore from the Mayurbhanj State in search of work. In thana Basta, which lies to the south of Jellasore and Baliapal, the population is practically stationary, while thana Balasore, which is to the south of Basta, shows a small decrease. This decrease is to be attributed mainly to emigration, which has brought down the excess of males over females by about 1,000. Soro, the remaining thana of the Sadar subdivision, has also a small loss, which is due to death by disease and emigration. It was the chief sufferer from the cholera epidemic already referred to, and emigration has been active. The Bhadrakh subdivision in the south is mainly responsible for the decrease of population in the district.

Every thana has lost ground, the percentage of decrease varying from 2·9 per cent. in Dhamnagar to 6·8 per cent. in Chandbali. This part of the district suffered most from scarcity, the floods of the Baitarani river having destroyed the crops over a great part of the subdivision in 1907 and 1908. Emigration is also more active than it is to the north, Chandbali being the only thana which does not contribute to the stream of emigrants. Though, however, it has not lost by emigration, this thana has the heaviest decrease owing to the fact that it suffered more than any other from floods, which caused widespread destruction of the crops and were followed by scarcity and virulent epidemics of cholera and other diseases.

PURI.

327. As in other districts of Orissa, the census of 1881 brought out a large increase of population in Puri amounting to 15·4 per cent. The growth of population was sustained during the ensuing ten years, at the close of which a further advance of 6·3 per cent. was recorded. Between 1891 and 1901 the district continued to develop in spite of adverse influences. The year 1891 witnessed a crop failure, floods and a cyclone; there was scarcity in 1897 necessitating relief measures in the neighbourhood of the Chilka lake and in parts of the Khurda subdivision; while the country round the Chilka was again subject to scarcity in 1900. The actual increase was 7·6 per cent., but part of this was due to the fact that a religious festival was in progress at Puri at the time when the census was taken, the foreign-born population exceeding that of 1891 by more than 13,000. The Sadar and Pipli thanas had the smallest rate of increase, while in the Gop thana and the two thanas of the Khurda subdivision the addition to population varied from 6 to 8½ per cent.

PURI.	Population 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**1,023,402**	+ **0·60**	+ **7·65**
Sadar Subdivision ...	**655,798**	− **0·34**	+ **7·25**
Puri	222,642	+ 2·54	+ 3·75
Puri Town	40,011	− 18·90	+ 71·33
Pipli	278,164	+ 0·33	+ 14·81
Gop	114,981	+ 0·55	− 14·69
Khurda Subdivision ...	**367,604**	+ **2·33**	+ **8·39**
Khurda	261,181	+ 2·71	+ 8·34
Banpur	106,423	+ 1·41	+ 8·53

328. Conditions during the ten years ending in 1901 were no more favourable than in the preceding decade. In 1901 there was a partial loss of the winter rice crop in parts of the Khurda subdivision and also in the country on the borders of the Chilka lake, necessitating relief measures in the latter tract. The crops were good in the following three years, *i.e.*, 1902 to 1904, but the outturn was much below the normal in the years 1905 and 1906, and there was a serious loss of crops in 1907 owing to the early cessation of the monsoon and heavy floods in the north-east. Throughout the year 1908 distress, deepening into famine conditions in limited areas, prevailed, and relief operations had to be carried on in the north-east of the district and in the neighbourhood of the Chilka lake. Fortunately, in the last two years of the decade the harvests were excellent and the prosperity of the people revived. The effect of these years of depression is reflected in the statistics of vital occurrences. In four years (1901, 1902, 1907 and 1908) the deaths outnumbered the births, and in the decade as a whole the excess of births over deaths was only 10,000. The mortality was especially great in 1908, a year of scarcity when the distress of the people was aggravated by an epidemic of cholera, which helped to bring the death-rate to the appalling figure of 59 per mille.

329. The combined effect of disease, scarcity and the stimulus given to emigration by poor harvests was to check the growth of the population, and the actual increase brought to light by the census is only 6,000 or only 0·6 per cent. The increase would have been greater had it not been for the adverse balance of migration. As compared with 1901, 6,000

PURI.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	506,570	516,832	506,839	510,445
Immigrants	17,667	25,773	20,919	32,114
Emigrants	16,123	19,397	11,487	17,504
Natural population	505,026	510,456	497,407	495,835

more of the native population were enumerated outside the district, and 9,000 less persons from other districts were present at the time of the census. As already explained, there was a large body of pilgrims in the district in 1901, as the date of the census was synchronous with the Gobind Dwadasi festival, the number in Puri town alone being over 17,085. Special care was taken at this census to separate the pilgrim population from others, and it was found that it only amounted to 7,139, of whom 5,293 were returned for Puri town. The decrease in the number of pilgrims alone may be taken as at least 10,000. This accounts for the apparently large loss returned for Puri town. If the pilgrims at both censuses are excluded, the town has grown at the rate of 7·7 per cent. There has been a small advance in the Khurda subdivision, which was comparatively free from seasonal calamities and should therefore *a priori* have grown more rapidly. The decline in the rate of growth is due to emigration; the addition to the female population in Khurda thana is more than double the addition to males, and in Banpur the male population is a little less than in 1901, whereas the females have added nearly 2,000 to their numbers. In the Puri subdivision there is a slight advance in the Puri thana (excluding the town), but the Pipli and Gop thanas are stationary.

CHOTA NAGPUR PLATEAU.

HAZARIBAGH.

330. The centre of the Hazaribagh district consists of a small plateau about 2,000 feet high, which slopes down abruptly on all sides except the south-west, where it stretches out into a long ridge as far as the boundary of the Palamau district. The central plateau, which corresponds to the Hazaribagh thana, is surrounded by a lower ring-shaped plateau, which forms the drainage area of a number of rivers. This lower plateau is broken up by small ranges of hills into several main valleys and numerous ravines. The rainfall and agricultural conditions vary greatly from valley to valley, but where the country is much broken by ravines, it is covered with forests which retain moisture, and suffers comparatively little from deficient rainfall. To the south-west is the Barkagaon thana, which comprises some broad valleys; Simaria and Chatra in the west consist partly of ridges and uplands, partly of ravines; Hunterganj thana to the north-west is a valley sloping down to the Gaya district, but the Pratappur police station in its extreme west is an area of ravines and jungle. To the south-east thanas Mandu and Gumian are full of ravines, but Bagodar and Ramgarh are broad valleys. In the Giridih subdivision Dumurhi lies below and receives the drainage from the Paresnath range, while Giridih, Kharagdiha, Dhanwar and Ganwan are undulating uplands.

331. The increase of population between 1881 (when the first reliable census was held) and 1891 amounted to 5·4 per cent., but this was not evenly distributed, for the whole of the north-west of the district lost ground, except the Hunterganj thana, while there was a large growth elsewhere, especially in the Giridih subdivision. During the next decade there was famine in 1897, when distress was general over a broad belt running north and south through the district, the thanas most affected being Barhi, Kodarma, Bagodar, Gumian, Ramgarh, Mandu and Hazaribagh. The addition to the population recorded in

HAZARIBAGH.	Population, 1911.	PERCENTAGE OF VARIATION.	
		1901—1911.	1891—1901.
DISTRICT TOTAL	**1,288,609**	**+ 9·39**	**+ 1·17**
Sadar Subdivision ...	**835,953**	**+ 9·97**	**- 0·31**
Barhi	52,635	+ 4·66	- 2·31
Hazaribagh	101,111	+ 8·81	- 8·31
Bagodar	72,168	+ 11·91	- 0·36
Barkagaon	61,478	+ 12·85	- 4·94
Ramgarh	95,125	+ 17·76	+ 1·41
Gumian	33,632	+ 19·52	- 2·67
Mandu	36,420	+ 21·53	+ 2·18
Petarbar	69,998	+ 21·28	+ 13·35
Chorparan	62,832	+ 3·24	- 3·26
Hunterganj	67,780	+ 6·42	+ 8·88
Chatra	35,191	— 4·48	- 5·88
Simaria	37,606	— 0·87	+ 3·89
Kodarma	87,977	+ 10·78	+ 2·22
Giridih Subdivision ...	**452,656**	**+ 8·34**	**+ 3·98**
Ganwan	64,121	+ 7·53	+ 4·08
Khargdiha	90,803	+ 3·66	+ 0·61
Dhanwar	73,538	+ 3·21	+ 2·26
Giridih	129,389	+ 10·60	+ 8·83
Dumurhi	94,805	+ 15·16	+ 2·55

1901 represented only 1·2 per cent., the smallness of the increase being attributable to the growing volume of emigration and also to the heavy death-rate following the famine of 1897.

332. Between 1901 and 1910 the health of the people was on the whole good. The birth-rate maintained a high level except in the years 1908 and 1909, and the births outnumbered the deaths by no less than 170,000. Good or fair crops were reaped until 1907, when the rainfall was unfavourably distributed, there being a heavy fall early in the season and a premature cessation in September. The rice crop was almost entirely a failure on high lands, and its outturn was diminished in the low lands. The yield both of *rabi* and of the important *mahua* crop was also only about a quarter of the normal. Distress ensued, which amounted to scarcity in some parts, viz., in the Barkagaon thana, the eastern half of the Hazaribagh thana, and the uplands of Simaria and Chatra. Gratuitous relief had to be given, mainly to old people, women and children left without means of support by the labourers and small cultivators, who migrated in large numbers to the coal-fields and elsewhere. Others were provided with loans, with the help of which they were able to tide over their difficulties. The rainfall next year was favourable, and good crops were reaped. By October 1909 all signs of scarcity had disappeared, and normal conditions were re-established.

333. The material condition of the people appears to have improved considerably during the last 20 years. It is reported: "Labourers and petty agriculturists formerly found it very difficult to get the bare necessaries of life. They have now got over the stage of actual want, and in many instances, after defraying all expenses, they are able to live up to a fairly high standard and to indulge in small luxuries previously unknown to them. This change is very noticeable among the aboriginals. Whereas formerly they depended solely upon the edible fruits and roots of the jungles for their subsistence for at least two or three months in the year, they now never, ordinarily, miss their food grain diet." The mica mines at Kodarma and the coal mines of Giridih furnish employment to many thousands, and the wages obtained there alleviate distress arising from the shortage of the crops in their neighbourhood. The demand for labour, it should be added, varies very largely according to the state of the market: the average labour force in the mica and coal mines was 18,000 in 1901, rose to 52,000 in 1906, and fell to under 7,000 in 1910. Except in the neighbourhood of the commercial and industrial centres, the people are poor, and the margin between sufficiency and exigency is often narrow.

334. The increase of population disclosed by the census is 110,648 or 9·4 per cent., which is due to natural growth among a people largely composed of prolific semi-aboriginals. It cannot be said that the movements of the people have affected the census results to any appreciable extent. The number of immigrants is almost exactly the same as in 1901, while the emigrant population has fallen off by only 6,000. The number of those born in the district who left it in order to find employment elsewhere reached a very high figure in 1901, and emigration has since been facilitated by the opening of the Grand Chord line. The drain on the available labour varies from year to year according to the outturn of the crops, the exodus being stimulated by bad harvests and checked by good crops; there is no doubt that it would have been greater at the time of the census had it not been for agricultural prosperity. A good index to the extent to which the latter influences emigration is afforded by the returns of coolies recruited for Assam, which show that in 1910-11 the number was only 684, whereas it was 3,465 in 1907-08, a year of scarcity. Every thana in the district has increased its population, except Chatra and Simaria in the west: acute distress was experienced in the upland tracts of these two thanas in

HAZARIBAGH.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population ...	629,103	659,506	570,122	607,839
Immigrants	21,375	20,256	19,469	22,414
Emigrants	77,496	67,045	83,372	66,784
Natural population ...	685,224	706,295	634,225	652,209

1908, and both have lost slightly. The greatest progress is observable in the south, where four thanas have grown rapidly, the ratio varying from 17·7 to 21·5 per cent. The advance has been slower in the north, where Barhi and Chorparan have a proportional growth of under 5 per cent. and also in the Giridih subdivision. Both the Giridih and Kodarma thanas have an increase of over 10 per cent., though at the time of the census the coal and mica mines were employing a smaller labour force than usual.

RANCHI.

335. In spite of the drain caused by increasing emigration, the population of Ranchi grew steadily between 1881 and 1901, the increase being 6·7 per cent. in the first and 5·2 per cent. in the second decade. The growth during the ten years ending in 1901 was retarded by several years of bad harvests and general depression, actual famine prevailing in some parts in 1897 and 1900, when the stream of emigration was greatly increased. After allowing for the effects of emigration, it was estimated that the true increase of population was about 13 per cent.

RANCHI.	Population, 1911.	PERCENTAGE OF VARIATION, 1901—1911.
DISTRICT TOTAL	**1,387,516**	**+ 16·80**
Sadar Subdivision ...	**526,172**	**+ 15·25**
Lohardaga	97,851	+ 17·87
Kuru	31,309	+ 13·16
Burmu	23,055	+ 5·30
Mandar	36,309	+ 7·59
Bero	39,781	+ 11·22
Lapung	23,249	+ 13·71
Ranchi	165,054	+ 18·36
Ormanjhi	20,134	+ 15·26
Angora	29,610	+ 14·73
Silli	39,820	+ 21·82
Khunti Subdivision ...	**349,633**	**+ 16·46**
Karra	38,811	+ 15·50
Torpa	35,318	+ 11·50
Khunti	77,079	+ 14·08
Bundu	33,023	+ 20·80
Sonahata	42,900	+ 13·85
Tamar	102,507	+ 21·44
Gumla Subdivision ...	**511,711**	**+ 18·68**
Bishunpur	18,142	+ 20·06
Chainpur	54,404	+ 17·29
Ghagra	35,511	+ 19·63
Sisai	61,468	+ 14·91
Gumla	46,551	+ 22·29
Raidih	26,180	+ 16·52
Palkot	26,781	+ 14·01
Basia	59,529	+ 12·67
Bano	31,915	+ 20·93
Kolebira	45,739	+ 22·21
Kochdega	69,734	+ 18·39
Kurdeg	35,757	+ 32·84

The decade ending in 1910 was a period of agricultural prosperity broken only by one bad year. In the first five years good crops enabled the people to recoup from the effects of the famine of 1900. A bumper harvest followed in 1906, leading to heavy exports, but the people, with the improvidence characteristic of aboriginals, squandered their gains and were left with few resources to tide them over the impending scarcity. The failure of the monsoon in 1907 led to a failure of the crops, the stock of food-grains was reduced to a very low limit, and distress became general. Famine was declared in the Kurdeg, Kochdega, Chainpur, Bishunpur, Gumla and Ghaghra thanas, relief works were opened in Sisai thana and part of Sonahatu, and test works in Burmu and part of Tamar; but elsewhere the people managed to hold out with the assistance of loans, which were freely granted. The crop of 1908 was fair, and made it possible to bring the famine operations to a close in September. The mortality in this year was heavy, for distress lowered the general vitality of the people and diminished their power of resistance to disease. Cholera and small-pox appeared in an epidemic form, causing nearly 10,000 deaths between them, and the death-roll from fever was also heavy. The two last years of the decade were, however, a period of agricultural prosperity, and the effects of the famine had disappeared by the time the census was taken. The birth-rate throughout the decade was very much higher than in the preceding ten years, rising from 38·3 to an average of 44·5 per mille. It was in excess of the death-rate in every year but 1908, and the net result was an excess of births over deaths amounting to 196,000.

336. The total addition to the population is 199,591 or 16·8 per cent., a very large increase, which must be attributed to the increased prosperity of the people, the high birth-rate natural to prolific aboriginal races, and, improved enumeration in tracts previously difficult of access. The balance of migration is heavily

RANCHI.	1911.		1901.	
	Male.	Female.	Male.	Female.
Actual population	676,753	710,763	577,180	610,745
Immigrants	17,168	16,475	17,030	15,026
Emigrants	160,188	145,121	142,910	132,341
Natural population ...	819,773	839,409	703,060	728,060

against the district, and the opening of the Purulia-Ranchi Railway has helped to swell the number of emigrants. Emigrants outnumber immigrants by 271,666 as against 220,517 in 1891, and the aggregate of the district-born who were enumerated elsewhere is now 30,058 more than it was ten years previously. The increase in the number of the latter is all the more noticeable, because 1910 was a year of good harvests, which should have reduced the stream of emigrants, whereas the census of 1901 took place after a year of scarcity when many thousands more than usual left their homes.* The security that has been assured to the raiyat by the settlement of the district is said to have brought back to their homes many cultivators who would otherwise have settled in Assam, but it does not appear to have checked in any way the general exodus.

337. The increase of population has been general throughout the district, the percentage being above 10 per cent. in all thanas except Burmu and Mandar. The thanas in which famine was declared have developed to an astonishing extent, the increase in them being nowhere below 17 per cent., while Kurdeg has added nearly one-third to the number of its inhabitants. Nowhere has the growth of population been so rapid as in this latter thana and the adjoining thanas of Kochdega, Kolebira and Bano. The average increase of 22·5 per cent. in these four thanas, writes the Deputy Commissioner, has been "the result of a tendency among aboriginal cultivators to escape from the more settled and landlord-ridden villages of the north and centre of the plateau to the jungles in the south. The movement used formerly to be a continuous advance southwards, but the abolition of *bethbegari* during the decade has rendered the position of the raiyat in Ranchi infinitely superior to that of his brethren in Gangpur, and has arrested further movement across the border." The only other thanas of the district with an increase exceeding 20 per cent. are Silli, Tamar, Bundu, Gumla and Bishunpur. The first three are among the most prosperous thanas of the district and reaped the greatest benefit from the rise in the price of lac during the first six years of the decade; the two last suffered severely from the famine of 1900, and many of their inhabitants must have left the district when the census of 1901 was taken.

PALAMAU.

338. Palamau added 8·3 per cent. to its population between 1881 and 1891, but in the next ten years the rate of progress was greatly diminished, the census of 1901 showing an increase of only 3·8 per cent. This result was, in a large measure, due to the fact that in 1896 and 1899 there were crop failures which culminated in famine. Two thanas, viz., Balumath and Latehar in the south-east, lost ground, while the north of the district displayed the greatest advance.

PALAMAU.	Population, 1911.	PERCENTAGE OF VARIATION. 1901—1911.	1891—1901.
DISTRICT TOTAL	**687,267**	**+ 10·92**	**+ 3·83**
Balumath	72,907	+ 12·66	- 3·91
Daltonganj	124,870	+ 18·28	+ 1·95
Garhwa	107,674	+ 15·46	+ 7·15
Ranka	47,255	+ 30·90	+ 5·66
Chhatarpur	48,637	+ 11·92	+ 5·82
Patan	91,480	+ 4·06	+ 6·13
Husainabad	110,589	- 7·65	+ 6·96
Latehar	51,608	+ 23·06	- 1·94
Mahuadanr	32,247	+ 19·44	+ 3·23

The first six years of the decade ending in 1910 were on the whole prosperous, and the people, whose condition had been reduced by a succession of bad or inferior crops, made a distinct recovery from the effects of the previous lean years. The crops were good, except in 1904-05, and the birth-rate high; in fact, in each of the last three of these six years it ranged above 50 per mille and was higher than in any other district of the province. Altogether, there was an excess in this period of 65,000 births over deaths, or thrice the excess of the preceding nine years. In the remaining four years of the decade the births were in excess by 3,000 only, as a consequence of the adverse conditions prevailing in the three years 1907-09, in each of which deaths outnumbered births. Short rainfall and

* In the famine year of 1907-1908 the number of coolies recruited for Assam rose to 4,148, the number in the preceding year being only 777. The return for 1901 shows a total of 4,474 recruited from both Ranchi and Palamau: separate figures for Palamau are not available, but the proportion of emigrants from that district was never very large.